

6

SURFACE INTEGRALS

6.1 Parametric representation of a surface

A deeper study of vector fields in 3-space requires the use of surface integrals. A surface integral can be thought of as a two-dimensional analog of a line integral where the region of integration is a surface rather than a curve. Before we can discuss surface integrals intelligently, we must agree on what we shall mean by a surface.

Roughly speaking, a surface is the locus of a point moving in space with two degrees of freedom. In our study of analytic geometry in Volume I we discussed two methods for describing such a locus by mathematical formulas. One is the *implicit representation* in which we describe a surface as a set of points (x, y, z) satisfying an equation of the form $F(x, y, z) = 0$. Sometimes we can solve such an equation for one of the coordinates in terms of the other two, say for z in terms of x and y . When this is possible we obtain an *explicit representation* given by one or more equations of the form $z = f(x, y)$. For example, a sphere of radius 1 and center at the origin has the implicit representation $x^2 + y^2 + z^2 - 1 = 0$. When this equation is solved for z it leads to two solutions, $z = \sqrt{1 - x^2 - y^2}$ and $z = -\sqrt{1 - x^2 - y^2}$. The first gives an explicit representation of the upper hemisphere and the second of the lower hemisphere.

A third method for describing surfaces is more useful in the study of surface integrals; this is the *parametric* or *vector* representation in which we have three equations expressing x , y , and z in terms of two parameters u and v :

$$(6.1) \quad x = X(u, v), \quad y = Y(u, v), \quad z = Z(u, v).$$

Here the point (u, v) is allowed to vary over some two-dimensional connected set T in the uv -plane, and the corresponding points (x, y, z) trace out a portion of a surface in xyz -space. This method for describing a surface is analogous to the representation of a space curve by three parametric equations involving one parameter. The presence of the two parameters in (6.1) makes it possible to transmit two degrees of freedom to the point (x, y, z) , as suggested by Figure 6.1. Another way of describing the same idea is to say that a surface is the image of a plane region T under the mapping defined by (6.1).

If we introduce the radius vector $\vec{r}$ from the origin to a general point (x, y, z) of the surface, we may combine the three parametric equations in (6.1) into one vector equation of the form

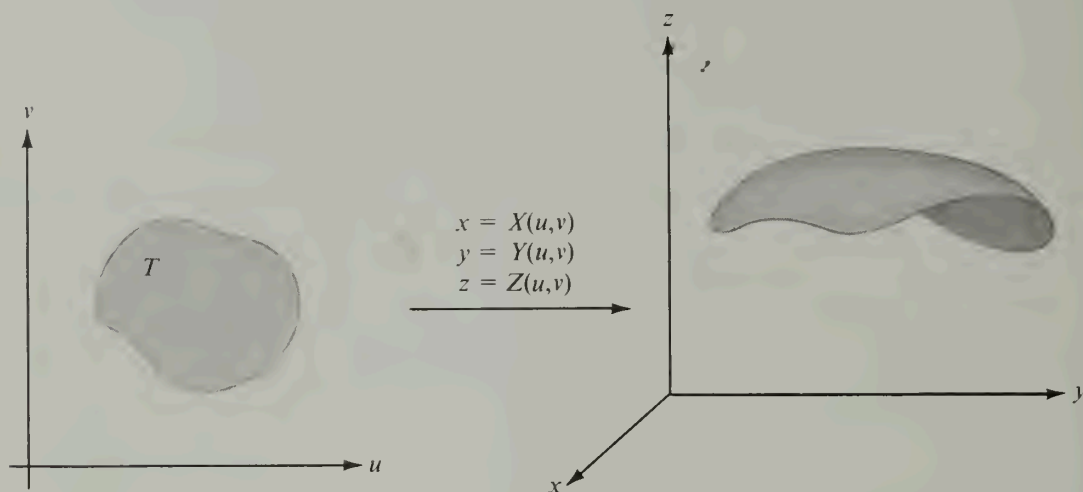


FIGURE 6.1 Parametric representation of a surface.

$$(6.2) \quad \vec{r}(u, v) = X(u, v) \vec{i} + Y(u, v) \vec{j} + Z(u, v) \vec{k}, \quad \text{where } (u, v) \in T.$$

This is called a *vector equation* for the surface.

There are, of course, many parametric representations for the same surface. One of these can always be obtained from an explicit form $z = f(x, y)$ by taking $X(u, v) = u$, $Y(u, v) = v$, $Z(u, v) = f(u, v)$. On the other hand, if it is possible to eliminate u and v from the parametric equations—for example, if we can solve the first two equations in (6.1) for u and v in terms of x and y and substitute in the third—we obtain an explicit representation $z = f(x, y)$.

Example 1: A parametric representation of a sphere. The three equations

$$(6.3) \quad x = a \cos u \cos v, \quad y = a \sin u \cos v, \quad z = a \sin v$$

serve as parametric equations for a sphere of radius a and center at the origin. If we square and add the three equations in (6.3) we find $x^2 + y^2 + z^2 = a^2$, and we see that every point (x, y, z) satisfying (6.3) lies on the sphere. The parameters u and v in this example may be interpreted geometrically as the angles shown in Figure 6.2. If we let the point (u, v) vary over the rectangle $T = [0, 2\pi] \times [-\frac{1}{2}\pi, \frac{1}{2}\pi]$, the points determined by (6.3) trace out the whole sphere. The upper hemisphere is the image of the rectangle $[0, 2\pi] \times [0, \frac{1}{2}\pi]$ and the lower hemisphere is the image of $[0, 2\pi] \times [-\frac{1}{2}\pi, 0]$. Figure 6.3 gives a concrete idea of how the rectangle $[0, 2\pi] \times [0, \frac{1}{2}\pi]$ is mapped onto the upper hemisphere. Imagine that the rectangle is made of a flexible plastic material capable of being stretched or shrunk. Figure 6.3 shows the rectangle being deformed into a hemisphere. The base AB eventually becomes the equator, the opposite edges AD and BC are brought into coincidence, and the upper edge DC shrinks to a point (the North Pole).

Example 2: A parametric representation of a cone. The vector equation

$$\vec{r}(u, v) = v \sin \alpha \cos u \vec{i} + v \sin \alpha \sin u \vec{j} + v \cos \alpha \vec{k}$$

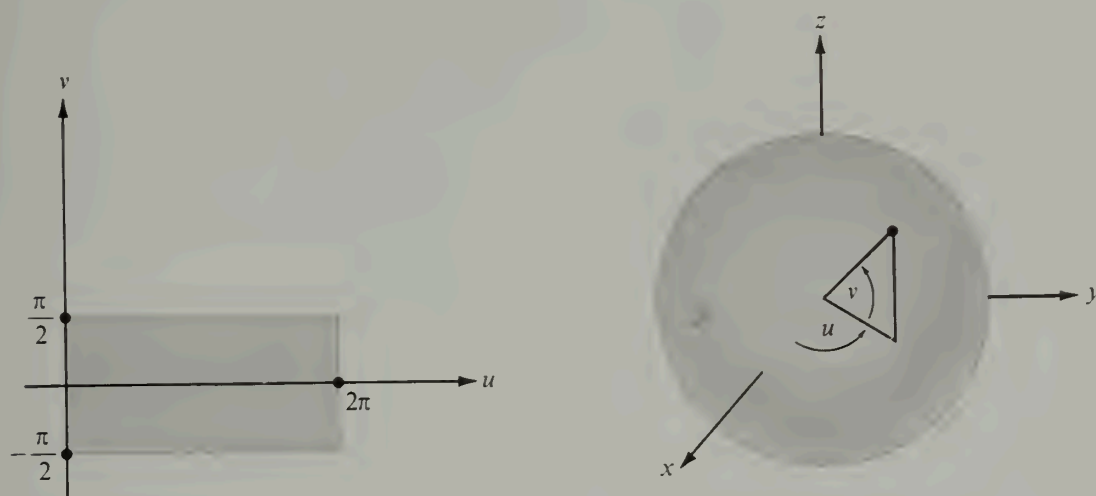


FIGURE 6.2 Parametric representation of a sphere.

represents the right circular cone shown in Figure 6.4, where α denotes half the vertex angle. Again, the parameters u and v may be given geometric interpretations; v is the distance from the vertex to the point (x, y, z) on the cone, and u is the polar-coordinate angle. When (u, v) is allowed to vary over a rectangle of the form $[0, 2\pi] \times [0, h]$, the corresponding points (x, y, z) trace out a cone of altitude $h \cos \alpha$. A plastic rectangle may be physically deformed into the cone by bringing the edges AD and BC into coincidence, as suggested by Figure 6.5, and letting the edge AB shrink to a point (the vertex of the cone). The surface in Figure 6.5 shows an intermediate stage of the deformation.

In the general study of surfaces, the functions X , Y , and Z that occur in the parametric equations (6.1) or in the vector equation (6.2) are assumed to be continuous on T . The image of T under the mapping $\vec{r}$ is called a *parametric surface* and will be denoted by the symbol $\vec{r}(T)$. In many of the examples we shall discuss, T will be a rectangle, a circular disk, or some other simply connected set bounded by a simple closed curve. If the function $\vec{r}$ is one-to-one on T , the image $\vec{r}(T)$ will be called a *simple parametric surface*. In

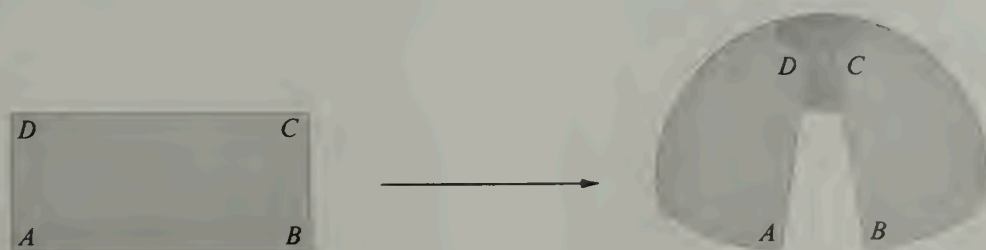


FIGURE 6.3 Deformation of a rectangle into a hemisphere.

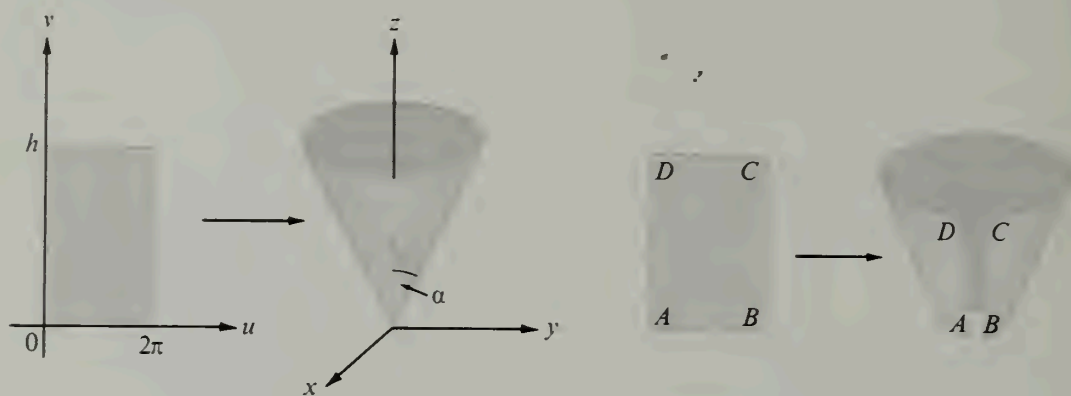


FIGURE 6.4 Parametric representation of a cone. FIGURE 6.5 Deformation of a rectangle into a cone.

such a case, distinct points of T map onto distinct points of the surface. In particular, every simple closed curve in R maps onto a simple closed curve lying on the surface.

A parametric surface $\vec{r}(T)$ may degenerate to a point or to a curve. For example, if all three functions X , Y , and Z are constant, the image $\vec{r}(T)$ is a single point. If X , Y , and Z are independent of v , the image $\vec{r}(T)$ is a curve. Another example of a degenerate surface occurs when $X(u, v) = u + v$, $Y(u, v) = (u + v)^2$, and $Z(u, v) = (u + v)^3$, where $T = [0, 1] \times [0, 1]$. If we write $t = u + v$ we see that the surface degenerates to the space curve having parametric equations $x = t$, $y = t^2$, and $z = t^3$, where $0 \leq t \leq 2$. Such degeneracies can be avoided by placing further restrictions on the mapping function $\vec{r}$, as described in the next section.

6.2 The fundamental vector product

Consider a surface described by the vector equation

$$\vec{r}(u, v) = X(u, v) \vec{i} + Y(u, v) \vec{j} + Z(u, v) \vec{k}, \quad \text{where } (u, v) \in T.$$

If X , Y , and Z are differentiable on T we may consider the two vectors

$$\frac{\partial \vec{r}}{\partial u} = \frac{\partial X}{\partial u} \vec{i} + \frac{\partial Y}{\partial u} \vec{j} + \frac{\partial Z}{\partial u} \vec{k}$$

and

$$\frac{\partial \vec{r}}{\partial v} = \frac{\partial X}{\partial v} \vec{i} + \frac{\partial Y}{\partial v} \vec{j} + \frac{\partial Z}{\partial v} \vec{k}.$$

The cross product of these two vectors $\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v$ will be referred to as the *fundamental vector product* of the representation $\vec{r}$. Its components may be expressed as Jacobian determinants. In fact, we have

$$\begin{aligned}
 \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial X}{\partial u} & \frac{\partial Y}{\partial u} & \frac{\partial Z}{\partial u} \\ \frac{\partial X}{\partial v} & \frac{\partial Y}{\partial v} & \frac{\partial Z}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\partial Y}{\partial u} & \frac{\partial Z}{\partial u} \\ \frac{\partial Y}{\partial v} & \frac{\partial Z}{\partial v} \end{vmatrix} \vec{i} + \begin{vmatrix} \frac{\partial Z}{\partial u} & \frac{\partial X}{\partial u} \\ \frac{\partial Z}{\partial v} & \frac{\partial X}{\partial v} \end{vmatrix} \vec{j} + \begin{vmatrix} \frac{\partial X}{\partial u} & \frac{\partial Y}{\partial u} \\ \frac{\partial X}{\partial v} & \frac{\partial Y}{\partial v} \end{vmatrix} \vec{k} \\
 (6.4) \quad &= \frac{\partial(Y, Z)}{\partial(u, v)} \vec{i} + \frac{\partial(Z, X)}{\partial(u, v)} \vec{j} + \frac{\partial(X, Y)}{\partial(u, v)} \vec{k}.
 \end{aligned}$$

If (u, v) is a point in T at which $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ are continuous and the fundamental vector product is nonzero, then the image point $\vec{r}(u, v)$ is called a *regular point* of $\vec{r}$. Points at which $\partial \vec{r}/\partial u$ or $\partial \vec{r}/\partial v$ fails to be continuous or $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v = \vec{0}$ are called *singular points* of $\vec{r}$. A surface $\vec{r}(T)$ is called *smooth* if all its points are regular points. Every surface has more than one parametric representation. Some of the examples discussed below show that a point of a surface may be a regular point for one representation but a singular point for some other representation. The geometric significance of regular and singular points may be explained as follows:

Consider a horizontal line segment in T . Its image under $\vec{r}$ is a curve (called a *u-curve*) lying on the surface $\vec{r}(T)$. The vector $\partial \vec{r}/\partial u$ is the velocity vector of this curve. (See Figure 6.6.) Likewise, $\partial \vec{r}/\partial v$ is the velocity vector of a *v-curve* obtained by setting $u = \text{constant}$. There is a *u-curve* and a *v-curve* passing through each point of the surface. The restriction $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v \neq \vec{0}$ means that the velocity vectors $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ are not collinear at this point. For a fixed v , let us think of the parameter u as representing "time." When u changes by an amount Δu , a point originally at $\vec{r}(u, v)$ moves along a *u-curve* a distance approximately equal to $|\partial \vec{r}/\partial u| \Delta u$ (since $|\partial \vec{r}/\partial u|$ represents the speed along the *u-curve*).

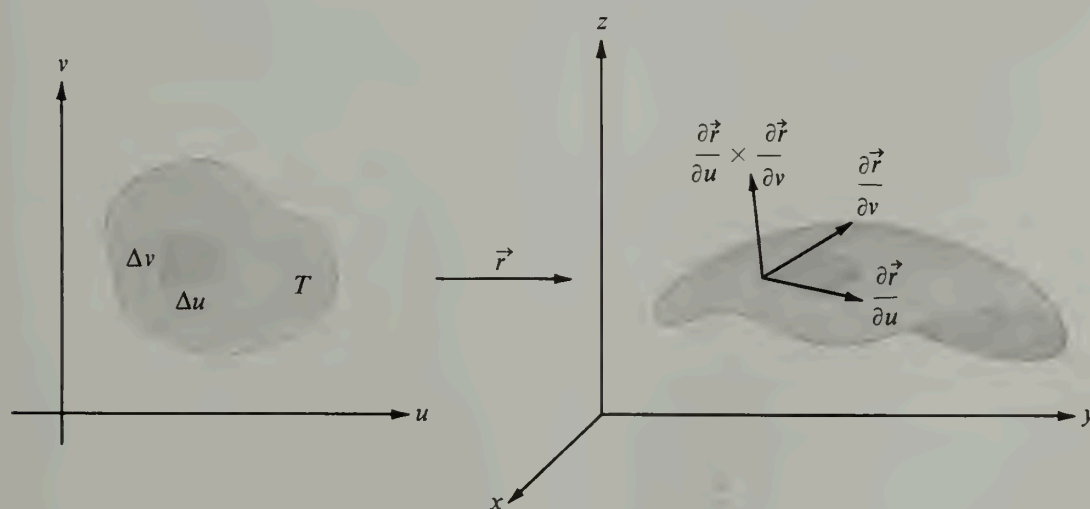


FIGURE 6.6 Geometric interpretation of the vectors $\frac{\partial \vec{r}}{\partial u}$, $\frac{\partial \vec{r}}{\partial v}$, and $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}$.

Similarly, for a fixed u a point of a v -curve moves in time Δv a distance nearly equal to $|\partial \vec{r}/\partial v| \Delta v$. A rectangle in T having area $\Delta u \Delta v$ is traced onto a portion of $\vec{r}(T)$ which we shall approximate by the parallelogram determined by the vectors $(\partial \vec{r}/\partial u) \Delta u$ and $(\partial \vec{r}/\partial v) \Delta v$. (See Figure 6.6.) The area of the parallelogram spanned by $(\partial \vec{r}/\partial u) \Delta u$ and $(\partial \vec{r}/\partial v) \Delta v$ is the magnitude of their cross product,

$$\left| \frac{\partial \vec{r}}{\partial u} \Delta u \times \frac{\partial \vec{r}}{\partial v} \Delta v \right| = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| \Delta u \Delta v.$$

Therefore the length of the fundamental vector product may be thought of as a local magnification factor for areas. At the points at which this vector product is zero the parallelogram collapses to a curve or point, and degeneracies occur. At each regular point the vectors $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ determine a plane having the vector $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v$ as a normal. Later in this section we shall prove that $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v$ is normal to every smooth curve on the surface; for this reason the plane determined by $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ is called the *tangent plane* of the surface. Continuity of $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ implies continuity of $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v$; this, in turn, means that the tangent plane turns continuously on a smooth surface. Thus we see that continuity of $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ prevents the occurrence of sharp edges or corners on the surface; the nonvanishing of $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v$ prevents degeneracies.

Example 1: Surfaces with an explicit representation, $z = f(x, y)$. For a surface with an explicit representation of the form $z = f(x, y)$, we can use x and y as the parameters, which gives us the vector equation

$$\vec{r}(x, y) = x\vec{i} + y\vec{j} + f(x, y)\vec{k}.$$

This representation always gives a simple parametric surface. The region T is called the projection of the surface on the xy -plane. (An example is shown in Figure 6.7.) To compute the fundamental vector product we note that

$$\frac{\partial \vec{r}}{\partial x} = \vec{i} + \frac{\partial f}{\partial x} \vec{k} \quad \text{and} \quad \frac{\partial \vec{r}}{\partial y} = \vec{j} + \frac{\partial f}{\partial y} \vec{k}.$$

This gives us

$$(6.5) \quad \frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{\partial f}{\partial x} \\ 0 & 1 & \frac{\partial f}{\partial y} \end{vmatrix} = -\frac{\partial f}{\partial x} \vec{i} - \frac{\partial f}{\partial y} \vec{j} + \vec{k}.$$

Since the z -component of $\partial \vec{r}/\partial x \times \partial \vec{r}/\partial y$ is 1, the fundamental vector product is never zero. Therefore the only singular points that can occur for this representation are points at which at least one of the partial derivatives $\partial f/\partial x$ or $\partial f/\partial y$ fails to be continuous.

A specific case is the equation $z = \sqrt{1 - x^2 - y^2}$, which represents a hemisphere of radius 1 and center at the origin, if $x^2 + y^2 \leq 1$. The vector equation

$$\vec{r}(x, y) = x\vec{i} + y\vec{j} + \sqrt{1 - x^2 - y^2} \vec{k}$$

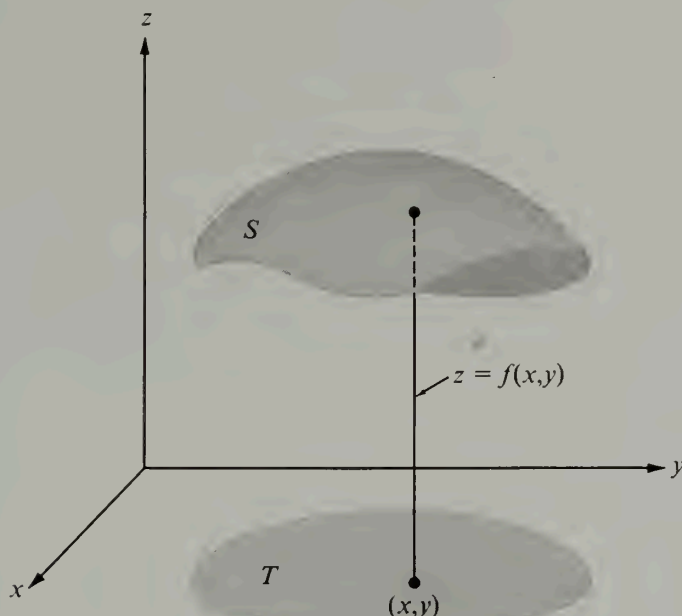


FIGURE 6.7 A surface S with an explicit representation, $z = f(x, y)$. The region T is the projection of S on the xy -plane.

maps the unit disk $T = \{(x, y) \mid x^2 + y^2 \leq 1\}$ onto the hemisphere in a one-to-one fashion. The partial derivatives $\partial \vec{r} / \partial x$ and $\partial \vec{r} / \partial y$ exist and are continuous everywhere in the interior of this disk, but they do not exist on the boundary. Therefore every point on the equator is a singular point of this representation.

Example 2. We consider the same hemisphere as in Example 1, but this time as the image of the rectangle $T = [0, 2\pi] \times [0, \frac{1}{2}\pi]$ under the mapping

$$\vec{r}(u, v) = a \cos u \cos v \vec{i} + a \sin u \cos v \vec{j} + a \sin v \vec{k}.$$

The vectors $\partial \vec{r} / \partial u$ and $\partial \vec{r} / \partial v$ are given by the formulas

$$\begin{aligned} \frac{\partial \vec{r}}{\partial u} &= -a \sin u \cos v \vec{i} + a \cos u \cos v \vec{j}, \\ \frac{\partial \vec{r}}{\partial v} &= -a \cos u \sin v \vec{i} - a \sin u \sin v \vec{j} + a \cos v \vec{k}. \end{aligned}$$

An easy calculation shows that their cross product is equal to

$$\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = a \cos v \vec{r}(u, v).$$

The image of T is not a simple parametric surface because this mapping is not one-to-one on T . In fact, every point on the line segment $v = \frac{1}{2}\pi$, $0 \leq u \leq 2\pi$, is mapped onto the point $(0, 0, a)$ (the North Pole). Also, because of the periodicity of the sine and

cosine, $\vec{r}$ takes the same values at the points $(0, v)$ and $(2\pi, v)$, so the right and left edges of T are mapped onto the same curve, a circular arc joining the North Pole to the point $(a, 0, 0)$ on the equator. (See Figure 6.3.) The vectors $\partial\vec{r}/\partial u$ and $\partial\vec{r}/\partial v$ are continuous everywhere in T . Since $|\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v| = a^2 \cos v$, the only singular points of this representation occur when $\cos v = 0$. The North Pole is the only such point.

We shall prove now that at each point P on a smooth parametric surface $\vec{r}(T)$, the vector $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$ is normal to every smooth curve on the surface passing through P . Let C be such a curve, and suppose that C is the image $\vec{r}(C^*)$ of a smooth curve C^* in T . If C^* is described by a function $\vec{\alpha}$ defined on an interval $[a, b]$, say

$$\vec{\alpha}(t) = U(t)\vec{i} + V(t)\vec{j},$$

the image curve C is described by the composite function

$$\vec{\rho}(t) = \vec{r}[\vec{\alpha}(t)] = X[\vec{\alpha}(t)]\vec{i} + Y[\vec{\alpha}(t)]\vec{j} + Z[\vec{\alpha}(t)]\vec{k}.$$

We wish to prove that the derivative $\vec{\rho}'(t)$ is perpendicular to the vector $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$ when the partial derivatives $\partial\vec{r}/\partial u$ and $\partial\vec{r}/\partial v$ are evaluated at $(U(t), V(t))$. To compute $\vec{\rho}'(t)$ we differentiate each component of $\vec{\rho}(t)$ by the chain rule (Theorem 4-8) to obtain

$$(6.6) \quad \vec{\rho}'(t) = \nabla X \cdot \vec{\alpha}'(t)\vec{i} + \nabla Y \cdot \vec{\alpha}'(t)\vec{j} + \nabla Z \cdot \vec{\alpha}'(t)\vec{k},$$

where the gradient vectors ∇X , ∇Y , and ∇Z are evaluated at $(U(t), V(t))$. Equation (6.6) can be rewritten as

$$\vec{\rho}'(t) = \frac{\partial\vec{r}}{\partial u} U'(t) + \frac{\partial\vec{r}}{\partial v} V'(t),$$

where the derivatives $\partial\vec{r}/\partial u$ and $\partial\vec{r}/\partial v$ are evaluated at $(U(t), V(t))$. Since $\partial\vec{r}/\partial u$ and $\partial\vec{r}/\partial v$ are each perpendicular to the cross product $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$, the same is true of $\vec{\rho}'(t)$. This proves that $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$ is normal to C , as asserted. For this reason, the vector $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$ is said to be *normal* to the surface $\vec{r}(T)$. At each regular point P of $\vec{r}(T)$ the vector $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$ is nonzero; the plane through P having this vector as a normal is called the *tangent plane* to the surface at P .

6.3 Exercises

In Exercises 1 through 6, eliminate the parameters u and v to obtain a Cartesian equation, thus showing that the given vector equation represents a portion of the surface named. Also, compute the fundamental vector product $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$ in terms of u and v .

1. Plane:

$$\vec{r}(u, v) = (x_0 + a_1u + b_1v)\vec{i} + (y_0 + a_2u + b_2v)\vec{j} + (z_0 + a_3u + b_3v)\vec{k}.$$

2. Elliptic paraboloid:

$$\vec{r}(u, v) = au \cos v \vec{i} + bu \sin v \vec{j} + u^2\vec{k}.$$

3. Ellipsoid:

$$\vec{r}(u, v) = a \sin u \cos v \vec{i} + b \sin u \sin v \vec{j} + c \cos u \vec{k}.$$

4. Surface of revolution:

$$\vec{r}(u, v) = u \cos v \vec{i} + u \sin v \vec{j} + f(u) \vec{k}.$$

5. *Cylinder:*

$$\vec{r}(u, v) = u\vec{i} + a \sin v \vec{j} + b \cos v \vec{k}.$$

6. *Torus:*

$\vec{r}(u, v) = (a + b \cos u) \sin v \vec{i} + (a + b \cos u) \cos v \vec{j} + b \sin u \vec{k}$, where $0 < b < a$. What are the geometric meanings of a and b ?

In Exercises 7 through 10 compute the magnitude of the fundamental vector product $\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v$ in terms of u and v .

7. $\vec{r}(u, v) = a \sin u \cosh v \vec{i} + b \cos u \cosh v \vec{j} + c \sinh v \vec{k}.$

8. $\vec{r}(u, v) = (u + v) \vec{i} + (u - v) \vec{j} + 4v^2 \vec{k}.$

9. $\vec{r}(u, v) = (u + v) \vec{i} + (u^2 + v^2) \vec{j} + (u^3 + v^3) \vec{k}.$

10. $\vec{r}(u, v) = u \cos v \vec{i} + u \sin v \vec{j} + \frac{1}{2} u^2 \sin 2v \vec{k}.$

6.4 Area of a parametric surface

Let $S = \vec{r}(T)$ be a parametric surface described by a vector-valued function $\vec{r}$ defined on a region T in the uv -plane. In Section 6.2 we found that the length of the fundamental vector product $\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v$ could be interpreted as a local magnification factor for areas. (See Figure 6.6.) A rectangle in T of area $\Delta u \Delta v$ is mapped by $\vec{r}$ onto a curvilinear parallelogram on S with area nearly equal to

$$\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| \Delta u \Delta v.$$

This observation suggests that we define the area of S by the following double integral:

$$(6.7) \quad \text{area of } S = \iint_T \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv.$$

In other words, to determine the area of S we first compute the fundamental vector product $\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v$ and then integrate its length over the region T . When $\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v$ is expressed in terms of its components, by means of Equation (6.4), we have

$$(6.8) \quad \text{area of } S = \iint_T \sqrt{\left(\frac{\partial(Y, Z)}{\partial(u, v)} \right)^2 + \left(\frac{\partial(Z, X)}{\partial(u, v)} \right)^2 + \left(\frac{\partial(X, Y)}{\partial(u, v)} \right)^2} du dv.$$

Written in this form, the integral for surface area resembles the integral for computing the arc length of a curve.†

If S is given explicitly by an equation of the form $z = f(x, y)$ we may use x and y as the parameters. The fundamental vector product is given by Equation (6.5), so we have

† Since the integral in (6.7) involves $\vec{r}$, the area of a surface will depend on the function used to describe the surface. When we discuss surface integrals we shall prove (in Section 6.7) that under certain general conditions the area is independent of the parametric representation. The result is analogous to Theorem 5-1, in which we discussed the invariance of line integrals under a change of parameter.

$$\left| \frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} \right| = \left| -\frac{\partial f}{\partial x} \vec{i} - \frac{\partial f}{\partial y} \vec{j} + \vec{k} \right| = \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}.$$

In this case the integral for surface area becomes

$$(6.9) \quad \text{area of } S = \iint_T \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dx dy,$$

where the region T is now the projection of S on the xy -plane, as illustrated in Figure 6.7.

When S lies in a plane parallel to the xy -plane, the function f is constant, so $\partial f/\partial x = \partial f/\partial y = 0$, and Equation (6.9) becomes

$$\text{area of } S = \iint_T dx dy.$$

This agrees with the usual formula for areas of plane regions.

Equation (6.9) can be written in another form that gives further insight into its geometric significance. At each point of S , let γ denote the angle between the normal vector $\vec{N} = \partial \vec{r}/\partial x \times \partial \vec{r}/\partial y$ and the unit coordinate vector $\vec{k}$. (See Figure 6.8.) Since the z -component of $\vec{N}$ is 1, we have

$$\cos \gamma = \frac{\vec{N} \cdot \vec{k}}{|\vec{N}| |\vec{k}|} = \frac{1}{|\vec{N}|} = \frac{1}{\left| \frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} \right|},$$

and hence $\left| \partial \vec{r}/\partial x \times \partial \vec{r}/\partial y \right| = 1/\cos \gamma$. Therefore Equation (6.9) becomes

$$(6.10) \quad \text{area of } S = \iint_T \frac{1}{\cos \gamma} dx dy.$$

Suppose now that S lies in a plane not perpendicular to the xy -plane. Then γ is constant and Equation (6.10) states that the area of $S = (\text{area of } T)/\cos \gamma$, or that

$$(6.11) \quad \text{area of } T = (\text{area of } S) \cos \gamma.$$

Equation (6.11) is sometimes referred to as the *area cosine principle*. It tells us that if a region S in one plane is projected onto a region T in another plane, making an angle γ with the first plane, the area of T is $\cos \gamma$ times that of S . This formula is obviously true when S is the rectangle shown in Figure 6.9, because distances in one direction are shortened by the factor $\cos \gamma$ while those in a perpendicular direction are unaltered by projection. Equation (6.11) extends this property to any plane region S having an area.

Suppose now that S is given by an implicit representation $F(x, y, z) = 0$. If S can be projected in a one-to-one fashion on the xy -plane, the equation $F(x, y, z) = 0$ defines z as a function of x and y , say $z = f(x, y)$, and the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ are related to those of F by the equations

$$\frac{\partial f}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} \quad \text{and} \quad \frac{\partial f}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}$$

for those points at which $\partial F/\partial z \neq 0$. Substituting these quotients in (6.9), we find

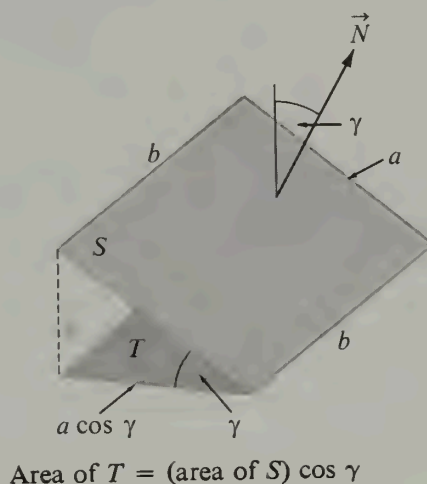
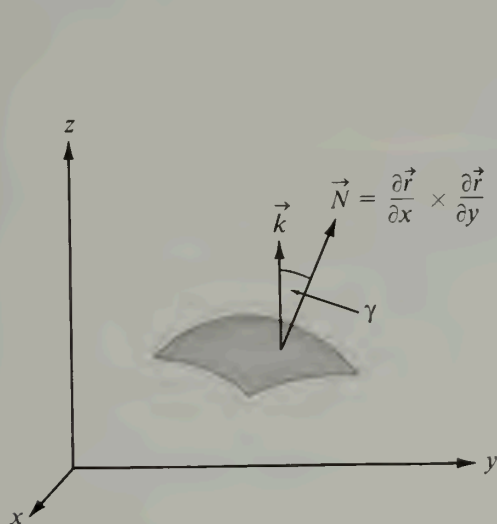


FIGURE 6.8 The length of $\frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y}$ is $1/\cos \gamma$.

FIGURE 6.9 The area cosine principle for a rectangle.

$$(6.12) \quad \text{area of } S = \iint_T \frac{\sqrt{(\partial F/\partial x)^2 + (\partial F/\partial y)^2 + (\partial F/\partial z)^2}}{|\partial F/\partial z|} dx dy.$$

Example 1: Area of a hemisphere. Consider a hemisphere S of radius a and center at the origin. We have at our disposal the implicit representation $x^2 + y^2 + z^2 = a^2$, $z \geq 0$; the explicit representation $z = \sqrt{a^2 - x^2 - y^2}$; and the parametric representation

$$(6.13) \quad \vec{r}(u, v) = a \cos u \cos v \vec{i} + a \sin u \cos v \vec{j} + a \sin v \vec{k}.$$

To compute the area of S from the implicit representation we refer to Equation (6.12) with

$$F(x, y, z) = x^2 + y^2 + z^2 - a^2.$$

The partial derivatives of F are $\partial F/\partial x = 2x$, $\partial F/\partial y = 2y$, $\partial F/\partial z = 2z$. The hemisphere S projects in a one-to-one fashion onto the circular disk $D = \{(x, y) \mid x^2 + y^2 \leq a^2\}$ in the xy -plane. We cannot apply Equation (6.12) directly because the partial derivative $\partial F/\partial z$ is zero on the boundary of D . However, the derivative $\partial F/\partial z$ is nonzero everywhere in the interior of D , so we can consider the smaller concentric disk $D(R)$ of radius R , where $R < a$. If $S(R)$ denotes the corresponding portion of the upper hemisphere, Equation (6.12) is now applicable and we find

$$\begin{aligned} \text{area of } S(R) &= \iint_{D(R)} \frac{\sqrt{(2x)^2 + (2y)^2 + (2z)^2}}{|2z|} dx dy \\ &= \iint_{D(R)} \frac{a}{z} dx dy = a \iint_{D(R)} \frac{1}{\sqrt{a^2 - x^2 - y^2}} dx dy. \end{aligned}$$

The last integral may be easily evaluated by the use of polar coordinates, giving us

$$\text{area of } S(R) = a \int_0^{2\pi} \left[\int_0^R \frac{1}{\sqrt{a^2 - r^2}} r dr \right] d\theta = 2\pi a(a - \sqrt{a^2 - R^2}).$$

When $R \rightarrow a$ this approaches the limit $2\pi a^2$.

We can avoid the limiting process in the foregoing calculation by using the parametric representation in (6.13). The calculations of Example 2 in Section 6.2 show that

$$\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| = |a \cos v \vec{r}(u, v)| = a^2 |\cos v|.$$

Therefore we may apply Equation (6.7), taking for the region T the rectangle $[0, 2\pi] \times [0, \frac{1}{2}\pi]$. We find

$$\text{area of } S = a^2 \iint_T |\cos v| du dv = a^2 \int_0^{2\pi} \left[\int_0^{\pi/2} \cos v dv \right] du = 2\pi a^2.$$

Example 2: Another theorem of Pappus. One of the theorems of Pappus states that a surface of revolution, obtained by rotating a plane curve of length L about an axis in the plane of the curve, has area $2\pi Lh$, where h is the distance from the centroid of the curve to the axis of rotation. We shall use Equation (6.7) to prove this theorem for a special surface of revolution.

Suppose a curve C , initially in the xz -plane, is rotated about the z -axis. Let its equation in the xz -plane be $z = f(x)$, where $a \leq x \leq b$, $a \geq 0$. The surface of revolution S so generated may be described by the vector equation

$$\vec{r}(u, v) = u \cos v \vec{i} + u \sin v \vec{j} + f(u) \vec{k},$$

where $(u, v) \in [a, b] \times [0, 2\pi]$. The parameters u and v can be interpreted as the radius and angle of polar coordinates, as illustrated in Figure 6.10. If $a \leq u \leq b$, all points (x, y, z) at a given distance u from the z -axis have the same z -coordinate, $f(u)$, so they all lie on the surface. The fundamental vector product of this representation is

$$\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos v & \sin v & f'(u) \\ -u \sin v & u \cos v & 0 \end{vmatrix} = -u f'(u) \cos v \vec{i} - u f'(u) \sin v \vec{j} + u \vec{k},$$

and hence

$$\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| = u \sqrt{1 + [f'(u)]^2}.$$

Therefore Equation (6.7) becomes

$$\text{area of } S = \int_0^{2\pi} \left[\int_a^b u \sqrt{1 + [f'(u)]^2} du \right] dv = 2\pi \int_a^b u \sqrt{1 + [f'(u)]^2} du.$$

The integral that remains can be expressed as $\int_C x ds$, a line integral with respect to arc length along the curve C . As such, it is equal to $\bar{x} L$, where $\bar{x}$ is the x -coordinate of the centroid of C and L is the length of C . (See Section 5.5.) Therefore the area of S is $2\pi L\bar{x}$. This proves the theorem of Pappus for the special surface considered.

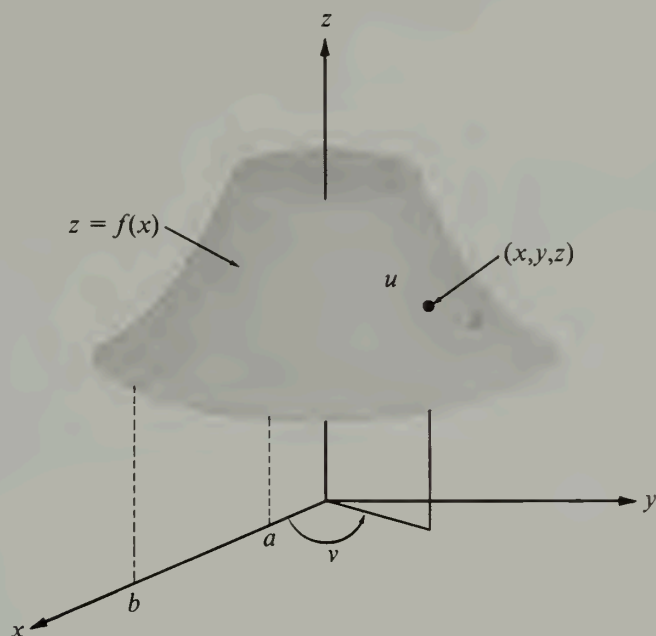


FIGURE 6.10 Area of a surface of revolution determined by Pappus' theorem.

6.5 Exercises

1. Let S be a parallelogram not parallel to any of the coordinate planes. Let S_1 , S_2 , and S_3 denote the areas of the projections of S on the three coordinate planes. Show that the area of S is $\sqrt{S_1^2 + S_2^2 + S_3^2}$.

2. Compute the area of the region cut from the plane $x + y + z = a$ by the cylinder $x^2 + y^2 = a^2$.

3. Compute the surface area of that portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying within the cylinder $x^2 + y^2 = ay$, where $a > 0$.

4. Compute the area of that portion of the surface $z^2 = 2xy$ which lies above the first quadrant of the xy -plane and is cut off by the planes $x = 2$ and $y = 1$.

5. A parametric surface S is described by the vector equation

$$\vec{r}(u, v) = u \cos v \vec{i} + u \sin v \vec{j} + u^2 \vec{k},$$

where $0 \leq u \leq 4$ and $0 \leq v \leq 2\pi$.

(a) Show that S is a portion of a quadric surface. Name this quadric, sketch it, and indicate the geometric meanings of the parameters u and v on the surface.

(b) Compute the fundamental vector product $\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v$ in terms of u and v .

(c) The area of S is $\pi(65\sqrt{65} - 1)/n$, where n is an integer. Compute the value of n .

6. Compute the area of that portion of the conical surface $x^2 + y^2 = z^2$ which lies above the xy -plane and is cut off by the sphere $x^2 + y^2 + z^2 = 2ax$.

7. Compute the area of that portion of the conical surface $x^2 + y^2 = z^2$ which lies between the two planes $z = 0$ and $x + 2z = 3$.

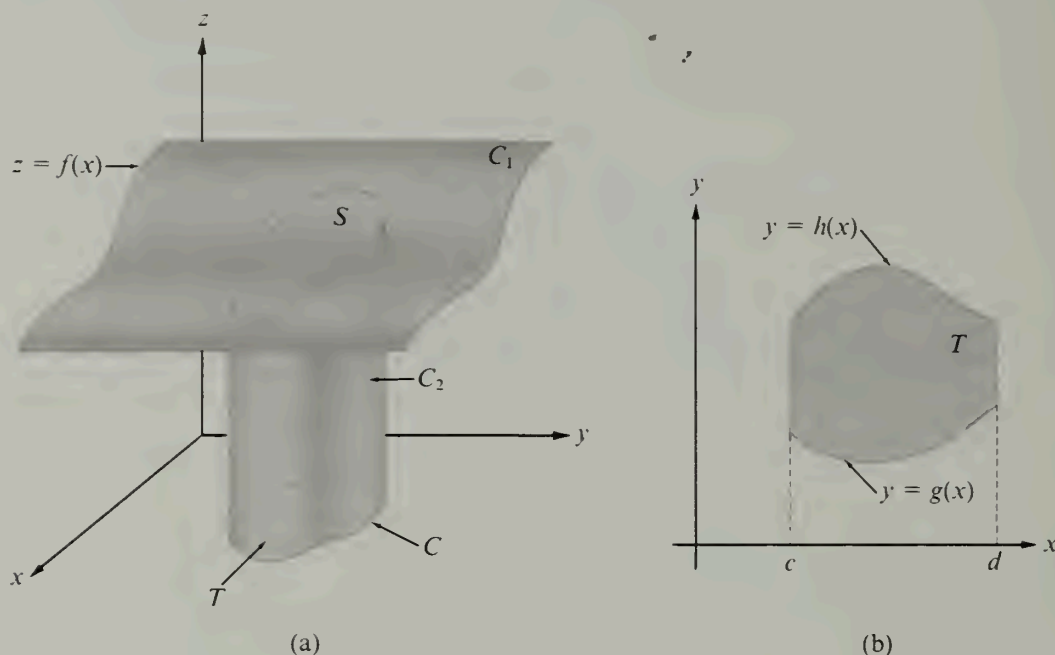


FIGURE 6.11 Exercise 10.

8. Compute the area of that portion of the paraboloid $x^2 + z^2 = 2ay$ which is cut off by the plane $y = a$.

9. Compute the area of the torus described by the vector equation

$$\vec{r}(u, v) = (a + b \cos u) \sin v \vec{i} + (a + b \cos u) \cos v \vec{j} + b \sin u \vec{k},$$

where $0 < b < a$ and $0 \leq u \leq 2\pi$, $0 \leq v \leq 2\pi$. Use the theorem of Pappus to check your answer.

10. A cylinder C_1 with generators parallel to the y -axis intersects the xz -plane along a curve with Cartesian equation $z = f(x)$, $a \leq x \leq b$. A cylinder C_2 with generators parallel to the z -axis traces out a simple closed curve C in the xy -plane forming the boundary of a plane region T as shown in Figure 6.11(a). The x -coordinate of each point of T satisfies $a < x < b$. Let

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dt.$$

[This is the arc-length function for the curve $z = f(x)$.] Let S be that portion of C_1 cut out by C_2 .

(a) Show that

$$\text{area of } S = \iint_T s'(x) dx dy.$$

(b) Show that

$$\text{area of } S = \oint_C s(x) dy,$$

where the integral is a line integral taken around C in a counterclockwise direction.

Assume now that the boundary curve C consists of portions of two curves $y = g(x)$ and $y = h(x)$ and (possibly) two line segments, as shown in Figure 6.11(b), where $g(x) < h(x)$ for $c < x < d$,

and $a < c < d < b$. If $c \leq x \leq d$, let $A(x)$ denote the area of that portion of T above the interval $[c, x]$.

(c) Show that

$$\text{area of } S = \int_c^d A'(x) s'(x) dx.$$

(d) If the curves $y = h(x)$ and $y = g(x)$ are parallel, show that

$$\text{area of } S = [h(c) - g(c)][s(d) - s(c)].$$

Justify this formula geometrically by referring to a suitable rectangle having the same area as S .

(e) If the curves $y = h(x)$ and $y = g(x)$ intersect when $x = c$ and when $x = d$, show that

$$\text{area of } S = \int_c^d [g'(x) - h'(x)] s(x) dx.$$

11. A sphere is inscribed in a right circular cylinder. The sphere is sliced by two parallel planes perpendicular to the axis of the cylinder. Show that the portions of the sphere and cylinder lying between these planes have equal surface areas.

6.6 Surface integrals

Surface integrals are, in many respects, analogous to line integrals; the integration takes place along a surface rather than along a curve. The definition of the line integral was suggested by a study of the concept of work. The definition of the surface integral will be motivated by a physical problem taken from the theory of fluid flow.

Suppose we have a fluid flowing in space with velocity vector $\vec{V}(x, y, z)$ and density $\rho(x, y, z)$ at each point (x, y, z) . As we observed earlier in connection with the physical interpretation of divergence (Section 5.22), the flux density vector

$$\vec{F}(x, y, z) = \rho(x, y, z) \vec{V}(x, y, z)$$

measures the rate of flow (mass per unit area) at (x, y, z) . We wish now to measure the total mass of fluid that flows through a given surface S in unit time. The following discussion is purely heuristic and is only intended to suggest a reasonable definition of the total mass flowing through S in unit time.

Let $S = \vec{r}(T)$ be a simple parametric surface given by an equation of the form

$$\vec{r}(u, v) = X(u, v) \vec{i} + Y(u, v) \vec{j} + Z(u, v) \vec{k},$$

where (u, v) ranges over a rectangular region T in the uv -plane. At each regular point of S let $\vec{n}$ denote the unit normal having the same direction as the fundamental vector product. That is, let

$$(6.14) \quad \vec{n} = \frac{\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}}{\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|}.$$

To measure the mass of fluid that flows through S in unit time in the direction indicated by $\vec{n}$, we partition the rectangle T into subrectangles $T_1, \dots, T_m$, and let $S_1, \dots, S_m$ be the corresponding surface patches on S . Let ΔS_k denote the area of the k th patch S_k . If the density ρ and the velocity $\vec{V}$ are constant on S_k and if S_k is nearly planar, the fluid flowing through S_k in unit time fills out a slanted cylindrical solid with base S_k and axis

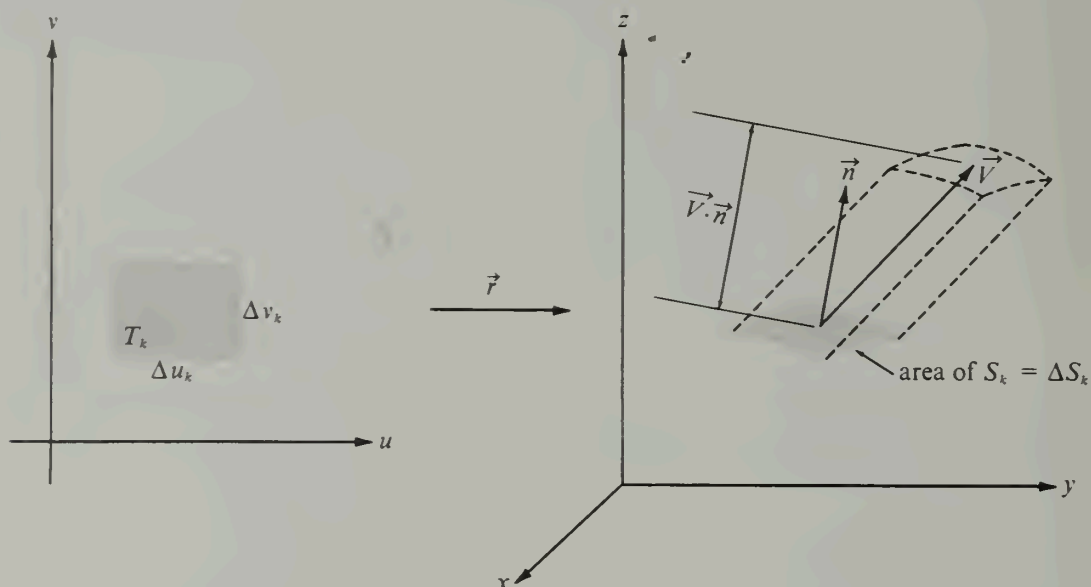


FIGURE 6.12 Fluid flowing through S_k in unit time fills out a cylindrical solid with volume $\vec{V} \cdot \vec{n} \Delta S_k$.

determined by the velocity vector $\vec{V}$. (See Figure 6.12.) The volume of this solid is the product of the area of its base ΔS_k with its altitude $\vec{V} \cdot \vec{n}$. Therefore the mass of fluid in this solid is $\rho \vec{V} \cdot \vec{n} \Delta S_k = \vec{F} \cdot \vec{n} \Delta S_k$. This suggests that the sum

$$(6.15) \quad \sum_{k=1}^m \vec{F} \cdot \vec{n} \Delta S_k$$

should be a reasonable approximation to the total mass of fluid flowing through S in unit time. Now we shall express this sum in terms of the rectangles T_k in the uv -plane. If the dimensions of the k th subrectangle T_k are Δu_k and Δv_k , the image S_k has an area ΔS_k approximately equal to $|\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v| \Delta u_k \Delta v_k$, since $|\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v|$ represents a local magnification factor for areas. Therefore, if we let $\phi = \vec{F} \cdot \vec{n}$, the sum in (6.15) is nearly equal to

$$(6.16) \quad \sum_{k=1}^m \phi \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| \Delta u_k \Delta v_k.$$

If $\phi |\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v|$ were constant on the open subrectangle T_k , the product $\phi |\partial \vec{r} / \partial u \times \partial \vec{r} / \partial v|$ would be a step function, and the sum in (6.16) would be the double integral of this step function over T . In this case the total mass of fluid flowing through S in unit time in the direction of $\vec{n}$ is

$$(6.17) \quad \iint_T \phi \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv = \iint_T \vec{F} \cdot \vec{n} \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv.$$

When the integrand is not a step function we shall use this integral as the *definition* of the total mass of fluid flowing through S in unit time in the direction of $\vec{n}$. This discussion also serves to motivate the following definition of a surface integral:

DEFINITION OF A SURFACE INTEGRAL. Let $S = \vec{r}(T)$ be a parametric surface described by a function $\vec{r}$ defined on a region T in the uv -plane, and let ϕ be a scalar field defined and bounded on S . The surface integral of ϕ over S is denoted by the symbol $\iint_{\vec{r}(T)} \phi \, dS$ [or by $\iint_S \phi(x, y, z) \, dS$], and is defined by the equation

$$(6.18) \quad \iint_{\vec{r}(T)} \phi \, dS = \iint_T \phi[\vec{r}(u, v)] \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du \, dv$$

whenever the double integral on the right exists.

Note. Since the double integral in (6.18) involves the function $\vec{r}$ in the integrand, the value of the surface integral $\iint_{\vec{r}(T)} \phi \, dS$ may depend on the particular representation used to describe the surface. In Section 6.7 we shall prove that under certain general conditions the value of the integral is independent of the representation.

The following examples illustrate some applications of surface integrals.

Example 1: Surface area. When $\phi = 1$, Equation (6.18) becomes

$$\iint_{\vec{r}(T)} dS = \iint_T \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du \, dv.$$

The double integral on the right is that used earlier in Section 6.4 to define surface area. Thus, the area of S is equal to the surface integral $\iint_{\vec{r}(T)} dS$. For this reason, the symbol dS is sometimes referred to as an “element of surface area,” and the surface integral $\iint_{\vec{r}(T)} \phi \, dS$ is said to be an integral of ϕ with respect to the element of surface area, extended over the surface $\vec{r}(T)$.

Example 2: Fluid flow. If $\vec{F}$ is the flux density of a fluid flow and if $\vec{n}$ is the unit normal to S , given by Equation (6.14), the surface integral $\iint_{\vec{r}(T)} \vec{F} \cdot \vec{n} \, dS$ represents the total mass of fluid flowing through S in unit time in the direction of $\vec{n}$. In fact, we have

$$\iint_{\vec{r}(T)} \vec{F} \cdot \vec{n} \, dS = \iint_T \vec{F} \cdot \vec{n} \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du \, dv,$$

so the surface integral $\iint_{\vec{r}(T)} \vec{F} \cdot \vec{n} \, dS$ is equal to the double integral in (6.17).

Example 3: Center of mass. Moment of inertia. If the scalar field ϕ is interpreted as the density (mass per unit area) of a thin material in the shape of the surface S , the total mass M of the surface is defined by the equation

$$M = \iint_S \phi(x, y, z) \, dS.$$

Its center of mass is the point $(\bar{x}, \bar{y}, \bar{z})$ determined by the equations

$$\bar{x}M = \iint_S x \phi(x, y, z) dS, \quad \bar{y}M = \iint_S y \phi(x, y, z) dS, \quad \bar{z}M = \iint_S z \phi(x, y, z) dS.$$

The moment of inertia I_L of S about an axis L is defined by the equation

$$I_L = \iint_S \delta^2(x, y, z) \phi(x, y, z) dS,$$

where $\delta(x, y, z)$ denotes the perpendicular distance from a general point (x, y, z) of S to the line L .

To illustrate, let us determine the center of mass of a uniform hemispherical surface of radius a . We use the parametric representation

$$\vec{r}(u, v) = a \cos u \cos v \vec{i} + a \sin u \cos v \vec{j} + a \sin v \vec{k},$$

where $(u, v) \in [0, 2\pi] \times [0, \frac{1}{2}\pi]$. This particular representation was discussed earlier in Example 2 of Section 6.2, where we found that the magnitude of the fundamental vector product is $a^2 |\cos v|$. In this example the density ϕ is constant, say $\phi = c$, and the mass M is $2\pi a^2 c$, the area of S times c . Because of symmetry, the coordinates $\bar{x}$ and $\bar{y}$ of the center of mass are 0. The coordinate $\bar{z}$ is given by

$$\begin{aligned} \bar{z}M &= c \iint_S z dS = c \iint_T a \sin v \cdot a^2 |\cos v| du dv \\ &= 2\pi a^3 c \int_0^{\pi/2} \sin v \cos v dv = \pi a^3 c = \frac{a}{2} M, \end{aligned}$$

so $\bar{z} = a/2$.

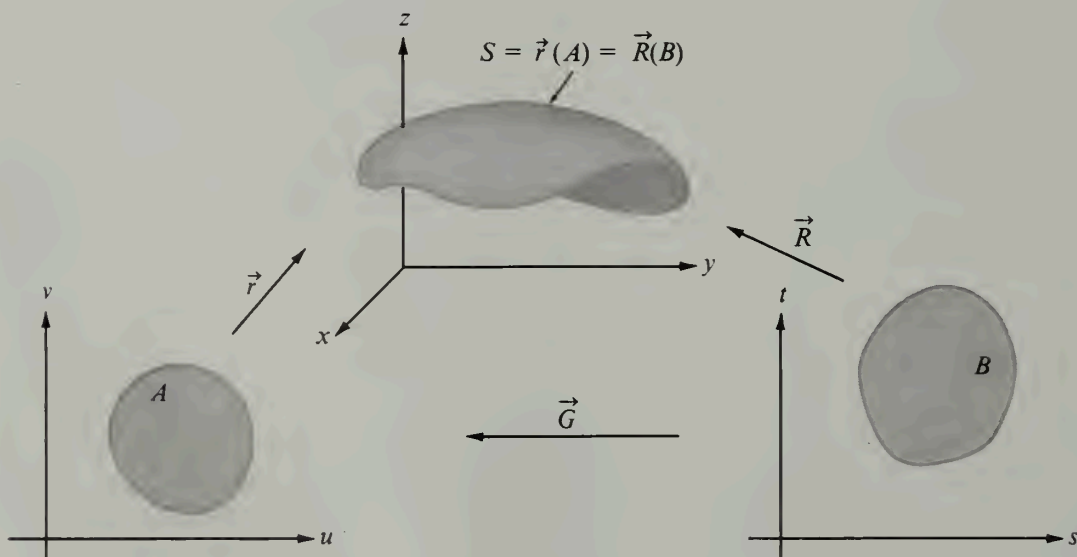


FIGURE 6.13 Two parametric representations of the same surface.

6.7 Change of parametric representation

We turn now to a discussion of the independence of surface integrals under a change of parametric representation. Suppose a function $\vec{r}$ maps a region A in the uv -plane onto a parametric surface $\vec{r}(A)$. Suppose also that A is the image of a region B in the st -plane under a one-to-one continuously differentiable mapping $\vec{G}$ given by

$$(6.19) \quad \vec{G}(s, t) = U(s, t) \vec{i} + V(s, t) \vec{j} \quad \text{if } (s, t) \in B.$$

Consider the function $\vec{R}$ defined on B by the equation

$$(6.20) \quad \vec{R}(s, t) = \vec{r}[\vec{G}(s, t)].$$

(See Figure 6.13.) Two functions $\vec{r}$ and $\vec{R}$ so related will be called *smoothly equivalent*. Smoothly equivalent functions describe the same surface. That is, $\vec{r}(A)$ and $\vec{R}(B)$ are identical as point sets. (This follows at once from the one-to-one nature of $\vec{G}$.) The next theorem describes the relationship between their fundamental vector products.

6-1 THEOREM. Let $\vec{r}$ and $\vec{R}$ be smoothly equivalent functions related by Equation (6.20), where $\vec{G} = U\vec{i} + V\vec{j}$ is a one-to-one continuously differentiable mapping of a region B in the st -plane onto a region A in the uv -plane given by Equation (6.19). Then we have

$$(6.21) \quad \frac{\partial \vec{R}}{\partial s} \times \frac{\partial \vec{R}}{\partial t} = \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \frac{\partial(U, V)}{\partial(s, t)},$$

where the partial derivatives $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ are to be evaluated at the point $(U(s, t), V(s, t))$. In other words, the fundamental vector product of $\vec{R}$ is equal to that of $\vec{r}$, times the Jacobian of the mapping $\vec{G}$.

Proof. The derivatives $\partial \vec{R}/\partial s$ and $\partial \vec{R}/\partial t$ may be computed by differentiation of Equation (6.20). If we apply the chain rule (Theorem 4-8) to each component of $\vec{R}$ and rearrange terms, we find that

$$\frac{\partial \vec{R}}{\partial s} = \frac{\partial \vec{r}}{\partial u} \frac{\partial U}{\partial s} + \frac{\partial \vec{r}}{\partial v} \frac{\partial V}{\partial s} \quad \text{and} \quad \frac{\partial \vec{R}}{\partial t} = \frac{\partial \vec{r}}{\partial u} \frac{\partial U}{\partial t} + \frac{\partial \vec{r}}{\partial v} \frac{\partial V}{\partial t},$$

where the derivatives $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ are evaluated at $(U(s, t), V(s, t))$. Now we cross multiply these two equations and, noting the order of the factors, we obtain

$$\frac{\partial \vec{R}}{\partial s} \times \frac{\partial \vec{R}}{\partial t} = \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) \left(\frac{\partial U}{\partial s} \frac{\partial V}{\partial t} - \frac{\partial U}{\partial t} \frac{\partial V}{\partial s} \right) = \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \frac{\partial(U, V)}{\partial(s, t)}.$$

This completes the proof.

The invariance of surface integrals under smoothly equivalent parametric representations is now an easy consequence of Theorem 6-1.

6-2 THEOREM. Let $\vec{r}$ and $\vec{R}$ be smoothly equivalent functions, as described in Theorem 6-1. If the surface integral $\iint_{\vec{r}(A)} \phi \, dS$ exists, the surface integral $\iint_{\vec{R}(B)} \phi \, dS$ also exists and we have

$$\iint_{\vec{r}(A)} \phi \, dS = \iint_{\vec{R}(B)} \phi \, dS.$$

Proof. By the definition of a surface integral we have

$$\iint_{\vec{r}(A)} \phi \, dS = \iint_A \phi[\vec{r}(u, v)] \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du \, dv.$$

Now we use the mapping $\vec{G}$ of Theorem 6-1 to transform this into a double integral over the region B in the st -plane. The transformation formula for double integrals states that

$$\iint_A \phi[\vec{r}(u, v)] \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du \, dv = \iint_B \phi\{\vec{r}[\vec{G}(s, t)]\} \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| \left| \frac{\partial(U, V)}{\partial(s, t)} \right| ds \, dt,$$

where the derivatives $\partial \vec{r}/\partial u$ and $\partial \vec{r}/\partial v$ on the right are to be evaluated at $(U(s, t), V(s, t))$. Because of Equation (6.21), the integral over B is equal to

$$\iint_B \phi[\vec{R}(s, t)] \left| \frac{\partial \vec{R}}{\partial s} \times \frac{\partial \vec{R}}{\partial t} \right| ds \, dt.$$

This, in turn, is the definition of the surface integral $\iint_{\vec{R}(B)} \phi \, dS$. The proof is now complete.

6.8 Other notations for surface integrals

If $S = \vec{r}(T)$ is a parametric surface, the fundamental vector product $\partial \vec{r}/\partial u \times \partial \vec{r}/\partial v$ is normal to S at each regular point of the surface. At each such point there are *two* unit normals, a unit normal $\vec{n}_1$ which has the same direction as the fundamental vector product, and a unit normal $\vec{n}_2$ which has the opposite direction. Thus,

$$\vec{n}_1 = \frac{\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}}{\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|} \quad \text{and} \quad \vec{n}_2 = - \frac{\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}}{\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|}.$$

Let $\vec{n}$ be one of the two normals $\vec{n}_1$ or $\vec{n}_2$. Let $\vec{F}$ be a vector field defined on S and assume the surface integral $\iint_S \vec{F} \cdot \vec{n} \, dS$ exists. Then we may write

$$\begin{aligned} (6.22) \quad \iint_S \vec{F} \cdot \vec{n} \, dS &= \iint_T \vec{F}[\vec{r}(u, v)] \cdot \vec{n}(u, v) \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du \, dv \\ &= \pm \iint_T \vec{F}[\vec{r}(u, v)] \cdot \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} du \, dv, \end{aligned}$$

where the $+$ sign is used if $\vec{n} = \vec{n}_1$ and the $-$ sign is used if $\vec{n} = \vec{n}_2$.

Suppose now we express $\vec{F}$ and $\vec{r}$ in terms of their components, say

$$\vec{F}(x, y, z) = P(x, y, z) \vec{i} + Q(x, y, z) \vec{j} + R(x, y, z) \vec{k}$$

and

$$\vec{r}(u, v) = X(u, v) \vec{i} + Y(u, v) \vec{j} + Z(u, v) \vec{k}.$$

Then the fundamental vector product of $\vec{r}$ is given by

$$\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = \frac{\partial(Y, Z)}{\partial(u, v)} \vec{i} + \frac{\partial(Z, X)}{\partial(u, v)} \vec{j} + \frac{\partial(X, Y)}{\partial(u, v)} \vec{k}.$$

If $\vec{n} = \vec{n}_1$, Equation (6.22) becomes

$$(6.23) \quad \iint_S \vec{F} \cdot \vec{n} \, dS = \iint_T P[\vec{r}(u, v)] \frac{\partial(Y, Z)}{\partial(u, v)} du \, dv \\ + \iint_T Q[\vec{r}(u, v)] \frac{\partial(Z, X)}{\partial(u, v)} du \, dv + \iint_T R[\vec{r}(u, v)] \frac{\partial(X, Y)}{\partial(u, v)} du \, dv;$$

if $\vec{n} = \vec{n}_2$, each double integral on the right must be replaced by its negative. The sum of the double integrals on the right is often written more briefly as

$$(6.24) \quad \iint_S P(x, y, z) \, dy \, dz + \iint_S Q(x, y, z) \, dz \, dx + \iint_S R(x, y, z) \, dx \, dy,$$

or even more briefly as

$$(6.25) \quad \iint_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy.$$

The integrals which appear in (6.24) and (6.25) are also referred to as surface integrals. Thus, for example, the surface integral $\iint_S P \, dy \, dz$ is defined by the equation

$$(6.26) \quad \iint_S P \, dy \, dz = \iint_T P[\vec{r}(u, v)] \frac{\partial(Y, Z)}{\partial(u, v)} du \, dv.$$

This notation is suggested by the formula for changing variables in a double integral.

Despite similarity in notation, the integral on the left of (6.26) is *not* a double integral. First of all, P is a function of three variables. Also, we must take into consideration the order in which the symbols dy and dz appear in the surface integral, because

$$\frac{\partial(Y, Z)}{\partial(u, v)} = - \frac{\partial(Z, Y)}{\partial(u, v)}$$

and hence

$$\iint_S P \, dy \, dz = - \iint_S P \, dz \, dy.$$

In this notation, formula (6.23) becomes

$$(6.27) \quad \iint_S \vec{F} \cdot \vec{n} \, dS = \iint_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy$$

if $\vec{n} = \vec{n}_1$. If $\vec{n} = \vec{n}_2$ the integral on the right must be replaced by its negative. This formula resembles the following formula for line integrals:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C P \, dx + Q \, dy + R \, dz.$$

The resemblance can be carried one step further if we introduce the symbolic "vector" $d\vec{S} = \vec{n} dS$ (sometimes called the vector element of surface area) and write $\iint_S \vec{F} \cdot d\vec{S}$ instead of $\iint_S \vec{F} \cdot \vec{n} dS$.

If the unit normal $\vec{n}$ is expressed in terms of its direction cosines, say

$$\vec{n} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k},$$

then $\vec{F} \cdot \vec{n} = P \cos \alpha + Q \cos \beta + R \cos \gamma$, and we may write

$$\iint_S \vec{F} \cdot \vec{n} dS = \iint_S (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS.$$

This equation holds when $\vec{n}$ is either $\vec{n}_1$ or $\vec{n}_2$. The direction cosines will depend on the choice of the normal. If $\vec{n} = \vec{n}_1$ we may use (6.27) to write

$$(6.28) \quad \iint_S (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS = \iint_S P dy dz + Q dz dx + R dx dy.$$

If $\vec{n} = \vec{n}_2$ we have, instead,

$$(6.29) \quad \iint_S (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS = - \iint_S P dy dz + Q dz dx + R dx dy.$$

6.9 Exercises

1. Let S denote the hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, and let $\vec{F}(x, y, z) = x\vec{i} + y\vec{j}$. Let $\vec{n}$ be the unit outward normal of S . Compute the value of the surface integral $\iint_S \vec{F} \cdot \vec{n} dS$, using:

(a) the vector representation

$$\vec{r}(u, v) = \sin u \cos v \vec{i} + \sin u \sin v \vec{j} + \cos u \vec{k},$$

(b) the explicit representation $z = \sqrt{1 - x^2 - y^2}$.

2. Show that the moment of inertia of a homogeneous spherical shell about a diameter is equal to $\frac{2}{3}Ma^2$, where M is the mass of the shell and a is its radius.

3. Find the center of mass of that portion of the homogeneous hemispherical surface $x^2 + y^2 + z^2 = a^2$ lying above the first quadrant in the xy -plane.

4. Let S denote the plane surface whose boundary is the triangle with vertices at $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, and let $\vec{F}(x, y, z) = x\vec{i} + y\vec{j} + z\vec{k}$. Let $\vec{n}$ denote the unit normal to S having a nonnegative z -component. Evaluate the surface integral $\iint_S \vec{F} \cdot \vec{n} dS$, using:

(a) the vector representation

$$\vec{r}(u, v) = (u + v) \vec{i} + (u - v) \vec{j} + (1 - 2u) \vec{k},$$

(b) an explicit representation of the form $z = f(x, y)$.

5. Let S be a parametric surface described by the explicit formula $z = f(x, y)$, where (x, y) varies over a plane region T , the projection of S in the xy -plane. Let $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ and let $\vec{n}$ denote the unit normal to S having a nonnegative z -component. Use the parametric representation $\vec{r}(x, y) = x\vec{i} + y\vec{j} + f(x, y)\vec{k}$ and show that

$$\iint_S \vec{F} \cdot \vec{n} dS = \iint_T \left(-P \frac{\partial f}{\partial x} - Q \frac{\partial f}{\partial y} + R \right) dx dy,$$

where each of P , Q , and R is to be evaluated at $(x, y, f(x, y))$.

6. Let S be as in Exercise 5, and let ϕ be a scalar field. Show that:

$$(a) \iint_S \phi(x, y, z) dS = \iint_T \phi[x, y, f(x, y)] \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dx dy.$$

$$(b) \iint_S \phi(x, y, z) dy dz = - \iint_T \phi[x, y, f(x, y)] \frac{\partial f}{\partial x} dx dy.$$

$$(c) \iint_S \phi(x, y, z) dz dx = - \iint_T \phi[x, y, f(x, y)] \frac{\partial f}{\partial y} dx dy.$$

7. Let S be the hemisphere $x^2 + y^2 + z^2 = 1, z \geq 0$, and let $\vec{n}$ denote the unit normal that points out of the sphere. If $\vec{F}(x, y, z) = (x^2 + xy - z^2) \vec{k}$, evaluate the surface integral

$$\iint_S (\text{curl } \vec{F}) \cdot \vec{n} dS.$$

8. Solve Exercise 7 if S also includes the planar base of the hemisphere. On the lower base the unit normal is $-\vec{k}$.

9. If S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$, compute the value of the surface integral

$$\iint_S xz dy dz + yz dz dx + x^2 dx dy.$$

Choose a representation for which the fundamental vector product points in the direction of the outward normal.

10. The cylinder $x^2 + y^2 = 2x$ cuts out a portion of a surface S from the upper nappe of the cone $x^2 + y^2 = z^2$. Compute the value of the surface integral

$$\iint_S (x^4 - y^4 + y^2 z^2 - z^2 x^2 + 1) dS.$$

11. Let S denote the portion of the plane $x + y + z = t$ cut off by the unit sphere $x^2 + y^2 + z^2 = 1$. Let $\phi(x, y, z) = 1 - x^2 - y^2 - z^2$ if (x, y, z) is inside this sphere, and let $\phi(x, y, z)$ be 0 otherwise. Show that

$$\iint_S \phi(x, y, z) dS = \begin{cases} \frac{\pi}{18} (3 - t^2)^2 & \text{if } |t| \leq \sqrt{3}, \\ 0 & \text{if } |t| > \sqrt{3}. \end{cases}$$

[Hint. Introduce new coordinates (x_1, y_1, z_1) with the z_1 -axis normal to the plane $x + y + z = t$. Then use the polar coordinates in the $x_1 y_1$ -plane as parameters for S .]

12. A homogeneous spherical shell of radius a is cut by one nappe of a right circular cone whose vertex is at the center of the sphere. If the vertex angle of the cone is α , where $0 < \alpha < \pi$, determine (in terms of a and α) the center of mass of the portion of the spherical shell that lies inside the cone.

6.10 The theorem of Stokes

Green's theorem in the plane asserts that

$$(6.30) \quad \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_C P dx + Q dy,$$

where S is a region bounded by a piecewise smooth simple closed curve C . There are two ways of expressing Green's theorem in vector form. If we introduce the vector field $\vec{F} = P\vec{i} + Q\vec{j}$ we can write Equation (6.30) in the form

$$\iint_S (\text{curl } \vec{F}) \cdot \vec{k} \, dx \, dy = \oint_C \vec{F} \cdot d\vec{r}.$$

If we let $\vec{V} = Q\vec{i} - P\vec{j}$ we can write

$$\iint_S \text{div } \vec{V} \, dx \, dy = \oint_C \vec{V} \cdot \vec{n} \, ds,$$

where $\vec{n}$ is the unit outer normal of C and the line integral is taken with respect to arc length. The extensions of these formulas to 3-space are known, respectively, as *Stokes' theorem*[†] and the *divergence theorem*. Stokes' theorem will be stated and proved in this section. The divergence theorem is discussed in Section 6.14.

6-3 THEOREM. *Stokes' theorem.* Let T be a region in the uv -plane bounded by a piecewise smooth Jordan curve Γ . Let $S = \vec{r}(T)$ be a smooth simple parametric surface described by a one-to-one mapping function $\vec{r}$ whose components have continuous second-order partial derivatives on some open set containing $T \cup \Gamma$. Let Γ be described by a function $\vec{\gamma}$ defined on an interval $[a, b]$. Since $\vec{r}$ is one-to-one, the image of Γ is a simple closed curve C described by the composite function $\vec{\rho}$ given by

$$\vec{\rho}(t) = \vec{r}[\vec{\gamma}(t)] \quad \text{for } a \leq t \leq b.$$

(See Figure 6.14.) Let $\vec{n}$ be the unit normal vector having the same direction as the fundamental vector product $\partial\vec{r}/\partial u \times \partial\vec{r}/\partial v$, and let $\vec{F}$ be a vector field that is continuously differentiable on S . Then we have the identity

$$(6.31) \quad \iint_S (\text{curl } \vec{F}) \cdot \vec{n} \, dS = \pm \oint_C \vec{F} \cdot d\vec{\rho},$$

where the $+$ or $-$ sign is used according to whether $\vec{\gamma}$ describes Γ in a positive (counterclockwise) or negative (clockwise) direction.

Note. If we write $\vec{F}(x, y, z) = P(x, y, z)\vec{i} + Q(x, y, z)\vec{j} + R(x, y, z)\vec{k}$, Stokes' theorem takes the form

$$(6.32) \quad \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy \, dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz \, dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy \\ = \pm \oint_C P \, dx + Q \, dy + R \, dz.$$

The curve C is traversed in the direction inherited from Γ through the mapping function $\vec{r}$.

Proof. To prove the theorem it suffices to establish the three identities

[†] In honor of G. G. Stokes (1819–1903), an Irish mathematician who made many fundamental contributions to hydrodynamics and optics.

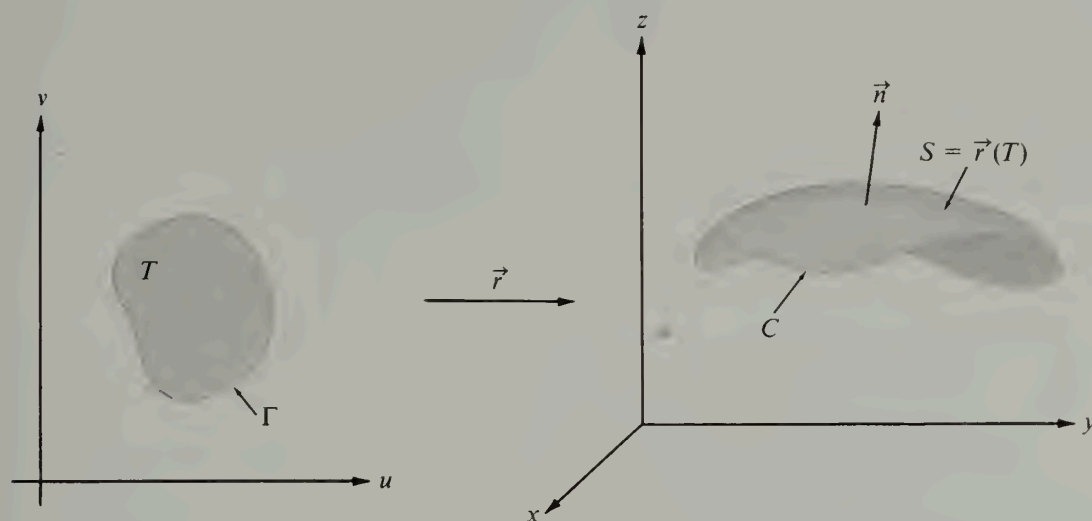


FIGURE 6.14 An example of a surface to which Stokes' theorem is applicable.

$$(6.33) \quad \pm \oint_C P \, dx = \iint_S \left(-\frac{\partial P}{\partial y} \, dx \, dy + \frac{\partial P}{\partial z} \, dz \, dx \right),$$

$$(6.34) \quad \pm \oint_C Q \, dy = \iint_S \left(-\frac{\partial Q}{\partial z} \, dy \, dz + \frac{\partial Q}{\partial x} \, dx \, dy \right),$$

$$(6.35) \quad \pm \oint_C R \, dz = \iint_S \left(-\frac{\partial R}{\partial x} \, dz \, dx + \frac{\partial R}{\partial y} \, dy \, dz \right),$$

with the same choice of sign in each formula. Addition of these three equations gives (6.32). In each case the idea of the proof is to express the line integral over C as a line integral over Γ , using the transformation law described in Theorem 5-11 (Section 5.28). Then we apply Green's theorem to the line integral over Γ , expressing it as a double integral over T . Finally, we show that this double integral is the same as that which defines the surface integral over S .

We begin with the line integral in (6.33). If we write

$$\vec{r}(u, v) = X(u, v) \vec{i} + Y(u, v) \vec{j} + Z(u, v) \vec{k}$$

and apply Theorem 5-11 [as extended in Equation (5.74)] we obtain

$$(6.36) \quad \oint_C P \, dx = \oint_{\Gamma} p \frac{\partial X}{\partial u} \, du + p \frac{\partial X}{\partial v} \, dv,$$

where p is the composite function given by

$$(6.37) \quad p(u, v) = P[\vec{r}(u, v)] = P[X(u, v), Y(u, v), Z(u, v)].$$

When we apply Green's theorem to the line integral over Γ , we find

$$\oint_{\Gamma} p \frac{\partial X}{\partial u} du + p \frac{\partial X}{\partial v} dv = \pm \iint_T \left\{ \frac{\partial}{\partial u} \left(p \frac{\partial X}{\partial v} \right) - \frac{\partial}{\partial v} \left(p \frac{\partial X}{\partial u} \right) \right\} du dv,$$

where the $+$ or $-$ sign is used according to whether γ describes Γ in the positive or negative direction. Combining this equation with (6.36) we find

$$(6.38) \quad \pm \oint_C P dx = \iint_T \left\{ \frac{\partial}{\partial u} \left(p \frac{\partial X}{\partial v} \right) - \frac{\partial}{\partial v} \left(p \frac{\partial X}{\partial u} \right) \right\} du dv.$$

Now we have

$$\frac{\partial}{\partial u} \left(p \frac{\partial X}{\partial v} \right) = \frac{\partial p}{\partial u} \frac{\partial X}{\partial v} + p \frac{\partial^2 X}{\partial u \partial v} \quad \text{and} \quad \frac{\partial}{\partial v} \left(p \frac{\partial X}{\partial u} \right) = \frac{\partial p}{\partial v} \frac{\partial X}{\partial u} + p \frac{\partial^2 X}{\partial v \partial u}.$$

When we subtract these equations the terms involving p cancel and we find

$$(6.39) \quad \frac{\partial}{\partial u} \left(p \frac{\partial X}{\partial v} \right) - \frac{\partial}{\partial v} \left(p \frac{\partial X}{\partial u} \right) = \frac{\partial p}{\partial u} \frac{\partial X}{\partial v} - \frac{\partial p}{\partial v} \frac{\partial X}{\partial u}.$$

To compute the derivatives $\partial p/\partial u$ and $\partial p/\partial v$ we differentiate Equation (6.37), using the chain rule, and we find

$$\frac{\partial p}{\partial u} = \frac{\partial P}{\partial x} \frac{\partial X}{\partial u} + \frac{\partial P}{\partial y} \frac{\partial Y}{\partial u} + \frac{\partial P}{\partial z} \frac{\partial Z}{\partial u} \quad \text{and} \quad \frac{\partial p}{\partial v} = \frac{\partial P}{\partial x} \frac{\partial X}{\partial v} + \frac{\partial P}{\partial y} \frac{\partial Y}{\partial v} + \frac{\partial P}{\partial z} \frac{\partial Z}{\partial v},$$

where the derivatives $\partial P/\partial x$, $\partial P/\partial y$, and $\partial P/\partial z$ are to be evaluated at $\vec{r}(u, v)$. Substituting these equations in (6.39) we obtain

$$\frac{\partial}{\partial u} \left(p \frac{\partial X}{\partial v} \right) - \frac{\partial}{\partial v} \left(p \frac{\partial X}{\partial u} \right) = - \frac{\partial P}{\partial y} \frac{\partial(X, Y)}{\partial(u, v)} + \frac{\partial P}{\partial z} \frac{\partial(Z, X)}{\partial(u, v)},$$

and hence Equation (6.38) becomes

$$\begin{aligned} \pm \oint_C P dx &= \iint_T \left\{ - \frac{\partial P}{\partial y} \frac{\partial(X, Y)}{\partial(u, v)} + \frac{\partial P}{\partial z} \frac{\partial(Z, X)}{\partial(u, v)} \right\} du dv \\ &= \iint_S \left(- \frac{\partial P}{\partial y} dx dy + \frac{\partial P}{\partial z} dz dx \right). \end{aligned}$$

This proves (6.33). The proofs of (6.34) and (6.35) are entirely analogous.

6.11 Applications of Stokes' theorem

We shall use Stokes' theorem to derive a three-dimensional analog of Theorem 5-7 (Section 5.16). We proved there that a two-dimensional continuously differentiable vector field $\vec{F} = P \vec{i} + Q \vec{j}$ is a gradient in an open simply connected set S in the plane if, and only if, $\partial P/\partial y = \partial Q/\partial x$ in S , that is, if and only if $\text{curl } \vec{F} = \vec{0}$ in S . We prove now that the theorem extends to 3-space if the set S is convex.

6- 4 THEOREM. Let K be an open convex set in 3-space, and let $\vec{F}$ be a vector field that is continuously differentiable on K . Then $\vec{F}$ is a gradient on K if and only if $\text{curl } \vec{F} = \vec{0}$.

Proof. If $\vec{F} = \nabla\phi$ for some ϕ , we have $\text{curl } \vec{F} = \text{curl}(\nabla\phi) = \vec{0}$, so the condition $\text{curl } \vec{F} = \vec{0}$ is necessary for $\vec{F}$ to be a gradient.

To prove that the condition is sufficient, choose a fixed point $\vec{A}$ in K and define a scalar field ϕ on K by the equation

$$\phi(\vec{X}) = \int_{\vec{A}}^{\vec{X}} \vec{F} \cdot d\vec{r},$$

where the integral is taken over the line segment joining $\vec{A}$ to $\vec{X}$. This segment lies in K because K is convex. Now let $\vec{Z}$ be any point in K . We shall prove that

$$(6.40) \quad \phi(\vec{Z}) - \phi(\vec{X}) = \int_{\vec{X}}^{\vec{Z}} \vec{F} \cdot d\vec{r},$$

where the path is again along a line segment. To prove (6.40) we consider the triangle with vertices $\vec{A}$, $\vec{X}$, $\vec{Z}$. Let S be the plane surface bounded by this triangle. Because of the convexity of K , S lies entirely in K , so $\text{curl } \vec{F} = \vec{0}$ on S . Therefore $\iint_S (\text{curl } \vec{F}) \cdot \vec{n} dS = 0$, where $\vec{n}$ is a unit normal to S . But, by Stokes' theorem, this surface integral is equal to the line integral of $\vec{F}$ around the boundary of S . Hence we have

$$\int_{\vec{A}}^{\vec{X}} \vec{F} \cdot d\vec{r} + \int_{\vec{X}}^{\vec{Z}} \vec{F} \cdot d\vec{r} + \int_{\vec{Z}}^{\vec{A}} \vec{F} \cdot d\vec{r} = 0,$$

where $\vec{r}$ is the function which traces out the triangular boundary. This equation implies

$$\int_{\vec{X}}^{\vec{Z}} \vec{F} \cdot d\vec{r} = \int_{\vec{A}}^{\vec{Z}} \vec{F} \cdot d\vec{r} - \int_{\vec{A}}^{\vec{X}} \vec{F} \cdot d\vec{r} = \phi(\vec{Z}) - \phi(\vec{X}),$$

which proves (6.40).

Now let $N(\vec{X}; r)$ be any neighborhood of $\vec{X}$ of radius r that lies entirely in the set K . If $\vec{Y}$ is a unit coordinate vector, the point $\vec{X} + h\vec{Y}$ lies in K for every real h satisfying $0 < |h| \leq r$. For such an h , let $\vec{Z} = \vec{X} + h\vec{Y}$. Then Equation (6.40) becomes

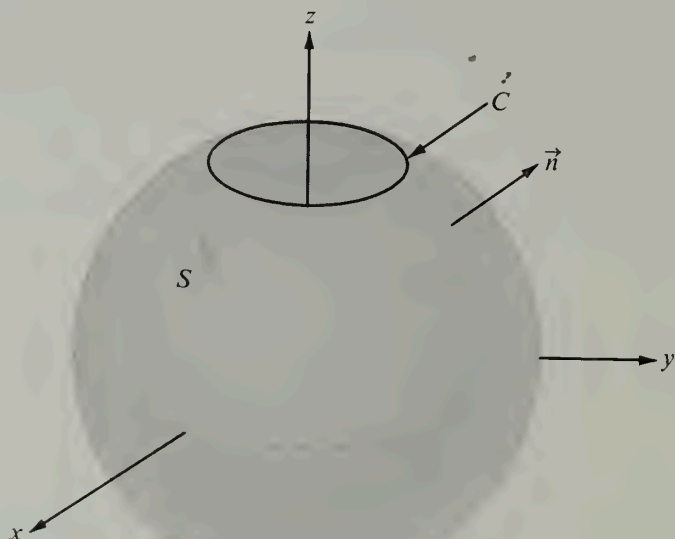
$$\phi(\vec{X} + h\vec{Y}) - \phi(\vec{X}) = \int_{\vec{X}}^{\vec{X}+h\vec{Y}} \vec{F} \cdot d\vec{r},$$

where the path is along the line segment joining $\vec{X}$ to $\vec{X} + h\vec{Y}$. But now we may repeat, word for word, the argument used in the proof of Theorem 5-3 [from Equation (5.23) on] to deduce that ϕ is differentiable at $\vec{X}$ and that $\nabla\phi(\vec{X}) = \vec{F}(\vec{X})$.

In Section 5.25 we proved that a vector field $\vec{V}$ is solenoidal (that is, $\text{div } \vec{V} = 0$) on an open rectangular parallelepiped if, and only if, $\vec{V}$ is the curl of another vector field. The following example shows that this statement is not true for *arbitrary* open sets.

Let D be the portion of 3-space between two concentric spheres with center at the origin and radii a and b , where $0 < a < b$. Let $\vec{V} = \vec{r}/r^3$, where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = |\vec{r}|$. It is easy to verify that $\text{div } \vec{V} = 0$ everywhere in D . In fact, we have the general formula

$$\text{div}(r^n \vec{r}) = (n + 3)r^n,$$

FIGURE 6.15 The surface S and curve C in Equation (6.41).

and in this example $n = -3$. We shall use Stokes' theorem to prove that this $\vec{V}$ is *not* a curl in D (although it is a curl on every open rectangular parallelepiped not containing the origin). To do this we assume there is a vector field $\vec{U}$ such that $\vec{V} = \text{curl } \vec{U}$ in D and obtain a contradiction. By Stokes' theorem we may write

$$(6.41) \quad \iint_S (\text{curl } \vec{U}) \cdot \vec{n} \, dS = \oint_C \vec{U} \cdot d\vec{r},$$

where S and C are the surface and curve shown in Figure 6.15. To construct S , we take a spherical surface of radius R concentric with the boundaries of D , where $a < R < b$, and we remove a small "polar cap," as indicated in the figure. The portion that remains is the surface S . The curve C is the circular edge shown. Let $\vec{n}$ denote the unit outer normal of S , so that $\vec{n} = \vec{r}/r$. Since $\text{curl } \vec{U} = \vec{V} = \vec{r}/r^3$, we have

$$(\text{curl } \vec{U}) \cdot \vec{n} = \frac{\vec{r}}{r^3} \cdot \frac{\vec{r}}{r} = \frac{1}{r^2}.$$

On the surface S this dot product has the constant value $1/R^2$. Therefore we have

$$\iint_S (\text{curl } \vec{U}) \cdot \vec{n} \, dS = \frac{1}{R^2} \iint_S dS = \frac{\text{area of } S}{R^2}.$$

When the polar cap shrinks to a point, the area of S approaches $4\pi R^2$ (the area of the whole sphere) and, therefore, the value of the surface integral in (6.41) approaches 4π .

Next we examine the line integral in (6.41). It is easy to prove that for any line integral $\int_C \vec{U} \cdot d\vec{r}$ we have the inequality

$$\left| \int_C \vec{U} \cdot d\vec{r} \right| \leq M \cdot (\text{length of } C),$$

where M is a constant depending on $\vec{U}$. (In fact, M can be taken to be the maximum value of $|\vec{U}|$ on C .) Therefore, as we let the polar cap shrink to a point, the length of C and the value of the line integral both approach zero. Thus we have a contradiction; the surface integral in (6.41) can be made arbitrarily close to 4π , and the corresponding line integral to which it is equal can be made arbitrarily close to 0. Therefore a function $\vec{U}$ whose curl is $\vec{V}$ cannot exist in the region D .

The difficulty here is caused by the geometric structure of the region D . Although this region is simply connected (that is, any simple closed curve in D is the edge of a parametric surface lying completely in D) there are closed *surfaces* in D that are not the complete boundaries of solids lying entirely in D . For example, no sphere about the origin is the complete boundary of a solid lying entirely in D . If the region D has the property that *every* closed surface in D is the boundary of a solid lying entirely in D , it can be shown that a vector field $\vec{U}$ exists such that $\vec{V} = \text{curl } \vec{U}$ in D if, and only if, $\text{div } \vec{V} = 0$ everywhere in D . The proof of this statement is difficult and will not be given here.

6.12 Exercises

In each of Exercises 1 through 4, transform the surface integral $\iint_S (\text{curl } \vec{F}) \cdot \vec{n} \, dS$ to a line integral by the use of Stokes' theorem, and then evaluate the line integral.

1. $\vec{F}(x, y, z) = y^2\vec{i} + xy\vec{j} + xz\vec{k}$, where S is the hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, and $\vec{n}$ is the unit normal with a nonnegative z -component.

2. $\vec{F}(x, y, z) = y\vec{i} + z\vec{j} + x\vec{k}$, where S is the portion of the paraboloid $z = 1 - x^2 - y^2$ with $z \geq 0$, and $\vec{n}$ is the unit normal with a nonnegative z -component.

3. $\vec{F}(x, y, z) = (y - z)\vec{i} + yz\vec{j} - xz\vec{k}$, where S consists of the five faces of the cube $0 \leq x \leq 2$, $0 \leq y \leq 2$, $0 \leq z \leq 2$ not in the xy -plane. The unit normal $\vec{n}$ is the outward normal.

4. $\vec{F}(x, y, z) = xz\vec{i} - y\vec{j} + x^2y\vec{k}$, where S consists of the three faces not in the xz -plane of the tetrahedron bounded by the three coordinate planes and the plane $3x + y + 3z = 6$. The normal $\vec{n}$ is the unit normal pointing out of the tetrahedron.

In Exercises 5 through 10, use Stokes' theorem to show that the line integrals have the values given. In each case, explain how to traverse C to arrive at the given answer.

5. $\int_C y \, dx + z \, dy + x \, dz = \pi a^2 \sqrt{3}$, where C is the curve of intersection of the sphere $x^2 + y^2 + z^2 = a^2$ and the plane $x + y + z = 0$.

6. $\int_C (y + z) \, dx + (z + x) \, dy + (x + y) \, dz = 0$, where C is the curve of intersection of the cylinder $x^2 + y^2 = 2y$ and the plane $y = z$.

7. $\int_C y^2 \, dx + xy \, dy + xz \, dz = 0$, where C is the curve of Exercise 6.

8. $\int_C (y - z) \, dx + (z - x) \, dy + (x - y) \, dz = 2\pi a(a + b)$, where C is the intersection of the cylinder $x^2 + y^2 = a^2$ and the plane $x/a + z/b = 1$, $a > 0$, $b > 0$.

9. $\int_C (y^2 + z^2) \, dx + (x^2 + z^2) \, dy + (x^2 + y^2) \, dz = 2\pi ab^2$, where C is the intersection of the hemisphere $x^2 + y^2 + z^2 = 2ax$, $z > 0$, and the cylinder $x^2 + y^2 = 2bx$, where $0 < b < a$.

10. $\int (y^2 - z^2) dx + (z^2 - x^2) dy + (x^2 - y^2) dz = 9a^3/2$, where C is the curve cut from the boundary of the cube $0 \leq x \leq a$, $0 \leq y \leq a$, $0 \leq z \leq a$ by the plane $x + y + z = 3a/2$.

11. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $P\vec{i} + Q\vec{j} + R\vec{k} = \vec{a} \times \vec{r}$, where $\vec{a}$ is a constant vector, show that $\int_C P dx + Q dy + R dz = 2 \iint_S \vec{a} \cdot \vec{n} dS$, where C is a curve bounding a parametric surface S and $\vec{n}$ is a suitable normal to S .

12. Let $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$, where $P = -y/(x^2 + y^2)$, $Q = x/(x^2 + y^2)$, $R = z$, and let D be the torus generated by rotating the circle $(x - 2)^2 + z^2 = 1$, $y = 0$, about the z -axis. Show that $\text{curl } \vec{F} = \vec{0}$ in D but that $\int_C P dx + Q dy + R dz$ is not zero if the curve C is the circle $x^2 + y^2 = 4$, $z = 0$.

13. Let S be a simple parametric surface of the type described in Stokes' theorem, and let C denote the edge of S . Let f and g be two continuously differentiable scalar fields defined on S , and let $\vec{n}$ denote a choice of a unit normal to S . If ∇f is perpendicular to $\nabla g \times \vec{n}$ everywhere on S , prove that the line integral of $f \nabla g$ around C is zero.

14. Let u and v be scalar fields that are continuously differentiable on an open rectangular parallelepiped R in 3-space.

(a) Show that a vector field $\vec{F}$ exists such that $\nabla u \times \nabla v = \text{curl } \vec{F}$ everywhere in R .

(b) Determine whether or not any of the following three vector fields may be used for $\vec{F}$ in part (a): (i) $\nabla(uv)$; (ii) $u \nabla v$; (iii) $v \nabla u$.

(c) If $u(x, y, z) = x^3 - y^3 + z^2$ and $v(x, y, z) = x + y + z$, evaluate the surface integral

$$\iint_S \nabla u \times \nabla v \cdot \vec{n} dS,$$

where S is the hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, and $\vec{n}$ is the unit normal with a nonnegative z -component.

*6.13 Extensions of Stokes' theorem

Stokes' theorem can be extended to more general simple smooth surfaces. If T is a multiply connected region like that shown in Figure 6.16 (with a finite number of holes), the one-to-one image $S = \vec{r}(T)$ will contain the same number of holes as T . To extend Stokes' theorem to such surfaces we use exactly the same type of argument as in the foregoing proof, except that we employ Green's theorem for multiply connected regions (Theorem 5-8). In place of the line integral $\oint_C \vec{F} \cdot d\vec{\rho}$ which appears in Equation (6.31) we need a sum of line integrals, with appropriate signs, taken over the images of the curves forming the boundary of T . For example, if T has two holes, as in Figure 6.16, and if the boundary curves Γ , Γ_1 , and Γ_2 are traversed in the directions shown, the identity in Stokes' theorem takes the form

$$\iint_S (\text{curl } \vec{F}) \cdot \vec{n} dS = \oint_C \vec{F} \cdot d\vec{\rho} + \oint_{C_1} \vec{F} \cdot d\vec{\rho}_1 + \oint_{C_2} \vec{F} \cdot d\vec{\rho}_2,$$

where C , C_1 , and C_2 are the images of Γ , Γ_1 , and Γ_2 , respectively, and $\vec{\rho}$, $\vec{\rho}_1$, and $\vec{\rho}_2$ are the composite functions $\vec{\rho}(t) = \vec{r}[\vec{\gamma}(t)]$, $\vec{\rho}_1(t) = \vec{r}[\vec{\gamma}_1(t)]$, $\vec{\rho}_2(t) = \vec{r}[\vec{\gamma}_2(t)]$. Here $\vec{\gamma}$, $\vec{\gamma}_1$, and $\vec{\gamma}_2$ are the functions that describe Γ , Γ_1 , and Γ_2 in the directions shown. The curves C , C_1 , and C_2 will be traversed in the directions inherited from Γ , Γ_1 , and Γ_2 through the mapping function $\vec{r}$.

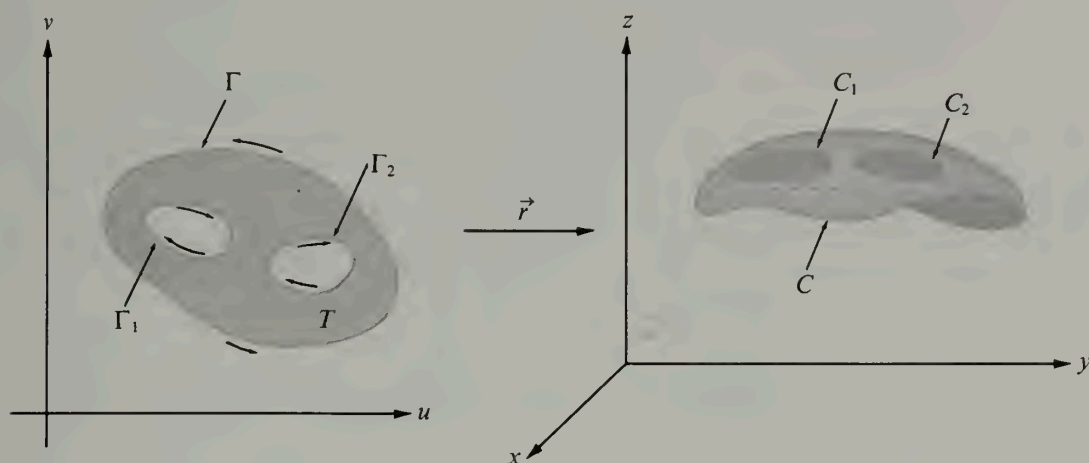


FIGURE 6.16 Extension of Stokes' theorem to surfaces that are one-to-one images of multiply connected regions.

Stokes' theorem can also be extended to some (but not all) smooth surfaces that are not simple. We shall illustrate a few of the possibilities with examples.

Consider first the cylinder shown in Figure 6.17. This is the union of two simple smooth parametric surfaces S_1 and S_2 , the images of two adjacent rectangles T_1 and T_2 , under mappings $\vec{r}_1$ and $\vec{r}_2$, respectively. If $\vec{\gamma}_1$ describes the positively oriented boundary Γ_1 of T_1 and $\vec{\gamma}_2$ describes the positively oriented boundary Γ_2 of T_2 , the functions $\vec{\rho}_1$ and $\vec{\rho}_2$ defined by

$$\vec{\rho}_1(t) = \vec{r}_1[\vec{\gamma}_1(t)], \quad \vec{\rho}_2(t) = \vec{r}_2[\vec{\gamma}_2(t)]$$

describe the images C_1 and C_2 of Γ_1 and Γ_2 , respectively. In this example the representations $\vec{r}_1$ and $\vec{r}_2$ can be chosen so that they agree on the intersection $\Gamma_1 \cap \Gamma_2$. If we apply Stokes' theorem to each piece S_1 and S_2 and add the two identities, we obtain

$$(6.42) \quad \iint_{S_1} (\text{curl } \vec{F}) \cdot \vec{n}_1 dS + \iint_{S_2} (\text{curl } \vec{F}) \cdot \vec{n}_2 dS = \int_{C_1} \vec{F} \cdot d\vec{\rho}_1 + \int_{C_2} \vec{F} \cdot d\vec{\rho}_2,$$



FIGURE 6.17 Extension of Stokes' theorem to a cylinder.

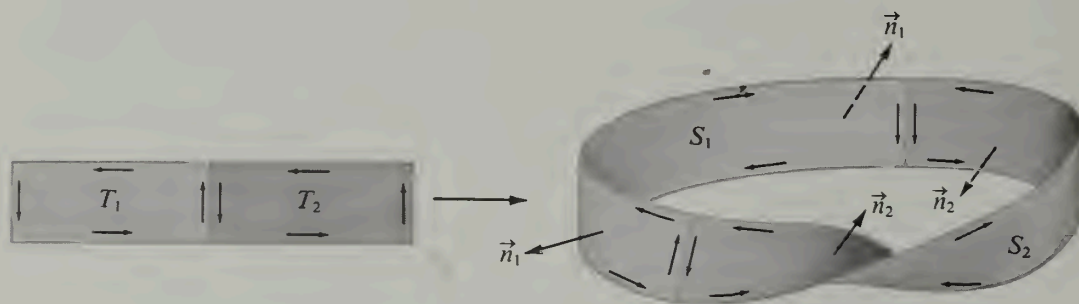


FIGURE 6.18 A Möbius band considered as the union of two simple parametric surfaces. Stokes' theorem does not extend to a Möbius band.

where $\vec{n}_1$ and $\vec{n}_2$ are the normals determined by the fundamental vector products of $\vec{r}_1$ and $\vec{r}_2$, respectively.

Now let $\vec{r}$ denote the mapping of $T_1 \cup T_2$ which agrees with $\vec{r}_1$ on T_1 and with $\vec{r}_2$ on T_2 , and let $\vec{n}$ be the corresponding unit normal determined by the fundamental vector product of $\vec{r}$. Since the normals $\vec{n}_1$ and $\vec{n}_2$ agree in direction on $S_1 \cap S_2$, the unit normal $\vec{n}$ is the same as $\vec{n}_1$ on S_1 and the same as $\vec{n}_2$ on S_2 . Therefore the sum of the surface integrals on the left of (6.42) is equal to

$$\iint_{S_1 \cup S_2} (\text{curl } \vec{F}) \cdot \vec{n} \, dS.$$

For this example, the representations $\vec{r}_1$ and $\vec{r}_2$ can be chosen so that $\vec{p}_1$ and $\vec{p}_2$ determine opposite directions on each arc of the intersection $C_1 \cap C_2$, as indicated by the arrows in Figure 6.17. The two line integrals on the right of (6.42) may be replaced by a sum of line integrals along the two circles C'_1 and C'_2 forming the upper and lower edges of $S_1 \cup S_2$, since the line integrals along each arc of the intersection $C_1 \cap C_2$ cancel. Therefore, Equation (6.42) may be written as

$$(6.43) \quad \iint_{S_1 \cup S_2} (\text{curl } \vec{F}) \cdot \vec{n} \, dS = \int_{C'_1} \vec{F} \cdot d\vec{p}_1 + \int_{C'_2} \vec{F} \cdot d\vec{p}_2,$$

where the line integrals are traversed in the directions inherited from Γ_1 and Γ_2 . The two circles C'_1 and C'_2 are said to form the complete boundary of $S_1 \cup S_2$. Equation (6.43) expresses the surface integral of $(\text{curl } \vec{F}) \cdot \vec{n}$ over $S_1 \cup S_2$ as a line integral over the complete boundary of $S_1 \cup S_2$. This equation is the extension of Stokes' theorem for a cylinder.

Suppose now we apply the same concepts to the surface shown in Figure 6.18. This surface is again the union of two smooth simple parametric surfaces S_1 and S_2 , the images of two adjacent rectangles T_1 and T_2 . This particular surface is called a *Möbius band*[†]; a model can easily be constructed from a long rectangular strip of paper by giving

[†] After A. F. Möbius (1790–1868), a pupil of Gauss. At the age of 26 he was appointed professor of astronomy at Leipzig, a position he held until his death. He made many contributions to celestial mechanics, but his most important researches were in geometry and in the theory of numbers.

one end a half-twist and then fastening the two ends together. We define $\vec{\rho}_1$, $\vec{\rho}_2$, C_1 , and C_2 for the Möbius band as we defined them for the cylinder above. The edge of $S_1 \cup S_2$ in this case is one simple closed curve C' , rather than two. This curve is called the complete boundary of the Möbius band.

If we apply Stokes' theorem to each piece S_1 and S_2 , as we did for the cylinder, we obtain Equation (6.42). But if we try to consolidate the two surface integrals and the two line integrals as we did above, we encounter two difficulties. First, the two normals $\vec{n}_1$ and $\vec{n}_2$ do not agree in direction everywhere on the intersection $C_1 \cap C_2$. (See Figure 6.18.) Therefore we cannot define a normal $\vec{n}$ for the whole surface by taking $\vec{n} = \vec{n}_1$ on S_1 and $\vec{n} = \vec{n}_2$ on S_2 , as we did for the cylinder. This is not serious, however, because we can define $\vec{n}$ to be $\vec{n}_1$ on S_1 and on $C_1 \cap C_2$, and then define $\vec{n}$ to be $\vec{n}_2$ everywhere else. This gives a discontinuous normal, but the discontinuities so introduced form a set of measure zero in the uv -plane and do not affect the existence or the value of the surface integral

$$\iint_{S_1 \cup S_2} (\text{curl } \vec{F}) \cdot \vec{n} \, dS.$$

A more serious difficulty is encountered when we try to consolidate the line integrals. In this example it is not possible to choose the mappings $\vec{r}_1$ and $\vec{r}_2$ in such a way that $\vec{\rho}_1$ and $\vec{\rho}_2$ determine opposite directions on each arc of the intersection $C_1 \cap C_2$. This is illustrated by the arrows in Figure 6.18; one of these arcs is traced twice in the same direction. On this arc the corresponding line integrals will not necessarily cancel as they did for the cylinder. Therefore the sum of the line integrals in (6.42) is not necessarily equal to the line integral over the complete boundary of $S_1 \cup S_2$, and Stokes' theorem cannot be extended to the Möbius band.

Note. The cylinder and the Möbius band are examples of *orientable* and *nonorientable* surfaces, respectively. We shall not attempt to define these terms precisely, but shall mention some of their differences. For an orientable surface $S_1 \cup S_2$ formed from two smooth simple parametric surfaces as described above, the mappings $\vec{r}_1$ and $\vec{r}_2$ can always be chosen so that $\vec{\rho}_1$ and $\vec{\rho}_2$ determine opposite directions on each arc of the intersection $C_1 \cap C_2$. For a nonorientable surface no such choice is possible. For a smooth orientable surface a unit normal vector can be defined in a continuous fashion over the entire surface. For a nonorientable surface no such definition of a normal is possible. A paper model of an orientable surface always has two sides that can be distinguished by painting them with two different colors. Nonorientable surfaces have only one side. For a rigorous discussion of these and other properties of orientable and nonorientable surfaces, see any book on combinatorial topology. Stokes' theorem can be extended to orientable surfaces by a procedure similar to that outlined above for the cylinder.

Another orientable surface is the sphere shown in Figure 6.19. This surface is the union of two simple parametric surfaces (hemispheres) S_1 and S_2 , which we may consider images of a circular disk in the xy -plane under mappings $\vec{r}_1$ and $\vec{r}_2$, respectively. We give $\vec{r}$, $\vec{\rho}_1$, $\vec{\rho}_2$, C_1 , C_2 the same meanings as in the above examples. In this case the curves C_1 and C_2 are completely matched by the mapping $\vec{r}$ (they intersect along the equator), and the surface $S_1 \cup S_2$ is said to be *closed*. Moreover, $\vec{r}_1$ and $\vec{r}_2$ can be chosen so that the directions determined by $\vec{\rho}_1$ and $\vec{\rho}_2$ are opposite on C_1 and C_2 , as suggested by the arrows in Figure 6.19. (This is why $S_1 \cup S_2$ is orientable.) If we apply Stokes' theorem to each hemisphere and add the results we obtain Equation (6.42), as before.

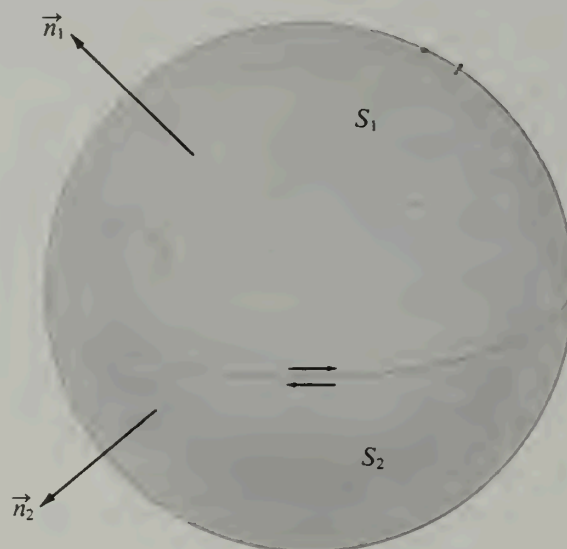


FIGURE 6.19 Extension of Stokes' theorem to a sphere.

The normals $\vec{n}_1$ and $\vec{n}_2$ agree on the intersection $C_1 \cap C_2$, and we can consolidate the integrals over S_1 and S_2 into one integral over the whole sphere. The two line integrals on the right of (6.42) cancel completely, leaving us with the formula

$$\iint_{S_1 \cup S_2} (\text{curl } \vec{F}) \cdot \vec{n} \, dS = 0.$$

This holds not only for a sphere, but for any orientable closed surface.

6.14 The divergence theorem (Gauss' theorem)

Stokes' theorem expresses a relationship between an integral extended over a surface and a line integral taken over the one or more curves forming the boundary of this surface. The divergence theorem expresses a relationship between a triple integral extended over a solid and a surface integral taken over the boundary of this solid.

6- 5 THEOREM. *Divergence theorem.* Let V be a solid in 3-space bounded by an orientable closed surface S , and let $\vec{n}$ be the unit outer normal to S . If $\vec{F}$ is a continuously differentiable vector field defined on V , we have

$$(6.44) \quad \iiint_V (\text{div } \vec{F}) \, dx \, dy \, dz = \iint_S \vec{F} \cdot \vec{n} \, dS.$$

Note. If we express $\vec{F}$ and $\vec{n}$ in terms of their components, say

$$\vec{F}(x, y, z) = P(x, y, z)\vec{i} + Q(x, y, z)\vec{j} + R(x, y, z)\vec{k}$$

and

$$\vec{n} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k},$$

then Equation (6.44) can be written as

$$(6.45) \quad \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz = \iint_S (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS.$$

Proof. It would suffice to establish the three equations

$$\iiint_V \frac{\partial P}{\partial x} dx dy dz = \iint_S P \cos \alpha dS,$$

$$\iiint_V \frac{\partial Q}{\partial y} dx dy dz = \iint_S Q \cos \beta dS,$$

$$\iiint_V \frac{\partial R}{\partial z} dx dy dz = \iint_S R \cos \gamma dS,$$

and add the results to obtain (6.45). We begin with the third of these formulas and prove it for solids of a very special type.

Assume V is a set of points (x, y, z) satisfying a relation of the form

$$g(x, y) \leq z \leq f(x, y) \quad \text{for } (x, y) \text{ in } T,$$

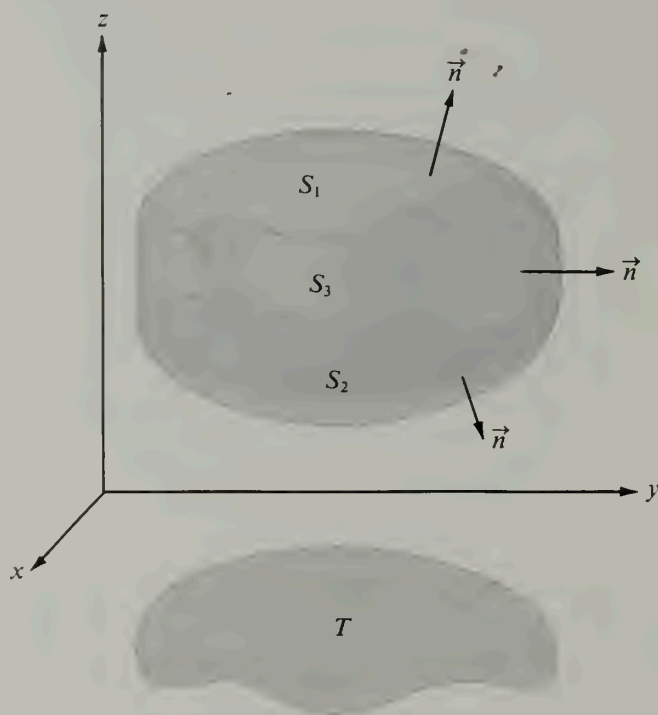
where T is a connected region in the xy -plane, and f and g are continuous functions on T , with $g(x, y) \leq f(x, y)$ for each (x, y) in T . Geometrically, this means that T is the projection of V on the xy -plane. Every line through T parallel to the z -axis intersects the solid V along a line segment connecting the surface $z = g(x, y)$ to the surface $z = f(x, y)$. The boundary surface S consists of an upper cap S_1 , given by the explicit formula $z = f(x, y)$; a lower part S_2 , given by $z = g(x, y)$; and (possibly) a portion S_3 of the cylinder generated by a line moving parallel to the z -axis along the boundary of T . The outer normal to S has a nonnegative z -component on S_1 , has a nonpositive component on S_2 , and is parallel to the xy -plane on S_3 . Solids of this type will be called “ xy -projectable.” (An example is shown in Figure 6.20.) They include all convex solids (for example, solid spheres, ellipsoids, cubes) and many solids that are not convex (for example, solid tori with axes parallel to the z -axis).

The idea of the proof is quite simple. We express the triple integral as a double integral extended over the projection T . Then we show that this double integral has the same value as the surface integral in question. We begin with the formula

$$\iiint_V \frac{\partial R}{\partial z} dx dy dz = \iint_T \left[\int_{g(x, y)}^{f(x, y)} \frac{\partial R}{\partial z} dz \right] dx dy.$$

The one-dimensional integral with respect to z may be evaluated by the second fundamental theorem of calculus, giving us

$$(6.46) \quad \iiint_V \frac{\partial R}{\partial z} dx dy dz = \iint_T \{ R[x, y, f(x, y)] - R[x, y, g(x, y)] \} dx dy.$$

FIGURE 6.20 An example of a solid that is xy -projectable.

For the surface integral we may write

$$(6.47) \quad \iint_S R \cos \gamma \, dS = \iint_{S_1} R \cos \gamma \, dS + \iint_{S_2} R \cos \gamma \, dS + \iint_{S_3} R \cos \gamma \, dS.$$

On S_3 the normal $\vec{n}$ is parallel to the xy -plane, so $\cos \gamma = 0$ and the integral over S_3 is zero. On the surface S_1 we use the representation

$$\vec{r}(x, y) = x\vec{i} + y\vec{j} + f(x, y)\vec{k},$$

and on S_2 we use the representation

$$\vec{r}(x, y) = x\vec{i} + y\vec{j} + g(x, y)\vec{k}.$$

On S_1 the normal $\vec{n}$ has the same direction as the fundamental vector product $\partial\vec{r}/\partial x \times \partial\vec{r}/\partial y$, so we may write [see Equation (6.28)]

$$\iint_{S_1} R \cos \gamma \, dS = \iint_{S_1} R \, dx \, dy = \iint_T R[x, y, f(x, y)] \, dx \, dy.$$

On S_2 the normal $\vec{n}$ has the direction opposite to that of $\partial\vec{r}/\partial x \times \partial\vec{r}/\partial y$ so, by Equation (6.29), we have

$$\iint_{S_2} R \cos \gamma \, dS = - \iint_{S_2} R \, dx \, dy = - \iint_T R[x, y, g(x, y)] \, dx \, dy.$$

Therefore Equation (6.47) becomes

$$\iint_S R \cos \gamma \, dS = \iint_T \{ R[x, y, f(x, y)] - R[x, y, g(x, y)] \} \, dx \, dy.$$

Comparing this with Equation (6.46) we see that

$$\iiint_V \frac{\partial R}{\partial z} \, dx \, dy \, dz = \iint_S R \cos \gamma \, dS.$$

In the foregoing proof the assumption that V is xy -projectable enabled us to express the triple integral over V as a double integral over its projection T in the xy -plane. It is clear that if V is yz -projectable we can use the same type of argument to prove the identity

$$\iiint_V \frac{\partial P}{\partial x} \, dx \, dy \, dz = \iint_S P \cos \alpha \, dS;$$

and if V is xz -projectable we obtain

$$\iiint_V \frac{\partial Q}{\partial y} \, dx \, dy \, dz = \iint_S Q \cos \beta \, dS.$$

Thus we see that the divergence theorem is valid for all solids projectable on all three coordinate planes. In particular the theorem holds for every convex solid.

A solid torus with its axis parallel to the z -axis is xy -projectable but not xz -projectable or yz -projectable. To extend the divergence theorem to such a solid we cut the torus into four equal parts by planes through its axis parallel to the xz - and yz -planes, respectively, and we apply the divergence theorem to each part. The triple integral over the whole torus is the sum of the triple integrals over the four parts. When we add the surface integrals over the four parts we find that the contributions from the faces common to adjacent parts cancel each other, since the outward normals have opposite directions on two such faces. Therefore the sum of the surface integrals over the four parts is equal to the surface integral over the entire torus. This example illustrates how the divergence theorem can be extended to certain nonconvex solids.

6.15 Applications of the divergence theorem

The concepts of curl and divergence of a vector field $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ were introduced in Chapter 5 by the formulas

$$(6.48) \quad \operatorname{div} \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

and

$$(6.49) \quad \operatorname{curl} \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}.$$

To compute $\operatorname{div} \vec{F}$ and $\operatorname{curl} \vec{F}$ from these formulas requires a knowledge of the components of $\vec{F}$. These components, in turn, depend on the choice of coordinate axes in 3-space. A

change in the position of the coordinate axes would mean a change in the components of $\vec{F}$ and, presumably, a corresponding change in the functions $\text{div } \vec{F}$ and $\text{curl } \vec{F}$. With the help of Stokes' theorem and the divergence theorem we can obtain formulas for the divergence and curl that do not involve the components of $\vec{F}$. These formulas show that the curl and divergence represent intrinsic properties of the vector field $\vec{F}$ and do not depend on the particular choice of coordinate axes. We discuss first the formula for the divergence.

6-6 THEOREM. Let $V(t)$ be a sphere of radius $t > 0$ with center at a point $\vec{A}$ in 3-space, and let $S(t)$ denote the boundary of $V(t)$. Let $\vec{F}$ be a vector field that is continuously differentiable on $V(t)$. Then if $|V(t)|$ denotes the volume of $V(t)$, and if $\vec{n}$ denotes the unit outer normal to S , we have

$$(6.50) \quad \text{div } \vec{F}(\vec{A}) = \lim_{t \rightarrow 0} \frac{1}{|V(t)|} \iint_{S(t)} \vec{F} \cdot \vec{n} \, dS.$$

Proof. Let $\phi = \text{div } \vec{F}$. If $\epsilon > 0$ is given we must find a $\delta > 0$ such that

$$(6.51) \quad \left| \phi(\vec{A}) - \frac{1}{|V(t)|} \iint_{S(t)} \vec{F} \cdot \vec{n} \, dS \right| < \epsilon \quad \text{whenever } 0 < t < \delta.$$

Since ϕ is continuous at $\vec{A}$, for the given ϵ there is a neighborhood $N(\vec{A}; h)$ such that

$$|\phi(\vec{X}) - \phi(\vec{A})| < \frac{\epsilon}{2} \quad \text{whenever } \vec{X} \in N(\vec{A}; h).$$

Therefore, if we write $\phi(\vec{A}) = \phi(\vec{X}) + [\phi(\vec{A}) - \phi(\vec{X})]$ and integrate both sides of this equation over a solid sphere $V(t)$ of radius $t < h$, we find

$$\phi(\vec{A}) |V(t)| = \iiint_{V(t)} \phi(\vec{X}) \, dx \, dy \, dz + \iiint_{V(t)} [\phi(\vec{A}) - \phi(\vec{X})] \, dx \, dy \, dz.$$

If we apply the divergence theorem to the first triple integral on the right and then transpose this term to the left, we obtain the relation

$$\left| \phi(\vec{A}) |V(t)| - \iint_{S(t)} \vec{F} \cdot \vec{n} \, dS \right| \leq \iiint_{V(t)} |\phi(\vec{A}) - \phi(\vec{X})| \, dx \, dy \, dz \leq \frac{\epsilon}{2} |V(t)| < \epsilon |V(t)|.$$

When we divide this inequality by $|V(t)|$ we see that (6.51) holds with $\delta = h$. This proves the theorem.

In the foregoing proof we made no special use of the fact that $V(t)$ was a sphere. The same theorem holds true if, instead of spheres, we use any set of solids $V(t)$ for which the divergence theorem is valid, provided these solids contain the point $\vec{A}$ and shrink to $\vec{A}$ as $t \rightarrow 0$. For example, each $V(t)$ could be a cube inscribed in a sphere of radius t about $\vec{A}$; exactly the same proof would apply.

Theorem 6-6 can also be justified on physical grounds. Suppose $\vec{F}$ represents the flux density vector of a steady flow. Then the surface integral $\iint_{S(t)} \vec{F} \cdot \vec{n} \, dS$ measures the total mass of fluid flowing through S in unit time in the direction of $\vec{n}$. The quotient

$\iint_{S(t)} \vec{F} \cdot \vec{n} \, dS / |V(t)|$ represents the mass per unit volume that flows through S in unit time in the direction of $\vec{n}$. As $t \rightarrow 0$, the limit of this quotient is the time rate of change per unit volume at the point $\vec{A}$. As pointed out in Section 5.22, this time rate of change is equal to the divergence of $\vec{F}$ at $\vec{A}$.

In some books on vector analysis, Equation (6.50) is taken as the *definition* of divergence. This makes it possible to assign a physical meaning to the divergence immediately. Also, formula (6.50) does not involve the components of $\vec{F}$. Therefore it holds true in any system of coordinates. If we choose for $V(t)$ a cube with its edges parallel to the xyz -coordinate axes and center at $\vec{A}$, we can use Equation (6.50) to deduce the formula in (6.48) which expresses $\text{div } \vec{F}$ in terms of the components of $\vec{F}$. This procedure is outlined in Exercise 14 of Section 6.16.

There is a formula analogous to (6.50) that is sometimes used as an alternative definition of the curl. It states that

$$(6.52) \quad \text{curl } \vec{F}(\vec{A}) = \lim_{t \rightarrow 0} \frac{1}{|V(t)|} \iint_{S(t)} \vec{n} \times \vec{F} \, dS,$$

where $V(t)$ and $S(t)$ have the same meanings as in Theorem 6-6. The surface integral that appears on the right has a vector-valued integrand. Such integrals have not been defined in this book, so we shall not attempt to prove (6.52).

There is another formula involving the curl that can be deduced from (6.52) or derived independently. It states that

$$(6.53) \quad \vec{n} \cdot \text{curl } \vec{F}(\vec{A}) = \lim_{t \rightarrow 0} \frac{1}{|S(t)|} \oint_{C(t)} \vec{F} \cdot d\vec{r}.$$

In this formula, $S(t)$ is a circular disk of radius t and center at $\vec{A}$, and $|S(t)|$ denotes its area. The vector $\vec{n}$ is a unit normal to $S(t)$, and $\vec{r}$ is the function that traces out $C(t)$ in a direction that appears counterclockwise when viewed from the tip of $\vec{n}$. The vector field $\vec{F}$ is assumed to be continuously differentiable on $S(t)$. A proof of (6.53) can be given by the same method we used to prove (6.50). We let $\phi(\vec{X}) = \vec{n} \cdot \text{curl } \vec{F}(\vec{X})$ and argue as before, except that we use surface integrals instead of triple integrals and Stokes' theorem instead of the divergence theorem.

If $\vec{F}$ is a velocity field, the line integral over $C(t)$ is called the circulation of $\vec{F}$ along $C(t)$; the limit in (6.53) represents the circulation per unit area at the point $\vec{A}$. Thus, $\vec{n} \cdot \text{curl } \vec{F}(\vec{A})$ can be regarded as a "circulation density" of $\vec{F}$ at point $\vec{A}$, with respect to a plane perpendicular to $\vec{n}$ at $\vec{A}$.

When $\vec{n}$ takes the successive values $\vec{i}$, $\vec{j}$, and $\vec{k}$, the dot products $\vec{i} \cdot \text{curl } \vec{F}$, $\vec{j} \cdot \text{curl } \vec{F}$, and $\vec{k} \cdot \text{curl } \vec{F}$ are the components of $\text{curl } \vec{F}$ in rectangular coordinates. When Equation (6.52) is taken as the starting point for the definition of curl, the formula in (6.49) for the rectangular components of $\text{curl } \vec{F}$ can be deduced from (6.53) in exactly this manner.

6.16 Exercises

1. Let S be the surface of the unit cube, $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$, and let $\vec{n}$ be the unit outer normal to S . If $\vec{F}(x, y, z) = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$, use the divergence theorem to evaluate the surface integral $\iint_S \vec{F} \cdot \vec{n} \, dS$. Verify the result by evaluating the surface integral directly.

2. The sphere $x^2 + y^2 + z^2 = 25$ is intersected by the plane $z = 3$. The smaller portion forms a solid V bounded by a closed surface S_0 made up of two parts, a spherical part S_1 and a planar part S_2 . If the unit outer normal of V is $\cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$, compute the value of the surface integral

$$\iint_S (xz \cos \alpha + yz \cos \beta + \cos \gamma) dS$$

if (a) S is the spherical cap S_1 , (b) S is the planar base S_2 , (c) S is the complete boundary S_0 . Solve part (c) by use of the results of parts (a) and (b), and also by use of the divergence theorem.

3. Let $\vec{n} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$ be the unit outer normal to a closed surface S which bounds a homogeneous solid V of the type described in the divergence theorem. Assume that the center of mass $(\bar{x}, \bar{y}, \bar{z})$ and the volume $|V|$ of V are known. Evaluate the following surface integrals in terms of $|V|$ and $\bar{x}, \bar{y}, \bar{z}$.

(a) $\iint_S (x \cos \alpha + y \cos \beta + z \cos \gamma) dS.$

(b) $\iint_S (xz \cos \alpha + 2yz \cos \beta + 3z^2 \cos \gamma) dS.$

(c) $\iint_S (y^2 \cos \alpha + 2xy \cos \beta - xz \cos \gamma) dS.$

(d) Express $\iint_S (x^2 + y^2)(x \vec{i} + y \vec{j}) \cdot \vec{n} dS$ in terms of the volume $|V|$ and a moment of inertia of the solid.

In Exercises 4 through 10, $\partial f / \partial n$ and $\partial g / \partial n$ denote directional derivatives of scalar fields f and g in the direction of the unit outer normal $\vec{n}$ to a closed surface S which bounds a solid V of the type described in the divergence theorem. That is, $\partial f / \partial n = \nabla f \cdot \vec{n}$ and $\partial g / \partial n = \nabla g \cdot \vec{n}$. In each of these exercises prove the given statement. You may assume continuity of all derivatives involved.

4. $\iint_S \frac{\partial f}{\partial n} dS = \iiint_V \nabla^2 f dx dy dz.$

5. $\iint_S \frac{\partial f}{\partial n} dS = 0$ whenever f is harmonic in V .

6. $\iint_S f \frac{\partial g}{\partial n} dS = \iiint_V f \nabla^2 g dx dy dz + \iiint_V \nabla f \cdot \nabla g dx dy dz.$

7. $\iint_S \left(f \frac{\partial g}{\partial n} - g \frac{\partial f}{\partial n} \right) dS = \iiint_V (f \nabla^2 g - g \nabla^2 f) dx dy dz.$

8. $\iint_S f \frac{\partial g}{\partial n} dS = \iint_S g \frac{\partial f}{\partial n} dS$ if both f and g are harmonic in V .

9. $\iint_S f \frac{\partial f}{\partial n} dS = \iiint_V |\nabla f|^2 dx dy dz$ if f is harmonic in V .

10. $\nabla^2 f(\vec{A}) = \lim_{t \rightarrow 0} \frac{1}{|V(t)|} \iint_{S(t)} \frac{\partial f}{\partial n} dS,$ where $V(t)$ is a sphere of radius t with center at $\vec{A}$,

$S(t)$ is the surface of $V(t)$, and $|V(t)|$ is the volume of $V(t)$.

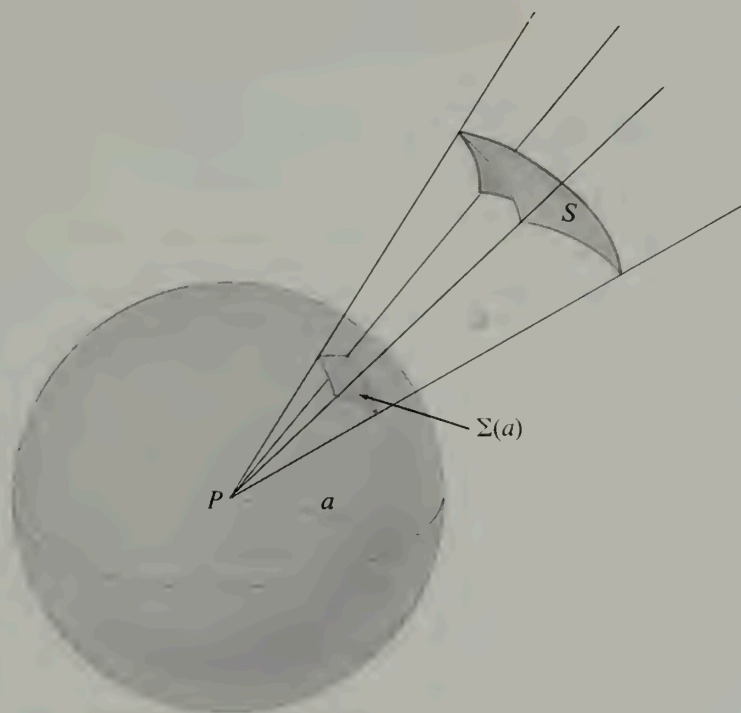


FIGURE 6.21 The solid angle $\Omega(S)$ with vertex P subtended by a surface S . It is measured by the quotient

$$|\Omega(S)| = \frac{\text{area of } \Sigma(a)}{a^2}.$$

11. Let V be a convex region in 3-space whose boundary is a closed surface S and let $\vec{n}$ be the unit outer normal to S . Let $\vec{F}$ and $\vec{G}$ be two continuously differentiable vector fields such that

$$\text{curl } \vec{F} = \text{curl } \vec{G} \quad \text{and} \quad \text{div } \vec{F} = \text{div } \vec{G} \quad \text{everywhere in } V,$$

and such that

$$\vec{G} \cdot \vec{n} = \vec{F} \cdot \vec{n} \quad \text{everywhere on } S.$$

Prove that $\vec{F} = \vec{G}$ everywhere in V [Hint. Let $\vec{H} = \vec{F} - \vec{G}$, find a scalar field f such that $\vec{H} = \nabla f$, and use a suitable identity to prove that $\iiint_V |\nabla f|^2 dx dy dz = 0$. From this deduce that $\vec{H} = \vec{0}$ in V .]

12. Given a vector field $\vec{G}$ and two scalar fields f and g , each continuously differentiable on a convex solid V bounded by a closed surface S . Let $\vec{n}$ denote the unit outer normal to S . Prove that there is at most one vector field $\vec{F}$ satisfying the following three conditions:

$$\text{curl } \vec{F} = \vec{G} \quad \text{and} \quad \text{div } \vec{F} = g \quad \text{in } V, \quad \vec{F} \cdot \vec{n} = f \quad \text{on } S.$$

13. Let S be a smooth parametric surface with the property that each line emanating from a point P intersects S once at most. Let $\Omega(S)$ denote the set of lines emanating from P and passing through S . (See Figure 6.21.) The set $\Omega(S)$ is called the *solid angle* with vertex P subtended by S . Let $\Sigma(a)$ denote the intersection of $\Omega(S)$ with the surface of the sphere of radius a centered at P .

The quotient

$$\frac{\text{area of } \Sigma(a)}{a^2}$$

is denoted by $|\Omega(S)|$ and is used as a measure of the solid angle $\Omega(S)$.

(a) Prove that this quotient is equal to the surface integral

$$\iint_S \frac{\vec{r} \cdot \vec{n}}{r^3} dS,$$

where $\vec{r}$ is the radius vector from P to an arbitrary point of S , and $r = |\vec{r}|$. The vector $\vec{n}$ is the unit normal to S directed away from P . This shows that the quotient for $|\Omega(S)|$ is independent of the radius a . Therefore the solid angle can be measured by the area of the intersection of $\Omega(S)$ and the unit sphere about P . [Hint. Apply the divergence theorem to the portion of $\Omega(S)$ lying between S and $\Sigma(a)$.]

(b) Two planes intersect along the diameter of a sphere with center at P . The angle of intersection is θ , where $0 < \theta < \pi$. Let S denote the smaller portion of the surface of the sphere intercepted by the two planes. Show that $|\Omega(S)| = 2\theta$.

14. Let $V(t)$ denote a cube of edge $2t$ and center at $\vec{A}$, and let $S(t)$ denote the boundary of the cube. Let $\vec{n}$ be the unit outer normal of $S(t)$ and let $|V(t)|$ denote the volume of the cube. For a given vector field $\vec{F}$ that is continuously differentiable at $\vec{A}$, assume that the following limit exists:

$$\lim_{t \rightarrow 0} \frac{1}{|V(t)|} \iint_{S(t)} \vec{F} \cdot \vec{n} dS,$$

and use this limit as the definition of the divergence, $\text{div } \vec{F}(\vec{A})$. Choose xyz -coordinate axes parallel to the edges of $V(t)$ and let P , Q , and R be the components of $\vec{F}$ relative to this coordinate system. Prove that $\text{div } \vec{F}(\vec{A}) = D_1 P(\vec{A}) + D_2 Q(\vec{A}) + D_3 R(\vec{A})$. [Hint. Express the surface integral as a sum of six double integrals taken over the faces of the cube. Then show that $1/|V(t)|$ times the sum of the two double integrals over the faces perpendicular to the z -axis approaches the limit $D_3 R(\vec{A})$ as $t \rightarrow 0$. Argue in a similar way for the remaining terms.]

15. A scalar field ϕ has the properties

$$|\nabla \phi|^2 = 4\phi \quad \text{and} \quad \text{div}(\phi \nabla \phi) = 10\phi.$$

Evaluate the surface integral

$$\iint_S \frac{\partial \phi}{\partial n} dS,$$

where S is the surface of a unit sphere with center at the origin, and $\partial \phi / \partial n$ is the directional derivative of ϕ in the direction of the unit outer normal to S .

LINEAR DIFFERENTIAL EQUATIONS

7.1 Historical introduction

Differential equations is the one branch of mathematics that, perhaps more than any other, has been directly inspired by mechanics, astronomy, and mathematical physics. Its history began in the 17th century when Newton, Leibniz, and the Bernoullis solved some simple differential equations of the first and second order arising from problems in geometry and mechanics. These early discoveries, beginning about 1690, seemed to suggest that the solutions of all differential equations based on geometric and physical problems could be expressed in terms of the familiar elementary functions of calculus. Therefore, much of the early work was aimed at developing ingenious techniques for solving differential equations by elementary means, that is to say, by addition, subtraction, multiplication, division, composition, and integration, applied only a finite number of times to the familiar functions of calculus.

In the introductory account of differential equations presented in Volume I (Chapter 4) we discussed such particular devices as separation of variables and the use of integrating factors. These and other special methods were devised more or less haphazardly before the end of the 17th century. During the 18th century, more systematic procedures were developed, primarily by Euler, Lagrange, and Laplace. It soon became apparent that relatively few differential equations could be solved by elementary means. Little by little, mathematicians began to realize that it was hopeless to try to discover methods for solving *all* differential equations. Instead, they found it more fruitful to ask whether or not a given differential equation has any solution at all and, when it has, to try to deduce properties of the solution from the differential equation itself. Within this framework, mathematicians began to think of differential equations as new sources of functions.

An important phase in the theory developed early in the 19th century, paralleling the general trend toward a more rigorous approach to the calculus. In the 1820's, Cauchy obtained the first "existence theorem" for differential equations. He proved that every first-order equation of the form

$$y' = f(x, y)$$

has a solution whenever the right member, $f(x, y)$, satisfies certain general conditions. One important example is the so-called *Ricatti equation*

$$(7.1) \quad y' = P(x)y^2 + Q(x)y + R(x),$$

where P , Q , and R are given functions. Cauchy's work implies the existence of a solution $y = f(x)$ satisfying Equation (7.1) in any open interval $(-r, r)$ about the origin, provided P , Q , and R have power-series expansions in $(-r, r)$. In 1841 Joseph Liouville (1809–1882) showed that in some cases this solution cannot be obtained by elementary means.

Experience has shown that it is difficult to obtain results of much generality about solutions of differential equations, except for a few types. Among these are the so-called *linear* differential equations which occur in a great variety of scientific problems. First-order linear equations were discussed in complete detail in Chapter 4 of Volume I. They have the form

$$(7.2) \quad y' + P(x)y = Q(x),$$

and their solutions are given by the explicit formula

$$(7.3) \quad y = e^{-\int P(x)dx} \left[\int Q(x)e^{\int P(x)dx} dx + C \right],$$

where C is an arbitrary constant. The Riccati equation, which is *not* of the form of Equation (7.2) (because of the presence of the term involving y^2), is an example of a first-order *nonlinear* equation.

Linear equations of the second order are those of the form

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = R(x).$$

If P_0 , P_1 , P_2 , and R are continuous on some interval I , and if P_0 is never zero on I , an existence theorem (discussed in Section 7.5) guarantees that solutions always exist over the interval I . Nevertheless, there is no general formula analogous to (7.3) for expressing these solutions in terms of P_0 , P_1 , P_2 , and R . Thus, even in this relatively simple generalization of (7.2), the theory is far from complete, except when P_0 , P_1 , and P_2 are constants, in which case the solutions can easily be determined by elementary means. (See Sections 7.6 and 7.8.)

Some second-order linear equations occur repeatedly in many applications; special names have been given to the equations and to their solutions. For example, the equation

$$x^2y'' + xy' + (x^2 - \alpha^2)y = 0,$$

where α is a constant, is called the *Bessel equation* of order α . Some of its solutions are known as *Bessel functions*. This equation is used in problems concerning vibrations of membranes, heat flow in cylinders, and propagation of electric currents in cylindrical conductors. It is named after the German astronomer F. W. Bessel (1784–1846), although the equation appeared earlier in the researches of Daniel Bernoulli (1732) and Euler (1764).

Another important example is the *Legendre equation*,

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0,$$

where α is a constant. This equation occurs in problems of attraction and in heat-flow problems with spherical symmetry. When α is a positive integer the equation has polynomial solutions called *Legendre polynomials*. The Legendre and Bessel equations will be studied in more detail in Sections 7.17 and 7.19.

7.2 Second-order equations solvable by special methods

This section describes some special methods for transforming certain second-order equations to first-order equations. Although these types of equations are encountered only rarely, some of them do occur naturally in problems in geometry and mechanics, so their study is of practical importance. Moreover, a study of these equations at this stage provides an opportunity to review some of the elementary techniques (discussed in Volume I) for solving first-order equations.

Consider a second-order equation of the form

$$y'' = f(x, y, y').$$

When the right member does not explicitly contain y , or when it does not explicitly contain x , the substitution $p = y'$ enables us to reduce the order of the equation. The following examples illustrate the technique.

Equations of the form $y'' = f(x, y')$. The substitution $p = y'$ gives us $p' = y''$, and the differential equation becomes $p' = f(x, p)$, a first-order equation for p as a function of x . If a solution of this equation can be expressed in the form $p = F(x, C_1)$, we may integrate to obtain the following solution of the original equation:

$$y = \int F(x, C_1) dx + C_2.$$

Example 1: $xy'' + y' = x$. We write $y' = p$ and $y'' = p'$, and the equation becomes $xp' + p = x$. This can be solved as a first-order equation, linear in p , or we can observe that the left member is the derivative of xp . In any case, integration yields $xp = x^2/2 + C_1$, or

$$p = \frac{x}{2} + \frac{C_1}{x} \quad \text{if } x \neq 0.$$

Since $p = dy/dx$ we may integrate this equation at once to obtain

$$y = \frac{x^2}{4} + C_1 \log|x| + C_2.$$

If $C_1 \neq 0$ the solution is valid if $x \neq 0$. Otherwise it is valid for all x .

Example 2. A plane curve with equation $y = f(x)$ has the property that its curvature at each point is equal to the cosine of the angle θ between the tangent line and the x -axis, where θ is measured so that $0 < \theta < \pi/2$. Find $f(x)$ if the curve passes through $(0, 1)$ with slope 0.

Solution. The vector $\vec{v} = \vec{i} + y' \vec{j}$ is tangent to the curve at (x, y) , and the cosine of θ is $1/|\vec{v}|$. The curvature at (x, y) is $|y''|/|\vec{v}|^3$. Since $\cos \theta > 0$, the curve satisfies the differential equation

$$\frac{y''}{|\vec{v}|^3} = \frac{1}{|\vec{v}|}, \quad \text{or} \quad y'' = |\vec{v}|^2 = 1 + (y')^2.$$

This is a nonlinear differential equation of second order. The substitution $p = y'$ reduces

it to a first-order equation for p , namely, $p' = 1 + p^2$. As such, it is separable and may be integrated to give

$$\arctan p = x + C_1, \quad \text{or} \quad p = \tan(x + C_1).$$

To make $p = 0$ when $x = 0$ we choose $C_1 = 0$. Therefore the first integration yields $y' = \tan x$. Integrating once more we find $y = -\log |\cos x| + C_2$. Since the curve passes through $(0, 1)$, we must have $C_2 = 1$, so the curve we seek has the equation

$$y = 1 - \log |\cos x|.$$

This formula is valid if x is not an odd integer multiple of $\pi/2$.

Equations of the form $y'' = f(y, y')$. An equation of this form can always be reduced to a pair of first-order differential equations. To determine these equations we assume that on some interval I there is a solution, say $y = Y(x)$, with a differentiable inverse ϕ . Then for x in I we may write $x = \phi(y)$ and

$$y' = Y'(x) = Y'[\phi(y)].$$

Therefore the derivative y' can be expressed in terms of y . If we let $P(y) = Y'[\phi(y)]$ we have

$$(7.4) \quad y' = P(y).$$

By the chain rule we obtain

$$y'' = P'(y)y' = P'(y)P(y).$$

The differential equation $y'' = f(y, y')$ now becomes

$$(7.5) \quad P'(y)P(y) = f[y, P(y)].$$

The two equations (7.4) and (7.5) are necessarily satisfied by any solution of the type described.

Now we may turn the argument around to describe a method for solving the original equation. First we try to find a function P satisfying the first-order equation (7.5). If we find such a P we take y as any solution of the first-order equation

$$\frac{dy}{dx} = P(y).$$

This latter equation is separable and can always be integrated. Because of the way P was chosen the resulting function y has the property that $y'' = f(y, y')$.

Example 3. The method works particularly well for the equation of simple harmonic motion, $y'' + k^2y = 0$, where $k > 0$. In this case Equation (7.5) becomes $P'(y)P(y) = -k^2y$, or

$$\frac{1}{2} \frac{d}{dy} (P^2(y)) = -k^2y.$$

Integrating we obtain $P^2(y) = -k^2y^2 + C$. Since C is a sum of squares we have $C \geq 0$ and we may write $C = k^2A^2$, where A is an arbitrary constant. The equation for P becomes $P^2(y) = k^2(A^2 - y^2)$. If we take the positive square root we find

$$P(y) = k \sqrt{A^2 - y^2}.$$

To determine y we solve the first-order equation

$$\frac{dy}{dx} = P(y) = k \sqrt{A^2 - y^2}.$$

Separating the variables and integrating we find

$$\arcsin \frac{y}{A} = kx + B, \quad \text{or} \quad y = A \sin(kx + B),$$

where B is another arbitrary constant. This particular form of the solution was obtained in Volume I (Section 4.12) by an entirely different method.

Example 4. To solve the nonlinear differential equation

$$y \frac{d^2y}{dx^2} - y^2 \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^2 = 0$$

by this method, we write $y' = P(y)$, $y'' = P'(y)P(y)$, and the equation becomes

$$yP(y)P'(y) - y^2P(y) - P^2(y) = 0.$$

One trivial solution is $P(y) = 0$; this leads to $y = C$. If $P(y)$ and y are nonzero we can rewrite the equation in the form

$$P'(y) - \frac{1}{y} P(y) = y,$$

a first-order linear equation for P . It has the solution $P(y) = y(y + C_1)$. To determine y we solve the first-order equation

$$\frac{dy}{dx} = y(y + C_1).$$

Separating the variables and integrating we obtain

$$C_1 x = \log \left| \frac{y}{y + C_1} \right| + C_2,$$

where C_1 and C_2 are arbitrary constants.

Example 5: Motion of a rocket with variable mass. A rocket is propelled by burning fuel in a combustion chamber, allowing the products of combustion to be expelled backward. Designate the position vector of the rocket at time t by $\vec{r}(t)$, the mass of the rocket (including fuel) by $m(t)$, and the velocity of the exhaust matter, relative to the rocket, by $\vec{c}(t)$. We shall prove that these quantities are related by the equation

$$(7.6) \quad m(t)\vec{r}''(t) - m'(t)\vec{c}(t) = \vec{F}(t),$$

where $\vec{F}(t)$ represents the vector sum of all external forces acting on the rocket at time t . In the examples to be considered here, $m(t)$, $\vec{c}(t)$, and $\vec{F}(t)$ are known or can be prescribed in terms of $\vec{r}(t)$ or its derivative $\vec{r}'(t)$ (the velocity of the rocket). Equation (7.6) then becomes a second-order differential equation for the position function $\vec{r}$.

Equation (7.6) can be written in the form

$$(7.7) \quad m(t)\vec{r}''(t) = m'(t)\vec{c}(t) + \vec{F}(t).$$

The term $m'(t)\check{c}(t)$ is the contribution to the accelerating force on the rocket caused by the thrust developed by the rocket engine; the term $\bar{F}(t)$ is the contribution caused by the external forces. Therefore, to prove (7.7) it suffices to show that *in the absence of external forces* we have

$$(7.8) \quad m(t)\ddot{r}(t) = m'(t)\check{c}(t).$$

For this purpose we consider first a rocket that fires its exhaust matter intermittently, like bullets from a gun. Specifically, we consider a time interval $[t, t + h]$, where h is a small positive number; we assume that some exhaust matter is expelled at time t , and that no further exhaust matter is expelled in the half-open interval $(t, t + h]$. On the basis of this assumption we obtain a formula whose limit, as $h \rightarrow 0$, is Equation (7.8).

Just before the exhaust material is expelled at time t , the rocket has mass $m(t)$ and velocity $\check{v}(t)$. At the end of the time interval $[t, t + h]$ the rocket has mass $m(t + h)$ and velocity $\check{v}(t + h)$. The mass of the expelled matter is $m(t) - m(t + h)$ and its velocity during the interval is $\check{v}(t) + \check{c}(t)$, since $\check{c}(t)$ is the velocity of the exhaust relative to the rocket. Just before the exhaust material is expelled at time t , the rocket is a system with momentum $m(t)\check{v}(t)$. At time $t + h$ this system consists of two parts, a rocket with momentum $m(t + h)\check{v}(t + h)$ and exhaust matter with momentum $[m(t) - m(t + h)][\check{v}(t) + \check{c}(t)]$. The law of conservation of momentum states that the momentum of the new system must be equal to that of the old. Therefore, we have

$$m(t)\check{v}(t) = m(t + h)\check{v}(t + h) + [m(t) - m(t + h)][\check{v}(t) + \check{c}(t)],$$

from which we obtain

$$(7.9) \quad m(t + h)[\check{v}(t + h) - \check{v}(t)] = [m(t + h) - m(t)]\check{c}(t).$$

Dividing by h and letting $h \rightarrow 0$ we find

$$(7.10) \quad m(t)\check{v}'(t) = m'(t)\check{c}(t),$$

which is equivalent to Equation (7.8).

Consider a special case in which the rocket starts from rest with an initial weight of w pounds (including b pounds of fuel) and moves vertically upward along a straight line. Assume the fuel is consumed at a constant rate of k pounds per second and that the products of combustion are discharged directly backward with a constant speed of c feet per second relative to the rocket. Assume the only external force acting on the rocket is the earth's gravitational attraction. We want to know how high the rocket will travel before all its fuel is consumed.

Since the motion is along a straight line we need not use vector notation. Let $r(t)$ denote the altitude at time t . Since all the fuel is consumed when $kt = b$, we restrict t to the interval $0 \leq t \leq b/k$. The only external force acting on the rocket is $-m(t)g$, so Equation (7.6) becomes

$$m(t)r''(t) + m'(t)c = -m(t)g.$$

The weight of the rocket at time t is $w - kt$, and its mass $m(t)$ is $(w - kt)/g$; hence $m'(t) = -k/g$ and we have

$$r''(t) = -g - \frac{m'(t)}{m(t)}c = -g + \frac{kc}{w - kt}.$$

Integrating, we find

$$r'(t) = -gt - c \log \frac{w - kt}{w}.$$

A second integration leads to the relation

$$r(t) = -\frac{1}{2}gt^2 + ct + \frac{c(w - kt)}{k} \log \frac{w - kt}{w}.$$

All the fuel is consumed when $t = b/k$. At that instant the altitude is

$$(7.11) \quad r\left(\frac{b}{k}\right) = -\frac{gb^2}{2k^2} + \frac{cb}{k} + \frac{c(w - b)}{k} \log \frac{w - b}{w}.$$

This formula is valid if $b < w$. For some rockets, the weight of the carrier is negligible compared to the weight of the fuel and it is of interest to consider the limiting case $b = w$. We cannot put $b = w$ in (7.11) because of the presence of the term $\log [(w - b)/w]$. However, if we let $b \rightarrow w$ the last term in (7.11) is an indeterminate form with limit 0. Therefore, when $b \rightarrow w$ the limiting value of the right member of (7.11) is

$$\lim_{b \rightarrow w} r\left(\frac{b}{k}\right) = -\frac{gw^2}{2k^2} + \frac{cw}{k} = -\frac{1}{2}gT^2 + cT,$$

where $T = w/k$ is the time required for the entire weight w to be consumed.

7.3 Exercises

Solve the differential equations in Exercises 1 through 10.

1. $y'' + k^2y^2 = 0, \quad k \neq 0.$

6. $xy'' - (1 + x^2)y' = 4 + 4x^2.$

2. $y'' - k^2y^2 = 0, \quad k \neq 0.$

7. $(1 - x^2)y'' + xy' = x.$

3. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0.$

8. $(1 + x^2)\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = -1.$

4. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1.$

9. $y'' + k^2y^3 = 0, \quad k \neq 0.$

5. $y'' + 2y = 2y^3.$

10. $y'' - k^2y^3 = 0, \quad k \neq 0.$

11. If y is a function of x , the change of variable $x = e^t$ transforms y into a function of t . Use the chain rule to show that

$$x \frac{dy}{dx} = \frac{dy}{dt} \quad \text{and} \quad x^2 \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}.$$

Use the change of variable suggested in Exercise 11 to solve the nonlinear differential equations in Exercises 12 through 15.

12. $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 - y \frac{dy}{dx} = 0.$

[Hint. Multiply by x .]

13. $(x^2 + y^2) \left(y - x \frac{dy}{dx}\right) + x^2y^2 \frac{d^2y}{dx^2} = 0.$

[Hint. Let $y = vx$.]

$$14. \quad x^3 \frac{d^2y}{dx^2} = (x^2 - 2xy) \frac{dy}{dx} + 2y^2 - xy. \quad [\text{Hint. Let } y = vx.]$$

$$15. \quad 2x^2y \frac{d^2y}{dx^2} + 4y^2 - x^2 \left(\frac{dy}{dx} \right)^2 = 2xy \frac{dy}{dx}. \quad [\text{Hint. Let } y = v^2.]$$

16. A plane curve with equation $y = f(x)$ has the property that its curvature at each point is equal to the sine of the angle θ between the tangent line and the x -axis, where θ is measured so that $0 < \theta < \pi/2$. Find $f(x)$ if the curve passes through the origin with slope 1.

17. A duck is swimming along a straight line with a constant speed. A dog, initially at a distance a from the duck, swims toward the duck with a constant speed k times that of the duck, where $k > 1$. (The arc length traversed by the dog in unit time is k times that traversed by the duck.) Place the coordinate axes so the dog is initially at the origin and the path of the duck is along the line $x = a$ in the direction of increasing y . Find a Cartesian equation for the curve of pursuit and show that the dog overtakes the duck at the point $(a, ka/(k^2 - 1))$. What is the curve of pursuit when $k = 1$?

18. A spaceship is returning to earth. Assume that the only external force acting on it is the action of gravity, and that it falls along a straight line toward the center of the earth. The effect of gravity is partly overcome by firing a rocket directly downward. The rocket fuel is consumed at a constant rate of k pounds per second and the exhaust material has a constant speed of c feet per second relative to the rocket. Find a formula for the distance the spaceship falls in time t if it starts from rest at time $t = 0$ with an initial weight of w pounds.

19. A rocket of initial weight w pounds starts from rest in free space (no external forces) and moves along a straight line. The fuel is consumed at a constant rate of k pounds per second and the products of combustion are discharged directly backward at a constant speed of c feet per second relative to the rocket. Find the distance traveled at time t .

20. Solve Exercise 19 if the initial speed of the rocket is v_0 and if the products of combustion are fired at such a speed that the discharged material remains at rest in space.

7.4 Linear differential equations

A differential equation of the form

$$(7.12) \quad P_0(x)y^{(n)} + P_1(x)y^{(n-1)} + \cdots + P_n(x)y = R(x)$$

is said to be a *linear equation of order n* . The functions $P_0, P_1, \dots, P_n$ multiplying the various derivatives of the unknown function y are called the *coefficients* of the equation. In our general discussion of the linear equation we shall assume that all the coefficients are continuous on some interval I . The word "interval" will refer either to a bounded interval of the form $[a, b]$, (a, b) , $[a, b)$, or $(a, b]$, with $a < b$, or to an unbounded interval. Unbounded intervals will be denoted by the symbols $(a, +\infty)$, $(-\infty, a)$, $[a, +\infty)$, $(-\infty, a]$. They are defined as follows:

$$\begin{aligned} (a, +\infty) &= \{x \mid x > a\}, & (-\infty, a) &= \{x \mid x < a\}, \\ [a, +\infty) &= \{x \mid x \geq a\}, & (-\infty, a] &= \{x \mid x \leq a\}. \end{aligned}$$

In addition, it is convenient to refer to the collection of *all* real numbers as the interval $(-\infty, +\infty)$. Thus, when we discuss a linear differential equation or its solution over an interval I , it will be understood that I is one of the nine types just described.

In the differential equation (7.12) the leading coefficient P_0 plays a special role, since it determines the order of the equation. Points at which $P_0(x) = 0$ are called *singular*

points of the equation. The presence of singular points sometimes introduces complications that require special investigation. To avoid these difficulties we assume that the function P_0 is never zero on I . Then we can divide both sides of Equation (7.12) by P_0 and rewrite the differential equation in a form with leading coefficient 1. Therefore, in our general discussion, we shall assume that the differential equation has the form

$$(7.13) \quad y^{(n)} + P_1(x)y^{(n-1)} + \cdots + P_n(x)y = R(x).$$

The discussion of linear equations may be simplified by the use of operator notation. Let $P_1, P_2, \dots, P_n$ be given functions defined on a common interval I . If f is any function with n derivatives on the interval I we may introduce an operator L which transforms f into another function $L(f)$ defined by the equation

$$L(f) = f^{(n)} + P_1 f^{(n-1)} + \cdots + P_n f.$$

The operator L itself is sometimes written symbolically as

$$L = D^n + P_1 D^{n-1} + \cdots + P_n,$$

where D^k denotes the k th derivative operator. In operator notation, the differential equation in (7.13) is written as

$$L(y) = R.$$

It is to be understood that the operator L depends on the coefficients $P_1, P_2, \dots, P_n$.

It is easy to verify that $L(y_1 + y_2) = L(y_1) + L(y_2)$, and that $L(cy) = c L(y)$ for every constant c . Therefore, for every pair of constants c_1 and c_2 we have

$$L(c_1 y_1 + c_2 y_2) = c_1 L(y_1) + c_2 L(y_2).$$

Thus we see that L is a *linear* operator. This is why the equation $L(y) = R$ is referred to as a linear equation. The linearity property can be extended by induction to give the formula

$$L\left(\sum_{k=1}^n c_k y_k\right) = \sum_{k=1}^n c_k L(y_k).$$

With each linear equation $L(y) = R$ we may associate the equation

$$(7.14) \quad L(y) = 0,$$

in which the right-hand side has been replaced by zero. This is called the *homogeneous* or *reduced* equation corresponding to $L(y) = R$. When R is not identically zero, the equation $L(y) = R$ is called the *nonhomogeneous* or *complete* equation associated with $L(y) = 0$. We shall find that we can always solve the nonhomogeneous equation whenever we can solve the corresponding homogeneous equation. Therefore, we begin our study with the homogeneous equation. Since most of the important features of the theory are exhibited in the study of the second-order equation, we discuss this case first.

7.5 Homogeneous linear equations of second order

We begin our study of linear equations of second order with two simple examples.

Example 1. Consider the equation of simple harmonic motion

$$(7.15) \quad y'' + k^2 y = 0,$$

where k is a nonzero constant. Two particular solutions are $y = \cos kx$ and $y = \sin kx$. By linearity, the combination

$$(7.16) \quad y = c_1 \cos kx + c_2 \sin kx$$

is also a solution for every choice of constants c_1 and c_2 . This differential equation was treated in Section 4.12 of Volume I, where we proved that *all* its solutions are given by formula (7.16). The equation was also solved in Section 7.2 of this chapter as an equation with x absent; the solution was expressed in the form $y = A \sin(kx + B)$, where A and B are arbitrary constants. The equivalence of this form of the solution with that in Equation (7.16) was established in Volume I.

In many physical problems it is necessary to select from the family of functions in (7.16) those solutions satisfying one or more additional conditions. These conditions often require the function and its derivative to have prescribed values at a given point. The problem of determining such a solution is called an *initial-value problem*. (This terminology originated in mechanics, where the prescribed values of y and y' represent displacement and velocity at some initial time.) For the differential equation of simple harmonic motion it is easy to verify that every initial-value problem has a solution. That is, for each point x_0 on the real axis and for every choice of real numbers k_0 and k_1 , there is a solution $y = f(x)$ satisfying the conditions

$$(7.17) \quad f(x_0) = k_0, \quad f'(x_0) = k_1.$$

To find such a solution we need only determine constants c_1 and c_2 satisfying the pair of linear algebraic equations

$$(7.18) \quad \begin{aligned} c_1 \cos kx_0 + c_2 \sin kx_0 &= k_0, \\ -kc_1 \sin kx_0 + kc_2 \cos kx_0 &= k_1. \end{aligned}$$

This system can always be solved for c_1 and c_2 if the determinant of the system is nonzero. The determinant in this case is

$$\begin{vmatrix} \cos kx_0 & \sin kx_0 \\ -k \sin kx_0 & k \cos kx_0 \end{vmatrix} = k(\cos^2 kx_0 + \sin^2 kx_0) = k \neq 0.$$

Therefore c_1 and c_2 satisfying (7.18) always exist. The pair c_1, c_2 is uniquely determined as soon as x_0, k_0 , and k_1 are specified, so there is only one solution satisfying the initial conditions (7.17).

Example 2. A related example is the differential equation

$$(7.19) \quad y'' - k^2 y = 0,$$

where k is a nonzero constant. One obvious solution is $y = e^{kx}$, and another is $y = e^{-kx}$. From these we can obtain further solutions by constructing linear combinations of the form

$$(7.20) \quad y = c_1 e^{kx} + c_2 e^{-kx},$$

where c_1 and c_2 are arbitrary constants.† We shall prove now that for each real x_0 a solution $y = f(x)$ exists for which $f(x_0)$ and $f'(x_0)$ have prescribed values. If we want $f(x_0) = k_0$ and $f'(x_0) = k_1$ we must find c_1 and c_2 to satisfy the following pair of linear algebraic equations:

$$\begin{aligned} c_1 e^{kx_0} + c_2 e^{-kx_0} &= k_0, \\ kc_1 e^{kx_0} - kc_2 e^{-kx_0} &= k_1. \end{aligned}$$

Again, this system can always be solved for c_1 and c_2 because the determinant of the system is nonzero; in fact, the determinant is

$$\begin{vmatrix} e^{kx_0} & e^{-kx_0} \\ ke^{kx_0} & -ke^{-kx_0} \end{vmatrix} = -ke^{kx_0} \cdot e^{-kx_0} - ke^{kx_0} \cdot e^{-kx_0} = -2k \neq 0.$$

The pair c_1, c_2 is uniquely determined as soon as x_0, k_0 , and k_1 are specified, so there is only one solution of the form (7.20) that satisfies the initial-value problem $L(y) = 0$, $f(x_0) = k_0, f'(x_0) = k_1$.

The foregoing examples illustrate two fundamental theorems in the theory of second-order linear differential equations—an *existence theorem*, which states that every initial-value problem has a solution, and a *uniqueness theorem*, which states that there is at most one solution to every initial-value problem. These theorems may be stated more explicitly as follows:

7- 1 THEOREM. *Existence theorem for second-order linear equations.* Let P_1 and P_2 be continuous functions on an open interval I , and let L denote the operator

$$(7.21) \quad L(y) = y'' + P_1 y' + P_2 y.$$

If x_0 is a point in I and if k_0 and k_1 are given real numbers, there exists a function f which is a solution of the differential equation $L(y) = 0$ on I and which satisfies the initial conditions

$$f(x_0) = k_0, \quad f'(x_0) = k_1.$$

7- 2 THEOREM. *Uniqueness theorem for second-order linear equations.* Let L be the linear operator of Theorem 7-1, let f and g be two functions that satisfy the differential equation $L(y) = 0$ on an open interval I , and assume that $f(x_0) = g(x_0)$ and $f'(x_0) = g'(x_0)$ for some x_0 in I . Then $f(x) = g(x)$ for all x in I .

Both these theorems will be deduced in Chapter 9 as corollaries of existence and uniqueness theorems concerning more general differential equations. In this chapter we shall use both the existence and uniqueness theorems to deduce properties of the solutions of linear equations.

The next theorem characterizes the set of solutions of a second-order linear equation.

7- 3 THEOREM. Let $L(y) = y'' + P_1 y' + P_2 y$, where P_1 and P_2 are continuous on an open interval I . Let u_1 and u_2 be two nonzero functions satisfying the equation

† It will be shown presently (in Theorem 7-3) that all solutions of (7.19) are included in (7.20).

$L(y) = 0$ on I , and assume that the quotient u_2/u_1 is not constant on I . Then for every choice of constants c_1 and c_2 , the linear combination $c_1u_1 + c_2u_2$ is a solution of $L(y) = 0$ on I . Conversely, if y is a solution of $L(y) = 0$ on I , there exist constants c_1 and c_2 such that

$$(7.22) \quad y = c_1u_1 + c_2u_2.$$

Proof. To verify that $c_1u_1 + c_2u_2$ is a solution for every choice of c_1 and c_2 , we note that $L(u_1) = L(u_2) = 0$, so by linearity we have

$$L(c_1u_1 + c_2u_2) = c_1 L(u_1) + c_2 L(u_2) = c_1 \cdot 0 + c_2 \cdot 0 = 0.$$

It remains to show that *all* solutions are to be found among those in (7.22). We shall deduce this as a consequence of the uniqueness theorem. The idea of the proof is this: Let f be any solution of $L(y) = 0$. Choose a point x_0 in I and consider $f(x_0)$ and $f'(x_0)$. If we can show that constants c_1 and c_2 exist satisfying the pair of equations

$$(7.23) \quad \begin{aligned} c_1u_1(x_0) + c_2u_2(x_0) &= f(x_0), \\ c_1u_1'(x_0) + c_2u_2'(x_0) &= f'(x_0), \end{aligned}$$

then both f and $c_1u_1 + c_2u_2$ have the same value and the same derivative at x_0 . By the uniqueness theorem it follows that $f = c_1u_1 + c_2u_2$, so f is one of the solutions in (7.22).

To show that the system (7.23) can be solved for c_1 and c_2 it suffices to show that there is an x_0 in I for which the determinant of the system is not zero. This determinant is

$$\begin{vmatrix} u_1(x_0) & u_2(x_0) \\ u_1'(x_0) & u_2'(x_0) \end{vmatrix} = u_1(x_0)u_2'(x_0) - u_2(x_0)u_1'(x_0).$$

For an arbitrary x in I define $W(x)$ to be the determinant

$$W(x) = \begin{vmatrix} u_1(x) & u_2(x) \\ u_1'(x) & u_2'(x) \end{vmatrix} = u_1(x)u_2'(x) - u_2(x)u_1'(x).$$

The determinant $W(x)$ is called the *Wronskian* of u_1 and u_2 , after J. M. H. Wronski (1778–1853). We wish to prove that $W(x_0) \neq 0$ for some x_0 in I . We shall prove this by contradiction. Suppose that $W(x)$ is zero for all x in I . Then the quotient u_2/u_1 must be constant on I because the derivative of this quotient is

$$\left(\frac{u_2}{u_1}\right)' = \frac{u_1u_2' - u_2u_1'}{u_1^2} = \frac{W}{u_1^2} = 0.$$

This contradicts the hypothesis that u_2/u_1 is not a constant. Therefore $W(x_0) \neq 0$ for at least one x_0 in I . As observed earlier, this implies the existence of constants c_1 and c_2 satisfying (7.23). Thus, Theorem 7-3 has been deduced as a consequence of the uniqueness theorem. (We remind the reader once more that we have not yet proved the uniqueness theorem.) The uniqueness theorem can also be deduced as a consequence of Theorem 7-3. That is, if it is known that every solution of the equation has the form $c_1u_1 + c_2u_2$, where u_1 and u_2 have nonconstant ratio on I , it can be shown that every initial-value problem has at most one solution on I . (See Exercise 14 of Section 7.7.)

In the course of the foregoing proof we showed that the Wronskian $W(x)$ is different

from 0 for at least one x in I . Actually, it is easy to prove that $W(x) \neq 0$ for all x in I . For this purpose we use the formula

$$(7.24) \quad W(x) = W(c)e^{-\int_c^x P_1(t) dt},$$

which is valid for every choice of c in I . [Equation (7.24) is known as Abel's formula for the Wronskian; a proof is outlined in Exercise 13 of Section 7.7.] When we choose c so that $W(c) \neq 0$ we find that $W(x) \neq 0$ for all x in I .

Since all solutions of the differential equation $L(y) = 0$ are contained in formula (7.22), the linear combination $c_1u_1 + c_2u_2$, with c_1 and c_2 arbitrary constants, is said to be the *general solution* of the equation. Theorem 7-3 tells us that we can find the general solution as soon as we know two particular nonzero solutions whose ratio is not constant.

If the ratio u_2/u_1 is constant, say $u_2/u_1 = c$, we have $u_2 = cu_1$, or $cu_1 - u_2 = 0$. This is a linear combination $c_1u_1 + c_2u_2$ that is identically zero without both c_1 and c_2 being zero. Two functions with this property are said to be *linearly dependent*. More precisely, we have the following definition:

DEFINITION. Two functions u_1 and u_2 are called linearly dependent on an interval I if there exist constants c_1 and c_2 , not both zero, such that $c_1u_1 + c_2u_2 = 0$ on I ; that is, if

$$(7.25) \quad c_1u_1(x) + c_2u_2(x) = 0$$

for every x in I . If no such identity holds except when $c_1 = c_2 = 0$, the functions u_1 and u_2 are said to be linearly independent on I .

Note. Two functions u_1 and u_2 cannot be linearly independent if either of them is zero. For example, if $u_1 = 0$ we have $1 \cdot u_1 + 0 \cdot u_2 = 0$ and Equation (7.25) is satisfied with $c_1 = 1$ and $c_2 = 0$.

In the language of linear independence, Theorem 7-3 states that if u_1 and u_2 are two linearly independent solutions of $L(y) = 0$, the general solution is $c_1u_1 + c_2u_2$. Thus, the problem of solving a homogeneous second-order linear equation is reduced to that of finding two linearly independent solutions. With the help of the existence theorem we can easily prove that such solutions always exist.

7-4 THEOREM. *Existence of linearly independent solutions.* Let $L(y) = y'' + P_1y' + P_2y$, where P_1 and P_2 are continuous on an open interval I . Let u_1 and u_2 be solutions of the equation $L(y) = 0$ on I satisfying the initial conditions

$$(7.26) \quad u_1(x_0) = 1, \quad u_1'(x_0) = 0,$$

and

$$(7.27) \quad u_2(x_0) = 0, \quad u_2'(x_0) = 1,$$

for some point x_0 in I . Then the solutions u_1 and u_2 are linearly independent on I .

Note. The existence of solutions u_1 and u_2 satisfying the initial conditions (7.26) and (7.27) is guaranteed by Theorem 7-1.

Proof. Suppose constants c_1 and c_2 exist such that

$$c_1u_1(x) + c_2u_2(x) = 0$$

for all x in I . We wish to prove that $c_1 = c_2 = 0$. Differentiation of the last equation gives us

$$c_1 u_1'(x) + c_2 u_2'(x) = 0$$

for all x in I . If we substitute $x = x_0$ in the last two equations and use the initial conditions (7.26) and (7.27) we find that $c_1 = c_2 = 0$. This proves that the solutions u_1 and u_2 are linearly independent on I .

Although a pair of linearly independent solutions always exists, it may not be possible to express these solutions in terms of elementary functions. However, when the coefficients of the equation are *constants*, the solutions can always be expressed in terms of exponential functions, sines, cosines, and polynomials. We turn now to this case.†

7.6 Homogeneous linear equations of second order with constant coefficients

A linear second-order homogeneous equation with constant coefficients can be put in the form

$$(7.28) \quad y'' + ay' + by = 0,$$

where a and b are constants. Since the coefficients are continuous everywhere, we seek solutions valid over the entire real axis. In this case we can determine all solutions of the equation without using the existence theorem or the uniqueness theorem stated in Section 7.5. First we shall exhibit two linearly independent solutions u_1 and u_2 . Then we prove that every solution of (7.28) can be expressed as a linear combination of these two.

The two examples $y'' + k^2y = 0$ and $y'' - k^2y = 0$ treated in the foregoing section are of this type. Their general solutions are

$$y = c_1 \cos kx + c_2 \sin kx \quad \text{and} \quad y = c_1 e^{kx} + c_2 e^{-kx},$$

respectively. These examples suggest that we try to find exponential or trigonometric functions as solutions of the general equation. If we try $y = e^{rx}$, where r is a constant, we have $y' = re^{rx}$, $y'' = r^2 e^{rx}$, so Equation (7.28) becomes

$$e^{rx}(r^2 + ar + b) = 0.$$

Since e^{rx} is never zero, the function $y = e^{rx}$ is a solution if and only if r satisfies the quadratic equation

$$r^2 + ar + b = 0.$$

This is called the *characteristic equation* associated with (7.28). Since it is a quadratic equation it has two roots r_1 and r_2 , which may be real or complex,‡ distinct or equal. If we want to confine the discussion to real-valued functions we must distinguish two cases, according to whether the roots r_1 and r_2 are real or complex.

† The homogeneous linear equation with constant coefficients was the first differential equation of a general type to be completely solved. A solution was first published by Euler in 1743. Apart from its historical interest, the linear equation with constant coefficients has important practical applications; because of the simplicity of its solution, it is of theoretical interest as well.

‡ To understand the discussion in this section, the reader must have some familiarity with complex numbers. He should know how to solve a quadratic equation when its roots are not real. A brief account of complex numbers is given in a supplement at the end of this chapter.

CASE 1. Real roots r_1 and r_2 . If the roots of the characteristic equation are real and distinct, the two functions

$$u_1(x) = e^{r_1 x} \quad \text{and} \quad u_2(x) = e^{r_2 x}$$

are linearly independent because their ratio is not constant.

Example 1. The equation $y'' - k^2 y = 0$, with k a nonzero constant, falls in this category. The characteristic equation is $r^2 - k^2 = 0$; its roots are $r_1 = k$, $r_2 = -k$, so $u_1(x) = e^{kx}$ and $u_2(x) = e^{-kx}$ are two linearly independent solutions.

If $k = 0$, the roots r_1 and r_2 are both equal to zero and the corresponding solutions are $u_1(x) = u_2(x) = e^{0 \cdot x} = 1$. Since u_1 and u_2 are identical they are not independent solutions. However, in this case the differential equation simplifies to $y'' = 0$, and it may be integrated at once to give us the general solution

$$y = c_1 + c_2 x.$$

This has the form

$$y = c_1 v_1(x) + c_2 v_2(x),$$

where $v_1(x) = 1$ and $v_2(x) = x$. The functions v_1 and v_2 are linearly independent solutions. The solution $y = v_1(x)$ originated from the characteristic equation; the second solution $y = v_2(x)$ is simply x times $v_1(x)$.

The foregoing example suggests how to treat the case of equal roots in the general equation (7.28). If $r_1 = r_2 = \alpha$, say, the characteristic equation leads to only one solution, $u_1(x) = e^{\alpha x}$. Guided by the foregoing example, we try to find a second solution by multiplying $u_1(x)$ by x . In this case the differential equation has the form $y'' - 2\alpha y' + \alpha^2 y = 0$; it is easy to verify that the equation is satisfied by $u_2(x) = x e^{\alpha x}$. Since $u_2(x)/u_1(x) = x$, the solutions u_1 and u_2 are independent.

CASE 2. Complex roots r_1 and r_2 . For the equation of simple harmonic motion, $y'' + k^2 y = 0$, the characteristic equation $r^2 + k^2 = 0$ has complex roots $r_1 = ik$ and $r_2 = -ik$. For this equation we have seen that the general solution is given by $y = c_1 \cos kx + c_2 \sin kx$. The use of trigonometric functions enables us to treat the general case of complex roots.

We first note that the roots in this case must be complex conjugates. That is, we have

$$r_1 = \alpha + i\beta \quad \text{and} \quad r_2 = \alpha - i\beta,$$

where α and β are real. Since

$$r^2 + ar + b = (r - r_1)(r - r_2) = r^2 - (r_1 + r_2)r + r_1 r_2,$$

the coefficients a and b are related to α and β by the equations

$$(7.29) \quad a = -(r_1 + r_2) = -2\alpha \quad \text{and} \quad b = r_1 r_2 = \alpha^2 + \beta^2.$$

When $\beta = 0$ we know that $y = e^{\alpha x}$ is a solution. When $\alpha = 0$ and $\beta \neq 0$ the equation is that of simple harmonic motion, and two independent solutions are given by $y = \cos \beta x$ and $y = \sin \beta x$. This suggests that for the general case of complex roots we try the functions

$$(7.30) \quad u_1(x) = e^{\alpha x} \cos \beta x \quad \text{and} \quad u_2(x) = e^{\alpha x} \sin \beta x.$$

These are linearly independent because their quotient $u_2(x)/u_1(x) = \tan \beta x$ is not constant. To show that they are indeed solutions we note that

$$u_1'(x) = e^{\alpha x}(-\beta \sin \beta x + \alpha \cos \beta x), \quad u_1''(x) = e^{\alpha x}[(\alpha^2 - \beta^2) \cos \beta x - 2\alpha\beta \sin \beta x],$$

and hence

$$u_1''(x) + au_1'(x) + bu_1(x) = e^{\alpha x}[(\alpha^2 - \beta^2) + \alpha\alpha + b] \cos \beta x + e^{\alpha x}[-2\alpha\beta - a\beta] \sin \beta x.$$

If we use the relations in (7.29) we see that each square bracket vanishes; hence u_1 is a solution. Similarly, we find that u_2 is a solution. Therefore the two functions in (7.30) are linearly independent solutions.

Example 2. The equation $y'' + 2y' + 5y = 0$ has characteristic equation $r^2 + 2r + 5 = 0$. The roots are

$$r_1 = -1 + 2i, \quad r_2 = -1 - 2i.$$

Therefore the independent solutions in (7.30) are given by

$$u_1(x) = e^{-x} \cos 2x, \quad u_2(x) = e^{-x} \sin 2x.$$

The results of this section are summarized in part (a) of the following theorem.

7- 5 THEOREM. (a) Every homogeneous second-order linear equation

$$y'' + ay' + by = 0$$

with constant coefficients has linearly independent solutions u_1 and u_2 which may be determined as follows: Let r_1 and r_2 be the roots of the corresponding characteristic equation

$$r^2 + ar + b = 0.$$

If r_1 and r_2 are real and distinct we may take

$$(7.31) \quad u_1(x) = e^{r_1 x}, \quad u_2(x) = e^{r_2 x}.$$

If the roots are real and equal, say $r_1 = r_2 = \alpha$, we may take

$$u_1(x) = e^{\alpha x}, \quad u_2(x) = xe^{\alpha x}.$$

If the roots are complex, with $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$, we may take

$$(7.32) \quad u_1(x) = e^{\alpha x} \cos \beta x, \quad u_2(x) = e^{\alpha x} \sin \beta x.$$

(b) Let u_1 and u_2 be the solutions described in part (a). Then every solution of $y'' + ay' + by = 0$ has the form

$$y = c_1 u_1(x) + c_2 u_2(x)$$

where c_1 and c_2 are constants.

Proof. Part (a) has already been proved. To prove part (b), let y be any solution of $L(y) = 0$, where $L(y) = y'' + ay' + by$. Let $v = y/u_1$, where u_1 is the solution in part (a). (The function v is defined at those points at which u_1 does not vanish.) Then we have

$$y = u_1 v, \quad y' = u_1 v' + u_1' v, \quad y'' = u_1 v'' + 2u_1' v' + u_1'' v,$$

and the differential equation $L(y) = 0$ becomes

$$v(u_1'' + au_1' + bu_1) + v'(au_1 + 2u_1') + v''u_1 = 0.$$

The coefficient of v is zero since $L(u_1) = 0$. Therefore v must satisfy the differential equation

$$v'' + v'(a + 2u_1'/u_1) = 0.$$

This is a first-order linear equation for v' ; when we solve it we find that v' has the form

$$v'(x) = Ae^{-ax}/[u_1(x)]^2$$

for some constant A . Now $-a = r_1 + r_2$, where r_1 and r_2 are the roots of the characteristic equation $r^2 + ar + b = 0$. Therefore the formula for $v'(x)$ becomes

$$v'(x) = A \frac{e^{(r_1+r_2)x}}{[u_1(x)]^2}.$$

From the formulas for u_1 and u_2 given in part (a) it is easy to verify that

$$\frac{e^{(r_1+r_2)x}}{[u_1(x)]^2} = C \frac{d}{dx} \left(\frac{u_2(x)}{u_1(x)} \right)$$

for some constant C (depending on r_1 and r_2). Therefore we may integrate the formula for $v'(x)$ to obtain

$$v(x) = c_2 \frac{u_2(x)}{u_1(x)} + c_1,$$

where c_1 and c_2 are constants. Multiplying by $u_1(x)$ we find that $y = c_1 u_1(x) + c_2 u_2(x)$, as required. This completes the proof of part (b).

Note. There is a more elegant way to determine the solutions u_1 and u_2 without distinguishing between real and complex roots of the characteristic equation. The definition of the exponential function can be extended so that $e^{r_1 x}$ and $e^{r_2 x}$ are meaningful when r_1 and r_2 are complex numbers. This extension can be made in such a way that the familiar properties of exponentials carry over to the complex case. The analysis given above for real distinct roots r_1 and r_2 remains valid for complex r_1 and r_2 as well. The details of this procedure are outlined in Section 7.24. It is shown there that the definition of the complex exponential function implies the relation

$$e^{(\alpha + i\beta)x} = e^{\alpha x} e^{i\beta x} = e^{\alpha x} (\cos \beta x + i \sin \beta x);$$

this accounts for the appearance of the trigonometric functions in (7.32).

7.7 Exercises

Solve the differential equations in Exercises 1 through 6.

1. $y'' - 2y' + 3y = 0.$

4. $y'' - 2y' + 5y = 0.$

2. $y'' + 2y' - 3y = 0.$

5. $y'' + 2y' + y = 0.$

3. $y'' - 2y' + 2y = 0.$

6. $y'' - 2y' + y = 0.$

In Exercises 7 through 10, find the particular solution satisfying the given initial conditions.

7. $2y'' + 3y' = 0$, with $y = 1$ and $y' = 1$ when $x = 0$.

8. $y'' + 25y = 0$, with $y = -1$ and $y' = 0$ when $x = 3$.

9. $y'' - 4y' - y = 0$, with $y = 2$ and $y' = -1$ when $x = 1$.

10. $y'' + 4y' + 5y = 0$, with $y = 2$ and $y' = y''$ when $x = 0$.

11. An integral curve $y = u(x)$ of the differential equation $y'' - 3y' - 4y = 0$ intersects an integral curve $y = v(x)$ of the differential equation $y'' + 4y' - 5y = 0$ at the origin. Determine the functions u and v if the two curves have equal slopes at the origin and if

$$\lim_{x \rightarrow \infty} \frac{[v(x)]^4}{u(x)} = \frac{5}{6}.$$

12. An integral curve $y = u(x)$ of the differential equation $y'' - 4y' + 29y = 0$ intersects an integral curve $y = v(x)$ of the differential equation $y'' + 4y' + 13y = 0$ at the origin. The two curves have equal slopes at the origin. Determine u and v if $u'(\pi/2) = 1$.

13. Let W denote the Wronskian of two functions u_1 and u_2 .

(a) Show that $W' = u_1 u_2'' - u_2 u_1''$.

(b) If u_1 and u_2 are solutions of the differential equation $y'' + P_1 y' + P_2 y = 0$ on an interval I , show that their Wronskian satisfies the first-order equation $W' + P_1 W = 0$. Solve this equation and derive Abel's formula for the Wronskian [Equation (7.24)].

14. Let u_1 and u_2 be linearly independent solutions of the differential equation $y'' + P_1(x)y' + P_2(x)y = 0$ on an interval I . In the proof of Theorem 7-3 we used the uniqueness theorem (Theorem 7-2) to prove that every solution of the equation has the form $y = c_1 u_1 + c_2 u_2$. Prove the converse. That is, assume it is known that every solution of this equation has the form $y = c_1 u_1 + c_2 u_2$, and use this to prove the uniqueness theorem.

15. Given that the differential equation $y'' + 4xy' + Q(x)y = 0$ has two solutions of the form $y_1 = u(x)$ and $y_2 = xu(x)$, where $u(0) = 1$. Determine both $u(x)$ and $Q(x)$ explicitly in terms of x .

16. Find all values of the constant k such that the differential equation $y'' + ky = 0$ has a non-zero solution $y = f_k(x)$ for which $f_k(0) = f_k(1) = 0$. For each permissible value of k , determine the corresponding solution $y = f_k(x)$. Consider both positive and negative values of k .

17. In each case, verify that u_1 and u_2 are linearly independent over every interval I , and find a linear differential equation of second order satisfied by u_1 and u_2 .

(a) $u_1(x) = e^x$, $u_2(x) = e^{-x}$.

(b) $u_1(x) = e^{2x}$, $u_2(x) = xe^{2x}$.

(c) $u_1(x) = e^{-x/2} \cos x$, $u_2(x) = e^{-x/2} \sin x$.

(d) $u_1(x) = \sin(2x + 1)$, $u_2(x) = \sin(2x + 2)$.

(e) $u_1(x) = \cosh x$, $u_2(x) = \sinh x$.

18. Let u_1 and u_2 be the linearly independent solutions of the equation $y'' + ay' + by = 0$ given in Theorem 7-5. If r_1 and r_2 are the roots of the characteristic equation $r^2 + ar + b = 0$, show that

$$\frac{e^{(r_1+r_2)x}}{[u_1(x)]^2} = C \frac{d}{dx} \left(\frac{u_2(x)}{u_1(x)} \right)$$

for some constant C .

7.8 Nonhomogeneous linear equations of second order. Variation of parameters

We turn now to a discussion of nonhomogeneous linear equations of second order. Consider the equation

$$(7.33) \quad y'' + P_1 y' + P_2 y = R,$$

where the functions P_1 , P_2 , and R are continuous on an interval I . This equation may be written as $L(y) = R$, where $L(y)$ denotes the left member of (7.33). Suppose y_1 and y_2 are any two solutions of this equation. Since $L(y_1) = L(y_2) = R$, linearity gives us

$$L(y_2 - y_1) = L(y_2) - L(y_1) = R - R = 0,$$

so $y_2 - y_1$ is a solution of the homogeneous equation $L(y) = 0$. Therefore, if u_1 and u_2 are two linearly independent solutions of the homogeneous equation, we must have $y_2 - y_1 = c_1 u_1 + c_2 u_2$, or

$$y_2 = c_1 u_1 + c_2 u_2 + y_1$$

for some choice of constants c_1 and c_2 . This equation must be satisfied by every pair of solutions y_1 and y_2 of the nonhomogeneous equation (7.33). Therefore, if we can determine *one particular solution* y_1 of the nonhomogeneous equation, *all* solutions are contained in the formula

$$(7.34) \quad y = c_1 u_1 + c_2 u_2 + y_1,$$

where c_1 and c_2 are arbitrary constants. Each such y is clearly a solution of (7.33) because $L(c_1 u_1 + c_2 u_2 + y_1) = L(c_1 u_1 + c_2 u_2) + L(y_1) = 0 + R = R$. Since all solutions of (7.33) are found in Equation (7.34), the linear combination $c_1 u_1 + c_2 u_2 + y_1$ is called the general solution of (7.33). Thus, we have proved the following theorem:

7-6 THEOREM. If y_1 is a particular solution of the nonhomogeneous equation $L(y) = R$, the general solution of this equation is obtained by adding to y_1 the general solution of the corresponding homogeneous equation $L(y) = 0$.

To use this theorem in practice we must solve two problems: (1) Find the general solution of the homogeneous equation $L(y) = 0$, and (2) find a particular solution of the nonhomogeneous equation $L(y) = R$. The next theorem shows that we can always solve problem (2) if we can solve problem (1).

7-7 THEOREM. Let u_1 and u_2 be linearly independent solutions of the homogeneous linear equation $L(y) = 0$ on an interval I , where $L(y) = y'' + P_1 y' + P_2 y$. Let $W(x) = u_1(x)u_2'(x) - u_2(x)u_1'(x)$ be the Wronskian of u_1 and u_2 . Then the nonhomogeneous equation $L(y) = R$ has a solution y_1 given by the formula

$$(7.35) \quad y_1(x) = v_1(x)u_1(x) + v_2(x)u_2(x),$$

where

$$(7.36) \quad v_1(x) = - \int u_2(x) \frac{R(x)}{W(x)} dx,$$

and

$$(7.37) \quad v_2(x) = \int u_1(x) \frac{R(x)}{W(x)} dx.$$

Proof. Let us try to find functions v_1 and v_2 such that the combination $y_1 = v_1 u_1 + v_2 u_2$ will satisfy the equation $L(y_1) = R$. We have

$$y_1' = v_1 u_1' + v_2 u_2' + (v_1' u_1 + v_2' u_2),$$

$$y_1'' = v_1 u_1'' + v_2 u_2'' + (v_1' u_1' + v_2' u_2') + (v_1'' u_1 + v_2'' u_2).$$

When we form the linear combination $L(y_1) = y_1'' + P_1 y_1' + P_2 y_1$, the terms involving v_1 and v_2 drop out because of the relations $L(u_1) = L(u_2) = 0$. The remaining terms give us the relation

$$L(y_1) = (v_1' u_1' + v_2' u_2') + (v_1'' u_1 + v_2'' u_2) + P_1 \cdot (v_1' u_1 + v_2' u_2).$$

We want to choose v_1 and v_2 so that $L(y_1) = R$. We can satisfy this equation if we choose v_1 and v_2 so that

$$v_1' u_1 + v_2' u_2 = 0 \quad \text{and} \quad v_1' u_1' + v_2' u_2' = R.$$

This is a pair of algebraic equations for v_1' and v_2' . The determinant of the system is the Wronskian of u_1 and u_2 . Since this is never zero on I , the system has a solution given by

$$v_1' = \frac{\begin{vmatrix} 0 & u_2 \\ R & u_2' \end{vmatrix}}{\begin{vmatrix} u_1 & u_2 \\ u_1' & u_2' \end{vmatrix}} = -u_2 R / W, \quad \text{and} \quad v_2' = \frac{\begin{vmatrix} u_1 & 0 \\ u_1' & R \end{vmatrix}}{\begin{vmatrix} u_1 & u_2 \\ u_1' & u_2' \end{vmatrix}} = u_1 R / W.$$

Integrating these relations we obtain Equations (7.36) and (7.37), respectively, thus completing the proof. The method by which we obtained the solution y_1 is sometimes called *variation of parameters* (or *variation of constants*). It was first used by Johann Bernoulli in 1697 to solve linear equations of first order, and then by Lagrange in 1774 to solve linear equations of second order.

Note. Since the functions v_1 and v_2 in Theorem 7-7 are expressed as indefinite integrals, each of them is determined only to within an additive constant. If we add a constant c_1 to v_1 and a constant c_2 to v_2 we change the function y_1 in (7.35) to a new function $y_2 = y_1 + c_1 u_1 + c_2 u_2$. By linearity, we have

$$L(y_2) = L(y_1) + L(c_1 u_1 + c_2 u_2) = L(y_1),$$

so the new function y_2 is also a particular solution of the nonhomogeneous equation.

Example 1. Find the general solution of the equation

$$y'' + y = \tan x.$$

Solution. We consider an interval in which the right-hand member is continuous, say the interval $[-a, a]$, where $0 < a < \pi/2$. The homogeneous equation has the linearly independent solutions

$$u_1(x) = \cos x, \quad u_2(x) = \sin x.$$

Their Wronskian is

$$W(x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1.$$

Equations (7.36) and (7.37) give us

$$v_1(x) = - \int \sin x \tan x \, dx = \sin x - \log |\sec x + \tan x|,$$

and

$$v_2(x) = \int \cos x \tan x \, dx = \int \sin x \, dx = -\cos x.$$

Therefore a particular solution of the nonhomogeneous equation is given by

$$\begin{aligned} y_1 &= v_1(x)u_1(x) + v_2(x)u_2(x) = \sin x \cos x - \cos x \log |\sec x + \tan x| - \sin x \cos x \\ &= -\cos x \log |\sec x + \tan x|; \end{aligned}$$

by Theorem 7-6, its general solution is

$$y = c_1 \cos x + c_2 \sin x - \cos x \log |\sec x + \tan x|.$$

Although Theorem 7-7 provides a general method for determining a particular solution of $L(y) = R$, special methods are available that are often easier to apply when the function R has certain special forms. This is especially true when the differential equation has constant coefficients. For example, suppose the equation is

$$(7.38) \quad y'' + ay' + by = R,$$

where a and b are constants, and suppose R is a polynomial of degree n . Then if $b \neq 0$ we can always find a polynomial of degree n that satisfies this equation.† We try a polynomial of the form

$$y_1(x) = \sum_{k=0}^n a_k x^k$$

with undetermined coefficients. Substituting in differential equation (7.38) and equating coefficients of like powers of x , we may determine $a_n, a_{n-1}, \dots, a_1, a_0$ in succession. The method is illustrated by the following example.

Example 2. Consider the equation

$$(7.39) \quad y'' + y = x^3.$$

Since the right member is a polynomial of degree 3, and since the coefficient of y is non-zero, we try to find a particular solution of the form $y_1(x) = Ax^3 + Bx^2 + Cx + D$.

† If $b = 0$ and $a \neq 0$ the equation cannot be satisfied by a polynomial of degree n , but it can be satisfied by a polynomial of degree $n + 1$. If both a and b are zero, the equation becomes $y'' = R$; its general solution is a polynomial of degree $n + 2$ obtained by two successive integrations.

Differentiating twice, we find $y_1''(x) = 6Ax + 2B$. The differential equation leads to the relation

$$(6Ax + 2B) + (Ax^3 + Bx^2 + Cx + D) = x^3.$$

Equating coefficients of like powers of x we obtain $A = 1$, $B = 0$, $C = -6$, and $D = 0$, so a particular solution of (7.39) is $y_1(x) = x^3 - 6x$. Since the homogeneous equation is the same as that in Example 1 [with linearly independent solutions $u_1(x) = \cos x$ and $u_2(x) = \sin x$] the general solution of (7.39) is

$$y = c_1 \cos x + c_2 \sin x + x^3 - 6x.$$

It may be of interest to compare this method with variation of parameters. Equations (7.36) and (7.37) give us

$$v_1(x) = - \int x^3 \sin x \, dx = -(3x^3 - 6) \sin x + (x^3 - 6x) \cos x$$

and

$$v_2(x) = \int x^3 \cos x \, dx = (3x^2 - 6) \cos x + (x^3 - 6x) \sin x.$$

When we form the combination $v_1 u_1 + v_2 u_2$ we find the particular solution $y_1(x) = x^3 - 6x$, as before. In this case, the use of variation of parameters required the evaluation of the integrals $\int x^3 \sin x \, dx$ and $\int x^3 \cos x \, dx$. With the method of undetermined coefficients no integration is required.

Similar methods are available when the right-hand member is a finite sum of terms of the form $p(x)e^{\alpha x} \sin \beta x$ or $p(x)e^{\alpha x} \cos \beta x$, where p is a polynomial. Some of the exercises in the next section describe these methods. These equations can also be handled by another method (the method of annihilators) described in Section 7.15.

Example 3. If we express the functions v_1 and v_2 in Theorem 7-7 as definite integrals we may write

$$v_1(x) = - \int_c^x u_2(t) \frac{R(t)}{W(t)} \, dt \quad \text{and} \quad v_2(x) = \int_c^x u_1(t) \frac{R(t)}{W(t)} \, dt,$$

where c is any fixed number in the interval I . When these equations are substituted in (7.35) the formula for y_1 becomes

$$(7.40) \quad y_1(x) = \int_c^x [u_1(t)u_2(x) - u_1(x)u_2(t)] \frac{R(t)}{W(t)} \, dt.$$

If the left member of the differential equation has constant coefficients, Equation (7.40) simplifies to an interesting form. Specifically, if the differential equation in question is

$$y'' + ay' + by = R,$$

where a and b are constants, Equation (7.40) can be written in the form

$$(7.41) \quad y_1(x) = \int_c^x f(x-t)R(t) \, dt,$$

where f is that particular solution of the homogeneous equation which satisfies the conditions

$$f(0) = 0 \quad \text{and} \quad f'(0) = 1.$$

We shall deduce (7.41) from (7.40) for the special case in which the characteristic equation $r^2 + ar + b = 0$ has distinct real roots. The proofs for the remaining cases are requested in Exercise 23 of the next section.

If the roots of the characteristic equation are r_1 and r_2 , the functions

$$u_1(x) = e^{r_1 x} \quad \text{and} \quad u_2(x) = e^{r_2 x}$$

are linearly independent solutions of the homogeneous equation. The Wronskian of these two solutions is

$$W(x) = \begin{vmatrix} e^{r_1 x} & e^{r_2 x} \\ r_1 e^{r_1 x} & r_2 e^{r_2 x} \end{vmatrix} = (r_2 - r_1)e^{(r_1 + r_2)x}.$$

Substituting these relations in Equation (7.40) we obtain

$$y_1(x) = \int_c^x \frac{e^{r_2(x-t)} - e^{r_1(x-t)}}{r_2 - r_1} R(t) dt.$$

This formula has the form of Equation (7.41) with

$$f(t) = \frac{e^{r_2 t} - e^{r_1 t}}{r_2 - r_1}.$$

Since f is a linear combination of u_1 and u_2 it is a particular solution of the homogeneous equation. Moreover, it has the two properties $f(0) = 0$ and $f'(0) = 1$.

Note. Integrals of the form $\int_c^x f(x-t)R(t) dt$ occur frequently in the theory of differential equations and elsewhere. They are known as *convolution integrals*.

Example 4. Let $L(y) = y'' + P_1 y' + P_2 y$. To solve the nonhomogeneous equation $L(y) = R$ by variation of parameters, we need to know two linearly independent solutions of the homogeneous equation. This example shows that if *one* solution u_1 of $L(y) = 0$ is known, and if u_1 is never zero on an interval I , a second solution u_2 of the homogeneous equation is given by the formula

$$(7.42) \quad u_2(x) = u_1(x) \int_c^x \frac{Q(t)}{[u_1(t)]^2} dt,$$

where $Q(x) = e^{-\int P_1(x) dx}$, and c is any point in I . These two solutions are linearly independent on I .

It is a straightforward exercise to show that the function u_2 does, indeed, satisfy $L(y) = 0$; we leave this verification to the reader. To prove that u_1 and u_2 are independent we note that $u_2(x)/u_1(x) = \int_c^x Q(t) [u_1(t)]^{-2} dt$. The ratio $u_2(x)/u_1(x)$ is not a constant because its derivative, $Q(x) [u_1(x)]^{-2}$, is never zero on I (since Q is an exponential).

It is of interest to know how the formula for u_2 might be discovered. Suppose we try to find a function v such that the product $y = u_1 v$ satisfies $L(y) = 0$. Differentiating, we have

$$y' = u_1 v' + u_1' v, \quad y'' = u_1 v'' + 2u_1' v' + u_1'' v,$$

and we find

$$L(y) = u_1 v'' + (2u_1' + P_1 u_1) v' + L(u_1) v$$

Since $L(u_1) = 0$, the function v must satisfy the equation

$$(7.43) \quad u_1 v'' + (2u_1' + P_1 u_1) v' = 0.$$

This is a first-order differential equation for v' . It may be solved as a first-order linear equation [using Equation (7.3)] or as a separable equation. By either method we obtain

$$(7.44) \quad v'(x) = A Q(x)/[u_1(x)]^2,$$

where A is an arbitrary constant. Since we seek only a particular solution we may choose $A = 1$. Integration of (7.44) leads to the formula

$$v(x) = \int_c^x \frac{Q(t)}{[u_1(t)]^2} dt.$$

Multiplying this by $u_1(x)$ we obtain the solution in (7.42). Knowing the two solutions u_1 and u_2 , we can find the general solution of $L(y) = R$ by variation of parameters.

An alternative procedure is to determine v so that the product vu_1 is a solution of the nonhomogeneous equation. Proceeding as above, we find that, in place of (7.43), v must satisfy the equation

$$(7.45) \quad u_1 v'' + (2u_1' + P_1 u_1) v' = R.$$

This may be solved as a first-order linear equation in v' . Knowing v' , we integrate to find v . If we use the general solution of (7.45) in determining v , the corresponding product $y = vu_1$ will be the general solution of $L(y) = R$. Specific examples occur in some of the exercises in the next section.

7.9 Exercises

Find the general solution of each of the differential equations in Exercises 1 through 11. If the solution is not valid over the entire real axis, describe an interval over which it is valid.

1. $y'' - y = x$.

6. $y'' + y' - 6y = 2x^3 + 5x^2 - 7x + 2$.

2. $y'' - 2y' + 3y = x^3$.

7. $y'' + 2y' + y = e^{-x}/x^2$.

3. $y'' - y' = x^2$.

8. $y'' + y = \cot^2 x$.

4. $y'' + y' = x^2 + 2x$.

9. $y'' - y = 2/(1 + e^x)$.

5. $y'' - 5y' + 4y = x^2 - 2x + 1$.

10. $y'' + y' - 2y = e^x/(1 + e^x)$.

11. $y'' + 6y' + 9y = f(x)$, where $f(x) = 1$ for $1 \leq x \leq 2$, and $f(x) = 0$ for all other values of x .

12. Consider the equation $y'' + ay' + by = p(x)e^{mx}$, where a , b , and m are constants, and p is a polynomial of degree n .

(a) Show that the change of variable $y = u(x)e^{mx}$ transforms this equation to

$$u'' + (2m + a)u' + (m^2 + am + b)u = p.$$

(b) Use part (a) to deduce that the equation $y'' + ay' + by = p(x)e^{mx}$ has a solution of the form $y = q(x)e^{mx}$, where q is a polynomial. What is the relation between the degrees of q and of p ?

Solve the equations in Exercises 13 through 16 either by variation of parameters or by the method suggested by Exercise 12.

13. $y'' - 4y = e^{2x}$.

15. $y'' + y' - 2y = e^x + e^{2x}$.

14. $y'' + 2y' = 3xe^x$.

16. $y'' - 2y' + y = x + 2xe^x$.

17. Consider the equation $y'' + ay' + by = p(x)e^{\alpha x} \cos \beta x$, where a , b , α , and β are constants, and p is a polynomial.

(a) Show that there is a particular solution of the form

$$y = e^{\alpha x}[q(x) \cos \beta x + r(x) \sin \beta x],$$

where q and r are polynomials. How are the degrees of q and r related to the degree of p ?

(b) Discuss, in a similar way, the equation $y'' + ay' + by = p(x)e^{\alpha x} \sin \beta x$.

Solve the equations in Exercises 18 through 21 either by variation of parameters or by the method suggested by Exercise 17.

18. $y'' + 4y = 3x \cos x$.

20. $y'' - 3y' = 2e^{2x} \sin x$.

19. $y'' + y = \sin x$.

21. $y'' + y = e^{2x} \cos 3x$.

22. The current $I(t)$ at time t flowing in an electric circuit obeys the differential equation

$$I''(t) + RI'(t) + I(t) = \sin \omega t,$$

where R and ω are positive constants. The solution can be expressed in the form $I(t) = F(t) + A \sin(\omega t + \alpha)$, where $F(t) \rightarrow 0$ as $t \rightarrow +\infty$, and A and α are constants depending on R and ω , with $A > 0$. If there is a value of ω which makes A as large as possible, then $\omega/(2\pi)$ is called a *resonance frequency* of the circuit.

(a) Find all resonance frequencies when $R = 1$.

(b) Find those values of R for which the circuit will have a resonance frequency.

23. If $L(y) = y'' + ay' + by$, where a and b are constants, let f be that particular solution of $L(y) = 0$ satisfying the conditions $f(0) = 0$ and $f'(0) = 1$. Show that a particular solution of $L(y) = R$ is given by the formula

$$y_1(x) = \int_c^x f(x-t)R(t) dt$$

for any choice of c . In particular, if the roots of the characteristic equation are equal, say $r_1 = r_2 = m$, show that the formula for $y_1(x)$ becomes

$$y_1(x) = e^{mx} \int_c^x (x-t)e^{-mt}R(t) dt.$$

24. Find the general solution of the equation

$$xy'' - 2(x+1)y' + (x+2)y = x^3e^{2x}$$

for $x > 0$, given that the homogeneous equation has a solution of the form $y = e^{mx}$.

25. Obtain one nonzero solution by inspection and then find the general solution of the differential equation

$$(y'' - 4y') + x^2(y' - 4y) = 0.$$

26. Find the general solution of the differential equation

$$4x^2y'' + 4xy' - y = 0,$$

given that there is a particular solution of the form $y = x^m$ for $x > 0$.

27. Find a solution of the homogeneous equation by trial, and then find the general solution of the equation

$$x(1-x)y'' - (1-2x)y' + (x^2-3x+1)y = (1-x)^3.$$

28. Find the general solution of the equation

$$(2x - 3x^3)y'' + 4y' + 6xy = 0,$$

given that it has a solution that is a polynomial in x .

29. Find the general solution of the equation

$$x^2(1 - x)y'' + 2x(2 - x)y' + 2(1 + x)y = x^2,$$

given that the homogeneous equation has a solution of the form $y = x^c$.

30. Let $g(x) = \int_1^x e^t/t \, dt$ if $x > 0$. (Do not attempt to evaluate this integral.) Find all values of the constant a such that the function f defined by

$$f(x) = \frac{1}{x} e^{ag(x)}$$

satisfies the linear differential equation

$$x^2y'' + (3x - x^2)y' + (1 - x - e^{2x})y = 0.$$

Use this information to determine the general solution of the equation on the interval $(0, +\infty)$.

7.10 Linear equations of order n

The principal results of the theory of linear equations of order two can be extended to linear equations of order n . This section indicates briefly how the extension takes place for the general linear equation. Section 7.12 treats in somewhat more detail the case of constant coefficients. We begin with the statements of the existence and uniqueness theorems.

7- 8 THEOREM. *Existence theorem for linear equations of order n .* Let $P_1, P_2, \dots, P_n$ be continuous functions on an open interval I , and let

$$L(y) = y^{(n)} + P_1y^{(n-1)} + \dots + P_ny.$$

If x_0 is a point in I and if $k_0, k_1, \dots, k_{n-1}$ are n given real numbers, there exists a function f which is a solution of the differential equation $L(y) = 0$ on I and which satisfies the initial conditions

$$f(x_0) = k_0, \quad f'(x_0) = k_1, \quad \dots, \quad f^{(n-1)}(x_0) = k_{n-1}.$$

7- 9 THEOREM. *Uniqueness theorem for linear equations of order n .* Let L be the linear operator of Theorem 7-8, let f and g be two functions that satisfy the differential equation $L(y) = 0$ on an open interval I , and assume that

$$f(x_0) = g(x_0), \quad f'(x_0) = g'(x_0), \quad \dots, \quad f^{(n-1)}(x_0) = g^{(n-1)}(x_0)$$

for some x_0 in I . Then $f(x) = g(x)$ for all x in I .

Both these theorems will be deduced as corollaries of more general theorems discussed in Chapter 9. We shall use both theorems to deduce properties of solutions of linear equations. As in the second-order case, the concept of linear independence plays a fundamental role. The definition for n functions is as follows:

DEFINITION. A collection of n functions $u_1, u_2, \dots, u_n$ is called linearly dependent on an interval I if there exist constants $c_1, c_2, \dots, c_n$, not all zero, such that

$$\sum_{i=1}^n c_i u_i(x) = 0$$

for every x in I . If no such identity holds except when $c_1 = c_2 = \cdots = c_n = 0$, the functions $u_1, u_2, \dots, u_n$ are said to be linearly independent on I .

Example 1. Let $u_1(x) = e^{r_1 x}$, $u_2(x) = e^{r_2 x}$, $\dots$, $u_n(x) = e^{r_n x}$, where $r_1, r_2, \dots, r_n$ are n distinct real numbers. (Functions of this type occur as solutions of homogeneous linear equations of order n with constant coefficients.) We shall prove that the functions $u_1, u_2, \dots, u_n$ are linearly independent over every interval I . The method of proof is by induction on the number n .

For $n = 2$ the result is obvious. Assume, then, that it is true for p exponential functions, and consider $p + 1$ functions, say $u_1, u_2, \dots, u_{p+1}$. Suppose constants $c_1, c_2, \dots, c_{p+1}$ exist such that

$$(7.46) \quad \sum_{k=1}^{p+1} c_k e^{r_k x} = 0$$

for all x in I . We wish to prove that $c_1 = c_2 = \cdots = c_{p+1} = 0$. If we multiply both sides of (7.46) by $e^{-r_{p+1}x}$ and then differentiate with respect to x , we obtain

$$\sum_{k=1}^p c_k (r_k - r_{p+1}) e^{(r_k - r_{p+1})x} = 0.$$

The p numbers $r_1 - r_{p+1}, r_2 - r_{p+1}, \dots, r_p - r_{p+1}$ are distinct. Therefore, because of the inductive hypothesis, the p exponential functions appearing in this sum are linearly independent on I . Hence we must have

$$c_k (r_k - r_{p+1}) = 0 \quad \text{for } k = 1, 2, \dots, p.$$

But since $r_k \neq r_{p+1}$ for each $k \leq p$, it follows that $c_k = 0$ for $k \leq p$. Using this in (7.46) we find also that $c_{p+1} = 0$. This proves, by induction, that the functions $u_1, u_2, \dots, u_n$ are linearly independent on every interval I .

Example 2. Let $u_1(x) = 1$, $u_2(x) = x$, $\dots$, $u_n(x) = x^{n-1}$. Then $u_1, u_2, \dots, u_n$ are linearly independent on every interval I . This is easily proved by induction on n . Details are requested in Exercise 3 of Section 7.11.

Example 3. If r is a constant, the n functions

$$u_1(x) = e^{rx}, \quad u_2(x) = x e^{rx}, \quad \dots, \quad u_n(x) = x^{n-1} e^{rx}$$

are linearly independent on every interval I . This follows at once from the result of Example 2. If constants $c_1, c_2, \dots, c_n$ exist such that $\sum c_k u_k = 0$ on I , we must have

$$e^{rx} \sum_{k=1}^n c_k x^{k-1} = 0$$

for all x in I . Since e^{rx} is not zero, the polynomial multiplying e^{rx} must be zero for all x in I , and the result of Example 2 shows that $c_1 = c_2 = \cdots = c_n = 0$.

Example 4. Let $r_1, r_2, \dots, r_n$ be n distinct real numbers, and let $Q_1, Q_2, \dots, Q_n$ be n polynomials, none of which is identically zero on an interval I . Then the n functions

$$u_1(x) = Q_1(x)e^{r_1x}, \quad u_2(x) = Q_2(x)e^{r_2x}, \quad \dots, \quad u_n(x) = Q_n(x)e^{r_nx}$$

are linearly independent on I . We can prove this by induction on n . For $n = 1$ or $n = 2$ the result is obvious. Therefore, assume the statement is true for $n = p$ and consider $p + 1$ distinct real numbers $r_1, r_2, \dots, r_{p+1}$ and $p + 1$ polynomials $Q_1, Q_2, \dots, Q_{p+1}$, none of which is identically zero on I . Let $c_1, c_2, \dots, c_{p+1}$ be $p + 1$ constants such that

$$\sum_{k=1}^{p+1} c_k Q_k(x) e^{r_k x} = 0$$

for all x in I . We multiply both sides of this equation by $e^{-r_{p+1}x}$ and rewrite it as

$$(7.47) \quad \sum_{k=1}^p c_k Q_k(x) e^{(r_k - r_{p+1})x} + c_{p+1} Q_{p+1}(x) = 0.$$

Differentiation of this relation gives us

$$\sum_{k=1}^p c_k [Q'_k(x) + Q_k(x)(r_k - r_{p+1})] e^{(r_k - r_{p+1})x} + c_{p+1} Q'_{p+1}(x) = 0.$$

Note that the polynomial multiplying $e^{(r_k - r_{p+1})x}$ has the same degree as Q_k . The degree of this polynomial is preserved under repeated differentiation of the equation. If we differentiate repeatedly until the coefficient of c_{p+1} is identically zero we obtain a relation of the form

$$\sum_{k=1}^p c_k P_k(x) e^{(r_k - r_{p+1})x} = 0,$$

where none of the polynomials P_k is identically zero on I . Since the p numbers $r_k - r_{p+1}$ are distinct, we may apply the induction hypothesis to deduce that $c_1 = c_2 = \dots = c_p = 0$. Substituting these relations in (7.47) and using the fact that Q_{p+1} is not identically zero we see that $c_{p+1} = 0$ as well. This proves, by induction, that the functions $u_1, u_2, \dots, u_n$ are linearly independent on every interval I .

Example 5. Let $m_1, m_2, \dots, m_k$ be k positive integers, let $r_1, r_2, \dots, r_k$ be k distinct real numbers, and let $n = m_1 + \dots + m_k$. For each pair of integers p, q satisfying $1 \leq p \leq k, 1 \leq q \leq m_p$, let

$$u_{q,p}(x) = x^{q-1} e^{r_p x}.$$

For example, when $p = 1$ the corresponding functions are

$$u_{1,1}(x) = e^{r_1 x}, \quad u_{2,1}(x) = x e^{r_1 x}, \quad \dots, \quad u_{m_1,1}(x) = x^{m_1-1} e^{r_1 x}.$$

We shall show that the n functions $u_{q,p}$ so defined are linearly independent over every interval. Consider n constants $c_{q,p}$ such that

$$(7.48) \quad \sum_{p=1}^k \sum_{q=1}^{m_p} c_{q,p} u_{q,p}(x) = 0$$

for all x on I . We wish to prove that all n constants $c_{q,p}$ are zero. Using the definition of $u_{q,p}(x)$ we may write Equation (7.48) in the form

$$(7.49) \quad \sum_{p=1}^k Q_p(x) e^{r_p x} = 0,$$

where each Q_p is a polynomial given by

$$Q_p(x) = \sum_{q=1}^{m_p} c_{q,p} x^{q-1}.$$

Because of the statement proved in Example 4, Equation (7.49) implies that at least one of these polynomials must be identically zero on I , say the polynomial Q_k . Therefore Equation (7.49) becomes

$$\sum_{p=1}^{k-1} Q_p(x) e^{r_p x} = 0$$

for all x in I . Again, by Example 4, at least one of the polynomials Q_p appearing in this sum must be identically zero on I . Continuing in this way we find that every polynomial Q_p is identically zero on I . Hence each coefficient $c_{q,p}$ must be zero.

With the help of the existence theorem we can easily prove that every linear equation of order n with continuous coefficients on an interval I has n linearly independent solutions on I . The proof is similar to that of Theorem 7-4 and is requested in Exercise 7 of Section 7.11. The extension of Theorem 7-3 takes the following form:

7-10 THEOREM. Let $P_1, P_2, \dots, P_n$ be continuous on an open interval I and let $L(y) = y^{(n)} + P_1 y^{(n-1)} + \dots + P_n y$. If $u_1, u_2, \dots, u_n$ are n linearly independent solutions of the equation $L(y) = 0$ on the interval I , then every solution of $L(y) = 0$ on I can be expressed in the form

$$(7.50) \quad y = \sum_{k=1}^n c_k u_k(x),$$

where $c_1, c_2, \dots, c_n$ are constants.

Note. Since all solutions of the differential equation are contained in (7.50), the linear combination $\sum c_i u_i$ (with arbitrary constants $c_1, c_2, \dots, c_n$) is called the *general solution* of the equation.

Proof. A proof of this theorem can be given by the same method used to prove Theorem 7-3. We shall carry out the details only for the case $n = 3$; the argument for $n > 3$ is entirely analogous.

In the proof of Theorem 7-3 we used the fact that the Wronskian of two independent solutions of a second-order equation is not identically zero. In this proof we need to know that the Wronskian of u_1, u_2, u_3 is not identically zero on I . The Wronskian of these functions is defined to be the determinant

$$W(x) = \begin{vmatrix} u_1(x) & u_2(x) & u_3(x) \\ u_1'(x) & u_2'(x) & u_3'(x) \\ u_1''(x) & u_2''(x) & u_3''(x) \end{vmatrix}.$$

We wish to prove that $W(x) \neq 0$ for some x in I . We shall do this by contradiction. Suppose $W(x) = 0$ for all x in I . For a fixed x in I , say $x = t$, consider the system of linear algebraic equations

$$\begin{aligned}a_1 u_1(t) + a_2 u_2(t) + a_3 u_3(t) &= 0 \\a_1 u_1'(t) + a_2 u_2'(t) + a_3 u_3'(t) &= 0 \\a_1 u_1''(t) + a_2 u_2''(t) + a_3 u_3''(t) &= 0,\end{aligned}$$

where a_1, a_2, a_3 are considered as “unknowns.” The determinant of this system is the Wronskian $W(t)$. Since this determinant is zero, the system has infinitely many solutions; in particular there is a solution with a_1, a_2, a_3 not all zero. (See Section 5.17 of Volume I.) Choose such a triple of numbers a_1, a_2, a_3 , and let g be the function defined by the equation

$$g(x) = a_1 u_1(x) + a_2 u_2(x) + a_3 u_3(x).$$

This g is a solution of the differential equation $L(y) = 0$ such that g and its first two derivatives vanish at the point $x = t$ (because of the way the a 's were chosen). By the uniqueness theorem it follows that g must be identically zero on I . Hence we have

$$a_1 u_1(x) + a_2 u_2(x) + a_3 u_3(x) = 0$$

for all x in I , contradicting the fact that u_1, u_2, u_3 are linearly independent on I . This contradiction shows that the Wronskian is not identically zero on I .

Now it is easy to carry out the proof of Theorem 7-10. First we note that, because of linearity, the function given by Equation (7.50) is a solution of the differential equation $L(y) = 0$. Let f be any solution of this equation. Choose a point x_0 in I such that $W(x_0) \neq 0$, and consider $f(x_0)$, $f'(x_0)$, and $f''(x_0)$. Since $W(x_0) \neq 0$, there exist constants c_1, c_2, c_3 satisfying the three linear algebraic equations

$$\begin{aligned}c_1 u_1(x_0) + c_2 u_2(x_0) + c_3 u_3(x_0) &= f(x_0) \\c_1 u_1'(x_0) + c_2 u_2'(x_0) + c_3 u_3'(x_0) &= f'(x_0) \\c_1 u_1''(x_0) + c_2 u_2''(x_0) + c_3 u_3''(x_0) &= f''(x_0).\end{aligned}$$

Therefore both f and $c_1 u_1 + c_2 u_2 + c_3 u_3$ have the same value and the same first and second derivatives at x_0 . By the uniqueness theorem the functions must be identical, so $f = c_1 u_1 + c_2 u_2 + c_3 u_3$. This shows that every solution of $L(y) = 0$ is contained in (7.50), completing the proof of Theorem 7-10 when $n = 3$.

Note. In the course of the foregoing proof we also showed that the Wronskian of three linearly independent solutions of a homogeneous linear equation of order three does not vanish identically on I . Actually, it can be shown that the Wronskian *never* vanishes on I . (See Exercise 8 in the next section.)

Let S denote the set of solutions of a linear equation $L(y) = 0$ on an interval I . If $u_1 \in S$ and $u_2 \in S$ then (by linearity) the linear combination $(c_1 u_1 + c_2 u_2) \in S$ for every choice of constants c_1 and c_2 . A set of functions with this property is called a *linear space* of functions. Theorem 7-10 tells us that every function in S can be expressed as a linear combination of n linearly independent members of S . This is analogous to the fact that every vector in n -space can be expressed as a linear combination of n linearly independent vectors. Some of the terminology of vector spaces is often used in discussions concerning linear spaces of functions. For example, the functions $u_1, u_2, \dots, u_n$ in Theorem 7-10 are said to form a *basis* for the linear space S ; the integer n is called the *dimension* of S .

7.11 Exercises

1. If $u_1, u_2, \dots, u_n$ are linearly independent functions on an interval I , prove that they are also linearly independent on every subinterval of I .
2. For each of the following statements, either give a proof or provide a counterexample.
 - (a) If $u_1, u_2, \dots, u_n$ are linearly independent on an interval I , and if $k < n$, then any k of these functions are linearly independent on I .
 - (b) If $u_1, u_2, \dots, u_n$ are linearly dependent on an interval I , and if $k < n$, then any k of these functions are linearly dependent on I .
3. If $u_k(x) = x^{k-1}$, use mathematical induction or some other method to prove that $u_1, u_2, \dots, u_n$ are linearly independent on every interval.
4. Let $u_k(x) = x^{k-1}f(x)$. Prove that $u_1, u_2, \dots, u_n$ are linearly independent on an interval I if there are at least n points in I at which the function f is not zero.
5. (a) Prove that any three linear polynomials are linearly dependent on $(-\infty, +\infty)$.
 (b) Prove that any four quadratic polynomials are linearly dependent on $(-\infty, +\infty)$.
 (c) Generalize to polynomials of degree n .
6. Given n real numbers $r_1 < r_2 < \dots < r_n$, let $u_k(x) = e^{r_k x}$. In Section 7.10 (Example 1) we used mathematical induction to prove that $u_1, u_2, \dots, u_n$ are linearly independent on every interval. Give an alternative proof by the following method. If constants $c_1, c_2, \dots, c_n$ exist such that $\sum c_k e^{r_k x} = 0$ for all real x , multiply this relation by $e^{-r_n x}$ and then let $x \rightarrow +\infty$ to deduce that $c_n = 0$. Repeat the process.
7. Let $L(y) = y^{(n)} + P_1 y^{(n-1)} + \dots + P_n y$, where $P_1, P_2, \dots, P_n$ are continuous on an interval I . Choose a point x_0 in I . For each $k = 0, 1, 2, \dots, n-1$, let u_k be that solution of $L(y) = 0$ satisfying the initial conditions

$$u_k^{(k)}(x_0) = 1, \quad u_k^{(j)}(x_0) = 0 \quad \text{for } j \neq k.$$

- (a) Prove that the functions $u_0, u_1, \dots, u_{n-1}$ are linearly independent on I .
- (b) Find, in terms of $u_0, u_1, \dots, u_{n-1}$, that particular solution of $L(y) = 0$ which satisfies the initial conditions

$$f(x_0) = k_0, \quad f'(x_0) = k_1, \quad \dots, \quad f^{(n-1)}(x_0) = k_{n-1}.$$

8. Let W denote the Wronskian of three functions u_1, u_2, u_3 having third-order derivatives.
 - (a) Show that the derivative of W is given by the determinant

$$W'(x) = \begin{vmatrix} u_1(x) & u_2(x) & u_3(x) \\ u_1'(x) & u_2'(x) & u_3'(x) \\ u_1''(x) & u_2''(x) & u_3''(x) \end{vmatrix}.$$

In other words, the derivative of W may be obtained by differentiating the elements in the last row of the determinant for W .

- (b) If u_1, u_2, u_3 are solutions of the linear equation

$$y''' + P_1 y'' + P_2 y' + P_3 y = 0$$

on an interval I , show that on the interval I the Wronskian satisfies the first-order linear equation $W' + P_1 W = 0$. Solve this equation to derive Abel's formula for the Wronskian,

$$W(x) = W(c)e^{-\int_c^x P_1(t) dt},$$

valid for each c in I . If there exists at least one point c such that $W(c) \neq 0$, this formula shows that $W(x) \neq 0$ for all x in I . In particular, such a c exists if u_1, u_2, u_3 are linearly independent solutions. (See the proof of Theorem 7-10.) Therefore, the Wronskian of three linearly independent solutions of a homogeneous linear equation of third order is never zero on I .

7.12 Linear equations of order n with constant coefficients

To solve a homogeneous linear differential equation of order n with constant coefficients, say

$$(7.51) \quad y^{(n)} + a_1 y^{(n-1)} + \dots + a_n y = 0,$$

we try to obtain solutions of the form $y = e^{rx}$. Since $y^{(k)} = r^k e^{rx}$, when we substitute in (7.51) we find that such a function will be a solution if and only if r satisfies the polynomial equation

$$r^n + a_1 r^{n-1} + \dots + a_n = 0.$$

This is called the *characteristic equation* of (7.51). The fundamental theorem of algebra tells us that this equation has exactly n roots, $r_1, r_2, \dots, r_n$, where each root is written as often as its multiplicity indicates. The roots may be real or complex. As in the second-order case, the form of the solution depends on the nature of the roots.

Real distinct roots. In this case the corresponding solutions are

$$(7.52) \quad u_1(x) = e^{r_1 x}, \quad u_2(x) = e^{r_2 x}, \quad \dots, \quad u_n(x) = e^{r_n x}.$$

In Example 1 of Section 7.10 we proved that these functions are linearly independent over every interval. Therefore, the general solution of (7.51), valid for all real x , is given by the formula

$$(7.53) \quad y = \sum_{k=1}^n c_k e^{r_k x}.$$

Example 1. $y''' - 7y' + 6 = 0$. The characteristic equation factors as follows:

$$r^3 - 7r + 6 = (r - 1)(r - 2)(r + 3) = 0.$$

The corresponding roots are $r_1 = 1$, $r_2 = 2$, $r_3 = -3$, so the general solution is

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{-3x}.$$

Real roots, some of which are repeated. If all the roots are real but not distinct, the functions in (7.52) are not independent and the formula in (7.53) does not give the general solution. If the distinct roots are $r_1, r_2, \dots, r_k$ and if they occur with respective multiplicities $m_1, m_2, \dots, m_k$, the part of the solution corresponding to the root m_p is given by the sum

$$(7.54) \quad (c_1 + c_2 x + c_3 x^2 + \dots + c_{m_p} x^{m_p-1}) e^{r_p x}.$$

For each $p \leq k$ and each $q \leq m_p$, the function $u_{q,p}(x) = x^{q-1} e^{r_p x}$ is a solution of the differential equation (see Exercise 15 of Section 7.13). In Section 7.10 (Example 5) we proved that the n functions $u_{q,p}$ so obtained are linearly independent on every interval. Therefore the general solution of the equation consists of a sum of k terms of the form (7.54); it is valid for all real x . The following examples illustrate some of the possibilities.

Example 2. $y''' - y'' - 8y' + 12y = 0$. The characteristic equation is

$$r^3 - r^2 - 8r + 12 = (r - 2)^2(r + 3) = 0.$$

The roots of this equation are 2, 2, -3 . The part of the solution corresponding to the double root 2 is $(c_1 + c_2x)e^{2x}$; the part corresponding to the simple root -3 is c_3e^{-3x} . Therefore the general solution is

$$y = (c_1 + c_2x)e^{2x} + c_3e^{-3x}.$$

Example 3. $y^{(6)} + 2y^{(5)} - 2y''' - y'' = 0$. The characteristic equation is

$$r^6 + 2r^5 - 2r^3 - r^2 = r^2(r - 1)(r + 1)^3 = 0.$$

The roots are 0, 0, 1, -1 , -1 , -1 . The part of the solution corresponding to the double root 0 is $c_1 + c_2x$; the part corresponding to the simple root 1 is c_3e^x ; and the part corresponding to the triple root -1 is $(c_4 + c_5x + c_6x^2)e^{-x}$. Therefore the general solution is

$$y = c_1 + c_2x + c_3e^x + (c_4 + c_5x + c_6x^2)e^{-x}.$$

Complex roots. If complex exponentials are used, there is no need to distinguish between real and complex roots of the characteristic equation. If real-valued solutions are desired, we introduce trigonometric functions as in the second-order case. Complex roots occur in conjugate pairs; the part of the solution corresponding to each pair $\alpha + i\beta$, $\alpha - i\beta$ has the form

$$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x).$$

If such a pair occurs with multiplicity m , each multiplier c_1 and c_2 must be replaced by a polynomial of degree $m - 1$. The following examples illustrate some of the possibilities.

Example 4. $y''' - 4y'' + 13y' = 0$. The characteristic equation, $r^3 - 4r^2 + 13r = 0$, has the roots 0, $2 \pm 3i$; the general solution is

$$y = c_1 + e^{2x}(c_2 \cos 3x + c_3 \sin 3x).$$

Example 5. $y''' - 2y'' + 4y' - 8y = 0$. The characteristic equation is

$$r^3 - 2r^2 + 4r - 8 = (r - 2)(r^2 + 4) = 0;$$

its roots are 2, $2i$, $-2i$, so the general solution of the differential equation is

$$y = c_1e^{2x} + c_2 \cos 2x + c_3 \sin 2x.$$

Example 6. $y^{(5)} - 9y^{(4)} + 34y''' - 66y'' + 65y' - 25y = 0$. The characteristic equation can be written as

$$(r - 1)(r^2 - 4r + 5)^2 = 0;$$

its roots are 1, $2 \pm i$, $2 \pm i$, so the general solution of the differential equation is

$$y = c_1e^x + e^{2x}[(c_2 + c_3x) \cos x + (c_4 + c_5x) \sin x].$$

7.13 Exercises

Find the general solution of each of the differential equations in Exercises 1 through 12.

1. $y''' - 2y'' - 3y' = 0$.
2. $y''' - y' = 0$.
3. $y''' + 4y'' + 4y' = 0$.
4. $y''' - 3y'' + 3y' - y = 0$.
5. $y^{(4)} + 4y''' + 6y'' + 4y' + y = 0$.
6. $y^{(4)} - 16y = 0$.
7. $y^{(4)} + 16y^3 = 0$.
8. $y''' - y = 0$.
9. $y^{(4)} + 4y''' + 8y'' + 8y' + 4y = 0$.
10. $y^{(4)} + 2y'' + y = 0$.
11. $y^{(6)} + 4y^{(4)} + 4y'' = 0$.
12. $y^{(6)} + 8y^{(4)} + 16y'' = 0$.

13. If m is a positive constant, find that particular solution $y = f(x)$ of the differential equation

$$y''' - my'' + m^2y' - m^3 = 0$$

which satisfies the conditions $f(0) = f'(0) = 0$, $f''(1) = 1$.

14. A linear differential equation with constant coefficients has characteristic equation $f(r) = 0$. If all the roots of the characteristic equation are negative, prove that every solution of the differential equation approaches zero as $x \rightarrow +\infty$. What can you conclude about the behavior of all solutions on the interval $[0, +\infty)$ if all the roots of the characteristic equation are nonpositive?

15. If $L(y) = y^{(n)} + a_1y^{(n-1)} + \cdots + a_ny$, where the a_i are constants, let $f(r) = r^n + a_1r^{n-1} + \cdots + a_n$.

(a) If α is a constant and if u has n derivatives, show that

$$L[e^{\alpha x}u(x)] = e^{\alpha x} \sum_{k=0}^n \frac{f^{(k)}(\alpha)}{k!} u^{(k)}(x).$$

[Hint. Use Leibniz's rule for the k th derivative of a product (see page 147 in Volume I).]

(b) If α is a root of multiplicity m of the characteristic equation $f(r) = 0$, show that $y = e^{\alpha x}u(x)$ will satisfy the differential equation $L(y) = 0$ if u is a polynomial of degree $m - 1$.

16. In each case, find a linear differential equation with constant coefficients satisfied by all the given functions.

- (a) $u_1(x) = e^x$, $u_2(x) = e^{-x}$, $u_3(x) = e^{2x}$, $u_4(x) = e^{-2x}$.
- (b) $u_1(x) = e^{-2x}$, $u_2(x) = xe^{-2x}$, $u_3(x) = x^2e^{-2x}$.
- (c) $u_1(x) = 1$, $u_2(x) = x$, $u_3(x) = e^x$, $u_4(x) = xe^x$.
- (d) $u_1(x) = x$, $u_2(x) = e^x$, $u_3(x) = xe^x$.
- (e) $u_1(x) = x^2$, $u_2(x) = e^x$, $u_3(x) = xe^x$.
- (f) $u_1(x) = e^{-2x}\cos 3x$, $u_2(x) = e^{-2x}\sin 3x$, $u_3(x) = e^{-2x}$, $u_4(x) = xe^{-2x}$.
- (g) $u_1(x) = \cosh x$, $u_2(x) = \sinh x$, $u_3(x) = x \cosh x$, $u_4(x) = x \sinh x$.
- (h) $u_1(x) = \cosh x \sin x$, $u_2(x) = \sinh x \cos x$, $u_3(x) = x$.

7.14 Nonhomogeneous linear equations of order n

The theory for nonhomogeneous equations is related to that of homogeneous equations exactly as in the second-order case. This relation is provided by the following theorem which extends Theorem 7-6 to n th order equations.

7-11 THEOREM. Let $P_1, P_2, \dots, P_n$, and R be functions continuous on an open interval I , and let

$$L(y) = y^{(n)} + P_1y^{(n-1)} + \cdots + P_ny.$$

If $u_1, u_2, \dots, u_n$ are n linearly independent solutions of the homogeneous equation $L(y) = 0$ on I , and if y_1 is any particular solution of the nonhomogeneous equation $L(y) = R$, then every solution of the nonhomogeneous equation can be expressed in the form

$$(7.55) \quad y = y_1 + \sum_{k=1}^n c_k u_k,$$

where $c_1, c_2, \dots, c_n$ are constants.

Note. For every choice of constants $c_1, c_2, \dots, c_n$ the function y determined by (7.55) is a solution of $L(y) = R$, since

$$L(y_1 + \sum_{k=1}^n c_k u_k) = L(y_1) + \sum_{k=1}^n c_k L(u_k) = L(y_1) = R.$$

Since all solutions of $L(y) = R$ are found in (7.55), the sum on the right of (7.55) (with arbitrary constants $c_1, c_2, \dots, c_n$) is called the *general solution* of the nonhomogeneous equation. Theorem 7-11 states that the general solution of the nonhomogeneous equation is obtained by adding to y_1 the general solution of the homogeneous equation.

Proof. Let y_2 be any solution of $L(y) = R$. Since we also have $L(y_1) = R$, linearity gives us $L(y_2 - y_1) = L(y_2) - L(y_1) = 0$, so $y_2 - y_1$ is a solution of the homogeneous equation. By Theorem 7-10 there exist constants $c_1, c_2, \dots, c_n$ such that

$$y_2 - y_1 = \sum_{k=1}^n c_k u_k.$$

Therefore, y_2 is one of the solutions in Equation (7.55). This completes the proof.

We turn now to the problem of determining one particular solution y_1 of the nonhomogeneous equation. If we know n linearly independent solutions $u_1, u_2, \dots, u_n$ of the homogeneous equation on an interval I , we can always find a particular solution of the nonhomogeneous equation by an extension of the method of variation of parameters. The method will be described in detail for $n = 3$; its extension to general n will then be clear.

The idea of the method is to determine functions v_1, v_2, v_3 in such a way that the sum

$$(7.56) \quad y_1 = \sum_{k=1}^3 u_k v_k$$

will satisfy $L(y) = R$. Differentiating (7.56) we obtain

$$y_1' = \sum_{k=1}^3 u_k' v_k + \sum_{k=1}^3 u_k v_k'.$$

We have three v 's to determine, so we should be able to put three conditions on them. If we make one condition be that the second sum should vanish in the foregoing equation, the formula for y_1' simplifies to

$$y_1' = \sum_{k=1}^3 u_k' v_k.$$

Differentiating this relation we find

$$(7.57) \quad y_1' = \sum_{k=1}^3 u_k'' v_k + \sum_{k=1}^3 u_k' v_k'.$$

If we can choose v_1, v_2, v_3 so as to make the second sum in (7.57) vanish also, the formula for y_1' simplifies to

$$y_1' = \sum_{k=1}^3 u_k'' v_k.$$

Differentiating once more we obtain

$$y_1'' = \sum_{k=1}^3 u_k''' v_k + \sum_{k=1}^3 u_k'' v_k'.$$

In this formula we try to make the second sum equal to R . If we can choose v_1, v_2, v_3 so that all the conditions stated are satisfied, we find

$$\begin{aligned} L(y_1) &= y_1''' + P_1 y_1'' + P_2 y_1' + P_3 y_1 \\ &= \left(\sum_{k=1}^3 u_k''' v_k + R \right) + P_1 \sum_{k=1}^3 u_k'' v_k + P_2 \sum_{k=1}^3 u_k' v_k + P_3 \sum_{k=1}^3 u_k v_k \\ &= \sum_{k=1}^3 (u_k''' + P_1 u_k'' + P_2 u_k' + P_3 u_k) v_k + R = \sum_{k=1}^3 L(u_k) v_k + R. \end{aligned}$$

Since $L(u_1) = L(u_2) = L(u_3) = 0$, the right member of the last equation reduces to R , and we see that y_1 satisfies $L(y_1) = R$.

To complete the proof we must show that v_1, v_2, v_3 can always be chosen so as to satisfy the conditions imposed on them. These conditions may be expressed as a system of linear algebraic equations for the derivatives v_1', v_2', v_3' :

$$(7.58) \quad \sum_{k=1}^3 u_k v_k' = 0, \quad \sum_{k=1}^3 u_k' v_k = 0, \quad \sum_{k=1}^3 u_k'' v_k' = R.$$

The determinant of this system is the Wronskian W of u_1, u_2, u_3 . Since u_1, u_2, u_3 are linearly independent solutions of $L(y) = 0$, their Wronskian is never zero on I (see Exercise 8 of Section 7.11); hence we can always solve the system (7.58) for v_1', v_2', v_3' . If the solution is obtained by Cramer's rule, it may be put in the form

$$v_1' = W_1/W, \quad v_2' = W_2/W, \quad v_3' = W_3/W,$$

where W_k is the determinant obtained by replacing the k th column of the Wronskian determinant by $(0, 0, R)$. Integrating these relations we obtain v_1, v_2, v_3 . Substituting these functions in (7.56) we find that the nonhomogeneous equation $L(y) = R$ has a solution y_1 given by the formula

$$y_1(x) = \sum_{k=1}^3 u_k(x) \int_c^x \frac{W_k(t)}{W(t)} dt,$$

where c is any point in I .

7.15 Special methods for determining a particular solution of the nonhomogeneous equation. The algebra of constant-coefficient operators

As in the second-order case, special methods may yield a particular solution y_1 of the nonhomogeneous equation more easily than the method of variation of parameters. For example, if the equation has the form

$$y^{(n)} + a_1 y^{(n-1)} + a_2 y^{(n-2)} + \cdots + a_n y = R,$$

where the a_i are constants and R is a polynomial, there always exists a polynomial solution y_1 which we can obtain by the method of undetermined coefficients. If a_n , the coefficient of y , is nonzero, the degree of y_1 is the same as that of R . If the coefficient of $y^{(k)}$ is nonzero but the coefficient of $y^{(j)}$ is zero for each $j < k$, the degree of y_1 exceeds that of R by k .

Example 1. Find a particular solution of the equation $y^{(4)} - 16y = x^4 + x + 1$.

Solution. We seek a polynomial solution y_1 of degree four, so we write

$$16y_1 = ax^4 + bx^3 + cx^2 + dx + e.$$

This gives us $16y_1^{(4)} = 24a$, so $y_1^{(4)} = 3a/2$. Substituting in the differential equation we must determine a, b, c, d, e to satisfy

$$\frac{3}{2}a - ax^4 - bx^3 - cx^2 - dx - e = x^4 + x + 1.$$

Equating coefficients of like powers of x we obtain

$$a = -1, \quad b = c = 0, \quad d = -1, \quad e = -\frac{5}{2},$$

so the particular solution y_1 is given by

$$y_1 = -\frac{1}{16}x^4 - \frac{1}{16}x - \frac{5}{32}.$$

There are other methods for solving nonhomogeneous equations that work especially well when the equation has constant coefficients. Some of these are best described in operator notation. Let A denote the linear operator defined by the equation

$$(7.59) \quad A(y) = a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y,$$

where $a_0, a_1, \dots, a_n$ are constants. Such an operator is called a *constant-coefficient operator*. If $a_0 \neq 0$ the operator is said to have order n . The function y which appears in this equation may be any real-valued function with n derivatives on some interval. We shall restrict ourselves to functions having derivatives of *every* order on $(-\infty, +\infty)$. The collection of all such functions y will be denoted by $\mathfrak{D}$ and will be referred to as the class of infinitely differentiable functions. When we apply an operator A to a function y in $\mathfrak{D}$, the result of the operation is another function $A(y)$ in class $\mathfrak{D}$. The value of the function $A(y)$ at a point x is $A(y)(x)$, sometimes written as $A[y(x)]$.

Next we develop an algebra of constant-coefficient operators; we discuss addition and

multiplication of operators, as well as multiplication of operators by scalars. Let A and B be two constant-coefficient operators, where A is given by (7.59) and B is given by

$$B(y) = b_0 y^{(m)} + b_1 y^{(m-1)} + \cdots + b_m y.$$

Since these operators are functions with domain $\mathfrak{D}$, we have $A = B$ if and only if $A(y) = B(y)$ for every y in $\mathfrak{D}$. The sum of A and B is a new operator C defined by the equation

$$C(y) = A(y) + B(y) \quad \text{for every } y \text{ in } \mathfrak{D}.$$

We write $A + B$ for C . The product of A and B (in that order) is denoted by AB and is the operator defined by the equation

$$(AB)(y) = A[B(y)] \quad \text{for every } y \text{ in } \mathfrak{D}.$$

If λ is a real number (a scalar) the operator λA is defined by the equation

$$(\lambda A)(y) = \lambda \cdot A(y) \quad \text{for every } y \text{ in } \mathfrak{D}.$$

It is clear that $A + B$, AB , and λA are also constant-coefficient operators. The definitions of the sum and product of operators may be extended by induction to any finite number of operators. Sums, products, and scalar multiples of operators satisfy the usual laws of algebra, such as the commutative, associative, and distributive laws. We could prove these laws directly from the definitions, but it is simpler and more instructive to proceed in another way.

With each constant-coefficient operator A we associate a polynomial p_A called the *characteristic polynomial* of A . If A is given by (7.59), p_A is that polynomial which has the same coefficients as A . That is, for every real r we have

$$p_A(r) = a_0 r^n + a_1 r^{n-1} + \cdots + a_n.$$

Conversely, given any polynomial p , there is a corresponding operator A whose coefficients are the same as those of p . The next theorem shows that this association between operators and polynomials is a one-to-one correspondence. Moreover, this correspondence associates with sums, products, and scalar multiples of operators the respective sums, products, and scalar multiples of their characteristic polynomials.

7-12 THEOREM. Let A and B denote constant-coefficient operators with characteristic polynomials p_A and p_B , respectively, and let λ be a real number. Then we have

- (a) $A = B$ if, and only if, $p_A = p_B$,
- (b) $p_{A+B} = p_A + p_B$,
- (c) $p_{AB} = p_A \cdot p_B$,
- (d) $p_{\lambda A} = \lambda \cdot p_A$.

Proof. We consider part (a) first. Assume $p_A = p_B$. We wish to prove that $A(y) = B(y)$ for every y in $\mathfrak{D}$. Since $p_A = p_B$, both polynomials have the same degree and the same coefficients. Therefore A and B have the same order and the same coefficients, so $A(y) = B(y)$ for every y in $\mathfrak{D}$.

Next we prove that $A = B$ implies $p_A = p_B$. The relation $A = B$ means that $A(y) = B(y)$ for every y in $\mathfrak{D}$. Take $y = e^{rx}$, where r is a constant. Since $y^{(k)} = r^k e^{rx}$ for every $k \geq 0$, we have

$$A(y) = p_A(r)e^{rx} \quad \text{and} \quad B(y) = p_B(r)e^{rx}.$$

The equation $A(y) = B(y)$ implies $p_A(r) = p_B(r)$. Since r is arbitrary we must have $p_A = p_B$. This completes the proof of part (a).

To prove part (b), let $C = A + B$, and let $y = e^{rx}$. Then we have

$$A(y) = p_A(r)e^{rx}, \quad B(y) = p_B(r)e^{rx}, \quad \text{and} \quad C(y) = p_C(r)e^{rx}.$$

Since $C = A + B$ we have $C(y) = A(y) + B(y)$, which gives us

$$p_C(r)e^{rx} = p_A(r)e^{rx} + p_B(r)e^{rx}.$$

Canceling e^{rx} we find $p_C(r) = p_A(r) + p_B(r)$ for every real r . This proves (b); the proofs of (c) and (d) are entirely analogous.

From Theorem 7-12 it follows that every algebraic relation involving sums, products, and scalar multiples of polynomials p_A and p_B also holds for the operators A and B . In particular, we have the *commutative laws*,

$$A + B = B + A, \quad AB = BA, \quad \lambda(\mu A) = \mu(\lambda A),$$

the *associative laws*,

$$A + (B + C) = (A + B) + C, \quad A(BC) = (AB)C, \quad \lambda(\mu A) = (\lambda\mu)A,$$

and the *distributive laws*,

$$A(B + C) = AB + AC, \quad \lambda(A + B) = \lambda A + \lambda B, \quad (\lambda + \mu)A = \lambda A + \mu A.$$

All constant-coefficient operators can be expressed as polynomials in the derivative operator D . If, as usual, we denote by D^k the k th derivative operator defined by $D^k(y) = y^{(k)}$ for every y in $\mathfrak{D}$, Equation (7.59) can be written in the form

$$A(y) = a_0 D^n(y) + a_1 D^{n-1}(y) + \cdots + a_0 y.$$

Therefore, by the definition of equality of operators, we have

$$A = a_0 D^n + a_1 D^{n-1} + \cdots + a_0.$$

If the characteristic polynomial p_A can be factored as a product of two or more polynomials, each factor must be the characteristic polynomial of some constant-coefficient operator, so, by Theorem 7-12, there is a corresponding factorization of the operator A . For example, if $p_A(r) = p_B(r)p_C(r)$, then $A = BC$. If $p_A(r)$ is factored as a product of n linear factors, say

$$p_A(r) = a_0(r - r_1)(r - r_2) \cdots (r - r_n),$$

the corresponding factorization of A takes the form

$$A = a_0(D - r_1)(D - r_2) \cdots (D - r_n).$$

Once we know the zeros of the characteristic polynomial p_A we can determine the general solution of the homogeneous equation $A(y) = 0$ without any integration. Thus, the problem of solving any homogeneous linear differential equation with constant coefficients is a purely algebraic problem.

In some cases, a particular solution of a nonhomogeneous equation $A(y) = R$ can also be obtained by algebraic means. For example, suppose the right-hand member R is itself a solution of a linear differential equation with constant coefficients, say $B(R) = 0$,

where B is a constant-coefficient operator. If we apply B to both sides of the equation $A(y) = R$ we find $BA(y) = 0$, so every solution of $A(y) = R$ is also a solution of the linear homogeneous equation $BA(y) = 0$. Since BA is a constant-coefficient operator, we can determine all the solutions of $BA(y) = 0$ by algebraic means. Now the problem remains of choosing from these solutions a particular function y_1 that satisfies $A(y_1) = R$. The following example illustrates how the process may be carried out in practice.

Example 2. Solve the differential equation $y'' - 5y' + 6y = xe^x$.

Solution. The differential equation has the form

$$(7.60) \quad A(y) = R,$$

where $R(x) = xe^x$ and $A = D^2 - 5D + 6$. The corresponding homogeneous equation can be written as

$$(D - 2)(D - 3)y = 0;$$

it has the linearly independent solutions $u_1(x) = e^{2x}$, $u_2(x) = e^{3x}$. Now we seek a particular solution y_1 of the nonhomogeneous equation. We recognize the function $R(x) = xe^x$ as a solution of the homogeneous equation

$$(D - 1)^2 y = 0.$$

Therefore, if we operate on both sides of (7.60) with the operator $(D - 1)^2$ we find that any function which satisfies (7.60) must also satisfy the equation

$$(D - 1)^2(D - 2)(D - 3)y = 0.$$

This differential equation has the characteristic roots 1, 1, 2, 3, so all its solutions are to be found in the linear combination

$$y = ae^x + bxe^x + ce^{2x} + de^{3x},$$

where a, b, c, d are constants. We want to choose a, b, c, d so that $A(ae^x + bxe^x + ce^{2x} + de^{3x}) = xe^x$. Since $A(ce^{2x} + de^{3x}) = 0$ for every choice of c and d , we need only choose a and b so that $A(ae^x + bxe^x) = xe^x$. If we put

$$y_1 = ae^x + bxe^x,$$

we have

$$D(y_1) = (a + b)e^x + bxe^x, \quad D^2(y_1) = (a + 2b)e^x + bxe^x,$$

so the equation $(D^2 - 5D + 6)y_1 = xe^x$ becomes

$$(2a - 3b)e^x + 2bxe^x = xe^x.$$

Canceling e^x and equating coefficients of like powers of x we find $a = \frac{3}{4}$, $b = \frac{1}{2}$. Therefore $y_1 = \frac{3}{4}e^x + \frac{1}{2}xe^x$ and the general solution of $A(y) = R$ is given by the formula

$$y = c_1e^{2x} + c_2e^{3x} + \frac{3}{4}e^x + \frac{1}{2}xe^x.$$

The method used in the foregoing example will always work if we can find a constant-coefficient operator B that makes $B(R)$ identically zero. Such an operator is said to *annihilate* R ; the method described in the example is called the *annihilator method*. From our knowledge of homogeneous linear differential equations with constant coefficients,

we know that the only real-valued functions annihilated by constant-coefficient operators are linear combinations of terms of the form

$$x^{m-1}e^{\alpha x}, \quad x^{m-1}e^{\alpha x}\cos \beta x, \quad x^{m-1}e^{\alpha x}\sin \beta x,$$

where m is a positive integer and α and β are real constants. The function $y = x^{m-1}e^{\alpha x}$ is a solution of a differential equation with a characteristic root α having multiplicity m . Therefore, this function has the annihilator $(D - \alpha)^m$. Each of the functions $y = x^{m-1}e^{\alpha x}\cos \beta x$ and $y = x^{m-1}e^{\alpha x}\sin \beta x$ is a solution of a differential equation with complex characteristic roots $\alpha \pm i\beta$, each occurring with multiplicity m , so they are annihilated by the operator $[D^2 - 2\alpha D + (\alpha^2 + \beta^2)]^m$. For ease of reference, we list these annihilators in Table I, along with some of their special cases.

Table I.

Function	Annihilator
$y = x^{m-1}$	D^m
$y = e^{\alpha x}$	$D - \alpha$
$y = x^{m-1}e^{\alpha x}$	$(D - \alpha)^m$
$y = \cos \beta x$ or $y = \sin \beta x$	$D^2 + \beta^2$
$y = x^{m-1}\cos \beta x$ or $y = x^{m-1}\sin \beta x$	$(D^2 + \beta^2)^m$
$y = e^{\alpha x}\cos \beta x$ or $y = e^{\alpha x}\sin \beta x$	$D^2 - 2\alpha D + (\alpha^2 + \beta^2)$
$y = x^{m-1}e^{\alpha x}\cos \beta x$ or $y = x^{m-1}e^{\alpha x}\sin \beta x$	$[D^2 - 2\alpha D + (\alpha^2 + \beta^2)]^m$

Although the annihilator method is very efficient when applicable, it is limited to equations whose right members R have a constant-coefficient annihilator. If $R(x)$ has the form e^{x^2} , $\log x$, or $\tan x$, the method will not work; we must then use variation of parameters or some other method to find a particular solution. The next example illustrates a method for reducing the equation to a succession of linear equations of first order.

Example 3. Find a particular solution of the equation

$$(D - 1)(D - 2)y = xe^{x+x^2}.$$

Solution. If we write

$$(7.61) \quad (D - 2)y = u,$$

the equation becomes

$$(D - 1)u = xe^{x+x^2}.$$

This is a first-order linear equation in u . As such it can be solved [using Equation (7.3)] to give

$$ue^{-x} = \int xe^{x^2}dx + C = \frac{1}{2}e^{x^2} + C.$$

Since we seek only a particular solution we take $C = 0$. This gives us

$$u = \frac{1}{2}e^{x+x^2}.$$

Substituting in (7.61) we obtain

$$(D - 2)y = \frac{1}{2}e^{x+x^2},$$

a first-order linear equation for y . Solving this by Equation (7.3) we find that a particular solution y_1 is given by

$$y_1 = \frac{1}{2}e^{2x} \int e^{x^2-x} dx.$$

Although the last integral cannot be evaluated in terms of elementary functions we consider the equation as having been solved, since the solution is expressed in terms of integrals of familiar functions.

7.16 Exercises

In each of Exercises 1 through 10, use the annihilator method to find a particular solution.

1. $y'' - y' = x^2$.
2. $y'' - 4y = e^{2x}$.
3. $y'' + 2y' = 3xe^x$.
4. $y'' + 4y = \sin x$.
5. $y'' - 2y' + y = e^x + e^{2x}$.
6. $y''' - y' = e^x$.
7. $y''' - y' = e^x + e^{-x}$.
8. $y''' + 3y'' + 3y' + y = xe^{-x}$.
9. $y'' + y = xe^x \sin 2x$.
10. $y^{(4)} - y = x^2e^{-x}$.

11. If a constant-coefficient operator A annihilates f and if a constant-coefficient operator B annihilates g , show that the product AB annihilates $f + g$.

12. Let A be a constant-coefficient operator with characteristic polynomial p_A .

(a) Use the annihilator method to prove that the differential equation $A(y) = e^{\alpha x}$ has a particular solution of the form

$$y_1 = \frac{e^{\alpha x}}{p_A(\alpha)}$$

if α is not a zero of the polynomial p_A .

(b) If α is a simple zero of p_A (multiplicity 1), prove that the equation $A(y) = e^{\alpha x}$ has the particular solution

$$y_1 = \frac{xe^{\alpha x}}{p_A'(\alpha)}.$$

(c) Generalize the results of (a) and (b) when α is a zero of p_A with multiplicity m .

13. Given two constant-coefficient operators A and B whose characteristic polynomials have no zeros in common. Let $C = AB$.

(a) Prove that every solution of the differential equation $C(y) = 0$ has the form $y = y_1 + y_2$, where $A(y_1) = 0$ and $B(y_2) = 0$.

(b) Prove that the functions y_1 and y_2 in part (a) are uniquely determined. That is, for a given y satisfying $C(y) = 0$ there is only one pair y_1, y_2 with the properties in part (a).

14. Let P and Q denote operators given by

$$P(y) = P_0y^{(n)} + P_1y^{(n-1)} + \cdots + P_ny, \quad Q(y) = Q_0y^{(m)} + Q_1y^{(m-1)} + \cdots + Q_my,$$

where $y \in \mathcal{D}$ and the P_i and Q_i are real-valued functions with derivatives of every order on an interval. Define addition $P + Q$ and multiplication PQ by the same formulas used for constant-coefficient operators. Show that, with the exception of the commutative law for multiplication, the algebraic laws for constant-coefficient operators hold also for these more general operators.

15. Let Q be the operator defined by $Q(y)(x) = x \cdot y(x)$ for each y in class $\mathfrak{D}$ and each real x . Let I denote the identity operator, defined by $I(y) = y$ for each y in $\mathfrak{D}$.

(a) Prove that $DQ - QD = I$.

(b) Show that $D^2Q - QD^2$ is a constant-coefficient operator of first order, and determine this operator explicitly as a linear polynomial in D .

(c) Show that $D^3Q - QD^3$ is a constant-coefficient operator of second order, and determine this operator explicitly as a quadratic polynomial in D .

(d) Guess the generalization suggested for the operator $D^nQ - QD^n$, and prove your result by induction.

In each of Exercises 16 through 20, find the general solution of the differential equation in the given interval.

16. $y'' - y = 1/x, \quad (0, +\infty).$

17. $y'' + 4y = \sec 2x, \quad \left(-\frac{\pi}{4}, \frac{\pi}{4}\right).$

18. $y'' - y = \sec^3 x - \sec x, \quad \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$

19. $y'' - 2y' + y = e^{e^x}(e^x - 1)^2, \quad (-\infty, +\infty).$

20. $y''' - 7y'' + 14y' - 8y = \log x, \quad (0, +\infty).$

7.17 The Legendre equation

Linearly independent solutions of a homogeneous linear equation can sometimes be obtained by the use of power series. One method was illustrated in Volume I (Section 9.24), where we solved the equation $(1 - x^2)y'' + 2y = 0$ by the method of undetermined coefficients. In this section we shall use the same method to solve the Legendre equation

$$(7.62) \quad (1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0,$$

where α is a constant. We try to find a power-series solution of the form

$$y = \sum_{n=0}^{\infty} a_n x^n$$

valid in some open interval $(-r, r)$ about the origin. Differentiating this series term by term we obtain

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{and} \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Therefore we have

$$2xy' = \sum_{n=1}^{\infty} 2n a_n x^n = \sum_{n=0}^{\infty} 2n a_n x^n,$$

and

$$\begin{aligned} (1 - x^2)y'' &= \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1) a_n x^n \\ &= \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=0}^{\infty} n(n-1) a_n x^n \end{aligned}$$

$$= \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} - n(n-1)a_n]x^n.$$

If we substitute these series in the differential equation (7.62), we see that the equation will be satisfied if, and only if, the coefficients satisfy the relation

$$(n+2)(n+1)a_{n+2} - n(n-1)a_n - 2na_n + \alpha(\alpha+1)a_n = 0$$

for all $n \geq 0$. This equation is the same as

$$(n+2)(n+1)a_{n+2} - (n-\alpha)(n+1+\alpha)a_n = 0,$$

or

$$(7.63) \quad a_{n+2} = -\frac{(\alpha-n)(\alpha+n+1)}{(n+1)(n+2)}a_n.$$

This relation enables us to determine $a_2, a_4, a_6, \dots$ successively in terms of a_0 . Similarly, we can compute $a_3, a_5, a_7, \dots$ in terms of a_1 . For the coefficients with even subscripts we have

$$a_2 = -\frac{\alpha(\alpha+1)}{1 \cdot 2}a_0,$$

$$a_4 = -\frac{(\alpha-2)(\alpha+3)}{3 \cdot 4}a_2 = (-1)^2 \frac{\alpha(\alpha-2)(\alpha+1)(\alpha+3)}{4!}a_0,$$

and, in general,

$$(7.64) \quad a_{2n} = (-1)^n \frac{\alpha(\alpha-2) \cdots (\alpha-2n+2) \cdot (\alpha+1)(\alpha+3) \cdots (\alpha+2n-1)}{(2n)!}a_0.$$

The validity of (7.64) for $n \geq 1$ can be established by induction. For the coefficients with odd subscripts we find

$$a_{2n+1} = (-1)^n \frac{(\alpha-1)(\alpha-3) \cdots (\alpha-2n+1) \cdot (\alpha+2)(\alpha+4) \cdots (\alpha+2n)}{(2n+1)!}a_1.$$

Therefore the series for y can be written as

$$(7.65) \quad y = a_0 u_1(x) + a_1 u_2(x),$$

where

$$(7.66) \quad u_1(x) = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{\alpha(\alpha-2) \cdots (\alpha-2n+2) \cdot (\alpha+1)(\alpha+3) \cdots (\alpha+2n-1)}{(2n)!} x^{2n}$$

and

$$(7.67) \quad u_2(x) = x + \sum_{n=1}^{\infty} (-1)^n \frac{(\alpha-1)(\alpha-3) \cdots (\alpha-2n+1) \cdot (\alpha+2)(\alpha+4) \cdots (\alpha+2n)}{(2n+1)!} x^{2n+1}.$$

The ratio test shows that each of these series converges for $|x| < 1$. Also, since the relation (7.63) is satisfied separately by the even and odd coefficients, each of u_1 and u_2 is a solution of the differential equation (7.62). These solutions satisfy the initial conditions

$$u_1(0) = 1, \quad u_1'(0) = 0, \quad u_2(0) = 0, \quad u_2'(0) = 1.$$

Since u_1 and u_2 are linearly independent, the general solution of the Legendre equation (7.62) over the open interval $(-1, 1)$ is given by the linear combination (7.65) with arbitrary constants a_0 and a_1 .

When α is 0 or a positive even integer, say $\alpha = 2m$, the series for $u_1(x)$ becomes a polynomial of degree $2m$ containing only even powers of x . Since we have

$$\alpha(\alpha - 2) \cdots (\alpha - 2n + 2) = 2m(2m - 2) \cdots (2m - 2n + 2) = \frac{2^m m!}{(m - n)!}$$

and

$$\begin{aligned} (\alpha + 1)(\alpha + 3) \cdots (\alpha + 2n - 1) &= (2m + 1)(2m + 3) \cdots (2m + 2n - 1) \\ &= \frac{(2m + 2n)!}{2^n (2m)!} \frac{m!}{(m + n)!} \end{aligned}$$

the formula for $u_1(x)$ in this case becomes

$$(7.68) \quad u_1(x) = 1 + \frac{(m!)^2}{(2m)!} \sum_{k=1}^m (-1)^k \frac{(2m + 2k)!}{(m - k)! (m + k)! (2k)!} x^{2k}.$$

For example, when $\alpha = 0, 2, 4, 6$ ($m = 0, 1, 2, 3$) the corresponding polynomials are

$$u_1(x) = 1, \quad 1 - 3x^2, \quad 1 - 10x^2 + \frac{35}{3}x^4, \quad 1 - 21x^2 + 63x^4 - \frac{231}{5}x^6.$$

The companion series for $u_2(x)$ is not a polynomial when α is even because none of its coefficients are zero.

When α is an *odd* positive integer, the roles of u_1 and u_2 are reversed; the series for $u_2(x)$ becomes a polynomial and the series for $u_1(x)$ is not a polynomial. Specifically, if $\alpha = 2m + 1$ we have

$$(7.69) \quad u_2(x) = x + \frac{(m!)^2}{(2m + 1)!} \sum_{k=1}^m (-1)^k \frac{(2m + 2k + 1)!}{(m - k)! (m + k)! (2k + 1)!} x^{2k+1}.$$

For example, when $\alpha = 1, 3, 5$ ($m = 0, 1, 2$), the corresponding polynomials are

$$u_2(x) = x, \quad x - \frac{5}{3}x^3, \quad x - \frac{14}{3}x^3 + \frac{21}{5}x^5.$$

Some of the polynomial solutions of the Legendre equation have a number of interesting properties which are important in other branches of mathematics. For example, the theory of polynomial approximation considers the problem of approximating a given continuous function f on the interval $[-1, 1]$ by a polynomial p_n of degree $\leq n$ in such a way that the "mean-square error"

$$\int_{-1}^1 |f(x) - p_n(x)|^2 dx$$

will be as small as possible. Polynomial solutions of the Legendre equation play a fundamental role in the solution of this problem. (See Section 8.10.) Some of the properties of these polynomial solutions can be deduced directly from the differential equation or from the formulas in (7.68) and (7.69). Others are more easily deduced from an alternative formula for these polynomials which we shall now derive.

First we shall obtain a single formula which contains (aside from constant factors) both the polynomials in (7.68) and (7.69). Let

$$P_n(x) = \frac{1}{2^n} \sum_{r=0}^{[n/2]} \frac{(-1)^r (2n-2r)!}{r! (n-r)! (n-2r)!} x^{n-2r},$$

where $[n/2]$ denotes the greatest integer $\leq n/2$. This is called the *Legendre polynomial* of degree n . When n is even, it is a constant multiple of the polynomial $u_1(x)$ in Equation (7.68); when n is odd, it is a constant multiple of the polynomial $u_2(x)$ in (7.69).† The first seven Legendre polynomials are given by the formulas

$$\begin{aligned} P_0(x) &= 1, & P_1(x) &= x, & P_2(x) &= \frac{1}{2}(3x^2 - 1), & P_3(x) &= \frac{1}{2}(5x^3 - 3x), \\ P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3), & P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x), \\ P_6(x) &= \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5). \end{aligned}$$

Figure 7.1 shows the graphs of the first five of these functions over the interval $[-1, 1]$.

In the sum defining $P_n(x)$ we note that

$$\frac{(2n-2r)!}{(n-2r)!} x^{n-2r} = \frac{d^n}{dx^n} x^{2n-2r} \quad \text{and} \quad \frac{1}{r! (n-r)!} = \frac{1}{n!} \binom{n}{r},$$

where $\binom{n}{r}$ is the binomial coefficient, and we write the sum in the form

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \sum_{r=0}^{[n/2]} (-1)^r \binom{n}{r} x^{2n-2r}.$$

When $[n/2] < r \leq n$, the term x^{2n-2r} has degree less than n , so its n th derivative is zero. Therefore we do not alter the sum if we allow r to run from 0 to n . This gives us

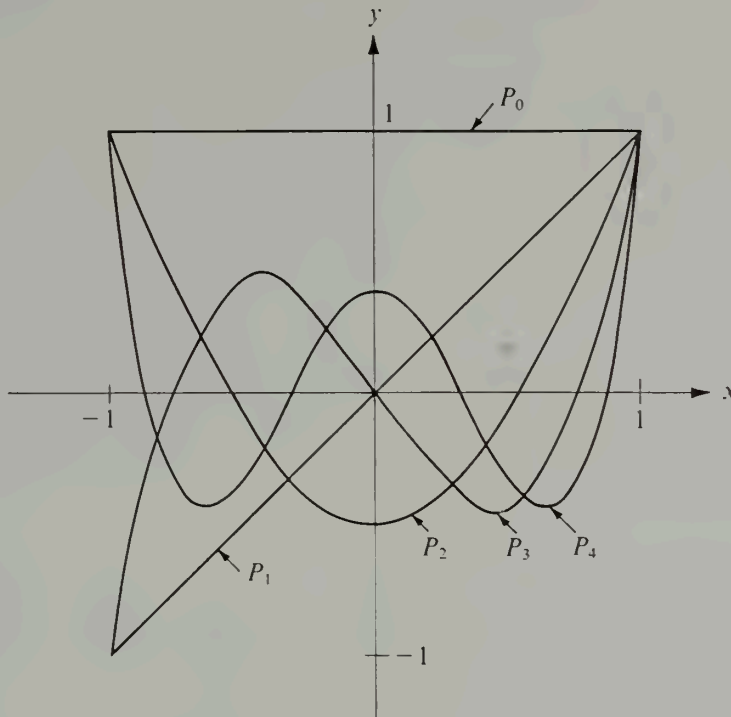
$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \sum_{r=0}^n (-1)^r \binom{n}{r} x^{2n-2r}.$$

Now we recognize the sum on the right as the binomial expansion of $(x^2 - 1)^n$. Therefore we have

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

This is known as *Rodrigues' formula* for the Legendre polynomials.

† When n is even, say $n = 2m$, we may replace the index of summation k in Equation (7.68) by a new index r , where $r = m - k$; we find that the sum in (7.68) is a constant multiple of $P_n(x)$. Similarly, when n is odd, a change of index transforms the sum in (7.69) to a constant multiple of $P_n(x)$.

FIGURE 7.1 Graphs of Legendre polynomials over the interval $[-1, 1]$.

Using Rodrigues' formula and the differential equation, we can derive a number of important properties of the Legendre polynomials. Some of these properties are listed below. Their proofs are outlined in the next set of exercises.

For each $n \geq 0$ we have

$$P_n(1) = 1.$$

Moreover, $P_n(x)$ is the only polynomial which satisfies the Legendre equation

$$(1 - x^2) y'' - 2xy' + n(n + 1)y = 0$$

and has the value 1 when $x = 1$.

For each $n \geq 0$ we have

$$P_n(-x) = (-1)^n P_n(x).$$

This shows that P_n is an even function when n is even, and an odd function when n is odd.

If $m \neq n$ we have

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0.$$

This is called the *orthogonality relation* for Legendre polynomials. When $m = n$ we have the relation

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}.$$

Every polynomial of degree n can be expressed as a linear combination of the Legendre polynomials $P_0, P_1, \dots, P_n$. In fact, if f is a polynomial of degree n we have

$$f(x) = \sum_{k=0}^n c_k P_k(x),$$

where

$$c_k = \frac{2k+1}{2} \int_{-1}^1 f(x) P_k(x) dx.$$

From the orthogonality relation it follows that

$$\int_{-1}^1 g(x) P_n(x) dx = 0$$

for every polynomial g of degree less than n . This property can be used to prove that the Legendre polynomial P_n has n distinct real zeros and that they all lie in the open interval $(-1, 1)$.

The method used to solve the Legendre equation can always be applied to every second-order equation of the form

$$(7.70) \quad y'' + P_1 y' + P_2 y = 0$$

if the coefficients P_1 and P_2 have power-series expansions. The theoretical justification for the method is provided by the following theorem.

7-13 THEOREM. Let P_1 and P_2 be functions having power-series expansions of the form

$$P_1(x) = \sum_{n=0}^{\infty} b_n(x - x_0)^n, \quad P_2(x) = \sum_{n=0}^{\infty} c_n(x - x_0)^n,$$

valid in some open interval $(x_0 - r, x_0 + r)$. Then the differential equation (7.70) has a power-series solution

$$(7.71) \quad y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$$

convergent in the same interval. Moreover, if k_0 and k_1 are given real numbers, there is exactly one power-series solution $y = f(x)$ satisfying the initial conditions

$$f(x_0) = k_0 \quad \text{and} \quad f'(x_0) = k_1.$$

For this solution we have $a_0 = k_0$ and $a_1 = k_1$. The remaining coefficients $a_2, a_3, a_4, \dots$ can be determined in terms of a_0 and a_1 by the method of undetermined coefficients.

Note. The coefficient of y'' in (7.70) is 1. The Legendre equation

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0$$

can be put in this form if we divide through by $1 - x^2$. Thus, for the Legendre equation we have

$$P_1(x) = -\frac{2x}{1 - x^2} \quad \text{and} \quad P_2(x) = \frac{\alpha(\alpha + 1)}{1 - x^2}$$

if $x^2 \neq 1$. Since $1/(1 - x^2) = \sum_{n=0}^{\infty} x^{2n}$ for $|x| < 1$, both P_1 and P_2 have power-series expansions in the open interval $(-1, 1)$, so Theorem 7-13 is applicable.

Proof. We shall not present all the details of the proof of Theorem 7-13 but shall merely sketch the principal steps. We substitute the given series for P_1 and P_2 in the differential equation and then try to determine coefficients a_n so that the equation will be satisfied by the series in (7.71). This leads to the recursion formula

$$(n + 2)(n + 1)a_{n+2} = - \sum_{k=0}^n [(k + 1)b_{n-k}a_{k+1} + c_{n-k}a_k],$$

expressing a_{n+2} in terms of $a_0, a_1, \dots, a_{n+1}$. We choose arbitrary values of a_0 and a_1 and use this formula to *define* the numbers $a_2, a_3, \dots$ in terms of a_0 and a_1 . This guarantees that the power series in (7.71) will satisfy the differential equation. To complete the proof it is necessary to show that the series so defined actually converges for every x in the interval $(x_0 - r, x_0 + r)$. This is done by dominating the series in (7.71) by another power series known to converge.

Choose a fixed point $x_1 \neq x_0$ in the interval $(x_0 - r, x_0 + r)$ and let $t = |x_1 - x_0|$. Since the series for P_1 and P_2 converge absolutely for $x = x_1$ the terms of these series are bounded, say

$$|b_k| t^k \leq M \quad \text{and} \quad |c_k| t^k \leq M$$

for some $M > 0$. The recursion formula implies the inequality

$$(n + 2)(n + 1) |a_{n+2}| \leq \frac{M}{t^n} \sum_{k=0}^n [(k + 1) |a_{k+1}| + |a_k|] t^k.$$

Now let $A_0 = |a_0|$, $A_1 = |a_1|$, and define $A_2, A_3, \dots$ successively by the recursion formula

$$(7.72) \quad (n + 2)(n + 1)A_{n+2} = \frac{M}{t^n} \sum_{k=0}^n [(k + 1)A_{k+1} + A_k] t^k + MA_{n+1}t$$

for $n \geq 1$. It follows that $|a_n| \leq A_n$ for all $n \geq 0$, so the series $\sum a_n(x - x_0)^n$ is dominated by the series $\sum A_n|x - x_0|^n$. Now we use the ratio test to show that $\sum A_n|x - x_0|^n$ converges if $|x - x_0| < t$. From (7.72) we can deduce the relation

$$(n + 2)(n + 1)A_{n+2} = \frac{1}{t} [(n + 1)n + M(n + 1)t + Mt^2]A_{n+1}$$

(by induction) and we have

$$\frac{A_{n+2} |x - x_0|^{n+2}}{A_{n+1} |x - x_0|^{n+1}} = \frac{(n + 1)n + M(n + 1)t + Mt^2}{(n + 2)(n + 1)t} |x - x_0| \rightarrow \frac{|x - x_0|}{t}$$

as $n \rightarrow \infty$. This limit is less than 1 if $|x - x_0| < t$. Hence $\sum a_n (x - x_0)^n$ converges if $|x - x_0| < t$. But since $t = |x_1 - x_0|$ and since x_1 was an arbitrary point in the interval $(x_0 - r, x_0 + r)$, the series $\sum a_n (x - x_0)^n$ must converge for all x in $(x_0 - r, x_0 + r)$.

7.18 Exercises

1. The Legendre equation (7.62) with $\alpha = 0$ has the polynomial solution $u_1(x) = 1$ and a solution u_2 , not a polynomial, given by the series in Equation (7.67).

(a) Show that the sum of the series for u_2 is given by

$$u_2(x) = \frac{1}{2} \log \frac{1+x}{1-x} \quad \text{for } |x| < 1.$$

(b) Verify directly that the function u_2 in part (a) is a solution of the Legendre equation when $\alpha = 0$.

2. Show that the function f defined by the equation

$$f(x) = 1 - \frac{x}{2} \log \frac{1+x}{1-x}$$

for $|x| < 1$ satisfies the Legendre equation (7.62) with $\alpha = 1$. Express this function as a linear combination of the solutions u_1 and u_2 given in Equations (7.66) and (7.67).

3. The Legendre equation (7.62) can be written in the form

$$[(x^2 - 1)y']' - \alpha(\alpha + 1)y = 0.$$

(a) If a, b, c are constants with $a > b$ and $4c + 1 > 0$, show that a differential equation of the type

$$[(x - a)(x - b)y']' - cy = 0$$

can be transformed to a Legendre equation by a change of variable of the form $x = At + B$, with $A > 0$. Determine A and B in terms of a and b .

(b) Use the method suggested in part (a) to transform the equation

$$(x^2 - x)y'' + (2x - 1)y' - 2y = 0$$

to a Legendre equation.

4. Find two linearly independent power-series solutions of the *Hermite equation*

$$y'' - 2xy' + 2\alpha y = 0$$

on an interval of the form $(-r, r)$. Show that one of these solutions is a polynomial when α is a nonnegative integer.

5. Use the method of undetermined coefficients to find a power-series solution of the differential equation

$$xy'' + (3 + x^3)y' + 3x^2y = 0$$

valid for all x . Find a second solution of the form $y = x^{-2} \sum a_n x^n$ valid for all $x \neq 0$.

6. Find a power-series solution of the differential equation

$$x^2 y'' + x^2 y' - (\alpha x + 2)y = 0$$

valid on an interval of the form $(-r, r)$.

7. This exercise shows how the Legendre equation arises when we seek solutions of Laplace's equation having a special form. Let f be a scalar field satisfying Laplace's equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0.$$

The introduction of spherical coordinates

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi,$$

transforms $f(x, y, z)$ to $F(\rho, \theta, \phi)$.

(a) Show that F satisfies the partial differential equation

$$\frac{\partial^2 F}{\partial \rho^2} + \frac{2}{\rho} \frac{\partial F}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 F}{\partial \phi^2} + \frac{\cot \phi}{\rho^2} \frac{\partial F}{\partial \phi} + \frac{1}{\rho^2 \sin^2 \phi} \frac{\partial^2 F}{\partial \theta^2} = 0.$$

(b) Suppose we seek solutions of this equation which are independent of θ and have the form $F(\rho, \theta, \phi) = \rho^n G(\phi)$. Show that the equation in (a) is satisfied if G satisfies the second-order equation

$$\frac{d^2 G}{d\phi^2} + \cot \phi \frac{dG}{d\phi} + n(n+1)G = 0.$$

(c) The change of variable $x = \cos \phi$ ($\phi = \arccos x$, $-1 \leq x \leq 1$) transforms $G(\phi)$ to $g(x)$. Show that g satisfies the Legendre equation

$$(1-x^2) \frac{d^2 g}{dx^2} - 2x \frac{dg}{dx} + n(n+1)g = 0.$$

In Exercises 8 through 14, $P_n(x)$ denotes the Legendre polynomial of degree n . These exercises outline proofs of the properties of the Legendre polynomials described in Section 7.17.

8. (a) Use Rodrigues' formula to show that

$$P_n(x) = \frac{1}{2^n} (x+1)^n + (x-1)Q_n(x),$$

where $Q_n(x)$ is a polynomial.

(b) Prove that $P_n(1) = 1$ and that $P_n(-1) = (-1)^n$.

(c) Prove that $P_n(x)$ is the only polynomial solution of Legendre's equation (with $\alpha = n$) having the value 1 when $x = 1$.

9. (a) Use the differential equations satisfied by P_n and P_m to show that

$$[(1-x^2)(P_n P'_m - P'_n P_m)]' = [n(n+1) - m(m+1)] P_n P_m.$$

(b) If $m \neq n$, integrate the equation in (a) from -1 to 1 to deduce the relation

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0.$$

10. (a) Let $f(x) = (x^2 - 1)^n$. Use integration by parts to show that

$$\int_{-1}^1 f^{(n)}(x) f^{(n)}(x) dx = - \int_{-1}^1 f^{(n+1)}(x) f^{(n-1)}(x) dx.$$

Apply this formula repeatedly to deduce that the integral on the left is equal to

$$2(2n)! \int_0^1 (1-x^2)^n dx.$$

(b) The substitution $x = \cos t$ transforms the integral $\int_0^1 (1-x^2)^n dx$ to $\int_0^{\pi/2} \sin^{2n+1} t dt$. Use the relation

$$\int_0^{\pi/2} \sin^{2n+1} t dt = \frac{2n(2n-2) \cdots 2}{(2n+1)(2n-1) \cdots 3 \cdot 1}$$

(see Exercise 10 on p. 258 of Volume I) and Rodrigues' formula to obtain

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}.$$

11. (a) Show that

$$P_n(x) = \frac{(2n)!}{2^n(n!)^2} x^n + Q_n(x),$$

where $Q_n(x)$ is a polynomial of degree less than n .

(b) Express the polynomial $f(x) = x^4$ as a linear combination of P_0, P_1, P_2, P_3 , and P_4 .

(c) Show that every polynomial f of degree n can be expressed as a linear combination of the Legendre polynomials $P_0, P_1, \dots, P_n$.

12. If f is a polynomial of degree n , write

$$f(x) = \sum_{k=0}^n c_k P_k(x).$$

[This is possible because of Exercise 11(c).] For a fixed m , $0 \leq m \leq n$, multiply both sides of this equation by $P_m(x)$ and integrate from -1 to 1 . Use Exercises 9(b) and 10(b) to deduce the relation

$$c_m = \frac{2m+1}{2} \int_{-1}^1 f(x) P_m(x) dx.$$

13. Use Exercises 9 and 11 to show that $\int_{-1}^1 g(x) P_n(x) dx = 0$ for every polynomial g of degree less than n .

14. (a) Use Rolle's theorem to show that P_n cannot have any multiple zeros in the open interval $(-1, 1)$. In other words, any zeros of P_n which lie in $(-1, 1)$ must be simple zeros.

(b) Assume P_n has m zeros in the interval $(-1, 1)$. If $m = 0$, let $Q_0(x) = 1$. If $m \geq 1$, let

$$Q_m(x) = (x - x_1)(x - x_2) \cdots (x - x_m),$$

where $x_1, x_2, \dots, x_m$ are the m zeros of P_n in $(-1, 1)$. Show that, at each point x in $(-1, 1)$, $Q_m(x)$ has the same sign as $P_n(x)$.

(c) Use part (b), along with Exercise 13, to show that the inequality $m < n$ leads to a contradiction. This shows that P_n has n distinct real zeros, all of which lie in the open interval $(-1, 1)$.

15. (a) Show that the value of the integral $\int_{-1}^1 P_n(x) P'_{n+1}(x) dx$ is independent of n .

(b) Evaluate the integral $\int_{-1}^1 x P_n(x) P_{n-1}(x) dx$.

7.19 The Bessel equation

In Section 7.17 we learned how to find a power-series solution of the differential equation

$$(7.73) \quad y'' + P_1 y' + P_2 y = 0$$

near a point x_0 whenever the coefficients P_1 and P_2 have power-series expansions near x_0 . If either P_1 or P_2 does not have a power-series expansion near x_0 , power-series solutions valid near x_0 may or may not exist. For example, suppose we try to find a power-series solution $y = \sum a_k x^k$ of the differential equation

$$(7.74) \quad x^2 y'' - y' - y = 0$$

near $x_0 = 0$. If we use the method of undetermined coefficients we are led to the recursion formula

$$a_{n+1} = \frac{n^2 - n - 1}{n + 1} a_n.$$

Although this gives us a power series $y = \sum a_k x^k$ which formally satisfies (7.74), the ratio test shows that this power series converges *only* for $x = 0$. Thus, there is no power-series solution of (7.74) valid in any open interval about $x_0 = 0$. This example does not violate Theorem 7-13 because when we put Equation (7.74) in the form (7.73) we find that the coefficients P_1 and P_2 are given by

$$P_1(x) = -\frac{1}{x^2} \quad \text{and} \quad P_2(x) = -\frac{1}{x^2}.$$

These functions do not have power-series expansions about the origin. The difficulty here is that the coefficient of y'' in (7.74) has the value 0 when $x = 0$; in other words, the differential equation has a singular point at $x = 0$.

A knowledge of the theory of functions of a complex variable is needed to appreciate the difficulties encountered in the investigation of differential equations near a singular point. However, some important special cases of equations with singular points can be treated by elementary methods. For example, suppose the differential equation in (7.73) is equivalent to an equation of the form

$$(7.75) \quad (x - x_0)^2 y'' + (x - x_0)P(x)y' + Q(x)y = 0,$$

where P and Q have power-series expansions in some open interval $(x_0 - r, x_0 + r)$. In this case we say that x_0 is a *regular* singular point of the equation. If we divide both sides of (7.75) by $(x - x_0)^2$ the equation becomes

$$y'' + \frac{P(x)}{x - x_0} y' + \frac{Q(x)}{(x - x_0)^2} y = 0$$

for $x \neq x_0$. If $P(x_0) \neq 0$ or $Q(x_0) \neq 0$, either the coefficient of y' or the coefficient of y will not have a power-series expansion about the point x_0 , so Theorem 7-13 will not be applicable. In 1873 the German mathematician Georg Frobenius (1849-1917) developed a useful method for treating such equations. He showed that the differential equation in (7.75) can always be satisfied by a function of the form

$$(7.76) \quad y = |x - x_0|^t \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

for some real exponent t , with $a_0 \neq 0$. The power series which multiplies $|x - x_0|^t$ will converge in the interval $(x_0 - r, x_0 + r)$. If the exponent t is zero or a positive even integer, the formula gives a power-series solution of the differential equation valid everywhere in the interval $(x_0 - r, x_0 + r)$. Otherwise, the solution is not a power series and the solution is valid for all x in the interval $(x_0 - r, x_0 + r)$ with the possible exception of the point x_0 itself.

We shall not discuss the method of Frobenius in all its ramifications but merely show how the method can be applied to solve the Bessel equation

$$x^2 y'' + xy' + (x^2 - \alpha^2)y = 0,$$

where α is a nonnegative constant. This equation has the form (7.75) with $x_0 = 0$, $P(x) = 1$, and $Q(x) = x^2 - \alpha^2$, so the point $x_0 = 0$ is a regular singular point. Since P and Q have power-series expansions over the entire real axis, we try to find a solution of the form

$$(7.77) \quad y = |x|^t \sum_{n=0}^{\infty} a_n x^n,$$

with $a_0 \neq 0$, valid for all real x with the possible exception of $x = 0$.

First we keep $x > 0$, so that $|x|^t = x^t$. Differentiation of (7.77) gives us

$$y' = t x^{t-1} \sum_{n=0}^{\infty} a_n x^n + x^t \sum_{n=0}^{\infty} n a_n x^{n-1} = x^{t-1} \sum_{n=0}^{\infty} (n+t) a_n x^n.$$

Similarly, we obtain

$$y'' = x^{t-2} \sum_{n=0}^{\infty} (n+t)(n+t-1) a_n x^n.$$

If $L(y) = x^2 y'' + x y' + (x^2 - \alpha^2) y$, we find

$$\begin{aligned} L(y) &= x^t \sum_{n=0}^{\infty} (n+t)(n+t-1) a_n x^n + x^t \sum_{n=0}^{\infty} (n+t) a_n x^n \\ &\quad + x^t \sum_{n=0}^{\infty} a_n x^{n+2} - x^t \sum_{n=0}^{\infty} \alpha^2 a_n x^n = x^t \sum_{n=0}^{\infty} [(n+t)^2 - \alpha^2] a_n x^n + x^t \sum_{n=0}^{\infty} a_n x^{n+2}. \end{aligned}$$

Now we put $L(y) = 0$, cancel x^t , and try to determine the a_n so that the coefficient of each power of x will vanish. For the constant term we need $(t^2 - \alpha^2)a_0 = 0$. Since we seek a solution with $a_0 \neq 0$, this requires that

$$(7.78) \quad t^2 - \alpha^2 = 0.$$

This quadratic equation in t is called the *indicial equation*. Its roots α and $-\alpha$ are the only possible values of t that can give us a solution of the desired type.

Consider first the choice $t = \alpha$. For this t the remaining equations for determining the coefficients become

$$(7.79) \quad [(1+\alpha)^2 - \alpha^2]a_1 = 0 \quad \text{and} \quad [(n+\alpha)^2 - \alpha^2]a_n + a_{n-2} = 0$$

for $n \geq 2$. Since $\alpha \geq 0$, the first of these implies that $a_1 = 0$. The second formula can be written as

$$(7.80) \quad a_n = -\frac{a_{n-2}}{(n+\alpha)^2 - \alpha^2} = -\frac{a_{n-2}}{n(n+2\alpha)},$$

so $a_3 = a_5 = a_7 = \dots = 0$. For the coefficients with even subscripts we have

$$a_2 = \frac{-a_0}{2(2+\alpha)} = \frac{-a_0}{2^2(1+\alpha)}, \quad a_4 = \frac{-a_2}{4(4+2\alpha)} = \frac{(-1)^2 a_0}{2^4 2!(1+\alpha)(2+\alpha)},$$

$$a_6 = \frac{-a_4}{6(6+2\alpha)} = \frac{(-1)^3 a_0}{2^6 3!(1+\alpha)(2+\alpha)(3+\alpha)},$$

and, in general,

$$a_{2n} = \frac{(-1)^n a_0}{2^{2n} n! (1+\alpha)(2+\alpha) \cdots (n+\alpha)}.$$

Therefore the choice $t = \alpha$ gives us the solution

$$y = a_0 x^\alpha \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} n! (1 + \alpha)(2 + \alpha) \cdots (n + \alpha)} \right).$$

The ratio test shows that the power series appearing in this formula converges for all real x .

In this discussion we assumed that $x > 0$. If $x < 0$ we can repeat the discussion with x^t replaced by $(-x)^t$. We again find that t must satisfy the equation $t^2 - \alpha^2 = 0$. Taking $t = \alpha$ we then obtain the same solution, except that the outside factor x^α is replaced by $(-x)^\alpha$. Therefore the function f_α given by the equation

$$(7.81) \quad f_\alpha(x) = a_0 |x|^\alpha \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} n! (1 + \alpha)(2 + \alpha) \cdots (n + \alpha)} \right)$$

is a solution of the Bessel equation valid for all real $x \neq 0$. For those values of α for which $f'_\alpha(0)$ and $f''_\alpha(0)$ exist the solution is also valid for $x = 0$.

Now consider the root $t = -\alpha$ of the indicial equation. We obtain, in place of (7.79), the equations

$$[(1 - \alpha)^2 - \alpha^2]a_1 = 0 \quad \text{and} \quad [(n - \alpha)^2 - \alpha^2]a_n + a_{n-2} = 0,$$

which become

$$(1 - 2\alpha)a_1 = 0 \quad \text{and} \quad n(n - 2\alpha)a_n + a_{n-2} = 0.$$

If 2α is not an integer these equations give us $a_1 = 0$ and

$$a_n = -\frac{a_{n-2}}{n(n - 2\alpha)}$$

for $n \geq 2$. Since this recursion formula is the same as (7.80), with α replaced by $-\alpha$, we are led to the solution

$$(7.82) \quad f_{-\alpha}(x) = a_0 |x|^{-\alpha} \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} n! (1 - \alpha)(2 - \alpha) \cdots (n - \alpha)} \right)$$

valid for all real $x \neq 0$.

The solution $f_{-\alpha}$ was obtained under the hypothesis that 2α is not a positive integer. However, the series for $f_{-\alpha}$ is meaningful even if 2α is a positive integer, so long as α is not a positive integer. It can be verified that $f_{-\alpha}$ satisfies the Bessel equation for such α . Therefore, for each $\alpha \geq 0$ we have the series solution f_α , given by Equation (7.81); and if α is not a positive integer we have found another solution $f_{-\alpha}$ given by Equation (7.82). We have not yet shown that the two solutions f_α and $f_{-\alpha}$ are linearly independent. Before we discuss their independence we shall simplify the form of the solutions. To do this we need some properties of Euler's gamma function, and we digress briefly to discuss these properties.

For each real $s > 0$ we define $\Gamma(s)$ by the improper integral

$$\Gamma(s) = \int_{0+}^{\infty} t^{s-1} e^{-t} dt.$$

It was shown in Volume I (Section 9.28) that this integral converges if $s > 0$ and diverges if $s \leq 0$. Integration by parts leads to the functional equation

$$(7.83) \quad \Gamma(s+1) = s \Gamma(s).$$

This implies that

$$\Gamma(s+2) = (s+1)\Gamma(s+1) = (s+1)s \Gamma(s),$$

$$\Gamma(s+3) = (s+2)\Gamma(s+2) = (s+2)(s+1)s \Gamma(s),$$

and, in general,

$$(7.84) \quad \Gamma(s+n) = (s+n-1) \cdots (s+1)s \Gamma(s)$$

for every positive integer n . Since $\Gamma(1) = \int_0^{\infty} e^{-t} dt = 1$, when we put $s = 1$ in (7.84) we find

$$\Gamma(n+1) = n!.$$

Thus, the gamma function is an extension of the factorial function from integers to positive real numbers.

The functional equation (7.83) can be used to extend the definition of $\Gamma(s)$ to negative values of s that are not integers. We write (7.83) in the form

$$(7.85) \quad \Gamma(s) = \frac{\Gamma(s+1)}{s}.$$

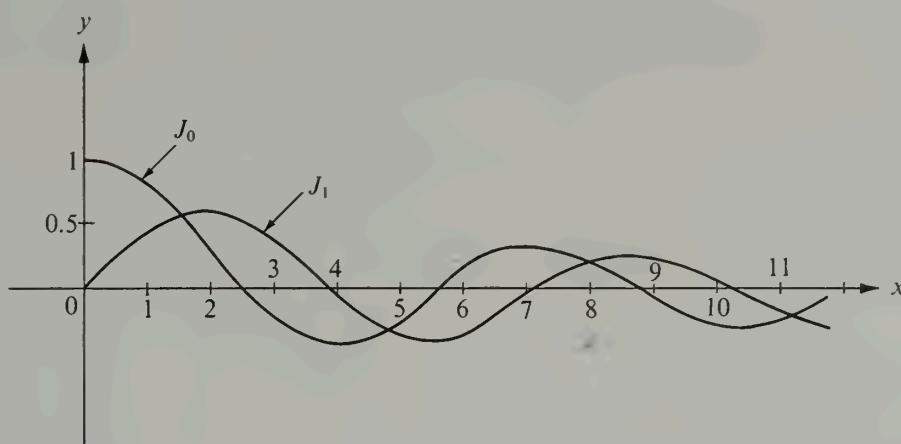
The right-hand member is meaningful if $s+1 > 0$ and $s \neq 0$. Therefore, we can use this equation to *define* $\Gamma(s)$ if $-1 < s < 0$. The right-hand member of (7.85) is now meaningful if $s+2 > 0$, $s \neq -1$, $s \neq 0$, and we can use this equation to define $\Gamma(s)$ for $-2 < s < -1$. Continuing in this manner, we can extend the definition of $\Gamma(s)$ by induction to every open interval of the form $-n < s < -n+1$, where n is a positive integer. The functional equation (7.83) and its extension in (7.84) are now valid for all real s for which both sides are meaningful.

We return now to the discussion of the Bessel equation. The series for f_α in Equation (7.81) contains the product $(1+\alpha)(2+\alpha)\cdots(n+\alpha)$. We can express this product in terms of the gamma function by taking $s = 1+\alpha$ in (7.84). This gives us

$$(1+\alpha)(2+\alpha)\cdots(n+\alpha) = \frac{\Gamma(n+1+\alpha)}{\Gamma(1+\alpha)}.$$

Therefore, if we choose $a_0 = 2^{-\alpha}/\Gamma(1+\alpha)$ in Equation (7.81) and denote the resulting function $f_\alpha(x)$ by $J_\alpha(x)$ when $x > 0$, the solution for $x > 0$ can be written as

$$(7.86) \quad J_\alpha(x) = \left(\frac{x}{2}\right)^\alpha \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(n+1+\alpha)} \left(\frac{x}{2}\right)^{2n}.$$

FIGURE 7.2 Graphs of the Bessel functions J_0 and J_1 .

The function J_α defined by this equation for $x > 0$ and $\alpha \geq 0$ is called the *Bessel function of the first kind of order α* . When α is a nonnegative integer, say $\alpha = p$, the Bessel function J_p is given by the power series

$$J_p(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! (n+p)!} \left(\frac{x}{2}\right)^{2n+p} \quad (p = 0, 1, 2, \dots).$$

Extensive tables of Bessel functions have been constructed. The graphs of the two functions J_0 and J_1 are shown in Figure 7.2.

We can define a new function $J_{-\alpha}$ by replacing α by $-\alpha$ in Equation (7.86), if α is such that $\Gamma(n+1-\alpha)$ is meaningful; that is, if α is not a positive integer. Therefore, if $x > 0$ and $\alpha > 0$, $\alpha \neq 1, 2, 3, \dots$, we define

$$J_{-\alpha}(x) = \left(\frac{x}{2}\right)^{-\alpha} \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(n+1-\alpha)} \left(\frac{x}{2}\right)^{2n}.$$

Taking $s = 1 - \alpha$ in (7.84) we obtain

$$\Gamma(n+1-\alpha) = (1-\alpha)(2-\alpha) \cdots (n-\alpha) \Gamma(1-\alpha)$$

and we see that the series for $J_{-\alpha}(x)$ is the same as that for $f_{-\alpha}(x)$ in Equation (7.82) with $a_0 = 2^\alpha/\Gamma(1-\alpha)$, $x > 0$. Therefore, if α is not a positive integer, $J_{-\alpha}$ is a solution of the Bessel equation for $x > 0$.

If α is not an integer, the two solutions $J_\alpha(x)$ and $J_{-\alpha}(x)$ are linearly independent on the positive real axis (since their ratio is not constant) and the general solution of the Bessel equation for $x > 0$ is

$$y = c_1 J_\alpha(x) + c_2 J_{-\alpha}(x).$$

If α is a nonnegative integer, say $\alpha = p$, we have found only the solution J_p and its constant multiples valid for $x > 0$. Another solution, independent of this one, can be found

by the method described in Example 4 of Section 7.8. We recall that if u_1 is a solution of $y'' + P_1y' + P_2y = 0$ that never vanishes on an interval I , a second solution u_2 independent of u_1 is given by the integral

$$u_2(x) = u_1(x) \int_c^x \frac{Q(t)}{[u_1(t)]^2} dt,$$

where $Q(x) = e^{-\int P_1(x) dx}$. For the Bessel equation we have $P_1(x) = 1/x$, so $Q(x) = 1/x$ and a second solution u_2 is given by the formula

$$(7.87) \quad u_2(x) = J_p(x) \int_c^x \frac{1}{t[J_p(t)]^2} dt,$$

if c and x lie in an interval I in which J_p does not vanish.

This second solution can be put in other forms. For example, from Equation (7.86) we may write

$$\frac{1}{[J_p(t)]^2} = \frac{1}{t^{2p}} g_p(t),$$

where $g_p(0) \neq 0$. In the interval I the function g_p has a power-series expansion

$$g_p(t) = \sum_{n=0}^{\infty} A_n t^n$$

which could be determined by equating coefficients in the identity $g_p(t) [J_p(t)]^2 = t^{2p}$. If we assume the existence of such an expansion, the integrand in (7.87) takes the form

$$\frac{1}{t[J_p(t)]^2} = \frac{1}{t^{2p+1}} \sum_{n=0}^{\infty} A_n t^n.$$

Integrating this formula term by term from c to x we obtain a logarithmic term $A_{2p} \log x$ (from the power t^{-1}) plus a series of the form $x^{-2p} \sum B_n x^n$. Therefore Equation (7.87) takes the form

$$u_2(x) = A_{2p} J_p(x) \log x + J_p(x) x^{-2p} \sum_{n=0}^{\infty} B_n x^n.$$

It can be shown that the coefficient $A_{2p} \neq 0$. If we multiply $u_2(x)$ by $1/A_{2p}$ the resulting solution is denoted by $K_p(x)$ and has the form

$$K_p(x) = J_p(x) \log x + x^{-p} \sum_{n=0}^{\infty} C_n x^n.$$

Having arrived at this formula, we can verify that a solution of this form actually exists by substituting the right-hand member in the Bessel equation and determining the coefficients C_n so as to satisfy the equation. The details of this calculation are lengthy and will be omitted. The final result can be expressed as

$$K_p(x) = J_p(x) \log x - \frac{1}{2} \left(\frac{x}{2} \right)^{-p} \sum_{n=0}^{p-1} \frac{(p-n-1)!}{n!} \left(\frac{x}{2} \right)^{2n} - \frac{1}{2} \left(\frac{x}{2} \right)^p \sum_{n=0}^{\infty} (-1)^n \frac{h_n + h_{n+p}}{n! (n+p)!} \left(\frac{x}{2} \right)^{2n},$$

where $h_0 = 0$ and $h_n = 1 + 1/2 + \cdots + 1/n$ for $n \geq 1$. The series on the right converges for all real x . The function K_p defined for $x > 0$ by this formula is called the *Bessel function of the second kind of order p* . Since K_p is not a constant multiple of J_p , the general solution of the Bessel equation in this case for $x > 0$ is

$$y = c_1 J_p(x) + c_2 K_p(x).$$

Further properties of the Bessel functions are discussed in the next set of exercises.

7.20 Exercises

1. (a) Let f be any solution of the Bessel equation of order α and let $g(x) = x^{1/2}f(x)$ for $x > 0$. Show that g satisfies the differential equation

$$y'' + \left(1 + \frac{1 - 4\alpha^2}{4x^2}\right)y = 0.$$

(b) When $4\alpha^2 = 1$ the differential equation in (a) becomes $y'' + y = 0$; its general solution is $y = A \cos x + B \sin x$. Use this information and the equation† $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ to show that, for $x > 0$,

$$J_{1/2}(x) = \left(\frac{2}{\pi x}\right)^{1/2} \sin x \quad \text{and} \quad J_{-1/2}(x) = \left(\frac{2}{\pi x}\right)^{1/2} \cos x.$$

(c) Deduce the formulas in part (b) directly from the series for $J_{1/2}(x)$ and $J_{-1/2}(x)$.

2. Use the series representation for Bessel functions to show that

$$(a) \frac{d}{dx} (x^\alpha J_\alpha(x)) = x^\alpha J_{\alpha-1}(x),$$

$$(b) \frac{d}{dx} (x^{-\alpha} J_\alpha(x)) = -x^{-\alpha} J_{\alpha+1}(x).$$

3. Let $F_\alpha(x) = x^\alpha J_\alpha(x)$ and $G_\alpha(x) = x^{-\alpha} J_\alpha(x)$ for $x > 0$. Note that each positive zero of J_α is a zero of F_α and is also a zero of G_α . Use Rolle's theorem and Exercise 2 to prove that the positive zeros of J_α and $J_{\alpha+1}$ interlace. That is, there is a zero of J_α between each pair of positive zeros of $J_{\alpha+1}$, and a zero of $J_{\alpha+1}$ between each pair of positive zeros of J_α . (See Figure 7.2.)

4. (a) From the relations in Exercise 2 deduce the recurrence relations

$$\frac{\alpha}{x} J_\alpha(x) + J'_\alpha(x) = J_{\alpha-1}(x) \quad \text{and} \quad \frac{\alpha}{x} J_\alpha(x) - J'_\alpha(x) = J_{\alpha+1}(x).$$

(b) Use the relations in part (a) to deduce the formulas

$$J_{\alpha-1}(x) + J_{\alpha+1}(x) = \frac{2\alpha}{x} J_\alpha(x) \quad \text{and} \quad J_{\alpha-1}(x) - J_{\alpha+1}(x) = 2J'_\alpha(x).$$

† The change of variable $t = u^2$ gives us

$$\Gamma(\tfrac{1}{2}) = \int_0^\infty t^{-1/2} e^{-t} dt = 2 \int_0^\infty e^{-u^2} du = \sqrt{\pi}.$$

(See Exercise 10 of Section 3.8 for a proof that $2 \int_0^\infty e^{-u^2} du = \sqrt{\pi}$.)

5. Use Exercise 1(b) and a suitable recurrence formula to show that

$$J_{3/2}(x) = \left(\frac{2}{\pi x}\right)^{1/2} \left(\frac{\sin x}{x} - \cos x\right).$$

Find a similar formula for $J_{-3/2}(x)$.

6. Prove that

$$\frac{1}{2} \frac{d}{dx} (J_{\alpha}^2(x) + J_{\alpha+1}^2(x)) = \frac{\alpha}{x} J_{\alpha}^2(x) - \frac{\alpha+1}{x} J_{\alpha+1}^2(x)$$

and

$$\frac{d}{dx} (x J_{\alpha}(x) J_{\alpha+1}(x)) = x (J_{\alpha}^2(x) - J_{\alpha+1}^2(x)).$$

7. (a) Use the identities in Exercise 6 to show that

$$J_0^2(x) + 2 \sum_{n=1}^{\infty} J_n^2(x) = 1 \quad \text{and} \quad \sum_{n=0}^{\infty} (2n+1) J_n(x) J_{n+1}(x) = \frac{1}{2} x.$$

(b) From part (a), deduce that $|J_0(x)| \leq 1$ and $|J_n(x)| \leq \frac{1}{2}\sqrt{2}$ for $n = 1, 2, 3, \dots$, and all $x \geq 0$.

8. Let $g_{\alpha}(x) = x^{1/2} f_{\alpha}(ax^b)$ for $x > 0$, where a and b are nonzero constants. Show that g_{α} satisfies the differential equation

$$x^2 y'' + (a^2 b^2 x^{2b} + \frac{1}{4} - \alpha^2 b^2) y = 0$$

if, and only if, f_{α} is a solution of the Bessel equation of order α .

9. Use Exercise 8 to express the general solution of each of the following differential equations in terms of Bessel functions for $x > 0$.

(a) $y'' + xy = 0$.

(c) $y'' + x^m y = 0$.

(b) $y'' + x^2 y = 0$.

(d) $x^2 y'' + (x^4 + \frac{1}{8}) y = 0$.

10. Generalize Exercise 8 when f_{α} and g_{α} are related by the equation $g_{\alpha}(x) = x^c f_{\alpha}(ax^b)$ for $x > 0$. Then find the general solution of each of the following equations in terms of Bessel functions for $x > 0$.

(a) $xy'' + 6y' + y = 0$.

(c) $xy'' + 6y' + x^4 y = 0$.

(b) $xy'' + 6y' + xy = 0$.

(d) $x^2 y'' - xy' + (x+1)y = 0$.

Supplement. Complex numbers

★7.21 Definitions and basic properties

The axioms for the real-number system imply that the square of a real number cannot be negative. (See Theorem 1-20 in Section 1.9, Volume I.) For example, there is no real number whose square is -1 , so the quadratic equation $x^2 + 1 = 0$ has no solution among the real numbers. New types of numbers, called *complex numbers*, have been introduced to provide solutions to such equations. In this section we shall define complex numbers and derive their basic properties.

Complex numbers are merely two-dimensional vectors endowed with a new type of multiplication. That is, a complex number is simply an ordered pair (x_1, x_2) of real numbers, with equality, addition, and multiplication by scalars (real numbers) defined as for two-dimensional vectors:

$$(x_1, x_2) = (y_1, y_2) \quad \text{means} \quad x_1 = y_1 \quad \text{and} \quad x_2 = y_2,$$

$$(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2),$$

$$a(x_1, x_2) = (ax_1, ax_2) \quad (\text{if } a \text{ is real}).$$

In addition, we have multiplication of two complex numbers defined by the equation

$$(7.88) \quad (x_1, x_2)(y_1, y_2) = (x_1y_1 - x_2y_2, x_1y_2 + x_2y_1).$$

It is easy to verify that this multiplication is commutative and associative (properties not possessed by the cross product) and that it is distributive with respect to addition. That is, if x, y, z are arbitrary complex numbers we have

$$xy = yx, \quad x(yz) = (xy)z, \quad \text{and} \quad x(y + z) = xy + xz.$$

For example, to prove the associative law we write $x = (x_1, x_2)$, $y = (y_1, y_2)$, $z = (z_1, z_2)$, and note that

$$\begin{aligned} x(yz) &= (x_1, x_2)(y_1z_1 - y_2z_2, y_1z_2 + y_2z_1) \\ &= (x_1(y_1z_1 - y_2z_2) - x_2(y_1z_2 + y_2z_1), x_1(y_1z_2 + y_2z_1) + x_2(y_1z_1 - y_2z_2)) \\ &= ((x_1y_1 - x_2y_2)z_1 - (x_1y_2 + x_2y_1)z_2, (x_1y_2 + x_2y_1)z_1 + (x_1y_1 - x_2y_2)z_2) \\ &= (x_1y_1 - x_2y_2, x_1y_2 + x_2y_1)(z_1, z_2) = (xy)z. \end{aligned}$$

The commutative and distributive laws may be similarly proved.

The zero complex number $(0, 0)$ is, of course, an identity element for addition. That is, $(x_1, x_2) + (0, 0) = (x_1, x_2)$ for all complex numbers (x_1, x_2) . Also, each complex number has a negative, since $(x_1, x_2) + (-x_1, -x_2) = (0, 0)$. We write $-(x_1, x_2)$ for $(-x_1, -x_2)$.

Multiplication also has an identity element. In fact, we have

$$(x_1, x_2)(1, 0) = (x_1, x_2)$$

so the complex number $(1, 0)$ leaves all complex numbers unchanged by multiplication. Each nonzero complex number has a reciprocal relative to this identity element. That is, if $(x_1, x_2) \neq (0, 0)$ there exists a complex number (y_1, y_2) such that

$$(x_1, x_2)(y_1, y_2) = (1, 0).$$

In fact, this equation is equivalent to the pair of equations

$$x_1y_1 - x_2y_2 = 1 \quad \text{and} \quad x_1y_2 + x_2y_1 = 0,$$

which has the unique solution

$$y_1 = \frac{x_1}{x_1^2 + x_2^2}, \quad y_2 = \frac{-x_2}{x_1^2 + x_2^2}.$$

The condition $(x_1, x_2) \neq (0, 0)$ ensures that $x_1^2 + x_2^2 \neq 0$, so the reciprocal is well defined. We write $(x_1, x_2)^{-1}$ or $1/(x_1, x_2)$ for the reciprocal of (x_1, x_2) . Thus, we have

$$(7.89) \quad \frac{1}{(x_1, x_2)} = \left(\frac{x_1}{x_1^2 + x_2^2}, \frac{-x_2}{x_1^2 + x_2^2} \right) \quad \text{if } (x_1, x_2) \neq (0, 0).$$

The foregoing discussion shows that the set of all complex numbers satisfies the *field properties* of the real-number system (Axioms 1 through 6 in Section 1.7 of Volume I). Therefore, all the laws of algebra deducible from the field axioms also hold for complex numbers. For example, we have cancellation laws for addition and multiplication. If we define $x - y$ to mean $x + (-y)$ we have the usual formulas, such as

$$a - b = -(b - a), \quad a(b - c) = ab - ac, \quad (-a)b = a(-b) = -(ab).$$

Also, we can define the quotient x/y of two complex numbers to be xy^{-1} (if y is nonzero) and we have algebraic formulas such as

$$(a/b)(c/d) = (ac)/(bd), \quad (a/b) + (c/d) = (ad + bc)/(ad),$$

whenever the denominators are nonzero.

The first component, x_1 , of a complex number (x_1, x_2) is called the *real part* of the complex number; the second component, x_2 , is called the *imaginary part*. When we perform algebraic operations on complex numbers with zero imaginary part we find

$$\begin{aligned} (x_1, 0) + (y_1, 0) &= (x_1 + y_1, 0), \\ a(x_1, 0) &= (ax_1, 0) \quad (a \text{ real}), \\ (x_1, 0)(y_1, 0) &= (x_1y_1, 0), \\ (x_1, 0)/(y_1, 0) &= (x_1/y_1, 0) \quad (y_1 \neq 0). \end{aligned}$$

This means that we can perform algebraic operations on complex numbers with zero imaginary part by performing the usual real-number operations on the real parts alone. For this reason we ordinarily make no distinction between the real number x and the complex number $(x, 0)$ whose real part is x ; we agree to identify x and $(x, 0)$ and we write $x = (x, 0)$. In particular, we have $0 = (0, 0)$, $1 = (1, 0)$, $-1 = (-1, 0)$, and so on. Thus, we can think of the complex-number system as an extension of the real-number system.

Complex numbers have some algebraic properties not possessed by real numbers. For example, the quadratic equation $x^2 + 1 = 0$, which has no solution among the real numbers, can now be solved with the use of complex numbers. In fact, the complex number $(0, 1)$ is a solution, since we have

$$(0, 1)^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1.$$

The complex number $(0, 1)$ is denoted by i and is called the *imaginary unit*. It has the property that its square is -1 , $i^2 = -1$. The reader can easily verify that $(-i)^2 = -1$, so $x = -i$ is another solution of the equation $x^2 + 1 = 0$.

In vector notation we have used the symbols $\vec{i}$ and $\vec{j}$ to represent the unit coordinate vectors $(1, 0)$ and $(0, 1)$, respectively. In terms of these symbols, every vector (a, b) can be expressed in the form $(a, b) = a\vec{i} + b\vec{j}$. In complex notation we use the symbols 1 and i (instead of $\vec{i}$ and $\vec{j}$) to represent the complex numbers $(1, 0)$ and $(0, 1)$, respectively. Since $(a, b) = (a, 0) + b(0, 1)$ we can express every complex number in the form†

$$(a, b) = a + bi.$$

† The use of $\vec{i}$ for $(1, 0)$ in vector notation and i for $(0, 1)$ in complex notation should cause no confusion. Some electrical engineers use the symbol j instead of i for the imaginary unit $(0, 1)$, since they wish to reserve i to represent electric current.

The advantage of this notation is that it aids us in algebraic manipulations of formulas involving addition and multiplication. For example, if we multiply $a + bi$ by $c + di$, using the distributive and associative laws, and replace i^2 by -1 , we find

$$(a + bi)(c + di) = ac - bd + (ad + bc)i,$$

which, of course, is in agreement with the definition in (7.88). Similarly, to compute the reciprocal of a nonzero complex number $a + bi$ we may write

$$\frac{1}{a + bi} = \frac{a - bi}{(a + bi)(a - bi)} = \frac{a - bi}{a^2 + b^2} = \frac{a}{a^2 + b^2} - \frac{bi}{a^2 + b^2}.$$

This formula is in agreement with that given in (7.89) for the reciprocal of $x_1 + x_2i$.

By the introduction of complex numbers we have gained much more than the ability to solve the simple quadratic equation $x^2 + 1 = 0$. Consider, for example, the quadratic equation $ax^2 + bx + c = 0$, where a, b, c are real and $a \neq 0$. By completing the square we may write this equation in the form

$$\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a^2} = 0.$$

If $4ac - b^2 \leq 0$ the equation has the real roots $(-b \pm \sqrt{b^2 - 4ac})/(2a)$. If $4ac - b^2 > 0$, the left member is positive for every real x and the equation has no real roots. In this case however, there are two complex roots, given by the formulas

$$(7.90) \quad r_1 = -\frac{b}{2a} + i\frac{\sqrt{4ac - b^2}}{2a} \quad \text{and} \quad r_2 = -\frac{b}{2a} - i\frac{\sqrt{4ac - b^2}}{2a}.$$

In 1799, Gauss proved that every polynomial equation of the form

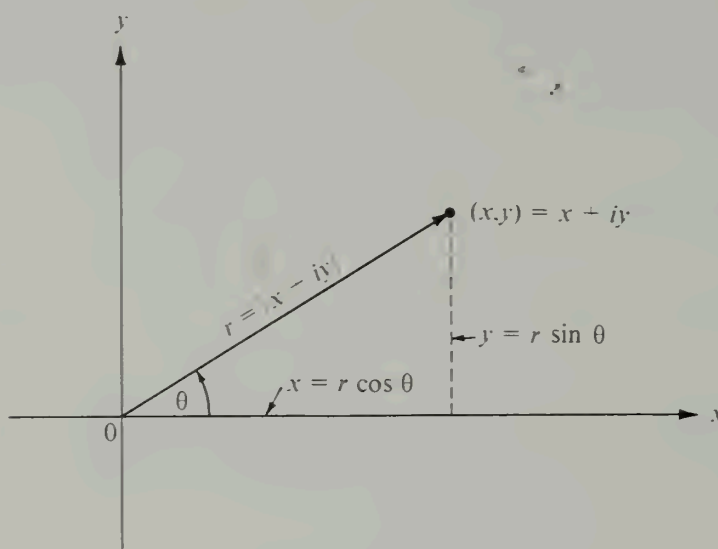
$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n = 0,$$

where $a_0, a_1, \dots, a_n$ are arbitrary real numbers, with $a_n \neq 0$, has a solution among the complex numbers if $n \geq 1$. Moreover, even if the coefficients $a_0, a_1, \dots, a_n$ are complex, a solution exists in the complex-number system. This fact is known as the *fundamental theorem of algebra*.† It shows that there is no need to construct numbers more general than complex numbers to solve polynomial equations with complex coefficients.

★7.22 Geometric interpretation. Modulus and argument

Since a complex number (x, y) is an ordered pair of real numbers it may be represented geometrically by a point in the plane, or by an arrow from the origin to the point (x, y) , as shown in Figure 7.3. In this context, the xy -plane is often referred to as the complex

† A proof of the fundamental theorem of algebra can be found in almost any book on the theory of functions of a complex variable. For example, see K. Knopp, *Theory of Functions*, Dover Publications, New York, 1945, or E. T. Copson, *An Introduction to the Theory of Functions of a Complex Variable*, Oxford Univ. Press, London, 1935. A more elementary proof is given in O. Schreier and E. Sperner, *Introduction to Modern Algebra and Matrix Theory*, Chelsea Publishing Company, New York, 1951.

FIGURE 7.3 Geometric representation of the complex number $x + iy$.

plane. The x -axis is called the real axis; the y -axis is the imaginary axis. If $(x, y) \neq (0, 0)$ we can express x and y in polar coordinates,

$$x = r \cos \theta, \quad y = r \sin \theta,$$

and we obtain

$$(7.91) \quad x + iy = r(\cos \theta + i \sin \theta).$$

The positive number r , which represents the distance of (x, y) from the origin, is called the *modulus* or *absolute value* of $x + iy$, and is denoted by $|x + iy|$. Thus, we have

$$|x + iy| = \sqrt{x^2 + y^2}.$$

The polar angle θ is called an *argument* of $x + iy$. We say *an* argument rather than *the* argument because for a given point (x, y) the angle θ is determined only up to multiples of 2π . Sometimes it is desirable to assign a unique argument to a complex number. This may be done by restricting θ to lie in a half-open interval of length 2π . The intervals $[0, 2\pi)$ and $(-\pi, \pi]$ are commonly used for this purpose. We shall use the interval $(-\pi, \pi]$ and refer to the corresponding θ as the *principal argument* of $x + iy$; we denote this θ by $\arg(x + iy)$. Thus, if $x + iy \neq 0$ and $r = |x + iy|$ we define $\arg(x + iy)$ to be the unique real θ satisfying the conditions

$$x = r \cos \theta, \quad y = r \sin \theta, \quad -\pi < \theta \leq \pi.$$

For the zero complex number we assign the modulus 0 and agree that any real θ may be used as an argument.

Since the absolute value of a complex number z is simply the length of the vector representing z , we have the usual properties of absolute values. For example,

$$|z| > 0 \quad \text{if } z \neq 0, \quad \text{and} \quad |z_1 - z_2| = |z_2 - z_1|.$$

Geometrically, the absolute value $|z_1 - z_2|$ represents the distance between the points z_1 and z_2 in the complex plane. The triangle inequality is also valid,

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

In addition, we have the following formulas for absolute values of products and quotients of complex numbers:

$$(7.92) \quad |z_1 z_2| = |z_1| |z_2|$$

and

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad \text{if } z_2 \neq 0.$$

If we write $z_1 = a + bi$ and $z_2 = c + di$ we obtain (7.92) at once from the identity

$$(ac - bd)^2 + (bc + ad)^2 = (a^2 + b^2)(c^2 + d^2).$$

The formula for $|z_1/z_2|$ follows from (7.92) if we write z_1 as a product,

$$z_1 = z_2 \frac{z_1}{z_2}.$$

If $z = x + iy$, the *complex conjugate* of z is the complex number $\bar{z} = x - iy$. Geometrically, $\bar{z}$ represents the reflection of z through the real axis. The definition of conjugate implies that

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{z_1/z_2} = \bar{z}_1/\bar{z}_2, \quad z\bar{z} = |z|^2.$$

The verification of these properties is left as an exercise for the reader.

If a quadratic equation with real coefficients has no real roots, its complex roots, given by (7.90), are conjugates. Conversely, if r_1 and r_2 are complex conjugates, say $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$, where α and β are real, then r_1 and r_2 are roots of a quadratic equation with real coefficients. In fact, we have

$$r_1 + r_2 = 2\alpha \quad \text{and} \quad r_1 r_2 = \alpha^2 + \beta^2,$$

so

$$(x - r_1)(x - r_2) = x^2 - (r_1 + r_2)x + r_1 r_2,$$

and the quadratic equation in question is

$$x^2 - 2\alpha x + \alpha^2 + \beta^2 = 0.$$

★7.23 Complex exponentials

We wish now to extend the definition of e^x so that it becomes meaningful when x is replaced by any complex number z . We wish this extension to be such that the law of exponents, $e^a e^b = e^{a+b}$, will be valid for all complex a and b . And, of course, we want e^z to agree with the usual exponential when z is real. There are several equivalent ways to carry out this extension. Probably the simplest is by means of the following definition.

DEFINITION. If $z = x + iy$, we define e^z to be the complex number given by the equation

$$(7.93) \quad e^z = e^x (\cos y + i \sin y).$$

Note that $e^z = e^x$ when $y = 0$, so this exponential agrees with the usual exponential when z is real. To justify this definition we shall use it to deduce the law of exponents.

If $a = x + iy$ and $b = u + iv$, we have

$$e^a = e^x(\cos y + i \sin y), \quad e^b = e^u(\cos v + i \sin v),$$

so

$$e^a e^b = e^x e^u [\cos y \cos v - \sin y \sin v + i(\cos y \sin v + \sin y \cos v)].$$

Now we use the addition formulas for $\cos(y + v)$ and $\sin(y + v)$ and the law of exponents for real exponentials and we see that the foregoing equation becomes

$$(7.94) \quad e^a e^b = e^{x+u} [\cos(y + v) + i \sin(y + v)].$$

Since $a + b = (x + u) + i(y + v)$, the right member of (7.94) is e^{a+b} . This proves that $e^a e^b = e^{a+b}$ for all complex a and b .

If we take $x = 0$ and $y = \theta$ in (7.93), we obtain the formula

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

valid for every real θ . Comparing this with Equation (7.91), we see that the polar-coordinate form of a complex number can be expressed in terms of complex exponentials. In fact, (7.91) becomes

$$x + iy = re^{i\theta}.$$

This shows that every complex number $x + iy \neq 0$ can be expressed in the form $re^{i\theta}$, with $r = |x + iy|$ and $\theta = \arg(x + iy) + 2n\pi$, n being any integer. This representation of complex numbers is especially useful in connection with multiplication and division of complex numbers. For example, if $z_1 = r_1 e^{i\theta}$ and $z_2 = r_2 e^{i\phi}$, we have

$$(7.95) \quad z_1 z_2 = r_1 e^{i\theta} r_2 e^{i\phi} = r_1 r_2 e^{i(\theta+\phi)}.$$

Therefore the product of the moduli, $r_1 r_2$, is the modulus of the product $z_1 z_2$, in agreement with Equation (7.92), and the sum of the arguments, $\theta + \phi$, is an admissible argument for the product $z_1 z_2$.

When $z = re^{i\theta}$, repeated application of (7.95) gives us the formula

$$z^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta),$$

valid for any nonnegative integer n . This formula is also valid for negative integers n if we define z^{-m} to be $(z^{-1})^m$ when m is a positive integer.

Similarly, we have

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta}}{r_2 e^{i\phi}} = \frac{r_1}{r_2} e^{i(\theta-\phi)},$$

so the modulus of z_1/z_2 is r_1/r_2 and the difference $\theta - \phi$ is an admissible argument for z_1/z_2 .

★7.24 Complex-valued functions

A function f whose values are complex numbers is called a complex-valued function. If the domain of f is a set of real numbers, f is called a complex-valued function of a real variable. If the domain is a set of complex numbers, f is called a complex-valued function of a complex variable, or more simply, a function of a complex variable. An example is the exponential function, defined by the equation

$$f(z) = e^z$$

for all complex z . Most of the familiar elementary functions of calculus, such as the exponential, the logarithm, and the trigonometric functions, can be extended to become functions of a complex variable. (See Exercises 11 and 12 in Section 7.25.) In this more general framework many new properties and interrelationships are often revealed. For example, the complex exponential function is periodic. In fact, if $z = x + iy$ and if n is any integer we have

$$e^{z+2n\pi i} = e^x[\cos(y + 2n\pi) + i \sin(y + 2n\pi)] = e^x(\cos y + i \sin y) = e^z.$$

Thus we see that $f(z + 2n\pi i) = f(z)$, so f has the period $2\pi i$. This property of the exponential function is revealed only when we study the exponential as a function of a complex variable.

The first systematic treatment of the differential and integral calculus of functions of a complex variable was given by Cauchy early in the 19th century. Since then the theory has developed into one of the most important and interesting branches of mathematics. It has become an indispensable tool for physicists and engineers and has connections in nearly every branch of pure mathematics. A discussion of this theory will not be given here. We shall discuss only one example of a complex-valued function. This is the function f of a real variable defined for all real x by the equation

$$(7.96) \quad f(x) = e^{tx},$$

where t is a fixed complex number. Functions of this type occur in the study of linear differential equations with constant coefficients. When t is real, the derivative of this function is given by the formula

$$(7.97) \quad f'(x) = te^{tx}.$$

We will show that, with a natural extension of the definition of derivative, this formula is also valid for complex t .

Suppose f is a complex-valued function defined for all real x in an open interval I . For a fixed x in I and real $h \neq 0$, we form the difference quotient

$$\frac{f(x+h) - f(x)}{h}.$$

If there is a complex number z such that

$$\lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x)}{h} - z \right| = 0,$$

we say that z is the derivative of f at x and we write $z = f'(x)$.

For each x in I , $f(x)$ is a complex number, so we can write

$$f(x) = u(x) + iv(x),$$

where $u(x)$ and $v(x)$ are real. This equation determines two real-valued functions u and v called, respectively, the real and imaginary parts of f . It is easy to prove that $f'(x)$ exists if, and only if, both derivatives $u'(x)$ and $v'(x)$ exist, in which case

$$f'(x) = u'(x) + iv'(x).$$

Let us apply this formula to compute the derivative of the function f given by (7.96). Suppose $t = \alpha + i\beta$, where α and β are real. By the definition of the complex exponential we have

$$f(x) = e^{tx} = e^{\alpha x + i\beta x} = e^{\alpha x} \cos \beta x + i e^{\alpha x} \sin \beta x.$$

Therefore the real and imaginary parts of f are given by

$$(7.98) \quad u(x) = e^{\alpha x} \cos \beta x \quad \text{and} \quad v(x) = e^{\alpha x} \sin \beta x.$$

These functions are differentiable for all x and their derivatives are given by the formulas

$$u'(x) = \alpha e^{\alpha x} \cos \beta x - \beta e^{\alpha x} \sin \beta x, \quad v'(x) = \alpha e^{\alpha x} \sin \beta x + \beta e^{\alpha x} \cos \beta x.$$

Since $f'(x) = u'(x) + iv'(x)$ we have

$$\begin{aligned} f'(x) &= \alpha e^{\alpha x} (\cos \beta x + i \sin \beta x) + i \beta e^{\alpha x} (\cos \beta x + i \sin \beta x) \\ &= (\alpha + i\beta) e^{(\alpha + i\beta)x} = t e^{tx}. \end{aligned}$$

Thus we see that the derivative formula (7.97) holds when t is complex. Differentiating once more we find $f''(x) = t^2 e^{tx}$.

Suppose now that t is a root of the quadratic equation

$$(7.99) \quad t^2 + at + b = 0.$$

Then we have

$$f''(x) + af'(x) + bf(x) = e^{tx}(t^2 + at + b) = 0,$$

so the function f is a complex-valued solution of the differential equation $L(y) = 0$, where

$$L(y) = y'' + ay' + by.$$

If a and b are real numbers, the real and imaginary parts of f will satisfy the differential equation $L(y) = 0$ separately. In fact, if we write $f(x) = u(x) + iv(x)$, then $f'(x) = u'(x) + iv'(x)$ and $f''(x) = u''(x) + iv''(x)$. Substituting $y = f(x)$ in the equation $L(y) = 0$ we find

$$u''(x) + au'(x) + bu(x) + i[v''(x) + av'(x) + bv(x)] = 0.$$

Since a complex number can be zero only if both its real and imaginary parts are zero, we see that u and v separately satisfy $L(y) = 0$. This shows that the functions u and v given by (7.98) are solutions of the differential equation $L(y) = 0$ if $\alpha + i\beta$ and $\alpha - i\beta$ are the roots of the quadratic equation (7.99).

★7.25 Exercises

1. Express the following complex numbers in the form $a + bi$.

- | | |
|--------------------------|--|
| (a) $(1 + i)^2$. | (e) $(1 + i)/(1 - 2i)$. |
| (b) $1/i$. | (f) $i^5 + i^{16}$. |
| (c) $1/(1 + i)$. | (g) $1 + i + i^2 + i^3$. |
| (d) $(2 + 3i)(3 - 4i)$. | (h) $\frac{1}{2}(1 + i)(1 + i^{-8})$. |

2. Compute the absolute values of the following complex numbers.

- | | |
|-------------------------|-----------------------------|
| (a) $1 + i$. | (d) $1 + i + i^2$. |
| (b) $3 + 4i$. | (e) $i^7 + i^{10}$. |
| (c) $(1 + i)/(1 - i)$. | (f) $2(1 - i) + 3(2 + i)$. |

3. Make a sketch showing the set of all z in the complex plane which satisfy each of the following conditions.

- | | |
|---------------------------|-------------------------------|
| (a) $ z < 1$. | (e) $ z - i = z + i $. |
| (b) $z + \bar{z} = 1$. | (f) $ z - 1 + z + 1 = 2$. |
| (c) $z - \bar{z} = i$. | (g) $ z - 1 - z + 1 = 2$. |
| (d) $ z - 1 = z + 1 $. | (h) $z + \bar{z} = z ^2$. |

4. Express in the form $re^{i\theta}$, with $-\pi < \theta \leq \pi$.

- | | |
|------------------------|--------------------------|
| (a) $2i$. | (f) $(1 + i)/\sqrt{2}$. |
| (b) $-3i$. | (g) $(-1 + i)^3$. |
| (c) -1 . | (h) $(-1 - i)^3$. |
| (d) 1 . | (i) $1/(1 + i)$. |
| (e) $-3 + \sqrt{3}i$. | (j) $1/(1 + i)^2$. |

5. (a) Given two distinct complex numbers z_1 and z_2 and a positive real number c . Describe the set of all z satisfying

$$|z - z_1| = c |z - z_2|$$

for $c = 1$, $c < 1$, and $c > 1$.

(b) Do the same as in part (a) for the inequality

$$|z - z_1| < c |z - z_2|.$$

6. Let f be a polynomial with real coefficients.

(a) Show that $f(\bar{z}) = \overline{f(z)}$ for every complex z .

(b) Use part (a) to deduce that the nonreal zeros of f (if any exist) must occur in pairs of conjugate complex numbers.

7. (a) If θ is real, show that

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

(b) Use the formulas in (a) to deduce the identities

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta), \quad \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta).$$

8. (a) Prove that $e^z \neq 0$ for all complex z .

(b) Find all complex z such that $e^z = 1$.

9. (a) Prove DeMoivre's theorem:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta,$$

valid for every real θ and every positive integer n .

(b) Take $n = 3$ in part (a) and deduce the trigonometric identities

$$\sin 3\theta = 3 \cos^2\theta \sin \theta - \sin^3\theta, \quad \cos 3\theta = \cos^3\theta - 3 \cos \theta \sin^2\theta.$$

10. Given a complex number $z \neq 0$. Write $z = re^{i\theta}$, where $\theta = \arg(z)$. Let $z_1 = Re^{i\alpha}$, where $R = r^{1/n}$ and $\alpha = \theta/n$, and let $\epsilon = e^{2\pi i/n}$, where n is a positive integer.

(a) Show that $z_1^n = z$. That is, z_1 is an n th root of z .

(b) Show that z has n distinct n th roots,

$$z_1, \epsilon z_1, \epsilon^2 z_1, \dots, \epsilon^{n-1} z_1,$$

and no others.

(c) Determine the three cube roots of i .

(d) Determine the four fourth roots of i .

(e) Determine the four fourth roots of $-i$.

11. The definitions of the sine and cosine functions can be extended to the complex plane as follows:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

When z is real, these formulas agree with the ordinary sine and cosine functions. (See Exercise 7.) Use these formulas to deduce the following properties of complex sines and cosines. Here u , v , and z denote complex numbers, with $z = x + iy$.

(a) $\sin(u + v) = \sin u \cos v + \cos u \sin v$.

(b) $\cos(u + v) = \cos u \cos v - \sin u \sin v$.

(c) $\sin^2 z + \cos^2 z = 1$.

(d) $\cos(iy) = \cosh y$, $\sin(iy) = i \sinh y$.

(e) $\cos z = \cos x \cosh y - i \sin x \sinh y$.

(f) $\sin z = \sin x \cosh y + i \cos x \sinh y$.

12. If z is a nonzero complex number, we define $\text{Log } z$, the complex logarithm of z , by the equation

$$\text{Log } z = \log |z| + i \arg(z).$$

When z is real and positive this formula agrees with the ordinary logarithm. Use this formula to deduce the following properties of complex logarithms.

(a) $\text{Log}(-1) = \pi i$, $\text{Log}(i) = \pi i/2$.

(b) $\text{Log}(z_1 z_2) = \text{Log } z_1 + \text{Log } z_2 + 2n\pi i$, where n is an integer.

(c) $\text{Log}(z_1/z_2) = \text{Log } z_1 - \text{Log } z_2 + 2n\pi i$, where n is an integer.

(d) $e^{\text{Log } z} = z$.

INTRODUCTION TO NUMERICAL ANALYSIS

8.1 Historical introduction

The planet Uranus was discovered in 1781 by a gifted amateur astronomer, William Herschel (1738–1822), with a homemade 10-ft telescope. With the use of Kepler's laws, the expected orbit of Uranus was quickly calculated from a few widely separated observations. It was found that the mean distance of Uranus from the sun was about twice that of Saturn and that one complete orbit would require 84 years. By 1830 the accumulated empirical data showed deviations from the scheduled orbit that could not be accounted for. Some astronomers felt that Newton's law of universal gravitation might not hold for distances as large as that of Uranus from the sun; others suspected that the perturbations were due to a hitherto undiscovered comet or more distant planet.

An undergraduate student at Cambridge University, John Couch Adams (1819–1892), was intrigued by the possibility of an undiscovered planet. He set himself the difficult task of calculating what the orbit of such a planet must be to account for the observed positions of Uranus, assuming the validity of Newton's law of gravitation. He completed his calculations in 1845 and asked the Royal Observatory at Greenwich to search for the hypothetical planet, but his request was not taken seriously.

A similar calculation was made independently and almost simultaneously by Jean Joseph Leverrier (1811–1877) of Paris, who asked Johann Galle, head of the Berlin Observatory, to confirm his prediction. The same evening that he received Leverrier's letter, Galle found the new planet, *Neptune*, almost exactly in its calculated position. This was another triumph for Newton's law of gravitation, and one of the first major triumphs of *numerical analysis*, the art and science of computation.

The history of numerical analysis goes back to ancient times. As early as 2000 B.C. the Babylonians were compiling mathematical tables. One clay tablet has been found containing the squares of the integers from 1 to 60. The Babylonians worshipped the heavenly bodies and kept elaborate astronomical records. The celebrated Alexandrian astronomer Claudius Ptolemy (circa 150 A.D.) possessed a Babylonian record of eclipses dating from 747 B.C.

In 220 B.C., Archimedes used regular polygons as approximations to a circle and deduced the inequalities $3\frac{10}{71} < \pi < 3\frac{1}{7}$. Numerical work from that time until the 17th century was centered principally around the preparation of astronomical tables. The advent of algebra in the 16th century brought about renewed activity in all branches of

mathematics, including numerical analysis. In 1614, Napier published the first table of logarithms. In 1620, the logarithms of the sine and tangent functions were tabulated to seven decimal places. By 1628, fourteen-place tables of the logarithms of the numbers from 1 to 100,000 had been computed.

Computations with infinite series began to flourish near the end of the 17th century, along with the development of the calculus. Early in the 18th century Jacob Stirling and Brook Taylor laid the foundations of the *calculus of finite differences*, which now plays a central role in numerical analysis. With the prediction of the existence and location of the planet Neptune by Adams and Leverrier in 1845, the scientific importance of numerical analysis became established once and for all.

Late in the 19th century the development of automatic calculating machinery further stimulated the growth of numerical analysis. This growth has been explosive since the end of World War II because of the progress in high-speed electronic computing devices. The new machines have made possible a great many outstanding scientific achievements which previously seemed unattainable.

The art of computation (as distinct from the science of computation) lays much stress on the detailed planning required in a particular calculation. It also deals with such matters as precision, accuracy, errors, and checking. This aspect of numerical analysis will not be discussed here; it is best learned by carrying out actual numerical calculations with specific problems. For valuable advice on practical methods and techniques the reader should consult the existing books on numerical analysis, some of which are listed in the bibliography at the end of this chapter. The bibliography also contains some of the standard mathematical tables; many of them also give practical information on how to carry out a specific calculation.

This chapter provides an introduction to the *science* of computation. It contains some of the basic mathematical principles that might be required of almost anyone who uses numerical analysis, whether he works with a desk calculator or with a large-scale high-speed computing machine. Aside from its practical value, the material in this chapter is of interest in its own right, and it is hoped that this brief introduction will stimulate the reader to learn more about this important and fascinating branch of mathematics.

8.2 Approximations by polynomials

A basic idea in numerical analysis is that of using simple functions, usually polynomials, to approximate a given function f . One type of polynomial approximation was discussed in Volume I in connection with Taylor's formula (Section 7.6). The problem there was to find a polynomial P which agrees with a given function f and some of its derivatives at a given point. We proved that if f is a function with a derivative of order n at a point x_0 , there is one and only one polynomial P of degree $\leq n$ which satisfies the $n + 1$ relations

$$P(x_0) = f(x_0), \quad P'(x_0) = f'(x_0), \quad \dots, \quad P^{(n)}(x_0) = f^{(n)}(x_0).$$

The solution is given by the *Taylor polynomial*,

$$P(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

We also discussed the error incurred in approximating $f(x)$ by $P(x)$ at points x other than x_0 . This error is defined to be the difference $R(x) = f(x) - P(x)$, so we may write

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R(x).$$

To make further statements about the error $R(x)$ we need more information about the function f . For example, suppose we know that f has a derivative of order $n + 1$ everywhere in an open interval (a, b) and that $f^{(n)}$ is continuous on the closed interval $[a, b]$. Then if x and x_0 are distinct points in $[a, b]$ the error term $R(x)$ can be expressed in the form

$$R(x) = (x - x_0)^{n+1} \frac{f^{(n+1)}(c)}{(n+1)!},$$

where c lies between x and x_0 . (See Theorem 7-6 in Volume I.)

There are many other ways to approximate a given function f by polynomials, depending on the use to be made of the approximation. For example, instead of asking for a polynomial that agrees with f and some of its derivatives at a given point, we may ask for a polynomial that takes the same values as f at a number of distinct points. Specifically, if the given distinct points are $x_0, x_1, \dots, x_n$ we may seek a polynomial P satisfying the conditions

$$(8.1) \quad P(x_0) = f(x_0), \quad P(x_1) = f(x_1), \quad \dots, \quad P(x_n) = f(x_n).$$

Since there are $n + 1$ conditions to be satisfied we try a polynomial of degree $\leq n$, say

$$P(x) = \sum_{k=0}^n a_k x^k,$$

with $n + 1$ coefficients $a_0, a_1, \dots, a_n$ to be determined. The $n + 1$ conditions (8.1) lead to a system of $n + 1$ linear equations for the coefficients. From the theory of linear equations it can be shown that this system has one and only one solution; hence such a polynomial always exists. If the equations are solved by Cramer's rule the coefficients $a_0, a_1, \dots, a_n$ are expressed as quotients of determinants. In practice, however, the polynomial P is seldom determined in this manner because the calculations are extremely laborious when n is large. Simpler methods have been developed to calculate the polynomial approximation. Some of these will be discussed in later sections. The polynomial which solves the foregoing problem is called an *interpolating polynomial*.

Another common type of polynomial approximation is the so-called *least-square approximation*. Here the given function f is defined and integrable on an interval $[a, b]$ and we seek a polynomial P of degree $\leq n$ such that the mean-square error

$$(8.2) \quad \int_a^b |f(x) - P(x)|^2 dx$$

will be as small as possible. In Section 8.10 we shall prove that for a continuous f such a polynomial exists and is uniquely determined. The Legendre polynomials introduced in Section 7.17 play a fundamental role in the solution of this problem.

Least-square approximation can be generalized by considering weighted least-square approximation. Instead of minimizing the integral in (8.2) we try to minimize the integral

$$(8.3) \quad \int_a^b |f(x) - P(x)|^2 w(x) dx,$$

where w is a nonnegative "weight" function. Another generalization may be obtained by using powers other than the second power in (8.2) or (8.3). For example, if p is any real number ≥ 1 we can seek a polynomial P which minimizes the integral

$$\int_a^b |f(x) - P(x)|^p dx,$$

or, more generally, a polynomial P which minimizes

$$\int_a^b |f(x) - P(x)|^p w(x) dx$$

with a nonnegative weight function w .

There is another type of polynomial approximation in which we begin with a function f that is continuous on an interval $[a, b]$ and consider the *absolute deviation*

$$A(x) = |f(x) - P(x)|,$$

where P is a polynomial of degree n . Since the function A is continuous on $[a, b]$, it attains its absolute maximum value somewhere on $[a, b]$. This maximum value depends on the choice of the polynomial P and we may denote it by $M(P)$. Thus we have

$$M(P) = \max_{a \leq x \leq b} |f(x) - P(x)|.$$

Now we can try to choose P so that this maximum will be as small as possible. It can be shown that for each given f such a polynomial P always exists and is uniquely determined. (A special case is considered in Section 8.12.) Some writers refer to this P as the *polynomial of uniform approximation*. It is usually very difficult to actually determine this P once f is given. In fact, no general procedure is known that will always yield P in a finite number of steps.

8.3 Polynomial approximation and normed linear spaces

All the different types of polynomial approximation described in the foregoing section are related by one central idea. To describe this idea we introduce the concept of a linear space of functions.

DEFINITION OF A LINEAR SPACE OF FUNCTIONS. Let S be a nonempty collection of real-valued functions defined on a given interval I . The set S is called a linear space of functions whenever it has the following two properties:

- (a) If $f \in S$, then $cf \in S$ for every real number c .
- (b) If $f \in S$ and $g \in S$, then $f + g \in S$.

Linear spaces of functions occur throughout mathematics. For example, in Chapter 7 we showed that the solutions of a homogeneous linear differential equation on an interval I form a linear space. Another example is the linear space consisting of all functions

defined and continuous on a given closed bounded interval I . This particular linear space will be denoted by $C(I)$, or simply by C . Much of the theory of polynomial approximation deals with the space C and some of its subspaces.† Among the subspaces of C are (a) the set of all polynomials, (b) the set of all polynomials of degree $\leq n$, (c) the set of all functions differentiable on I , (d) the set of all continuous functions having a derivative of order n at a fixed point x_0 in I , (e) the set of all continuous functions having a power-series expansion in a neighborhood of a point x_0 in I . It is easy to verify that each of these sets of functions is a linear space. Proofs are requested in Exercise 1 of Section 8.4.

When we speak of “approximating” one function f in C by another function g in C , we consider the difference $f - g$, which we call the *error* of the approximation, and then we decide on a way to measure the size of this error. Since the error is a member of C , we need a measure of the “size” of the functions in C . Now the members of a linear space of functions are analogous to vectors. To extend this analogy we shall require the “size” of a function to have the same algebraic properties as the length of a vector. Measures with these properties are called *norms*; they are defined as follows:

DEFINITION OF A NORM. Let S be a linear space of functions. A real-valued function N defined on S is called a norm if it has the following properties:

- (a) $N(f) \geq 0$ for each f in S .
- (b) $N(cf) = |c|N(f)$ for each f in S and each real number c .
- (c) $N(f + g) \leq N(f) + N(g)$ for all f and g in S .
- (d) $N(f) = 0$ implies $f = 0$.

A linear space with a norm assigned to it is called a normed linear space.

The norm of f is sometimes written $\|f\|$ instead of $N(f)$. In this notation, the fundamental properties become:

- (a) $\|f\| \geq 0$,
- (b) $\|cf\| = |c| \|f\|$,
- (c) $\|f + g\| \leq \|f\| + \|g\|$,
- (d) $\|f\| = 0$ implies $f = 0$.

These properties are entirely analogous to the corresponding properties of the length of a vector. It is helpful to carry this analogy further and to refer to $\|f - g\|$ as the “distance” between two functions f and g . Thus, when we approximate one function by another, the error in the approximation is measured by the distance between the two functions relative to the chosen norm.

A function N that satisfies properties (a), (b), and (c), but *not* (d), is called a *seminorm*. Some problems in the theory of approximation deal with seminormed linear spaces; others with normed linear spaces. The following examples will be discussed in this chapter.

Example 1: Taylor seminorm. For a fixed integer $n \geq 1$, let S denote the set of all functions having a derivative of order n at a fixed point x_0 . The set S is a linear space. If $f \in S$, let

† A *subspace* of C is a subset of C that is itself a linear space.

$$N(f) = \sum_{k=0}^n |f^{(k)}(x_0)|.$$

It is easy to verify that the function N so defined is a seminorm. It is not a norm because $N(f) = 0$ if and only if

$$f(x_0) = f'(x_0) = \cdots = f^{(n)}(x_0) = 0,$$

and these equations can be satisfied by a nonzero function. For example, $N(f) = 0$ when $f(x) = (x - x_0)^{n+1}$.

Example 2: Interpolation seminorm. Let S denote the linear space of all real-valued functions defined on an interval $[a, b]$. For a fixed set of $n + 1$ distinct points $x_0, x_1, \dots, x_n$ in $[a, b]$, let N be defined by the equation

$$N(f) = \sum_{k=0}^n |f(x_k)|$$

if $f \in S$. This function N is a seminorm on S . It is not a norm because $N(f) = 0$ if and only if $f(x_0) = f(x_1) = \cdots = f(x_n) = 0$, and it is clear that these equations can be satisfied by a function f that is not zero everywhere on $[a, b]$.

Example 3: Square norm. Let C denote the linear space of functions continuous on an interval $[a, b]$. If $f \in C$ define

$$(8.4) \quad N(f) = \left(\int_a^b |f(x)|^2 dx \right)^{1/2}.$$

The function N is a norm on C . Properties (a) and (b) are clearly satisfied. A proof of property (c) is outlined in Exercise 4 of Section 8.4. To prove that N satisfies property (d), let f be any function in C that is not everywhere zero on $[a, b]$. Then there is at least one point x_0 in $[a, b]$ at which $f(x_0) \neq 0$. By continuity of f , there is a subinterval $[c, d]$ of $[a, b]$ containing x_0 in which $|f(x)| > \frac{1}{2} |f(x_0)|$. Since

$$N(f) = \left(\int_a^b |f(x)|^2 dx \right)^{1/2} \geq \left(\int_c^d |f(x)|^2 dx \right)^{1/2} > \frac{(d - c)^{1/2} |f(x_0)|}{2} > 0,$$

we have $N(f) > 0$. This shows that $N(f) = 0$ only if $f(x) = 0$ for all x in $[a, b]$, so property (d) is satisfied.

Note. Let S denote the set of functions f that are integrable on $[a, b]$. The set S is a linear space, and the function N defined by (8.4) is a seminorm on S . It is not a norm because we can have $N(f) = 0$ without f being identically zero on $[a, b]$.

Example 4: Maximum norm. Let C denote the linear space of functions continuous on an interval $[a, b]$. If $f \in C$, define

$$N(f) = \max_{a \leq x \leq b} |f(x)|,$$

where the symbol on the right stands for the absolute maximum value of $|f|$ on $[a, b]$. The verification of all four norm properties is requested in Exercise 5 of Section 8.4.

Let C be the space of functions continuous on a given interval $[a, b]$, and let S be a linear subspace of C . For a given integer $n \geq 1$, assume S contains all polynomials of

degree $\leq n$. Assume also that a norm or seminorm has been defined on S . Choose a function f in S . If there is a polynomial P of degree $\leq n$ such that

$$\|f - P\| \leq \|f - Q\|$$

for all polynomials Q of degree $\leq n$, we say that P is a *best polynomial approximation* to f with the specified degree. The term "best" is, of course, relative to the given norm (or seminorm). The best polynomial for one choice of norm need not be best for another choice of norm.

Once a norm or seminorm has been chosen, three problems immediately suggest themselves.

(1) *Existence*: Given f in S , is there a best polynomial approximation to f with the specified degree?

(2) *Uniqueness*: If a best polynomial approximation to f exists with the specified degree, is it uniquely determined?

(3) *Construction*: If a best polynomial approximation to f exists with the specified degree, how can it be determined?

There are, of course, many other problems that can be considered. For example, if a unique best polynomial P_n of degree $\leq n$ exists, we may wish to obtain upper bounds for $\|f - P_n\|$ that can be used to satisfy practical requirements. Or we may ask whether $\|f - P_n\| \rightarrow 0$ as $n \rightarrow \infty$ for the given norm or possibly for some other norm. If so, we say that the polynomial approximations converge to f in this norm. In such a case arbitrarily close approximations exist relative to this norm if n is sufficiently large. These examples illustrate some of the types of problems considered in the general theory of polynomial approximation. In this introductory treatment we restrict our attention primarily to the three problems of existence, uniqueness, and construction, as described above.

For approximation by Taylor polynomials these three problems can be completely solved. If f has a derivative of order n at a point x_0 , it is easy to prove that the best polynomial approximation of degree $\leq n$ relative to the Taylor seminorm for this n is the Taylor polynomial

$$(8.5) \quad P(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

In fact, for this polynomial we have

$$\|f - P\| = \sum_{k=0}^n |f^{(k)}(x_0) - P^{(k)}(x_0)| = 0,$$

so the inequality $\|f - P\| \leq \|f - Q\|$ is trivially satisfied for all polynomials Q . Therefore P is a best polynomial approximation relative to this seminorm. To establish uniqueness, we consider any polynomial Q of degree $\leq n$ such that $\|f - Q\| = 0$. This equation implies that

$$Q(x_0) = f(x_0), \quad Q'(x_0) = f'(x_0), \quad \dots, \quad Q^{(n)}(x_0) = f^{(n)}(x_0).$$

From the discussion in Section 7.6 of Volume I we know that the Taylor polynomial in (8.5) is the only polynomial satisfying all these equations. Therefore $Q = P$. Equation (8.5) also solves the problem of construction.

The corresponding problems for the interpolation seminorm will be treated next in Section 8.5. In later sections we discuss polynomial approximation relative to the square norm and relative to the maximum norm.

8.4 Exercises

1. Prove that each of the following collections of functions is a linear space.
 - (a) All polynomials.
 - (b) All polynomials of degree $\leq n$.
 - (c) All functions continuous on an interval I .
 - (d) All functions having a derivative at each point of I .
 - (e) All functions having a derivative of order n at each point of I .
 - (f) All functions having a derivative of order n at a fixed point x_0 .
 - (g) All functions having power-series expansions in a neighborhood of a given point x_0 .
2. Determine whether or not each of the following collections of real-valued functions is a linear space.
 - (a) All polynomials of degree n .
 - (b) All functions defined and bounded on an interval $[a, b]$.
 - (c) All step functions defined on an interval $[a, b]$.
 - (d) All functions monotonic on an interval $[a, b]$.
 - (e) All functions integrable on an interval $[a, b]$.
 - (f) All functions that are piecewise monotonic on an interval $[a, b]$.
 - (g) All functions that can be expressed in the form $f - g$, where f and g are monotonic increasing on an interval $[a, b]$.

3. Let C denote the linear space of real-valued functions continuous on an interval $[a, b]$. A function N is defined on C by the equation given. In each case, determine which of the four properties of a norm are satisfied by N , and determine thereby whether N is a norm, a seminorm, or neither.

- | | |
|---------------------------------|--|
| (a) $N(f) = f(a)$. | (e) $N(f) = \left \int_a^b f(x) dx \right $. |
| (b) $N(f) = f(a) $. | (f) $N(f) = \int_a^b f(x) dx$. |
| (c) $N(f) = f(b) - f(a) $. | (g) $N(f) = \int_a^b f(x) ^2 dx$. |
| (d) $N(f) = \int_a^b f(x) dx$. | (h) $N(f) = \left \int_a^b f(x) dx \right ^2$. |

4. (a) If f and g are integrable on $[a, b]$, we have

$$\int_a^b [x f(t) + g(t)]^2 dt \geq 0$$

for all real x . This inequality can be written in the form $Ax^2 + 2Bx + C \geq 0$, where

$$A = \int_a^b [f(t)]^2 dt, \quad B = \int_a^b f(t)g(t)dt, \quad C = \int_a^b [g(t)]^2 dt.$$

Prove that $B^2 \leq AC$ or, in other words, that

$$\left(\int_a^b f(t)g(t) dt \right)^2 \leq \left(\int_a^b [f(t)]^2 dt \right) \left(\int_a^b [g(t)]^2 dt \right).$$

This is called the *Cauchy-Schwarz inequality* for integrals.

(b) Let

$$N(f) = \left(\int_a^b [f(t)]^2 dt \right)^{1/2}.$$

Show that the Cauchy-Schwarz inequality implies $\int_a^b f(t)g(t) dt \leq N(f)N(g)$ and deduce that $N(f+g) \leq N(f) + N(g)$.

(c) Show that N is a norm for the linear space of all functions continuous on $[a, b]$.

(d) Show that N is a seminorm for the linear space of all functions integrable on $[a, b]$.

5. Let C be the linear space of functions continuous on an interval $[a, b]$. If $f \in C$, define

$$N(f) = \max_{a \leq x \leq b} |f(x)|.$$

Show that N is a norm for C .

6. Let B denote the linear space of all real-valued functions that are defined and bounded on an interval $[a, b]$. If $f \in B$, define

$$N(f) = \text{lub}_{a \leq x \leq b} |f(x)|,$$

where the symbol on the right stands for the least upper bound of the set of all numbers $|f(x)|$ for x in $[a, b]$. Show that N is a norm for B .

7. Refer to Exercise 3. Determine which of the given functions N have the property that $N(fg) \leq N(f)N(g)$ for all f and g in C .

8. For a fixed integer $n \geq 1$, let S be the set of all functions having a derivative of order n at a fixed point x_0 . If $f \in S$, let

$$N(f) = \sum_{k=0}^n \frac{1}{k!} |f^{(k)}(x_0)|.$$

(a) Show that N is a seminorm on S .

(b) Show that $N(fg) \leq N(f)N(g)$ for all f, g in S . Prove also that the Taylor seminorm does not have this property.

8.5 Interpolating polynomials

We turn now to approximation by interpolation polynomials. The values of a function f are known at $n+1$ distinct points $x_0, x_1, \dots, x_n$ and we seek a polynomial P of degree $\leq n$ that satisfies the conditions

$$(8.6) \quad P(x_0) = f(x_0), \quad P(x_1) = f(x_1), \quad \dots, \quad P(x_n) = f(x_n).$$

First we prove that if such a polynomial exists it is unique. Then we prove it exists by explicit construction. This polynomial minimizes the distance from f to P , measured in the interpolation seminorm for this n ,

$$\|f - P\| = \sum_{k=0}^n |f(x_k) - P(x_k)|.$$

Since this distance is 0 if P satisfies (8.6), the interpolating polynomial P is the best approximation relative to this seminorm.

8-1 THEOREM. *Uniqueness theorem.* Given $n+1$ distinct points $x_0, x_1, \dots, x_n$, let P and Q be two polynomials of degree $\leq n$ such that

$$P(x_k) = Q(x_k)$$

for each $k = 0, 1, 2, \dots, n$. Then $P(x) = Q(x)$ for all x .

Proof. Let $R(x) = P(x) - Q(x)$. The function R is a polynomial of degree $\leq n$ which has $n + 1$ distinct zeros at the points $x_0, x_1, \dots, x_n$. The only polynomial with this property is the zero polynomial. Therefore $R(x) = 0$ for all x , so $P(x) = Q(x)$ for all x .

The interpolating polynomial P can be constructed in many ways. We describe first a method of Lagrange. Let $A(x)$ be the polynomial given by the equation

$$(8.7) \quad A(x) = (x - x_0)(x - x_1) \cdots (x - x_n) = \prod_{j=0}^n (x - x_j).$$

This polynomial has a simple zero at each of the points x_j . Let $A_k(x)$ denote the polynomial of degree n obtained from $A(x)$ by deleting the factor $x - x_k$. That is, let

$$(8.8) \quad A_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n (x - x_j).$$

The polynomial $A_k(x)$ has a simple zero at each point $x_j \neq x_k$. At the point x_k itself we have

$$(8.9) \quad A_k(x_k) = \prod_{\substack{j=0 \\ j \neq k}}^n (x_k - x_j).$$

This is nonzero since no factor in the product is zero. Therefore the polynomial $A_k(x)/A_k(x_k)$ has the value 1 when $x = x_k$ and the value 0 when $x = x_j$ for $x_j \neq x_k$. Now let

$$P(x) = \sum_{k=0}^n \frac{f(x_k)A_k(x)}{A_k(x_k)}.$$

When $x = x_j$, each term in this sum vanishes except the j th term, which has the value $f(x_j)$. Therefore $P(x_j) = f(x_j)$ for each j . Since each term of this sum is a polynomial of degree n , the sum itself is a polynomial of degree $\leq n$. Thus, we have found a polynomial satisfying the required conditions. These results may be summarized by the following theorem:

8- 2 THEOREM. Given $n + 1$ distinct points $x_0, x_1, \dots, x_n$ and $n + 1$ real numbers $f(x_0), f(x_1), \dots, f(x_n)$, not necessarily distinct, there exists one and only one polynomial P of degree $\leq n$ such that $P(x_j) = f(x_j)$ for each $j = 0, 1, 2, \dots, n$. This polynomial is given by the formula

$$(8.10) \quad P(x) = \sum_{k=0}^n \frac{f(x_k)A_k(x)}{A_k(x_k)},$$

where $A_k(x)$ is the polynomial defined by (8.8).

Formula (8.10) for $P(x)$ is called *Lagrange's interpolation formula*. We may write it in the form

$$P(x) = \sum_{k=0}^n f(x_k)L_k(x),$$

where $L_k(x)$ is a polynomial of degree n given by

$$(8.11) \quad L_k(x) = \frac{A_k(x)}{A_k(x_k)}.$$

Thus, for each fixed x , $P(x)$ is a linear combination of the prescribed values $f(x_0), f(x_1), \dots, f(x_n)$. The multipliers $L_k(x)$ depend only on the points $x_0, x_1, \dots, x_n$ and not on the prescribed values. They are called *Lagrange interpolation coefficients*. If we use the formulas in (8.8) and (8.9) we may write Equation (8.11) in the form

$$(8.12) \quad L_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{x - x_j}{x_k - x_j}.$$

This product formula provides an efficient method for evaluating the number $L_k(x)$ for a given x .

Note. The Lagrange coefficients $L_k(x)$ are often expressed in the form

$$L_k(x) = \frac{A_k(x)}{A'(x_k)},$$

where A' is the derivative of the polynomial in (8.7). To prove this formula it suffices to show that $A'(x_k) = A_k(x_k)$. Differentiating the relation

$$A(x) = (x - x_k)A_k(x)$$

we obtain $A'(x) = (x - x_k)A'_k(x) + A_k(x)$. When $x = x_k$ this gives us $A'(x_k) = A_k(x_k)$.

Example. Determine the polynomial of degree ≤ 3 that takes the values y_0, y_1, y_2, y_3 at the points $-2, -1, 1, 2$, respectively.

Solution. We take $x_0 = -2, x_1 = -1, x_2 = 1, x_3 = 2$. The polynomials $L_k(x)$ in (8.12) are given by the formulas

$$L_0(x) = \frac{(x+1)(x-1)(x-2)}{(-2+1)(-2-1)(-2-2)} = -\frac{1}{12}(x+1)(x-1)(x-2),$$

$$L_1(x) = \frac{(x+2)(x-1)(x-2)}{(-1+2)(-1-1)(-1-2)} = \frac{1}{6}(x+2)(x-1)(x-2),$$

$$L_2(x) = \frac{(x+2)(x+1)(x-2)}{(1+2)(1+1)(1-2)} = -\frac{1}{6}(x+2)(x+1)(x-2),$$

$$L_3(x) = \frac{(x+2)(x+1)(x-1)}{(2+2)(2+1)(2-1)} = \frac{1}{12}(x+2)(x+1)(x-1).$$

Therefore the required polynomial is

$$\begin{aligned} P(x) &= y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x) \\ &= -\frac{y_0}{12}(x+1)(x-1)(x-2) + \frac{y_1}{6}(x+2)(x-1)(x-2) \\ &\quad -\frac{y_2}{6}(x+2)(x+1)(x-2) + \frac{y_3}{12}(x+2)(x+1)(x-1). \end{aligned}$$

To compute the value of $P(x)$ for a specific x it is usually better to leave the polynomial in this form rather than to rewrite it in increasing powers of x . For example, if $y_0 = -5$, $y_1 = 1$, $y_2 = 1$, and $y_3 = 7$, the value of $P(x)$ for $x = \frac{3}{2}$ is given by

$$\begin{aligned} P\left(\frac{3}{2}\right) &= \frac{5}{12}\left(\frac{5}{2}\right)\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \frac{1}{6}\left(\frac{7}{2}\right)\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) - \frac{1}{6}\left(\frac{7}{2}\right)\left(\frac{5}{2}\right)\left(-\frac{1}{2}\right) + \frac{7}{12}\left(\frac{7}{2}\right)\left(\frac{5}{2}\right)\left(\frac{1}{2}\right) \\ &= -\frac{25}{96} - \frac{7}{48} + \frac{35}{48} + \frac{245}{96} = \frac{276}{96} = 2\frac{7}{8}. \end{aligned}$$

In the foregoing discussion the interpolation points $x_0, x_1, \dots, x_n$ were assumed to be distinct but otherwise arbitrary. Now we assume they are equally spaced and show that the Lagrange coefficients $L_k(x)$ can be considerably simplified. Suppose $x_0 < x_1 < x_2 < \dots < x_n$, and let h denote the distance between adjacent points. Then we may write

$$x_j = x_0 + jh$$

for $j = 0, 1, 2, \dots, n$. Since $x_k - x_j = (k - j)h$, Equation (8.12) becomes

$$(8.13) \quad L_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{x - x_0 - jh}{(k - j)h} = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{t - j}{k - j},$$

where

$$t = \frac{x - x_0}{h}.$$

In the last term on the right of (8.13) the product of the factors independent of t is

$$\begin{aligned} (8.14) \quad \prod_{\substack{j=0 \\ j \neq k}}^n \frac{1}{k - j} &= \left(\prod_{j=0}^{k-1} \frac{1}{k - j} \right) \left(\prod_{j=k+1}^n \frac{1}{k - j} \right) = \frac{1}{k!} \prod_{j=k+1}^n \frac{(-1)}{j - k} \\ &= \frac{(-1)^{n-k}}{k! (n - k)!} = \frac{(-1)^{n-k}}{n!} \binom{n}{k}, \end{aligned}$$

where $\binom{n}{k}$ is the binomial coefficient. Since $x = x_0 + th$, Equation (8.13) now becomes

$$(8.15) \quad L_k(x_0 + th) = \frac{(-1)^{n-k}}{n!} \binom{n}{k} \prod_{\substack{j=0 \\ j \neq k}}^n (t - j).$$

For each fixed n , the right member of (8.15) is a function of k and t that can be tabulated. Extensive tables of the Lagrangian coefficients for equally spaced interpolation points have been prepared by the National Bureau of Standards. (See Reference 11 in the bibliography at the end of this chapter.) If x and h are chosen so that the number $t = (x - x_0)/h$ is one for which the Lagrangian coefficients $L_k(x_0 + th)$ are tabulated,

the actual calculation of $P(x_0 + th)$ is reduced to a multiplication of the $f(x_k)$ by the tabulated $L_k(x_0 + th)$, followed by addition.

8.6 Error analysis in polynomial interpolation

Let f be a function defined on an interval $[a, b]$ containing the $n + 1$ distinct points $x_0, x_1, \dots, x_n$, and let P be the interpolation polynomial of degree $\leq n$ which agrees with f at these points. If we alter the values of f at points other than the interpolation points we do not alter the polynomial P . This shows that the function f and the polynomial P may differ considerably at points other than the interpolation points. If the given function f has certain qualities of "smoothness" throughout the interval $[a, b]$ we can expect that the interpolating polynomial P will be a good approximation to f at points other than the x_k . The next theorem gives a useful expression that enables us to study the error in polynomial interpolation when the given function has a derivative of order $n + 1$ throughout $[a, b]$.

8- 3 THEOREM. Let $x_0, x_1, \dots, x_n$ be $n + 1$ distinct points in the domain of a function f , and let P be the interpolation polynomial of degree $\leq n$ that agrees with f at these points. Choose a point x in the domain of f and let $[\alpha, \beta]$ be any closed interval containing the points $x_0, x_1, \dots, x_n$, and x . If f has a derivative of order $n + 1$ in the interval $[\alpha, \beta]$ there is at least one point c in the open interval (α, β) such that

$$(8.16) \quad f(x) - P(x) = \frac{A(x)}{(n+1)!} f^{(n+1)}(c),$$

where

$$A(x) = (x - x_0)(x - x_1) \cdots (x - x_n).$$

Note. Point c depends on both x and n .

Proof. If x is one of the interpolation points x_k , then $A(x_k) = 0$ and Equation (8.16) is trivially satisfied for any choice of c in (α, β) . Suppose, then, that x is not one of the interpolation points. Keep x fixed and define a new function F on $[\alpha, \beta]$ by the equation

$$(8.17) \quad F(t) = A(x)[f(t) - P(t)] - A(t)[f(x) - P(x)].$$

The right-hand side of this equation, as a function of t , has a derivative of order $n + 1$; hence the same is true of the left-hand side. Since $P(t)$ is a polynomial in t of degree $\leq n$, its $(n + 1)$ st derivative is identically zero. The polynomial $A(t)$ has degree $n + 1$, the term of highest degree being t^{n+1} , and we have $A^{(n+1)}(t) = (n + 1)!$. Therefore, if we differentiate Equation (8.17) $n + 1$ times with respect to t we obtain the formula

$$(8.18) \quad F^{(n+1)}(t) = A(x)f^{(n+1)}(t) - (n + 1)! [f(x) - P(x)].$$

From the definition in Equation (8.17) we see that F has the value zero at the $n + 1$ interpolation points $x_0, x_1, \dots, x_n$ and also at the point x . Therefore $F(t) = 0$ at $n + 2$ distinct points in the interval $[\alpha, \beta]$. These points determine $n + 1$ adjacent subintervals of $[\alpha, \beta]$ and the function F vanishes at both endpoints of each of these subintervals. By Rolle's theorem, the derivative $F'(t)$ must be zero for at least one t interior to each sub-

interval. If we choose exactly one such t from each subinterval we obtain $n + 1$ distinct points in the open interval (α, β) at which $F'(t) = 0$. These points, in turn, determine n subintervals at whose endpoints we have $F'(t) = 0$. Applying Rolle's theorem to F' we find that the second derivative $F''(t)$ is zero for at least n distinct points in (α, β) . After applying Rolle's theorem $n + 1$ times in this manner we finally find that there is at least one point c in (α, β) at which $F^{(n+1)}(c) = 0$. Substituting this value of c in Equation (8.18) we obtain

$$(n + 1)! [f(x) - P(x)] = A(x) f^{(n+1)}(c),$$

which is the same as (8.16). This completes the proof.

It should be noted that, as with approximation by Taylor polynomials, the error term involves the $(n + 1)$ st derivative $f^{(n+1)}(c)$ evaluated at an unknown point c . If the extreme values of $f^{(n+1)}$ in $[a, b]$ are known, useful upper and lower bounds for the error can be obtained.

Suppose now that the interpolation points are equally spaced and that $x_0 < x_1 < x_2 < \dots < x_n$. If h denotes the spacing we may write

$$x_j = x_0 + jh \quad \text{and} \quad x = x_0 + th,$$

where $t = (x - x_0)/h$. Since $x - x_j = (t - j)h$, the polynomial $A(x)$ may be written as

$$A(x) = \prod_{j=0}^n (x - x_j) = h^{n+1} \prod_{j=0}^n (t - j).$$

Formula (8.16) now becomes

$$(8.19) \quad f(x) - P(x) = \frac{f^{(n+1)}(c)}{(n + 1)!} h^{n+1} \prod_{j=0}^n (t - j),$$

with $t = (x - x_0)/h$.

Example: Error in linear interpolation. Suppose a function f with a second derivative is tabulated and we wish to estimate its value at a point x intermediate to two consecutive entries x_0 and $x_0 + h$. If we use linear interpolation we approximate the graph of f over the interval $[x_0, x_0 + h]$ by a straight line, as shown in Figure 8.1. If P denotes the linear interpolating polynomial, the error estimate in (8.19) becomes

$$(8.20) \quad f(x) - P(x) = \frac{f''(c)}{2!} h^2 t(t - 1),$$

where $t = (x - x_0)/h$. When x lies between x_0 and $x_0 + h$ we have $0 < t < 1$ and the maximum value of $|t(t - 1)|$ in this interval is $1/4$. Therefore (8.20) gives us the estimate

$$|f(x) - P(x)| \leq \frac{|f''(c)| h^2}{8}.$$

The point c is an unknown point in the interval $(x_0, x_0 + h)$. If the second derivative f'' is bounded in this interval, say $|f''(x)| \leq M$, the error estimate becomes

$$|f(x) - P(x)| \leq \frac{Mh^2}{8}.$$

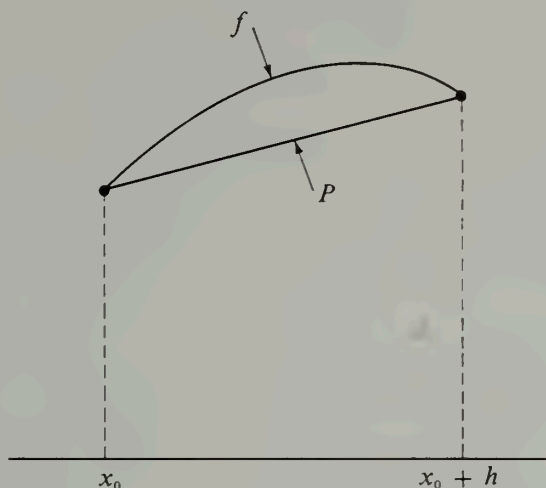


FIGURE 8.1 Linear interpolation.

In particular, if f is a sine or cosine, then $|f''(x)| \leq 1$ for all x and we have $|f(x) - P(x)| \leq h^2/8$. If a table of sines or cosines has entries for every degree (one degree = $\pi/180$ radians) we have $h = \pi/180$, so

$$\frac{h^2}{8} = \frac{\pi^2}{8(180)^2} < \frac{10}{259,200} < \frac{1}{25,000} = 0.00004.$$

Since this error does not exceed $1/2$ in the fourth decimal place, linear interpolation would be satisfactory in a four-place table. The error estimate can be improved in portions of the table where $|f''(c)|$ is considerably less than 1.

8.7 Exercises

- In each case find the polynomial P of lowest possible degree satisfying the given conditions.
 - $P(-1) = 0$, $P(0) = 2$, $P(2) = 7$.
 - $P(1) = 1$, $P(2) = 0$, $P(3) = 0$, $P(4) = 1$.
 - $P(1) = 1$, $P(2) = 2$, $P(3) = 3$, $P(0) = 1$.
 - $P(0) = -2$, $P(1) = 0$, $P(-1) = -2$, $P(2) = 16$.
 - $P(-2) = 11$, $P(-1) = -11$, $P(0) = -5$, $P(1) = -1$.
- Let $f(x) = \cos(\pi x/4)$. Find the polynomial of smallest possible degree that takes the same values as f at the points -2 , $-4/3$, 0 , $4/3$, 2 .
- Let P be a polynomial of degree $\leq n$ and let $A(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$, where $x_0, x_1, \dots, x_n$ are $n + 1$ distinct points.
 - Show that for any polynomial B the polynomial Q given by $Q(x) = P(x) + A(x)B(x)$ agrees with P at the points $x_0, x_1, \dots, x_n$.
 - Prove also the converse. That is, if Q is any polynomial that agrees with P at the points $x_0, x_1, \dots, x_n$, then $Q(x) = P(x) + A(x)B(x)$ for some polynomial B .

4. (a) Find the polynomial Q of lowest possible degree that satisfies the conditions

$$Q(-2) = -5, \quad Q(-1) = -1, \quad Q(1) = 1, \quad Q'(0) = -1.$$

[Hint. First find a polynomial P that takes the prescribed values at $-2, -1, 1$, and then use Exercise 3 to determine Q .]

- (b) Find the polynomial Q of lowest possible degree that satisfies the conditions in part (a) with $Q'(0) = -3$ instead of $Q'(0) = -1$.

5. Let $f(x) = \log_4 x$ for $x > 0$. Compute $P(32)$, where P is the polynomial of lowest possible degree that agrees with f at the points:

(a) $x = 1, 64$.

(c) $x = 4, 16, 64$.

(b) $x = 1, 16, 256$.

(d) $x = 1, 4, 16, 64, 256$.

In each case compute the difference $f(32) - P(32)$. These examples show that the accuracy in polynomial interpolation is not necessarily improved by increasing the number of interpolation points.

6. The Lagrange interpolation coefficients $L_k(x)$ given by Equation (8.12) depend not only on x but also on the interpolation points $x_0, x_1, \dots, x_n$. We can indicate this dependence by writing $L_k(x) = L_k(x; \vec{X})$, where $\vec{X}$ denotes the vector in $(n+1)$ -space given by $\vec{X} = (x_0, x_1, \dots, x_n)$. For a given real number b , let $\vec{b}$ denote the vector in $(n+1)$ -space all of whose components are equal to b . If $a \neq 0$, show that

$$L_k(ax + b; a\vec{X} + \vec{b}) = L_k(x; \vec{X}).$$

This is called the *invariance property* of the Lagrange interpolation coefficients. The next exercise shows how this property can be used to help simplify calculations in practice.

7. Let P denote the polynomial of degree ≤ 4 that has the values

$$P(2.4) = 72, \quad P(2.5) = 30, \quad P(2.7) = 18, \quad P(2.8) = 24, \quad P(3.0) = 180.$$

- (a) Introduce new interpolation points u_j related to the given points x_j by the equation $u_j = 10x_j - 24$. The u_j are integers. For each $k = 0, 1, 2, 3, 4$, determine the Lagrange interpolation coefficients $L_k(x)$ in terms of u , where $u = 10x - 24$.

- (b) Use the invariance property of Exercise 6 to compute $P(2.6)$.

8. A table of the function $f(x) = \log x$ contains entries for $x = 1$ to $x = 10$ at intervals of 0.001. Values intermediate to each pair of consecutive entries are to be computed by linear interpolation. Assume the entries in the table are exact.

- (a) Show that the error in linear interpolation will not exceed $1/8$ in the sixth decimal place.

- (b) For what values of x will linear interpolation be satisfactory for a seven-place table?

- (c) What should be the spacing of the entries in the interval $1 \leq x \leq 2$ so that linear interpolation will be satisfactory in a seven-place table?

In Exercises 9 through 13, $x_0, x_1, \dots, x_n$ are distinct points and

$$A(x) = \prod_{j=0}^n (x - x_j), \quad A_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n (x - x_j), \quad L_k(x) = \frac{A_k(x)}{A_k(x_k)}.$$

9. Derive the formula

$$A'(x) = \sum_{k=0}^n A_k(x)$$

by use of (a) logarithmic differentiation; (b) Lagrange's interpolation formula.

10. Prove each of the following formulas:

$$(a) \sum_{k=0}^n L_k(x) = 1 \quad \text{and} \quad \sum_{k=0}^n \frac{A'_k(x)}{A'(x_k)} = 0 \quad \text{for all } x.$$

$$(b) \sum_{k=0}^n \frac{1}{A'(x_k)} = 0. \quad [\text{Hint. Use part (a) with suitable values of } x.]$$

11. Let P be any polynomial of degree $\leq n$. Show that the coefficient of x^n is equal to

$$\sum_{k=0}^n \frac{P(x_k)}{A'(x_k)}.$$

12. Assume $x_0, x_1, \dots, x_n$ are integers satisfying $x_0 < x_1 < \dots < x_n$.

(a) Prove that $|A'(x_k)| \geq k! (n-k)!$ and deduce that

$$\sum_{k=0}^n \frac{1}{|A'(x_k)|} \leq \frac{2^n}{n!}.$$

(b) Let P be any polynomial of degree n , with the term of highest degree equal to x^n . Let M denote the largest of the numbers $|P(x_0)|, |P(x_1)|, \dots, |P(x_n)|$. Prove that $M \geq n!/2^n$. [Hint. Use part (a) and Exercise 11.]

13. Prove the following formulas. In parts (a) and (b), x is any point different from $x_0, x_1, \dots, x_n$.

$$(a) \frac{A'(x)}{A(x)} = \sum_{j=0}^n \frac{1}{x - x_j}.$$

$$(b) \frac{A''(x)}{A'(x)} = \frac{A_k(x)}{A'(x)} \sum_{\substack{j=0 \\ j \neq k}}^n \frac{1}{x - x_j} + \sum_{\substack{j=0 \\ j \neq k}}^n \frac{1}{x - x_j} - \frac{A(x)}{A'(x)} \sum_{\substack{j=0 \\ j \neq k}}^n \frac{1}{(x - x_j)^2}.$$

$$(c) \frac{A''(x_k)}{A'(x_k)} = 2 \sum_{\substack{j=0 \\ j \neq k}}^n \frac{1}{x_k - x_j}.$$

14. (a) Given $n+1$ distinct points $x_0, x_1, \dots, x_n$, let P and Q be two polynomials of degree $\leq n$ satisfying the $n+1$ conditions

$$P(x_0) = Q(x_0), \quad P'(x_1) = Q'(x_1), \quad P''(x_2) = Q''(x_2), \quad \dots, \quad P^{(n)}(x_n) = Q^{(n)}(x_n).$$

Prove that $P(x) = Q(x)$ for all x .

(b) Let $B_0(x) = 1$, and for $n \geq 1$ define

$$B_n(x) = \frac{x(x-n)^{n-1}}{n!}.$$

Show that $B'_n(x) = B_{n-1}(x-1)$ for $n \geq 1$ and deduce that

$$B_n(0) = B'_n(1) = B''_n(2) = \dots = B_n^{(n-1)}(n-1) = 0 \quad \text{and} \quad B_n^{(n)}(n) = 1.$$

(c) Show that the one and only polynomial of degree $\leq n$ satisfying the conditions

$$P(0) = c_0, \quad P'(1) = c_1, \quad P''(2) = c_2, \quad \dots, \quad P^{(n)}(n) = c_n$$

is given by

$$P(x) = \sum_{k=0}^n c_k B_k(x).$$

(d) If $x_k = x_0 + kh$ for $k = 0, 1, 2, \dots, n$, where $h > 0$, generalize the results in (b) and (c).

8.8 Polynomial approximation by least squares

Let C denote the linear space consisting of all functions that are continuous on an interval $[a, b]$. If $f \in C$, let $\|f\|$ denote the square norm, defined by

$$\|f\| = \left(\int_a^b [f(x)]^2 dx \right)^{1/2}.$$

We wish to consider the problem of best polynomial approximation relative to this norm.

Choose a function f in C and an integer $n \geq 1$ and consider any polynomial T of degree $\leq n$, say

$$T(x) = \sum_{i=0}^n a_i x^i.$$

A polynomial that minimizes $\|f - T\|$ also minimizes $\|f - T\|^2$, so we consider

$$\|f - T\|^2 = \int_a^b [f(x) - T(x)]^2 dx = \int_a^b \left(f(x) - \sum_{i=0}^n a_i x^i \right)^2 dx.$$

The integral on the right is a function of the $n + 1$ coefficients, which we denote as $g(a_0, a_1, \dots, a_n)$ or as $g(\vec{A})$, where $\vec{A} = (a_0, a_1, \dots, a_n)$. We seek those points $\vec{A}$ in $(n + 1)$ -space at which the scalar field g will have a minimum. If such points exist they must satisfy $\nabla g(\vec{A}) = \vec{0}$. Since

$$g(\vec{A}) = \int_a^b [f(x)]^2 dx - 2 \sum_{i=0}^n a_i \int_a^b x^i f(x) dx + \sum_{i=0}^n \sum_{j=0}^n a_i a_j \int_a^b x^{i+j} dx,$$

the k th partial derivative $D_k g(\vec{A})$ exists for each $\vec{A}$ and is given by the formula

$$\begin{aligned} D_k g(\vec{A}) &= -2 \int_a^b x^k f(x) dx + \sum_{i=0}^n a_i \int_a^b x^{i+k} dx + \sum_{j=0}^n a_j \int_a^b x^{k+j} dx \\ &= -2 \int_a^b x^k f(x) dx + 2 \sum_{i=0}^n a_i \int_a^b x^{i+k} dx. \end{aligned}$$

When we put $D_k g(\vec{A}) = 0$ for $k = 0, 1, 2, \dots, n$ we obtain a system of $n + 1$ linear equations that must be satisfied by the coefficients $a_0, a_1, \dots, a_n$ of a minimizing polynomial, if it exists. It can be shown that this system always has exactly one solution, and that this solution does, indeed, minimize g . In this way it is possible to establish the existence and uniqueness of the best polynomial approximation relative to the least-square norm. However, because of the labor involved, the coefficients are rarely determined by solving this system of equations.

The problem of least-square polynomial approximation can be expressed in a more general form that is somewhat easier to deal with. Instead of writing the polynomial T

in powers of x , we consider any set of $n + 1$ polynomials $\phi_0(x), \phi_1(x), \dots, \phi_n(x)$, where $\phi_0(x) = 1$ and $\phi_k(x)$ has degree k , and we write T in the form

$$T(x) = \sum_{i=0}^n c_i \phi_i(x).$$

(For a proof that such a representation is always possible, see Exercise 1 in Section 8.11.) With this representation of T the square of the norm $\|f - T\|^2$ defines a function of the $n + 1$ coefficients $c_0, c_1, \dots, c_n$, say

$$h(c_0, c_1, \dots, c_n) = \|f - T\|^2 = \int_a^b \left(f(x) - \sum_{i=0}^n c_i \phi_i(x) \right)^2 dx.$$

Arguing as above, we are led to consider the $n + 1$ equations $D_k h(c_0, c_1, \dots, c_n) = 0$. In this case we have

$$D_k h(c_0, c_1, \dots, c_n) = -2 \int_a^b \phi_k(x) f(x) dx + 2 \sum_{j=0}^n c_j \int_a^b \phi_j(x) \phi_k(x) dx,$$

so the coefficients in question must satisfy the system of $n + 1$ equations

$$(8.21) \quad \sum_{j=0}^n c_j \int_a^b \phi_j(x) \phi_k(x) dx = \int_a^b \phi_k(x) f(x) dx, \quad k = 0, 1, 2, \dots, n.$$

Now we choose polynomials ϕ_k which make this system easy to solve. The best choice is that which makes the coefficient of c_k equal to 1 in the k th equation and equal to zero in the remaining equations. This means we wish to choose the polynomials ϕ_k so that

$$(8.22) \quad \int_a^b \phi_j(x) \phi_k(x) dx = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

If we can do this the k th equation in (8.21) gives us

$$(8.23) \quad c_k = \int_a^b \phi_k(x) f(x) dx.$$

Any set of polynomials satisfying (8.22) is called an *orthonormal set*. In the next section we show that orthonormal sets of polynomials can always be constructed. Then we prove that in the space of continuous functions the coefficients given by (8.23) lead to the best approximating polynomial relative to the square norm.

8.9 The Gram-Schmidt process

We shall construct an orthonormal set of polynomials by a method known as the *Gram-Schmidt process*. This method is applicable not only to polynomials but, more generally, to any linearly independent set† of continuous functions. It is best described in terms of the *inner product* of two functions, which is defined as follows.

DEFINITION. Let f and g be two functions that are integrable on an interval $[a, b]$. The inner product of f and g , denoted by (f, g) , is defined by the equation

† The concept of linear independence was defined in Section 7.10.

$$(8.24) \quad (f, g) = \int_a^b f(x) g(x) dx.$$

Equation (8.24) is analogous to the formula

$$\vec{X} \cdot \vec{Y} = \sum_{k=1}^n x_k y_k$$

for the dot product of two vectors $\vec{X} = (x_1, \dots, x_n)$ and $\vec{Y} = (y_1, \dots, y_n)$ in n -space. The inner product of two functions satisfies the same algebraic laws as the dot product of two vectors. It is clearly commutative, $(f, g) = (g, f)$, and it has the following properties which follow at once from the linearity property of the integral:

$$(8.25) \quad (cf, g) = (f, cg) = c(f, g) \quad \text{for every constant } c,$$

and

$$(8.26) \quad (f_1 + f_2, g) = (f_1, g) + (f_2, g), \quad (f, g_1 + g_2) = (f, g_1) + (f, g_2).$$

By analogy with vectors we call two functions f and g *orthogonal* on $[a, b]$ if $(f, g) = 0$. If $n + 1$ functions $f_0, f_1, \dots, f_n$ have the property that

$$(f_i, f_k) = \begin{cases} 0 & \text{if } i \neq k \\ 1 & \text{if } i = k \end{cases}$$

the collection $\{f_0, f_1, \dots, f_n\}$ is said to be *orthonormal* on $[a, b]$.

The equation

$$(8.27) \quad \|f\| = (f, f)^{1/2} = \left(\int_a^b [f(x)]^2 dx \right)^{1/2}$$

defines a seminorm on the linear space S of functions that are integrable on $[a, b]$. The collection C of functions continuous on $[a, b]$ is a linear subspace of S and Equation (8.27) defines a norm on C . In the following discussion we need to know that $\|f\| \neq 0$ when $f \neq 0$, so we confine our attention to the space C .

Suppose $g_0, g_1, \dots, g_n$ are $n + 1$ linearly independent functions in C . From these functions we shall construct an orthonormal set $\{f_0, f_1, \dots, f_n\}$. The f_k will be constructed in succession so that each is orthogonal to all the earlier f_j . To start the process we choose

$$f_0(x) = c_0 g_0(x).$$

Since $\|f_0\| = |c_0| \|g_0\|$ we can satisfy the condition $\|f_0\| = 1$ by choosing $c_0 = \|g_0\|^{-1}$, provided the norm of g_0 is nonzero. However, $g_0 \neq 0$ because g_0 is a member of a linearly independent set of functions. Since g_0 is continuous we have $\|g_0\| \neq 0$, so the specified choice of c_0 is always possible.

Next we wish to choose f_1 so that $\|f_1\| = 1$ and $(f_1, f_0) = 0$. There are two conditions to be satisfied. Therefore, we try a linear combination of g_0 and g_1 with two constants to be determined. Since $f_0 = c_0 g_0$ we may write this combination in the form

$$(8.28) \quad f_1(x) = c_1 [g_1(x) - a_{1,0} f_0(x)].$$

The inner product of f_1 and f_0 is given by

$$\begin{aligned} (f_1, f_0) &= (c_1 [g_1 - a_{1,0} f_0], f_0) = c_1 (g_1 - a_{1,0} f_0, f_0) \\ &= c_1 [(g_1, f_0) - a_{1,0} (f_0, f_0)] = c_1 [(g_1, f_0) - a_{1,0}]. \end{aligned}$$

The choice $a_{1,0} = (g_1, f_0)$ will make $(f_1, f_0) = 0$. Equation (8.28) now becomes

$$f_1(x) = c_1 [g_1(x) - (g_1, f_0)f_0(x)].$$

Let $h = g_1 - (g_1, f_0)f_0$. Since h is a linear combination of g_0 and g_1 , h is not identically zero (because the coefficient of g_1 is nonzero). Therefore $\|h\| \neq 0$ and the choice $c_1 = \|h\|^{-1}$ will make $\|f_1\| = 1$.

Now suppose we have chosen k functions $f_0, f_1, \dots, f_{k-1}$ so that each f_j is a linear combination of $g_0, g_1, \dots, g_j$ and so that $\{f_0, f_1, \dots, f_{k-1}\}$ is an orthonormal set. To enlarge this to an orthonormal set of $k + 1$ functions $\{f_0, f_1, \dots, f_k\}$ we consider a linear combination of the form

$$(8.29) \quad f_k(x) = c_k \left[g_k(x) - \sum_{i=0}^{k-1} a_{k,i} f_i(x) \right].$$

If $j < k$ the inner product (f_k, f_j) is given by

$$(f_k, f_j) = c_k \left[(g_k, f_j) - \sum_{i=0}^{k-1} a_{k,i} (f_i, f_j) \right] = c_k [(g_k, f_j) - a_{k,j}].$$

We shall make this zero by choosing

$$(8.30) \quad a_{k,j} = (g_k, f_j) \quad \text{for } j = 1, 2, \dots, k-1.$$

The factor which multiplies c_k in (8.29) is not identically zero, so if we choose c_k to be the reciprocal of the norm of this factor we have $\|f_k\| = 1$. Therefore the set $\{f_0, f_1, \dots, f_k\}$ is orthonormal. This proves, by induction, that the Gram-Schmidt process gives an orthonormal set $\{f_0, f_1, \dots, f_n\}$, as required.

Note. From Equation (8.29) we see that each function g_k is a linear combination of $f_0, f_1, \dots, f_k$. Therefore each function f_m with $m > k$ is orthogonal to g_k . Also, if g_k is a polynomial of degree k for $k = 0, 1, 2, \dots, n$, then f_k has degree k , and the coefficients of x^k in $g_k(x)$ and in $f_k(x)$ have the same sign.

8.10 Orthonormal sets of polynomials

To obtain an orthonormal set $\{f_0, f_1, \dots, f_n\}$ of $n + 1$ polynomials we may begin with the linearly independent polynomials $g_0, g_1, \dots, g_n$ given by

$$g_k(x) = x^k \quad \text{for } k = 0, 1, 2, \dots, n$$

and use the Gram-Schmidt process. The first polynomial g_0 has norm $\|g_0\| = \sqrt{b-a}$, so we take

$$f_0(x) = \frac{1}{\sqrt{b-a}}.$$

Now we apply Equation (8.29) repeatedly, with $a_{k,j}$ given by (8.30). For $k = 1$ we have

$$a_{1,0} = (g_1, f_0) = \int_a^b x f_0(x) dx = \frac{1}{\sqrt{b-a}} \frac{b^2 - a^2}{2}.$$

Equation (8.29) now becomes

$$f_1(x) = c_1 [x - a_{1,0}f_0(x)] = c_1 \left(x - \frac{b+a}{2} \right).$$

To make $\|f_1\| = 1$ we must choose c_1 so that

$$c_1^2 \int_a^b \left(x - \frac{b+a}{2} \right)^2 dx = 1.$$

The change of variable $t = x - h$, where $h = (b+a)/2$, transforms the integral to

$$\int_{a-h}^{b-h} t^2 dt = 2 \int_0^{(b-a)/2} t^2 dt = \frac{(b-a)^3}{12}.$$

Therefore, we choose $c_1 = 2\sqrt{3}(b-a)^{-3/2}$, and we obtain

$$f_1(x) = \frac{2\sqrt{3}}{(b-a)^{3/2}} \left(x - \frac{a+b}{2} \right).$$

Further polynomials $f_2, f_3, \dots$ can be determined in the same manner.

The Legendre polynomials, introduced in Section 7.17, satisfy the relation

$$\int_{-1}^1 P_n(x)P_m(x) dx = \begin{cases} 0 & \text{if } m \neq n, \\ \frac{2}{2n+1} & \text{if } m = n. \end{cases}$$

Therefore the polynomials $\phi_0, \phi_1, \dots, \phi_n$ given by

$$(8.31) \quad \phi_k(x) = \sqrt{\frac{2k+1}{2}} P_k(x)$$

form an orthonormal set on the interval $[-1, 1]$. We can prove that these polynomials are the same as those constructed from $1, x, x^2, \dots, x^n$ by the Gram-Schmidt process over the interval $[-1, 1]$. This follows at once from the following uniqueness theorem.

8-4 THEOREM. Let $\{f_0, f_1, \dots, f_n\}$ and $\{\phi_0, \phi_1, \dots, \phi_n\}$ be two sets of polynomials that are orthonormal over an interval $[a, b]$. Assume that f_k and ϕ_k each have degree k . If the coefficients of x^k in $f_k(x)$ and $\phi_k(x)$ have the same sign, then $f_k = \phi_k$.

Proof. We shall prove that $\|f_k - \phi_k\| = 0$. We have

$$\|f_k - \phi_k\|^2 = (f_k - \phi_k, f_k - \phi_k) = (f_k, f_k) - 2(f_k, \phi_k) + (\phi_k, \phi_k) = 2 - 2(f_k, \phi_k).$$

Therefore the theorem will be proved if we show that $(f_k, \phi_k) = 1$.

Since f_k and ϕ_k each have degree k we may write

$$f_k(x) = a_k x^k + \dots \quad \text{and} \quad \phi_k(x) = b_k x^k + \dots,$$

where $a_k \neq 0$, $b_k \neq 0$, and $+\dots$ indicates a sum of lower powers of x . Let $g_k(x) = x^k$. The polynomial g_k can be expressed as a linear combination of $f_0, f_1, \dots, f_k$ and also as a linear combination of $\phi_0, \phi_1, \dots, \phi_k$. Therefore $(f_k, g_j) = (\phi_k, g_j) = 0$ for each $j < k$.

Since $(f_k, f_k) = 1$ we have

$$1 = \int_a^b [f_k(x)]^2 dx = \int_a^b f_k(x) (a_k x^k + \dots) dx = a_k \int_a^b f_k(x) x^k dx,$$

so

$$\int_a^b f_k(x) x^k dx = \frac{1}{a_k}.$$

Similarly,

$$\int_a^b \phi_k(x) x^k dx = \frac{1}{b_k}.$$

Therefore we have

$$(8.32) \quad (f_k, \phi_k) = \int_a^b f_k(x) (b_k x^k + \dots) dx = \frac{b_k}{a_k}$$

and

$$(\phi_k, f_k) = \int_a^b \phi_k(x) (a_k x^k + \dots) dx = \frac{a_k}{b_k}.$$

Since $(f_k, \phi_k) = (\phi_k, f_k)$ we obtain $b_k/a_k = a_k/b_k$, or $a_k^2 = b_k^2$. But a_k and b_k have the same sign, so they are equal. Therefore (8.32) states that $(f_k, \phi_k) = 1$ and, as we have already noted, this proves that $\|f_k - \phi_k\| = 0$. Since f_k and ϕ_k are continuous we must have $f_k = \phi_k$, as asserted.

We return now to the problem of least-square approximation in the space of continuous functions. Let $\{\phi_0, \phi_1, \dots, \phi_n\}$ be an orthonormal set of polynomials on $[a, b]$ and let f be any function that is continuous on $[a, b]$. In Section 8.8 we proved that if a polynomial of the form

$$(8.33) \quad T(x) = \sum_{k=0}^n c_k \phi_k(x)$$

minimizes the square norm $\|f - T\|$, the coefficients c_k in (8.33) must necessarily be given by the equations

$$(8.34) \quad c_k = \int_a^b \phi_k(x) f(x) dx$$

for $k = 0, 1, 2, \dots, n$. Therefore, if such a polynomial exists it is unique.

The foregoing discussion suggests a method for establishing the existence of a best T . Choose an orthonormal set of polynomials $\{\phi_0, \phi_1, \dots, \phi_n\}$ on $[a, b]$ where ϕ_k has degree k . (Except for sign, each ϕ_k is uniquely determined.) If f is given, define numbers $c_0, c_1, \dots, c_n$ by Equation (8.34), and let T be the polynomial given by (8.33). We shall prove that

$$(8.35) \quad \|f - T\| \leq \|f - Q\|$$

for all polynomials Q of degree $\leq n$. This will prove that T is, indeed, the best polynomial approximation relative to the square norm.

To prove (8.35) we write Q in the form

$$Q(x) = \sum_{k=0}^n b_k \phi_k(x)$$

and consider the number

$$(8.36) \quad \|f - Q\|^2 = (f - Q, f - Q) = (f, f) - 2(f, Q) + (Q, Q).$$

Since we have

$$(f, Q) = \left(f, \sum_{k=0}^n b_k \phi_k\right) = \sum_{k=0}^n b_k (f, \phi_k) = \sum_{k=0}^n b_k c_k$$

and

$$(Q, Q) = \left(\sum_{k=0}^n b_k \phi_k, \sum_{m=0}^n b_m \phi_m\right) = \sum_{k=0}^n \sum_{m=0}^n b_k b_m (\phi_k, \phi_m) = \sum_{k=0}^n b_k^2,$$

Equation (8.36) becomes

$$\|f - Q\|^2 = \|f\|^2 - 2 \sum_{k=0}^n b_k c_k + \sum_{k=0}^n b_k^2 = \|f\|^2 - \sum_{k=0}^n c_k^2 + \sum_{k=0}^n (b_k - c_k)^2.$$

Now it is clear that among all possible choices of coefficients $b_0, b_1, \dots, b_n$ the choice $b_k = c_k$ for each k makes the sum

$$\sum_{k=0}^n (b_k - c_k)^2$$

as small as possible. Therefore $\|f - Q\|$ has its smallest value when $Q = T$. This proves (8.35). Incidentally, we have also shown that for the best polynomial T we have

$$\|f - T\|^2 = \|f\|^2 - \sum_{k=0}^n c_k^2.$$

The foregoing discussion is summarized by the following theorem.

8- 5 THEOREM. Let f be continuous on $[a, b]$ and let $\phi_0, \phi_1, \dots, \phi_n$ be an orthonormal set of polynomials on $[a, b]$ where ϕ_k has degree k . Define numbers $c_0, c_1, \dots, c_n$ by the equation

$$(8.37) \quad c_k = \int_a^b \phi_k(x) f(x) dx.$$

Then among all polynomials T of degree $\leq n$ there is exactly one that minimizes the square norm $\|f - T\|$, namely

$$T(x) = \sum_{k=0}^n c_k \phi_k(x).$$

For this T we have

$$\|f - T\|^2 = \|f\|^2 - \sum_{k=0}^n c_k^2.$$

Example: Least-square approximation over the interval $[-1, 1]$. Let f be a continuous function on $[-1, 1]$. The polynomial T of degree $\leq n$ that approximates f best over the interval $[-1, 1]$ relative to the square norm can be written in the form

$$(8.38) \quad T(x) = \sum_{k=0}^n a_k P_k(x),$$

where $P_0, P_1, \dots, P_n$ are the Legendre polynomials. To determine the coefficients we introduce the normalized Legendre polynomials $\phi_k(x) = P_k(x)/\|P_k\|$ of Equation (8.31) and write (8.38) in the form

$$T(x) = \sum_{k=0}^n c_k \phi_k(x),$$

where $c_k = \|P_k\| a_k$. Since the set $\{\phi_0, \phi_1, \dots, \phi_n\}$ is orthonormal on $[-1, 1]$ the coefficients c_k are given by Equation (8.37). Hence

$$c_k = \int_{-1}^1 \phi_k(t) f(t) dt = \frac{1}{\|P_k\|} \int_{-1}^1 P_k(t) f(t) dt.$$

Since $a_k = c_k/\|P_k\|$ the coefficients in (8.38) are given by

$$a_k = \frac{1}{\|P_k\|^2} \int_{-1}^1 P_k(t) f(t) dt = \frac{2k+1}{2} \int_{-1}^1 P_k(t) f(t) dt.$$

The first three coefficients are

$$a_0 = \frac{1}{2} \int_{-1}^1 f(t) dt, \quad a_1 = \frac{3}{2} \int_{-1}^1 t f(t) dt, \quad a_2 = \frac{5}{2} \int_{-1}^1 \frac{3t^2 - 1}{2} f(t) dt.$$

8.11 Exercises

1. Let $P_0, P_1, \dots, P_n$ be $n+1$ polynomials, where P_k has degree k and $P_0 \neq 0$.

(a) Prove that $P_0, P_1, \dots, P_n$ are linearly independent over every interval.

(b) Prove that every polynomial T of degree $\leq n$ can be expressed in the form

$$T(x) = \sum_{k=0}^n c_k P_k(x),$$

where the c_k are constants. Show also that for a given T the coefficients c_k are unique.

2. Let $\{P_0, P_1, \dots, P_n\}$ be a set of $n+1$ polynomials orthonormal on $[a, b]$. Prove that these polynomials are linearly independent over any interval.

3. Let $\{P_0, P_1, \dots, P_n\}$ be a set of polynomials orthonormal on $[a, b]$, and assume that P_k has degree k .

(a) If $k \geq 1$ prove that P_k has at least one zero in the interval $[a, b]$. [Hint. Use the fact that $(P_0, P_k) = 0$.]

(b) If $k \geq 1$, let $c_1, c_2, \dots, c_m$ be the real zeros of P_k of odd multiplicity which lie in $[a, b]$, and let $A(x) = (x - c_1) \cdots (x - c_m)$. Show that $(A, P_k) = 0$ if $m < k$. From this relation deduce that $k = m$. This proves that all the zeros of P_k are real and distinct and lie in the interval $[a, b]$.

4. Let $P_0, P_1, \dots, P_n$ be $n+1$ polynomials orthonormal on $[a, b]$, and assume that P_k has degree k .

(a) Prove that any three consecutive polynomials in this set are connected by a recurrence relation of the form

$$P_{k+1}(x) = (a_k x + b_k)P_k(x) + c_k P_{k-1}(x)$$

for $1 \leq k \leq n-1$, where a_k, b_k, c_k are constants.

(b) Determine this recurrence relation explicitly when the polynomials are the orthonormal Legendre polynomials.

5. Refer to Exercise 4, and let p_k denote the coefficient of x^k in $P_k(x)$.

(a) Show that $a_k = p_{k+1}/p_k$.

(b) Use the recurrence relation in Exercise 4 to derive the formula

$$\sum_{k=0}^m P_k(x)P_k(y) = \frac{p_m}{p_{m+1}} \frac{P_{m+1}(x)P_m(y) - P_m(x)P_{m+1}(y)}{x-y},$$

valid for $x \neq y$. Discuss also the limiting case $x = y$.

6. Given a function f continuous on $[-1, 1]$. Let T_k denote the polynomial of degree $\leq k$ that minimizes the square norm $\|f - T_k\|$.

(a) Write $T_k(x) = A_0 + A_1x + \cdots + A_kx^k$ and determine the coefficients explicitly in terms of f when $k = 1, 2, 3$.

(b) Show that

$$\|f - T_1\|^2 = \int_{-1}^1 [f(t)]^2 dt - 2A_0^2 - \frac{2}{3}A_1^2,$$

and obtain similar formulas for $\|f - T_2\|^2$ and $\|f - T_3\|^2$.

7. Let $f(x) = |x|$ for $-1 \leq x \leq 1$. Find the polynomial of degree ≤ 4 that approximates f best over the interval $[-1, 1]$ relative to the square norm.

8. This exercise shows how to construct a polynomial S_n of degree n such that

$$(8.39) \quad \int_{-1}^1 S_n(x)Q(x) dx = Q(1)$$

for every polynomial Q of degree $\leq n$.

(a) Show that there is at most one polynomial S_n with this property. [Hint. If T_n is another, consider $\|S_n - T_n\|^2 = (S_n, S_n - T_n) - (T_n, S_n - T_n)$.]

(b) Let P_k denote the Legendre polynomial of degree k . If

$$S_n(x) = \sum_{k=0}^n s_k P_k(x) \quad \text{and} \quad Q(x) = \sum_{k=0}^n q_k P_k(x),$$

show that the relation (8.39) implies

$$\sum_{k=0}^n \frac{2}{2k+1} s_k q_k = \sum_{k=0}^n q_k.$$

Use this formula and the uniqueness property in (a) to deduce that the one and only polynomial S_n of degree n that satisfies (8.39) for all Q of degree $\leq n$ is

$$S_n(x) = \sum_{k=0}^n \frac{2k+1}{2} P_k(x).$$

(c) Let $S_n(x) = c_0 + c_1x + \cdots + c_nx^n$. Determine the coefficients explicitly when $n = 0, 1, 2, 3$.

9. Refer to Exercise 8. Write

$$(1-x)S_n(x) = \sum_{k=0}^{n+1} a_k P_k(x)$$

and use the relation (8.39) to prove that

$$(1-x)S_n(x) = \frac{n+1}{2} (P_n(x) - P_{n+1}(x)).$$

This implies the relation

$$\sum_{k=0}^n \frac{2k+1}{2} P_k(x) = \frac{n+1}{2} \frac{P_n(x) - P_{n+1}(x)}{1-x}$$

if $x \neq 1$. Show that this formula holds also for $x = 1$.

10. Let w be a nonnegative function that is continuous on $[a, b]$. Let C denote the space of all functions continuous on $[a, b]$. If f and g are in C , define the inner product (f, g) with respect to the weight function w by the equation

$$(f, g) = \int_a^b f(x)g(x)w(x) dx.$$

(a) Show that this inner product is commutative and has the linearity properties expressed in (8.25) and (8.26).

(b) If w is never zero on $[a, b]$ show that the equation $\|f\| = (f, f)^{1/2}$ defines a norm on C .

(c) If $n+1$ functions $f_0, f_1, \dots, f_n$ satisfy the relations

$$(f_i, f_k) = \begin{cases} 0 & \text{if } i \neq k \\ 1 & \text{if } i = k \end{cases}$$

the set $\{f_0, f_1, \dots, f_n\}$ is said to be orthonormal on $[a, b]$ relative to w . Prove that, relative to the weight w , a set of $n+1$ orthonormal polynomials always exists.

(d) Show that for a given function f in C there is one and only one best polynomial approximation to f relative to the norm defined in part (b).

11. Let $f(x) = 1/x$ for $x \neq 0$.

(a) Show that the constant polynomial P that best approximates f over the interval $[1, n]$ relative to the square norm is $P(x) = (\log n)/(n-1)$. Compute $\|P - f\|^2$ for this P .

(b) Find the linear polynomial P that best approximates f over the interval $[1, n]$ relative to the square norm. Compute $\|P - f\|^2$ for this P when $n = 2$.

12. Let $f(x) = e^x$.

(a) Show that the constant polynomial P that best approximates f over the interval $[0, n]$ relative to the square norm is $P(x) = (e^n - 1)/n$. Compute $\|P - f\|^2$ for this P .

(b) Find the linear polynomial P that best approximates f over the interval $[0, 1]$ relative to the square norm. Compute $\|P - f\|^2$ for this P .

8.12 Polynomial approximation relative to the maximum norm. Chebyshev polynomials

Let C denote the linear space of functions continuous on an interval $[a, b]$. If $f \in C$ let $\|f\|$ denote the maximum norm, defined by

$$\|f\| = \max_{a \leq x \leq b} |f(x)|.$$

The problem of best polynomial approximation relative to this norm is more difficult than in the cases considered up to now. Although it can be shown that for each f in C there exists a unique best polynomial approximation P of degree $\leq n$, the known proofs use more advanced mathematical tools than we have developed in this book and will not be discussed here.† In approximation by the Taylor seminorm, the interpolation seminorm, and the least-square norm, we have specific formulas that enable us to construct the optimum polynomial P once f is given. In approximation by the maximum norm no such formulas are known, except when f has certain special forms. In the absence of general formulas the existence of an optimum P is more difficult to establish.

In this section we consider an important special problem in polynomial approximation relative to the maximum norm. This problem arises naturally from the theory of polynomial interpolation. In Theorem 8-3 (Section 8.6) we derived the error formula

$$(8.40) \quad f(x) - P(x) = \frac{A(x)}{(n+1)!} f^{(n+1)}(c),$$

where

$$A(x) = (x - x_0)(x - x_1) \cdots (x - x_n).$$

Here P is the unique polynomial of degree $\leq n$ that agrees with f at $n+1$ distinct points $x_0, x_1, \dots, x_n$ in $[a, b]$. The function f is assumed to have a derivative of order $n+1$ on $[a, b]$, and c is an unknown point lying somewhere in $[a, b]$. To estimate the error in (8.40) we need bounds for the $(n+1)$ st derivative $f^{(n+1)}$ and for the product $A(x)$. Since A is a polynomial, its absolute value has a maximum, $\|A\|$, somewhere in the interval $[a, b]$. This maximum will depend on the choice of the points $x_0, x_1, \dots, x_n$, and it is natural to try to choose these points so the norm of A will be as small as possible.

The formula for $A(x)$ may be written as

$$A(x) = g(x) - Q(x),$$

where $g(x) = x^{n+1}$ and $Q(x)$ has degree $\leq n$. Thus, we have $\|A\| = \|g - Q\|$, so $\|A\|$ will be a minimum if and only if Q is the polynomial of degree $\leq n$ which best approximates g . Therefore the problem of minimizing $\|A\|$ is a polynomial approximation problem relative to the maximum norm. The general theory tells us that a unique solution exists. For this special case the optimum A can be constructed explicitly. The problem was first solved by Chebyshev; its solution leads to an interesting class of polynomials that also occur in other connections. We begin the discussion with a brief account of these polynomials and then return to the approximation problem in question.

DEFINITION. The polynomials $T_0, T_1, T_2, \dots$ defined by the recursion formula

$$(8.41) \quad T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x) \quad \text{for } n \geq 1,$$

with $T_0(x) = 1$ and $T_1(x) = x$, are called Chebyshev polynomials of the first kind.

The next four polynomials are

† A proof may be found in John Todd's *Introduction to the Constructive Theory of Functions*, Academic Press, Inc., New York, N.Y., 1963, Chapter 8.

$$T_2(x) = 2x^2 - 1, \quad T_3(x) = 4x^3 - 3x, \quad T_4(x) = 8x^4 - 8x^2 + 1,$$

$$T_5(x) = 16x^5 - 20x^3 + 5x.$$

From the recursion formula we see at once that T_n is a polynomial of degree n ; the coefficient of x^n is equal to 2^{n-1} if $n \geq 1$.

Further properties of these polynomials are easily deduced from the following theorem.

8- 6 THEOREM. If $-1 \leq x \leq 1$ we have

$$(8.42) \quad T_n(x) = \cos(n \arccos x).$$

Proof. Let $f_n(x) = \cos(n \arccos x)$ for $-1 \leq x \leq 1$. Taking $n = 0$ and $n = 1$ we find

$$f_0(x) = \cos 0 = 1 \quad \text{and} \quad f_1(x) = \cos(\arccos x) = x,$$

so Equation (8.42) holds for $n = 0$ and $n = 1$. If we prove that f_n satisfies the recursion formula (8.41) the theorem follows by induction. Using the trigonometric identity

$$\cos(n+1)\theta + \cos(n-1)\theta = 2 \cos \theta \cos n\theta$$

with $\theta = \arccos x$ we find $f_{n+1}(x) + f_{n-1}(x) = 2x f_n(x)$, which is the same recursion as (8.41). This completes the proof.

The zeros of T_n for $n \geq 1$ are easily determined from the formula in (8.42). Since $\cos n\theta = 0$ only if $n\theta$ is an odd multiple of $\pi/2$, we have $T_n(x) = 0$ for x in $[-1, 1]$ only if $n \arccos x = (2k+1)\pi/2$ for some integer k . Therefore the zeros of T_n in the interval $[-1, 1]$ are to be found among the numbers

$$(8.43) \quad x_k = \cos \frac{2k+1}{n} \frac{\pi}{2}, \quad k = 0, \pm 1, \pm 2, \dots$$

The values $k = 0, 1, 2, \dots, n-1$ give n distinct zeros $x_0, x_1, \dots, x_{n-1}$, all lying in the open interval $(-1, 1)$. Since a polynomial of degree n cannot have more than n zeros, these must be *all* the zeros of T_n . The remaining x_k in (8.43) are repetitions of these n . We can now write

$$T_n(x) = 2^{n-1} (x - x_0)(x - x_1) \cdots (x - x_{n-1}) = 2^{n-1} \prod_{k=0}^{n-1} \left(x - \cos \frac{(2k+1)\pi}{2n} \right).$$

By Rolle's theorem, the relative maxima and minima of T_n must occur between successive zeros; there are $n-1$ such points in the open interval $(-1, 1)$. From the cosine formula for T_n we see that the extreme values, ± 1 , are taken at the $n-1$ interior points $\cos(k\pi/n)$, $k = 1, 2, \dots, n-1$, and also at the two endpoints $x = 1$ and $x = -1$. Therefore in the closed interval $[-1, 1]$ the extreme values ± 1 are taken *alternately* at the $n+1$ points $t_0, t_1, \dots, t_n$ given by $t_k = \cos(k\pi/n)$ for $k = 0, 1, 2, \dots, n$.

We return now to the problem of finding the polynomial Q of degree $\leq n$ which best approximates $g(x) = x^{n+1}$ relative to the maximum norm. For the interval $[-1, 1]$ this problem is solved by the following theorem.

8- 7 THEOREM. Let $p_{n+1}(x) = x^{n+1} - Q(x)$, where $Q(x)$ is a polynomial of degree $\leq n$, and let

$$\|p_{n+1}\| = \max_{-1 \leq x \leq 1} |p_{n+1}(x)|.$$

Then we have the inequality

$$(8.44) \quad \|p_{n+1}\| \geq \frac{1}{2^n}.$$

Moreover, $\|p_{n+1}\| = 1/2^n$ if and only if $p_{n+1}(x) = 2^{-n} T_{n+1}(x)$, where T_{n+1} is the Chebyshev polynomial of degree $n + 1$.

Proof. Let $S(x) = 2^{-n} T_{n+1}(x)$. In the interval $[-1, 1]$ the polynomial S takes its extreme values, 2^{-n} and -2^{-n} , alternately at the $n + 2$ distinct points

$$(8.45) \quad t_k = \cos \frac{k\pi}{n+1}, \quad k = 0, 1, 2, \dots, n+1.$$

Therefore $\|S\| = 2^{-n}$ and the inequality (8.44) is satisfied (as an equality) when $p_{n+1} = S$.

We show next that the inequality

$$(8.46) \quad \|p_{n+1}\| < 2^{-n}$$

leads to a contradiction. Assume, then, that p_{n+1} satisfies (8.46) and consider the difference

$$r(x) = S(x) - p_{n+1}(x).$$

At the points t_k given by (8.45) we have

$$r(t_k) = (-1)^k 2^{-n} - p_{n+1}(t_k) = (-1)^k [2^{-n} - (-1)^k p_{n+1}(t_k)].$$

Because of (8.46) the factor in square brackets is always positive. Therefore $r(t_k)$ has alternating signs at the $n + 2$ points $t_0, t_1, \dots, t_{n+1}$. Since r is continuous it must vanish at least once between consecutive sign changes. Therefore r has at least $n + 1$ distinct zeros. But since r is a polynomial of degree $\leq n$, this means that r is identically zero. Therefore $p_{n+1} = S$, contradicting (8.46).

Thus, we have shown that (8.44) holds for all polynomials p_{n+1} of the specified type. We have also shown that equality holds in (8.44) when $p_{n+1} = S$. Because of the uniqueness theorem, S is the only polynomial that gives equality. (We remind the reader that we have not proved the uniqueness theorem.)

Although Theorem 8-7 refers to the interval $[-1, 1]$ it can be used to deduce a corresponding result for an arbitrary interval $[a, b]$.

8- 8 THEOREM. Let $p_{n+1}(x) = x^{n+1} - Q(x)$, where Q is a polynomial of degree $\leq n$, and let

$$\|p_{n+1}\| = \max_{a \leq x \leq b} |p_{n+1}(x)|.$$

Then we have the inequality

$$\|p_{n+1}\| \geq \frac{(b-a)^{n+1}}{2^{2n+1}}$$

with equality holding if and only if

$$p_{n+1}(x) = \frac{(b-a)^{n+1}}{2^{2n+1}} T_{n+1}\left(\frac{2x-a-b}{b-a}\right).$$

Proof. Consider the transformation

$$t = \frac{2x-a-b}{b-a}.$$

This maps the interval $a \leq x \leq b$ in a one-to-one fashion onto the interval $-1 \leq t \leq 1$. Since

$$x = \frac{b-a}{2} t + \frac{b+a}{2}$$

we have

$$p_{n+1}(x) = \left(\frac{b-a}{2} t + \frac{b+a}{2}\right)^{n+1} - Q(x) = \left(\frac{b-a}{2}\right)^{n+1} [t^{n+1} - Q^*(t)],$$

where $Q^*(t)$ is a polynomial in t with degree $\leq n$. If we apply Theorem 8-7 to the polynomial $p_{n+1}^*(t) = t^{n+1} - Q^*(t)$ and note that

$$\left(\frac{2}{b-a}\right)^{n+1} p_{n+1}(x) = p_{n+1}^*(t)$$

we immediately obtain Theorem 8-8.

We return now to the error formula (8.40) for polynomial interpolation. If we choose the interpolation points $x_0, x_1, \dots, x_n$ to be the $n+1$ zeros of the Chebyshev polynomial T_{n+1} we may write (8.40) in the form

$$f(x) - P(x) = \frac{T_{n+1}(x)}{2^n(n+1)!} f^{(n+1)}(c).$$

The points $x_0, x_1, \dots, x_n$ all lie in the open interval $(-1, 1)$ and are given by

$$x_k = \cos\left(\frac{2k+1}{n+1} \frac{\pi}{2}\right) \quad \text{for } k = 0, 1, 2, \dots, n.$$

If x is in the interval $[-1, 1]$ we have $|T_{n+1}(x)| \leq 1$ and the error is estimated by the inequality

$$|f(x) - P(x)| \leq \frac{1}{2^n(n+1)!} |f^{(n+1)}(c)|.$$

If the interpolation takes place in an interval $[a, b]$ with the points

$$t_k = \frac{b-a}{2} x_k + \frac{b+a}{2},$$

as interpolation points, the product

$$A(x) = (x - t_0)(x - t_1) \cdots (x - t_n)$$

satisfies the inequality $|A(x)| \leq (b-a)^{n+1}/2^{2n+1}$ for all x in $[a, b]$. The corresponding estimate for $f(x) - P(x)$ is

$$|f(x) - P(x)| \leq \frac{(b-a)^{n+1}}{2^{2n+1}(n+1)!} |f^{(n+1)}(c)|.$$

*8.13 Error analysis in polynomial approximation relative to the maximum norm

If a function f has a derivative of order $n+1$ we can obtain an explicit formula for the deviation between f and its polynomial approximations relative to the maximum norm. This formula will be deduced as a consequence of the following theorem.

8-9 THEOREM. Let f be continuous on $[a, b]$ and assume T is a polynomial of degree $\leq n$ that best approximates f on $[a, b]$ relative to the maximum norm. Let

$$R(x) = f(x) - T(x)$$

denote the error in this approximation, and let $D = \|f - T\|$. Then:

- (a) If $D = 0$ the function R is identically zero on $[a, b]$.
- (b) If $D > 0$ the function R has at least $n+1$ sign changes in $[a, b]$.

Proof. For each x in $[a, b]$ we have the inequality

$$(8.47) \quad 0 \leq |R(x)| = |f(x) - T(x)| \leq \max_{a \leq x \leq b} |f(x) - T(x)| = D.$$

The statement in part (a) follows at once from this inequality.

To prove part (b) we argue by induction on the number of sign changes of R . First we show that R has at least one sign change in $[a, b]$. Then we prove that if R has k sign changes, where $k \leq n$, it also has $k+1$ sign changes.

The first step in the induction is proved by contradiction. Assume R has no sign changes in $[a, b]$. Then either $R(x) \geq 0$ for all x in $[a, b]$ or $R(x) \leq 0$ throughout $[a, b]$. Consider first the case $R(x) \geq 0$. Then from (8.47) we have $0 \leq R(x) \leq D$, so that

$$0 \leq |R(x) - \frac{1}{2}D| \leq \frac{1}{2}D$$

for all x in $[a, b]$. Now let $Q(x) = T(x) + \frac{1}{2}D$. This is a polynomial of degree $\leq n$ which differs from f at each point by an amount

$$|f(x) - Q(x)| = |f(x) - T(x) - \frac{1}{2}D| = |R(x) - \frac{1}{2}D| \leq \frac{1}{2}D.$$

Since this is true for all x in $[a, b]$ we also have the inequality

$$\|f - Q\| \leq \frac{1}{2}D < D = \|f - T\|,$$

which contradicts the assumption that T is a best approximation to f . Therefore the

assumption that $R(x) \geq 0$ throughout $[a, b]$ leads to a contradiction. In a similar way we can show that we cannot have $R(x) \leq 0$ throughout $[a, b]$. This proves that R has at least one sign change in $[a, b]$.

Now assume that R changes sign at least k times in $[a, b]$, where $k \leq n$. We wish to prove that it changes sign at least $k + 1$ times. To do this we assume R changes sign *exactly* k times and arrive at a contradiction. Since R is a continuous function it must have at least one zero in each interval in which there is a sign change (by Bolzano's theorem). Therefore R has at least k zeros in the open interval (a, b) .[†] From these zeros we can choose k distinct points $x_1 < x_2 < \cdots < x_k$ such that the function R changes its sign in every neighborhood of each x_i . Now let A_k be the polynomial

$$A_k(x) = (x - x_1)(x - x_2) \cdots (x - x_k)$$

and let $F_k(x) = a_k A_k(x)$, where $a_k = \pm 1/\|A_k\|$. The choice of sign will be specified presently. Let S denote the set of points x in $[a, b]$ at which $R(x) \neq 0$. The functions R and F_k have sign changes at the points x_i and *at no other points* in $[a, b]$. Therefore R and F_k have the same sign everywhere in S or the opposite sign everywhere in S . We choose the sign of a_k which will make R and F_k agree in sign everywhere in S .

Now let Q be the polynomial given by

$$Q(x) = T(x) + \frac{D}{2} F_k(x).$$

Since F_k has degree $k \leq n$, the polynomial Q has degree $\leq n$. The difference between f and Q is given by

$$(8.48) \quad |f(x) - Q(x)| = \left| f(x) - T(x) - \frac{D}{2} F_k(x) \right| = \left| R(x) - \frac{D}{2} F_k(x) \right|.$$

If we can prove that

$$(8.49) \quad \left| R(x) - \frac{D}{2} F_k(x) \right| < D$$

for all x in $[a, b]$, we will obtain $|f(x) - Q(x)| < D$ from (8.48) and it will follow that

$$\|f - Q\| < D = \|f - T\|.$$

This contradicts the assumption that T is the best approximation to f . The contradiction completes the inductive step needed to prove the theorem.

We still need to prove that (8.49) holds for all x in $[a, b]$. If x is not in S , (8.49) holds trivially, since $R(x) = 0$ and $\|F_k\| = 1$. Therefore, suppose $x \in S$. We know that R and F_k agree in sign everywhere in S . If x is a point at which $F_k(x) > 0$ and $R(x) > 0$, we have

$$R(x) - \frac{D}{2} F_k(x) < R(x) \leq D$$

and

$$R(x) - \frac{D}{2} F_k(x) > -\frac{D}{2} F_k(x) \geq -\frac{D}{2} \|F_k\| = -\frac{D}{2},$$

[†] The function R may have other zeros, such as double zeros, at which it does not change sign.

so (8.49) holds. On the other hand, if x is a point at which $F_k(x) < 0$ and $R(x) < 0$, we have

$$R(x) - \frac{D}{2} F_k(x) < -\frac{D}{2} F_k(x) = \frac{D}{2} |F_k(x)| \leq \frac{D}{2} \|F_k\| = \frac{D}{2}$$

and

$$R(x) - \frac{D}{2} F_k(x) > R(x) = -|R(x)| \geq -D,$$

so (8.49) holds for such x as well. Therefore (8.49) holds for all x in $[a, b]$, and the proof is complete.

Now we assume f has a derivative of order $n + 1$ and find an explicit formula for the error.

8-10 THEOREM. Assume the derivative $f^{(n+1)}$ exists on $[a, b]$ and let T be the polynomial of degree $\leq n$ that best approximates f on $[a, b]$ relative to the maximum norm. Then there are $n + 1$ distinct points $x_0, x_1, \dots, x_n$ in the open interval (a, b) such that for each x in $[a, b]$ we have

$$(8.50) \quad f(x) - T(x) = \frac{A(x)}{(n+1)!} f^{(n+1)}(c),$$

where

$$A(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$$

and

$$\min\{x_0, x_1, \dots, x_n, x\} < c < \max\{x_0, x_1, \dots, x_n, x\}.$$

Proof. Let $R(x) = f(x) - T(x)$. By Theorem 8-9 there are $n + 1$ distinct points $x_0, x_1, \dots, x_n$ in (a, b) at which R vanishes. At these points the polynomial T agrees with f , so T must be the interpolating polynomial relative to these interpolation points. Applying the error formula in Theorem 8-3 we obtain (8.50).

It should be noted that we cannot use the Lagrange interpolation formula to construct T explicitly in this case because the exact location of the x_i is not known.

8.14 Exercises

In this set of exercises T_n denotes the Chebyshev polynomial of degree n .

1. Prove that $T_n(-x) = (-1)^n T_n(x)$. This shows that T_n is an even function when n is even and an odd function when n is odd.

2. (a) Prove that in the open interval $(-1, 1)$ the derivative T'_n is given by the formula

$$T'_n(x) = \frac{n \sin n\theta}{\sin \theta}, \quad \text{where } \theta = \arccos x.$$

(b) Compute $T'_n(1)$ and $T'_n(-1)$.

3. Use the trigonometric identity

$$2 \sin \theta \sin n\theta = \cos(n-1)\theta - \cos(n+1)\theta$$

and the result of Exercise 2 to prove that

$$(1 - x^2)T'_n(x) = n T_{n-1}(x) - nx T_n(x)$$

for $-1 < x < 1$. Then show that the formula holds for all real x .

4. The Chebyshev polynomial T_n satisfies the differential equation

$$(1 - x^2)y'' - xy' + n^2y = 0$$

over the entire real axis. Prove this by each of the following methods:

(a) Use the result of Exercise 3 and the recursion formula (8.41).

(b) Introduce the change of variable $x = \cos \theta$ in the differential equation

$$\frac{d^2(\cos n\theta)}{d\theta^2} = -n^2 \cos n\theta.$$

5. The Chebyshev polynomials satisfy the following orthogonality relations:

$$\int_{-1}^1 \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}} dx = \begin{cases} 0 & \text{if } n \neq m, \\ \pi & \text{if } n = m = 0, \\ \frac{\pi}{2} & \text{if } n = m > 0. \end{cases}$$

Prove this by each of the following methods:

(a) From the differential equation in Exercise 4 deduce that

$$T_m(x) \frac{d}{dx} \left(\sqrt{1-x^2} T'_n(x) \right) + n^2 \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}} = 0.$$

Write a corresponding formula with n and m interchanged, subtract the two equations, and integrate from -1 to 1 .

(b) Use the orthogonality relations

$$\int_0^\pi \cos m\theta \cos n\theta d\theta = \begin{cases} 0 & \text{if } n \neq m, \\ \pi & \text{if } n = m = 0, \\ \frac{\pi}{2} & \text{if } n = m > 0, \end{cases}$$

and introduce the change of variable $x = \cos \theta$.

6. Find the polynomial of degree ≤ 4 that best approximates the function $f(x) = x^5$ in the interval $[0, 1]$, relative to the maximum norm.

7. Prove that for $-1 < x < 1$ we have

$$\frac{T_n(x)}{\sqrt{1-x^2}} = (-1)^n \frac{2^n n!}{(2n)!} \frac{d^n}{dx^n} (1-x^2)^{n-1/2}.$$

8. Let $y_1, y_2, \dots, y_n$ be n real numbers, and let

$$x_k = \cos \frac{(2k-1)\pi}{2n} \quad \text{for } k = 1, 2, \dots, n.$$

Let P be the polynomial of degree $\leq n-1$ that takes the value y_k at x_k for $1 \leq k \leq n$. If x is not one of the x_k show that

$$P(x) = \frac{1}{n} \sum_{k=1}^n (-1)^{k-1} y_k \sqrt{1-x_k^2} \frac{T_n(x)}{x-x_k}.$$

9. Let P be a polynomial of degree $\leq n - 1$ such that

$$\sqrt{1 - x^2} |P(x)| \leq 1$$

for $-1 \leq x \leq 1$. Prove that $\|P\| \leq n$, where $\|P\|$ is the maximum of $|P|$ on the interval $[-1, 1]$. [Hint. Use Exercise 8. Consider three cases: $x_1 \leq x \leq 1$; $-1 \leq x \leq x_n$; $x_n \leq x \leq x_1$; in the first two use Exercise 13(a) of Section 8.7. In the third case note that $\sqrt{1 - x^2} \geq \sin(\pi/2n) > 1/n$.]

10. A polynomial P is called *primary* if the coefficient of the term of highest degree is 1. For a given interval $[a, b]$ let $\|P\|$ denote the maximum of $|P|$ on $[a, b]$. Prove each of the following statements:

(a) If $b - a < 4$, for every $\epsilon > 0$ there exists a primary polynomial P with $\|P\| < \epsilon$.

(b) If for every $\epsilon > 0$ there exists a primary polynomial P with $\|P\| < \epsilon$, then $b - a < 4$.

In other words, primary polynomials with arbitrarily small norm exist if, and only if, the interval $[a, b]$ has length less than 4.

In Exercises 11 through 15, $U_n(x) = T'_{n+1}(x)/(n+1)$ for $n = 0, 1, 2, \dots$.

11. (a) Prove that $U_n(x) = 2x U_{n-1}(x) - U_{n-2}(x)$ for $n \geq 2$.

(b) Determine the explicit form of the polynomials $U_0, U_1, \dots, U_5$.

(c) Prove that $|U_n(x)| \leq n + 1$ if $-1 \leq x \leq 1$.

12. Show that U_n satisfies the differential equation

$$(1 - x^2)y'' - 3xy' + n(n+2)y = 0.$$

13. Derive the orthogonality relations

$$\int_{-1}^1 \sqrt{1 - x^2} U_m(x) U_n(x) dx = \begin{cases} 0 & \text{if } m \neq n, \\ \frac{\pi}{2} & \text{if } m = n. \end{cases}$$

14. Prove that

$$\sqrt{1 - x^2} U_n(x) = (-1)^n \frac{2^n(n+1)!}{(2n+1)!} \frac{d^n}{dx^n} (1 - x^2)^{n+1/2}.$$

15. Let $y_1, y_2, \dots, y_n$ be n real numbers and let

$$x_k = \cos \frac{k\pi}{n+1} \quad \text{for } k = 1, 2, \dots, n.$$

Let P be the polynomial of degree $\leq n - 1$ that takes the value y_k at x_k for $1 \leq k \leq n$. If x is not one of the x_k show that

$$P(x) = \frac{1}{n+1} \sum_{k=1}^n (-1)^{k-1} (1 - x_k^2) y_k \frac{U_n(x)}{x - x_k}.$$

8.15 Newton's interpolation polynomial with divided differences

Let P_n denote the interpolation polynomial of degree $\leq n$ that agrees with a given function f at $n + 1$ distinct points $x_0, x_1, \dots, x_n$. Lagrange's interpolation formula tells us that

$$(8.51) \quad P_n(x) = \sum_{k=0}^n L_k(x) f(x_k),$$

where $L_k(x)$ is a polynomial of degree n (the Lagrange interpolation coefficient) given by the product formula

$$(8.52) \quad L_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{x - x_j}{x_k - x_j}, \quad \text{for } k = 0, 1, 2, \dots, n.$$

Suppose we adjoin a new interpolation point x_{n+1} to the given points $x_0, x_1, \dots, x_n$. To determine the corresponding polynomial P_{n+1} by Lagrange's formula it is necessary to compute a new interpolation coefficient L_{n+1} and to recompute all the earlier coefficients $L_0, L_1, \dots, L_n$, each of which is now a polynomial of degree $n + 1$. In practice this involves considerable labor. Therefore it is desirable to have another formula for determining P_n that provides an easier transition from P_n to P_{n+1} . One such formula was discovered by Newton; to discuss it we introduce the concept of divided differences.

DEFINITION OF DIVIDED DIFFERENCES. Let $x_0, x_1, \dots, x_n$ be $n + 1$ distinct points at which a function f is defined. For each x in the domain of f the divided differences of order 1, 2, $\dots$, $n + 1$ are defined, respectively, by the following formulas:

$$\begin{aligned} f(x_0, x) &= \frac{f(x_0) - f(x)}{x_0 - x} & \text{if } x \neq x_0, \\ f(x_0, x_1, x) &= \frac{f(x_0, x_1) - f(x_0, x)}{x_1 - x} & \text{if } x \neq x_0, x \neq x_1, \\ &\vdots \\ f(x_0, x_1, \dots, x_n, x) &= \frac{f(x_0, x_1, \dots, x_{n-1}, x_n) - f(x_0, x_1, \dots, x_{n-1}, x)}{x_n - x} \\ && \text{if } x \neq x_0, x \neq x_1, \dots, x \neq x_n. \end{aligned}$$

Note. The notations $[x_0, x_1, \dots, x_r, x]$ and $f[x_0, x_1, \dots, x_r, x]$ are also used for the divided difference of order $r + 1$.

The function value $f(x)$ can be expressed in terms of divided differences. To begin with, the equation for $f(x_0, x)$ may be rewritten as

$$(8.53) \quad f(x) = f(x_0) + (x - x_0)f(x_0, x) \quad \text{if } x \neq x_0.$$

Also, we may write the equation for $f(x_0, x_1, x)$ in the form

$$f(x_0, x) = f(x_0, x_1) + (x - x_1)f(x_0, x_1, x)$$

and substitute in (8.53) to obtain

$$f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x) \quad \text{if } x \neq x_0, x \neq x_1.$$

Continuing in this manner we obtain the formula

$$(8.54) \quad f(x) = f(x_0) + \sum_{r=0}^{k-1} (x - x_0) \cdots (x - x_r)f(x_0, x_1, \dots, x_{r+1}) \\ + (x - x_0) \cdots (x - x_k)f(x_0, x_1, \dots, x_k, x),$$

which is valid if $x \neq x_0, x \neq x_1, \dots, x \neq x_k$, where $k \leq n$. The first $k + 1$ terms on the right of (8.54) determine a polynomial of degree $\leq k$ which we denote by Q_k . Thus, we may write (8.54) in the form

$$(8.55) \quad f(x) = Q_k(x) + (x - x_0) \cdots (x - x_k)f(x_0, x_1, \dots, x_k, x)$$

for $x \neq x_0, \dots, x \neq x_k$, where

$$(8.56) \quad Q_k(x) = f(x_0) + \sum_{r=0}^{k-1} (x - x_0) \cdots (x - x_r)f(x_0, x_1, \dots, x_{r+1}).$$

The polynomial Q_k is called *Newton's divided-difference polynomial*. From (8.56) we see that

$$(8.57) \quad Q_k(x) = Q_{k-1}(x) + (x - x_0) \cdots (x - x_{k-1})f(x_0, x_1, \dots, x_k)$$

for every x . This property of the Q_k enables us to prove the following theorem.

8-11 THEOREM. The Newton divided-difference polynomial Q_n agrees with f at the points $x_0, x_1, \dots, x_n$. Therefore, Q_n is identical to the Lagrange interpolation polynomial given by (8.51).

Proof. We argue by induction on n . For $n = 0$ we have $Q_0(x) = f(x_0)$; in particular, $Q_0(x_0) = f(x_0)$, so the theorem holds in this case. Assume, therefore, that the theorem has been proved for $n - 1$, and consider the polynomial Q_n . We wish to prove that

$$(8.58) \quad Q_n(x_j) = f(x_j) \quad \text{for } j = 0, 1, 2, \dots, n.$$

For $j < n$ we use Equation (8.57) to obtain $Q_n(x_j) = Q_{n-1}(x_j)$. By the inductive hypothesis, we have $Q_{n-1}(x_j) = f(x_j)$ for $j < n$, so (8.58) holds for these j . Now we prove that (8.58) also holds for $j = n$. From Equation (8.57) we obtain

$$(8.59) \quad Q_n(x_n) = Q_{n-1}(x_n) + (x_n - x_0) \cdots (x_n - x_{n-1})f(x_0, x_1, \dots, x_n).$$

But if we use (8.55) with $k = n - 1$ and $x = x_n$ we see that the right member of (8.59) is equal to $f(x_n)$. This completes the proof.

Theorem 8-11 gives us a new method for determining the interpolation polynomial P_n . We compute the n divided differences $f(x_0, x_1), f(x_0, x_1, x_2), \dots, f(x_0, x_1, \dots, x_n)$ and we have

$$(8.60) \quad P_n(x) = f(x_0) + \sum_{r=0}^{n-1} (x - x_0) \cdots (x - x_r)f(x_0, x_1, \dots, x_{r+1}).$$

This is called *Newton's interpolation formula with divided differences*. Since

$$P_{n+1}(x) = P_n(x) + (x - x_0) \cdots (x - x_n)f(x_0, x_1, \dots, x_{n+1}),$$

we can calculate P_{n+1} simply by adding one new term to P_n . This property of Newton's interpolation formula is not possessed by Lagrange's formula.

The practical usefulness of Newton's formula depends, of course, on the ease with which the divided differences $f(x_0, x_1, \dots, x_{r+1})$ can be computed. Although it is possible to compute each of these differences directly from its definition, it is preferable to use an alternate method that we shall describe in the next section. The method is based on the following *symmetry property* of divided differences.

8-12 THEOREM. The divided difference $f(x_0, x_1, \dots, x_n)$ is a symmetric function of $x_0, x_1, \dots, x_n$. That is, if k is a one-to-one function whose domain and range is the set of integers $\{0, 1, 2, \dots, n\}$, we have

$$f(x_0, x_1, \dots, x_n) = f(x_{k(0)}, x_{k(1)}, \dots, x_{k(n)}).$$

Proof. From Equation (8.60) we note that $f(x_0, x_1, \dots, x_n)$ is the coefficient of x^n in the interpolation polynomial of degree $\leq n$ that agrees with f at the interpolation points $x_0, x_1, \dots, x_n$. Let $y_j = x_{k(j)}$. Then $f(y_0, y_1, \dots, y_n)$ is the coefficient of x^n in the interpolation polynomial of degree $\leq n$ that agrees with f at the interpolation points $y_0, y_1, \dots, y_n$. Since the sets $\{x_0, x_1, \dots, x_n\}$ and $\{y_0, y_1, \dots, y_n\}$ are equal the corresponding interpolation polynomials are identical, so the coefficients of x^n must be equal. This proves the theorem.

The symmetric property of divided differences can also be deduced from the following theorem which expresses the n th divided difference as a linear combination of the function values $f(x_0), \dots, f(x_n)$.

8-13 THEOREM. If $x_0, x_1, \dots, x_n$ are distinct, we have

$$(8.61) \quad f(x_0, x_1, \dots, x_n) = \sum_{k=0}^n \frac{f(x_k)}{A_k(x_k)},$$

where

$$A_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n (x - x_j).$$

Proof. Let P_n be the interpolation polynomial of degree $\leq n$ that agrees with f at $x_0, x_1, \dots, x_n$. By Lagrange's formula we have

$$P_n(x) = \sum_{k=0}^n L_k(x) f(x_k),$$

where $L_k(x)$ is the polynomial of degree n given by (8.52). Since the coefficient of x^n in $L_k(x)$ is $1/A_k(x_k)$, the coefficient of x^n in $P_n(x)$ is the sum appearing on the right of (8.61). On the other hand, Newton's formula shows that the coefficient of x^n in $P_n(x)$ is equal to $f(x_0, x_1, \dots, x_n)$. This completes the proof.

8.16 Divided-difference tables

The symmetry property in Theorem 8-12 leads to an alternative formula for calculating divided differences that is especially useful in connection with tabular data. The formula in question is given by the following theorem.

8-14 THEOREM. If $x_k, x_{k+1}, \dots, x_{k+r}$ are distinct, we have

$$f(x_k, x_{k+1}, \dots, x_{k+r}) = \frac{f(x_k, x_{k+1}, \dots, x_{k+r-1}) - f(x_{k+1}, x_{k+2}, \dots, x_{k+r})}{x_k - x_{k+r}}.$$

Proof. Let $y_j = x_{k+j}$. We wish to prove that

$$(8.62) \quad f(y_0, y_1, \dots, y_r) = \frac{f(y_0, y_1, \dots, y_{r-1}) - f(y_1, y_2, \dots, y_r)}{y_0 - y_r}.$$

From the definition of $f(y_0, y_1, \dots, y_r)$ we have

$$f(y_0, y_1, \dots, y_r) = \frac{f(y_0, y_1, \dots, y_{r-2}, y_{r-1}) - f(y_0, y_1, \dots, y_{r-2}, y_r)}{y_{r-1} - y_r}.$$

Because of the symmetry property we can interchange y_0 and y_{r-1} without changing the value of $f(y_0, y_1, \dots, y_r)$. If we make this interchange in the right member, the foregoing equation becomes

$$f(y_0, y_1, \dots, y_r) = \frac{f(y_{r-1}, y_1, \dots, y_{r-2}, y_0) - f(y_{r-1}, y_1, \dots, y_{r-2}, y_r)}{y_0 - y_r}.$$

In the numerator on the right we use the symmetry property in each term and obtain (8.62) at once.

The diagram in Table 8.1, called a *divided-difference table*, shows how divided differences can be systematically calculated from a tabulation of the values of f at the points $x_0, x_1, \dots, x_n$ by successive applications of Theorem 8-14. For brevity we have written f_i for $f(x_i)$, f_{ij} for $f(x_i, x_j)$, f_{ijk} for $f(x_i, x_j, x_k)$, and so on. The arrows are shown to help indicate how the table is constructed. In each column, the uppermost entry gives the

TABLE 8.1

x	$f(x)$	First divided differences	Second divided differences	Third divided differences	Fourth divided differences
x_0	f_0				
		$\frac{f_0 - f_1}{x_0 - x_1} = f_{01}$			
x_1	f_1		$\frac{f_{01} - f_{12}}{x_0 - x_2} = f_{012}$		
		$\frac{f_1 - f_2}{x_1 - x_2} = f_{12}$		$\frac{f_{012} - f_{123}}{x_0 - x_3} = f_{0123}$	
x_2	f_2		$\frac{f_{12} - f_{23}}{x_1 - x_3} = f_{123}$		$\frac{f_{0123} - f_{1234}}{x_0 - x_4} = f_{01234}$
		$\frac{f_2 - f_3}{x_2 - x_3} = f_{23}$		$\frac{f_{123} - f_{234}}{x_1 - x_4} = f_{1234}$	
x_3	f_3		$\frac{f_{23} - f_{34}}{x_2 - x_4} = f_{234}$		
		$\frac{f_3 - f_4}{x_3 - x_4} = f_{34}$			
x_4	f_4				

divided difference required in Newton's formula. This particular table provides the necessary divided differences for the interpolation polynomial of degree ≤ 4 . If another interpolation point x_5 is added, the new divided difference f_{012345} can be computed by calculating a new entry at the bottom of each existing column and adding a new column for the fifth-order difference.

Since interpolating polynomials are unique, the error estimates given for Lagrange's formula also apply to Newton's formula. From (8.55) we have

$$(8.63) \quad f(x) - P_n(x) = (x - x_0) \cdots (x - x_n) f(x_0, x_1, \dots, x_n, x).$$

If f has a derivative of order $n + 1$ in an interval $[\alpha, \beta]$ containing the points $x_0, x_1, \dots, x_n, x$, Theorem 8-3 tells us that

$$f(x) - P_n(x) = (x - x_0) \cdots (x - x_n) \frac{f^{(n+1)}(c)}{(n+1)!}$$

for some c in (α, β) . Comparing this with (8.63) we see that

$$f(x_0, x_1, \dots, x_n, x) = \frac{f^{(n+1)}(c)}{(n+1)!}.$$

Therefore, if upper and lower bounds for $f^{(n+1)}$ in (α, β) are known we can obtain corresponding bounds for $f(x_0, x_1, \dots, x_n, x)$.

8.17 Exercises

1. A function f has the values given in the following table:

x	0	2	3	5
$f(x)$	0.000	20.134	30.452	52.110

Let P be the polynomial of degree ≤ 3 that agrees with f at the tabulated values. Construct a divided-difference table and use Newton's interpolation formula to evaluate $P(2.5)$.

2. A function f has the values given in the following table:

x	-2	0	1	3
$f(x)$	-216	-12	0	-6

(a) Let P be the polynomial of degree ≤ 3 that agrees with f at the tabulated values. Construct a divided-difference table and use Newton's interpolation formula to evaluate $P(2)$.

(b) Let Q be the polynomial of degree ≤ 4 that agrees with f at the tabulated values and assume that $Q(5) = -54$. Extend the difference table in part (a) and compute $Q(2)$.

3. Let $f(x) = c_0 + c_1x + \cdots + c_nx^n$, where $c_n \neq 0$. If $x_0, x_1, \dots, x_k$ are $k + 1$ distinct numbers, prove that

$$f(x_0, x_1, \dots, x_k) = \begin{cases} c_n & \text{if } k = n, \\ 0 & \text{if } k > n. \end{cases}$$

4. Prove each of the following properties of divided differences. In each case assume $x_0 \neq x_1$.

(a) If $f(x) = u(x) + v(x)$, then

$$\tilde{f}(x_0, x_1) = u(x_0, x_1) + v(x_0, x_1).$$

(b) If $f(x) = u(x)v(x)$, then

$$f(x_0, x_1) = u(x_0)v(x_0, x_1) + u(x_0, x_1)v(x_1).$$

(c) If $f(x) = u(x)/v(x)$ whenever $v(x) \neq 0$, then

$$f(x_0, x_1) = \frac{v(x_0)u(x_0, x_1) - u(x_0)v(x_0, x_1)}{v(x_0)v(x_1)},$$

provided $v(x_0) \neq 0, v(x_1) \neq 0$.

(d) If $f(x) = u[v(x)]$, then

$$f(x_0, x_1) = u[v(x_0), v(x_1)] v(x_0, x_1),$$

provided $v(x_0) \neq v(x_1)$.

5. (a) Show that the relation in Exercise 4(b) generalizes as follows:

$$f(x_0, x_1, x_2) = u(x_0)v(x_0, x_1, x_2) + u(x_0, x_1)v(x_1, x_2) + u(x_0, x_1, x_2)v(x_2).$$

(b) Guess the generalization suggested for $f(x_0, x_1, \dots, x_n)$ and prove it by induction.

6. Up to now, the divided difference $f(x_0, x)$ has been defined only for $x \neq x_0$. If f has a derivative at x_0 , prove that

$$\lim_{x \rightarrow x_0} f(x_0, x) = f'(x_0).$$

Because of this relation it is reasonable to define $f(x, x)$ to be $f'(x)$ at those points x for which $f'(x)$ exists. [We do not define $f(x, x)$ at those x for which $f'(x)$ does not exist.]

7. Assume f has a derivative everywhere in an interval (a, b) . Choose a fixed x_0 in (a, b) and define a new function g on (a, b) by the equation $g(x) = f(x_0, x)$.

(a) Use Exercise 6 to prove that g is continuous everywhere on (a, b) .

(b) If $x \neq x_0$, prove that $g'(x)$ exists and is given by the formula

$$g'(x) = \frac{f'(x) - f(x_0, x)}{x - x_0}.$$

(c) If f'' exists everywhere in (a, b) and if $x \neq x_0$, prove that $g'(x) = \frac{1}{2}f''(c)$ for some c between x and x_0 . [Hint. Use part (b) and Taylor's formula with remainder.]

(d) If f'' is continuous in (a, b) , define $g'(x_0) = \frac{1}{2}f''(x_0)$. Prove that g' is continuous everywhere on (a, b) .

8. Let $x_0, x_1, \dots, x_n, x$ be $n + 2$ distinct points.

(a) Prove that

$$(x - x_0)(x - x_1) \cdots (x - x_n)f(x_0, x_1, \dots, x_n, x) = \sum_{k=0}^n L_k(x)[f(x) - f(x_k)],$$

where $L_k(x)$ is the Lagrange interpolation coefficient given by (8.52). [Hint. Use Equation (8.63) and Lagrange's interpolation formula.]

(b) Use part (a) to deduce the formula

$$(8.64) \quad f(x_0, x_1, \dots, x_n, x) = \sum_{k=0}^n \frac{f(x_k, x)}{A_k(x_k)},$$

where

$$A_k(x_k) = \prod_{\substack{j=0 \\ j \neq k}}^n (x_k - x_j).$$

Equation (8.64) expresses a divided difference of order $n + 1$ as a linear combination of divided differences of first order.

9. In Exercise 6 we assigned a meaning to $f(x, x)$ for those x at which $f'(x)$ exists. Therefore, the right member of Equation (8.64) is meaningful if $x = x_j$, provided $f'(x_j)$ exists. If $x_0, x_1, \dots, x_n$ are distinct, and if $f'(x_j)$ exists, we define $f(x_0, x_1, \dots, x_n, x_j)$ by Equation (8.64) with $x = x_j$.

(a) Use this definition to prove that

$$f(x_0, x_1, \dots, x_n, x_j) = \lim_{x \rightarrow x_j} f(x_0, x_1, \dots, x_n, x).$$

(b) Let g be the function defined on (a, b) by the equation $g(x) = f(x_0, x_1, \dots, x_n, x)$. If f' exists on (a, b) prove that g is continuous everywhere on (a, b) .

8.18 Ordinary differences. Equally spaced interpolation points

When the interpolation points $x_0, x_1, \dots, x_n$ are equally spaced the calculation of divided differences can be considerably simplified. To discuss this case we introduce the so-called *forward difference operator* Δ .

DEFINITION. Let h be a fixed real number and let f be a given function. The function Δf defined by the equation

$$\Delta f(x) = f(x + h) - f(x)$$

is called the first forward difference of f . It is defined at those points x for which both x and $x + h$ are in the domain of f . Higher order differences $\Delta^2 f, \Delta^3 f, \dots$ are defined inductively as follows:

$$\Delta^{k+1}f = \Delta(\Delta^k f) \quad \text{for } k = 1, 2, 3, \dots$$

Note. The notations $\Delta_h f(x)$ and $\Delta f(x; h)$ are also used for $\Delta f(x)$ when it is desirable to indicate the dependence on h . It is convenient to define $\Delta^0 f = f$.

Now suppose f is defined at $n + 1$ equally spaced points $x_0 < x_1 < x_2 < \dots < x_n$, where $x_j = x_0 + jh$ for $j = 0, 1, 2, \dots, n$. Here h is a positive number that represents the spacing between adjacent points. The next theorem expresses the divided differences in terms of forward differences.

8-15 THEOREM. If $h > 0$ and $x_j = x_0 + jh$ for $j = 0, 1, 2, \dots, n$, we have

$$f(x_0, x_1, \dots, x_n) = \frac{\Delta^n f(x_0)}{n! h^n}.$$

Proof. The proof proceeds by induction on the number of points. For two points ($n = 1$) we have

$$f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_0 + h) - f(x_0)}{h} = \frac{\Delta f(x_0)}{h},$$

so the theorem holds. Therefore, assume the theorem holds for n points. From Theorem 8-14 we have

$$\begin{aligned} f(x_0, x_1, \dots, x_n) &= \frac{f(x_0, x_1, \dots, x_{n-1}) - f(x_1, x_2, \dots, x_n)}{x_0 - x_n} \\ &= \frac{f(x_1, x_2, \dots, x_n) - f(x_0, x_1, \dots, x_{n-1})}{nh}. \end{aligned}$$

From the inductive hypothesis we have

$$f(x_1, x_2, \dots, x_n) = \frac{\Delta^{n-1}f(x_1)}{(n-1)! h^{n-1}} \quad \text{and} \quad f(x_0, x_1, \dots, x_{n-1}) = \frac{\Delta^{n-1}f(x_0)}{(n-1)! h^{n-1}}.$$

Since $\Delta^{n-1}f(x_1) = \Delta^{n-1}f(x_0 + h)$, we have

$$f(x_0, x_1, \dots, x_n) = \frac{\Delta^{n-1}f(x_0 + h) - \Delta^{n-1}f(x_0)}{n! h^n} = \frac{\Delta^n f(x_0)}{n! h^n}.$$

This completes the proof.

Newton's interpolation formula (8.60) now becomes

$$(8.65) \quad P_n(x) = f(x_0) + \sum_{r=0}^{n-1} \frac{\Delta^{r+1}f(x_0)}{(r+1)! h^{r+1}} \prod_{j=0}^r (x - x_j).$$

If we write

$$\prod_{j=0}^r (x - x_j) = \prod_{j=0}^r (x - x_0 - jh) = h^{r+1} \prod_{j=0}^r \left(\frac{x - x_0}{h} - j \right) = h^{r+1} \prod_{j=0}^r (t - j),$$

where

$$t = \frac{x - x_0}{h},$$

Equation (8.65) becomes

$$P_n(x) = f(x_0) + \sum_{r=0}^{n-1} \frac{\Delta^{r+1}f(x_0)}{(r+1)!} \prod_{j=0}^r (t - j).$$

The differences $\Delta^{r+1}f(x_0)$ are easily computed in succession from an ordinary difference table, as illustrated in Table 8.2. Again, we have written f_i for $f(x_i)$.

TABLE 8.2

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
x_0	f_0			
	$\swarrow$			
	f_1	$f_1 - f_0 = \Delta f(x_0)$		
x_1	$\swarrow$			
	f_2	$f_2 - f_1 = \Delta f(x_1)$	$\Delta f(x_1) - \Delta f(x_0) = \Delta^2 f(x_0)$	
	$\swarrow$			
	f_3	$f_3 - f_2 = \Delta f(x_2)$	$\Delta f(x_2) - \Delta f(x_1) = \Delta^2 f(x_1)$	$\Delta^2 f(x_1) - \Delta^2 f(x_0) = \Delta^3 f(x_0)$
x_2	$\swarrow$			
	f_4			
x_3				

Theorem 8-13 shows that the divided difference $f(x_0, x_1, \dots, x_n)$ is a linear combination of the function values $f(x_0), \dots, f(x_n)$, given by

$$(8.66) \quad f(x_0, x_1, \dots, x_n) = \sum_{k=0}^n \frac{f(x_k)}{A_k(x_k)},$$

where

$$A_k(x_k) = \prod_{\substack{j=0 \\ j \neq k}}^n (x_k - x_j).$$

For equally spaced points with $x_k - x_j = h(k - j)$ the product multiplying $f(x_k)$ in (8.66) can be simplified. Using Equation (8.14) (page 398) we obtain

$$(8.67) \quad \frac{1}{A_k(x_k)} = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{1}{x_k - x_j} = \frac{1}{h^n} \prod_{\substack{j=0 \\ j \neq k}}^n \frac{1}{k - j} = \frac{(-1)^{n-k}}{n! h^n} \binom{n}{k}.$$

Since we also have $\Delta^n f(x_0) = n! h^n f(x_0, x_1, \dots, x_n)$ for equally spaced points, Equations (8.66) and (8.67) give us the formula

$$\Delta^n f(x_0) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(x_k),$$

which expresses the n th order difference $\Delta^n f(x_0)$ as a linear combination of $f(x_0), f(x_1), \dots, f(x_n)$.

Factorial polynomials. If n is a positive integer, the product $t(t - 1) \cdots (t - n + 1)$ is a polynomial in t of degree n called the n th *factorial polynomial*, or the *factorial n th power* of t . It is denoted by the symbol $t^{(n)}$. Thus, by definition,

$$t^{(n)} = \prod_{j=0}^{n-1} (t - j).$$

We also define $t^{(0)} = 1$. If we consider the forward difference operator Δ with $h = 1$, that is, $\Delta f(x) = f(x + 1) - f(x)$, we find that

$$\Delta t^{(n)} = n t^{(n-1)} \quad \text{for } n \geq 1.$$

This is analogous to the differentiation formula $Dt^n = n t^{n-1}$ for ordinary powers. Thus, the factorial power $t^{(n)}$ is related to differences much in the same way that the ordinary power t^n is related to derivatives.

With the use of factorial polynomials, Newton's interpolation formula (8.65) becomes

$$P_n(x_0 + th) = \sum_{k=0}^n \frac{\Delta^k f(x_0)}{k!} t^{(k)}.$$

Expressed in this form, Newton's formula resembles the Taylor formula for the polynomial of degree $\leq n$ that agrees with f and its first n derivatives at x_0 . If we write

$$\binom{t}{k} = \frac{t^{(k)}}{k!} = \frac{t(t-1) \cdots (t-k+1)}{k!},$$

Newton's formula takes the form

$$P_n(x_0 + th) = \sum_{k=0}^n \binom{n}{k} \Delta^k f(x_0).$$

Further properties of factorial polynomials are developed in the following exercises.

8.19 Exercises

1. Let $\Delta f(x) = f(x+h) - f(x)$. If f is a polynomial of degree n , say

$$f(x) = \sum_{r=0}^n a_r x^r$$

with $a_n \neq 0$, show that (a) $\Delta^k f(x)$ is a polynomial of degree $n-k$ if $k \leq n$; (b) $\Delta^n f(x) = n! h^n a_n$; (c) $\Delta^k f(x) = 0$ for $k > n$.

2. Let $\Delta f(x) = f(x+h) - f(x)$. If $f(x) = \sin(ax+b)$, prove that

$$\Delta^n f(x) = \left(2 \sin \frac{ah}{2}\right)^n \sin\left(ax+b + \frac{nah + n\pi}{2}\right).$$

3. Let $\Delta f(x) = f(x+h) - f(x)$.

(a) If $f(x) = a^x$, where $a > 0$, show that $\Delta^k f(x) = (a^h - 1)^k a^x$.

(b) If $g(x) = (1+a)^{x/h}$, where $a > 0$, show that $\Delta^k g(x) = a^k g(x)$.

(c) Show that the polynomial P_n of degree n that takes the values $P_n(k) = (1+a)^k$ for $k = 0, 1, 2, \dots, n$ is given by

$$P_n(x) = \sum_{k=0}^n \frac{a^k}{k!} x^{(k)}.$$

4. Let $x^{(n)}$ be the factorial n th power of x . Since $x^{(n)}$ is a polynomial in x of degree n with the value 0 when $x = 0$, we may write

$$x^{(n)} = \sum_{k=1}^n S_{k,n} x^k.$$

The numbers $S_{k,n}$ are called *Stirling numbers of the first kind*. From the definition of $x^{(n)}$ it is clear that $S_{n,n} = 1$ for $n \geq 0$.

(a) Show that $S_{n-1,n} = -n(n-1)/2$ and that $S_{1,n} = (-1)^{n-1}(n-1)!$ for $n \geq 1$.

(b) Prove that $S_{k,n+1} = S_{k-1,n} - n S_{k,n}$. Use this relation to verify the entries in Table 8.3, a table of Stirling numbers of the first kind, and construct the next three rows of the table.

TABLE 8.3

n	$S_{1,n}$	$S_{2,n}$	$S_{3,n}$	$S_{4,n}$	$S_{5,n}$	$S_{6,n}$	$S_{7,n}$
1	1						
2	-1	1					
3	2	-3	1				
4	-6	11	-6	1			
5	24	-50	35	-10	1		
6	-120	274	-225	85	-15	1	
7	720	-1764	1624	-735	175	-21	1

(c) Express the polynomial $x^{(4)} + 3x^{(3)} + 2x^{(1)} + 1$ as a linear combination of powers of x .

5. (a) Prove that

$$x = x^{(1)}, \quad x^2 = x^{(1)} + x^{(2)}, \quad x^3 = x^{(1)} + 3x^{(2)} + x^{(3)},$$

and that, in general,

$$x^n = \sum_{k=1}^n \frac{\Delta^k f(0)}{k!} x^{(k)},$$

where $f(x) = x^n$ and $\Delta f(x) = f(x+1) - f(x)$. The numbers $T_{k,n} = \Delta^k f(0)/k!$ are called *Stirling numbers of the second kind*.

(b) Prove that

$$\Delta^k x^{n+1} = (x+k) \Delta^k x^n + k \Delta^{k-1} x^n$$

and use this to deduce that $T_{k,n+1} = T_{k-1,n} + k T_{k,n}$.

(c) Use the recursion formula in part (b) to verify the entries in Table 8.4, a table of Stirling numbers of the second kind, and construct the next three rows of the table.

TABLE 8.4

n	$T_{1,n}$	$T_{2,n}$	$T_{3,n}$	$T_{4,n}$	$T_{5,n}$	$T_{6,n}$	$T_{7,n}$
1	1						
2	1	1					
3	1	3	1				
4	1	7	6	1			
5	1	15	25	10	1		
6	1	31	90	65	15	1	
7	1	63	301	350	140	21	1

(d) Express the polynomial $x^4 + 3x^3 + 2x - 1$ as a linear combination of factorial polynomials.

6. (a) If p is a positive integer and if a and b are integers with $a < b$, prove that

$$\sum_{k=a}^{b-1} k^{(p)} = \frac{b^{(p+1)} - a^{(p+1)}}{p+1}.$$

This formula is analogous to the integration formula for $\int_a^b x^p dx$. It should be noted, however, that the upper limit in the sum is $b-1$, not b .

(b) Verify that $k(k+3) = 4k^{(1)} + k^{(2)}$. Then use part (a) to show that

$$\sum_{k=1}^n k(k+3) = 4 \frac{(n+1)^{(2)}}{2} + \frac{(n+1)^{(3)}}{3} = \frac{n(n+1)(n+5)}{3}.$$

(c) If $f(k)$ is a polynomial in k of degree r , prove that

$$\sum_{k=1}^n f(k)$$

is a polynomial in n of degree $r+1$.

7. Use the method suggested in Exercise 6 to express each of the following sums as a polynomial in n .

$$(a) \sum_{k=1}^n (4k^2 + 7k + 6).$$

$$(c) \sum_{k=1}^n k(k+1)(k+2).$$

$$(b) \sum_{k=1}^n k^2(k+1).$$

$$(d) \sum_{k=1}^n k^4.$$

8. Let A denote the linear operator defined by the equation

$$A(f) = a_0 \Delta^n f + a_1 \Delta^{n-1} f + \cdots + a_{n-1} \Delta f + a_n f,$$

where $a_0, a_1, \dots, a_n$ are constants. This is called a *constant-coefficient difference operator*. It is analogous to the constant-coefficient derivative operator described in Section 7.15. With each such A we can associate the characteristic polynomial p_A defined by

$$p_A(r) = a_0 r^n + a_1 r^{n-1} + \cdots + a_{n-1} r + a_n.$$

Conversely, with every polynomial p we can associate an operator A having this polynomial as its characteristic polynomial. If A and B are constant-coefficient difference operators and if λ is a real number, define $A + B$, AB , and λA by the same formulas used in Section 7.15 for derivative operators. Then prove that Theorem 7-12 is valid for constant-coefficient difference operators.

8.20 Approximate integration. The trapezoidal rule

Many problems in both pure and applied mathematics lead to new functions whose properties have not been studied or whose values have not been tabulated. To satisfy certain practical needs of applied science it often becomes necessary to obtain quantitative information about such functions, either in graphical or numerical form. Many of these functions occur as integrals of the type

$$F(x) = \int_a^x f(t) dt,$$

where the integrand f is given by an explicit analytic formula or is known in part by tabular data. The remainder of this chapter describes some of the most elementary methods for finding numerical approximations to such integrals. The basic idea is very simple. We approximate the integrand f by another function P whose integral is easily computed, and then we use the integral of P as an approximation to the integral of f .

If f is nonnegative the integral $\int_a^b f(x) dx$ represents the area of the ordinate set of f over $[a, b]$. This geometric interpretation of the integral immediately suggests certain procedures for approximate integration. Figure 8.2 shows an example of a function f with known values at $n+1$ equally spaced points $a, a+h, a+2h, \dots, a+nh = b$, where $h = (b-a)/n$. Let $x_k = a + kh$. For each $k = 0, 1, 2, \dots, n-1$ the graph of f over the interval $[x_k, x_{k+1}]$ has been approximated by a linear function that agrees with f at the endpoints x_k and x_{k+1} . Let P denote the corresponding piecewise linear interpolating function defined over the full interval $[a, b]$. Then we have

$$(8.68) \quad P(x) = \frac{x_{k+1} - x}{h} f(x_k) + \frac{x - x_k}{h} f(x_{k+1}) \quad \text{if } x_k \leq x \leq x_{k+1}.$$

Integrating over the interval $[x_k, x_{k+1}]$ we find that

$$\int_{x_k}^{x_{k+1}} P(x) dx = h \frac{f(x_k) + f(x_{k+1})}{2}.$$

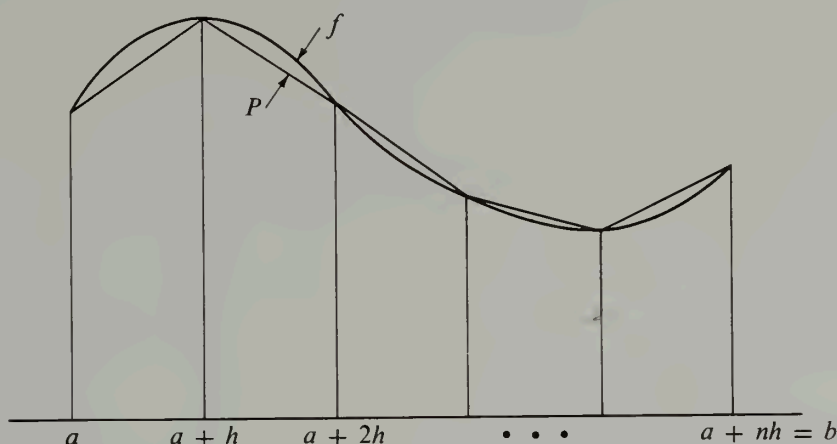


FIGURE 8.2 The trapezoidal rule obtained by piecewise linear interpolation.

When f is positive this is the area of the trapezoid determined by the graph of P over $[x_k, x_{k+1}]$. The formula holds, of course, even if f is not positive everywhere. Adding the integrals over all subintervals $[x_k, x_{k+1}]$ we obtain

$$\begin{aligned}
 (8.69) \quad \int_a^b P(x) \, dx &= \frac{h}{2} \sum_{k=0}^{n-1} [f(x_k) + f(x_{k+1})] \\
 &= \frac{h}{2} \left(f(a) + 2 \sum_{k=1}^{n-1} f(a + kh) + f(b) \right).
 \end{aligned}$$

To use this sum as an approximation to the integral $\int_a^b f(x) \, dx$ we need an estimate for the error, $\int_a^b f(x) \, dx - \int_a^b P(x) \, dx$. If f has a continuous second derivative on $[a, b]$ this error is given by the following theorem.

8-16 THEOREM. *Trapezoidal rule.* Assume f has a continuous second derivative f'' on $[a, b]$. If n is a positive integer, let $h = (b - a)/n$. Then we have

$$(8.70) \quad \int_a^b f(x) \, dx = \frac{b-a}{2n} \left(f(a) + 2 \sum_{k=1}^{n-1} f(a + kh) + f(b) \right) - \frac{(b-a)^3}{12n^2} f''(c)$$

for some c in $[a, b]$.

Note. Equation (8.70) is known as the *trapezoidal rule*. The term $-f''(c)(b-a)^3/12n^2$ represents the error in approximating $\int_a^b f(x) \, dx$ by $\int_a^b P(x) \, dx$. Once the maximum value of f'' on $[a, b]$ is known we can approximate the integral of f to any desired degree of accuracy by taking n sufficiently large. Note that no knowledge of the interpolating function P is required to use this formula. It is only necessary to know the values of f at the points $a, a+h, \dots, a+nh$, and to have an estimate for $|f''(c)|$.

Proof. Let P be the interpolating function given by (8.68), where $x_k = a + kh$. In each subinterval $[x_k, x_{k+1}]$ we apply the error estimate for linear interpolation given by Theorem 8-3 and we find

$$(8.71) \quad f(x) - P(x) = (x - x_k)(x - x_{k+1}) \frac{f''(c_k)}{2!}$$

for some c_k in (x_k, x_{k+1}) . Let M_2 and m_2 denote the maximum and minimum, respectively, of f'' on $[a, b]$, and let

$$B(x) = (x - x_k)(x_{k+1} - x)/2.$$

Then $B(x) \geq 0$ in the interval $[x_k, x_{k+1}]$, and from (8.71) we obtain the inequalities

$$m_2 B(x) \leq P(x) - f(x) \leq M_2 B(x)$$

in this interval. Integrating, we have

$$(8.72) \quad m_2 \int_{x_k}^{x_{k+1}} B(x) dx \leq \int_{x_k}^{x_{k+1}} [P(x) - f(x)] dx \leq M_2 \int_{x_k}^{x_{k+1}} B(x) dx.$$

The integral of B is given by

$$\int_{x_k}^{x_{k+1}} B(x) dx = \frac{1}{2} \int_{x_k}^{x_{k+1}} (x - x_k)(x_{k+1} - x) dx = \frac{1}{2} \int_0^h t(h - t) dt = \frac{h^3}{12}.$$

Therefore the inequalities (8.72) give us

$$m_2 \leq \frac{12}{h^3} \int_{x_k}^{x_{k+1}} [P(x) - f(x)] dx \leq M_2.$$

Adding these inequalities for $k = 0, 1, 2, \dots, n - 1$ and dividing by n , we obtain

$$m_2 \leq \frac{12}{nh^3} \int_a^b [P(x) - f(x)] dx \leq M_2.$$

Since the function f'' is continuous on $[a, b]$, it assumes every value between its minimum m_2 and its maximum M_2 somewhere in $[a, b]$. In particular, we have

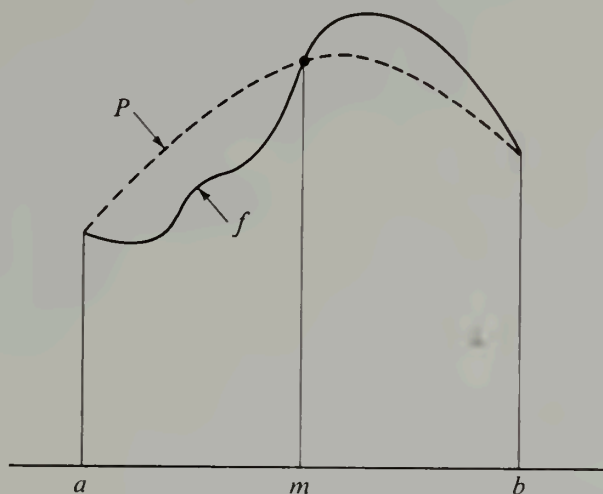
$$f''(c) = \frac{12}{nh^3} \int_a^b [P(x) - f(x)] dx$$

for some c in $[a, b]$. In other words,

$$\int_a^b f(x) dx = \int_a^b P(x) dx - \frac{nh^3}{12} f''(c).$$

Using (8.69) and the relation $h = (b - a)/n$ we obtain (8.70).

To derive the trapezoidal rule we used a linear polynomial to interpolate between each adjacent pair of values of f . More accurate formulas can be obtained by interpolating with polynomials of higher degree. In the next section we consider an important special case that is remarkable for its simplicity and accuracy.

FIGURE 8.3 Interpolation by a quadratic polynomial P .

8.21 Simpson's rule

The solid curve in Figure 8.3 is the graph of a function f over an interval $[a, b]$. The mid-point of the interval, $(a + b)/2$, is denoted by m . The dotted curve is the graph of a quadratic polynomial P that agrees with f at the three points a , m , and b . If we use the integral $\int_a^b P(x) dx$ as an approximation to $\int_a^b f(x) dx$ we are led to an approximate integration formula known as *Simpson's rule*.

Instead of determining P explicitly, we introduce a linear transformation that carries the interval $[a, b]$ onto the interval $[0, 2]$. If we write

$$t = \frac{x - a}{m - a}, \quad \text{or} \quad x = a + (m - a)t,$$

we see that t takes the values 0, 1, 2 when x takes the values a , m , b . Now let

$$\phi(t) = P[a + (m - a)t].$$

Then $\phi(t)$ is a quadratic polynomial in t that takes the values $P(a)$, $P(m)$, $P(b)$ at the points $t = 0, 1, 2$, respectively. Also, we have

$$\int_0^2 \phi(t) dt = \int_0^2 P[a + (m - a)t] dt = \frac{1}{m - a} \int_a^b P(x) dx;$$

hence

$$(8.73) \quad \int_a^b P(x) dx = (m - a) \int_0^2 \phi(t) dt = \frac{b - a}{2} \int_0^2 \phi(t) dt.$$

Now we use Newton's interpolation formula to construct ϕ . We have

$$\phi(t) = \phi(0) + t \Delta\phi(0) + t(t-1) \frac{\Delta^2\phi(0)}{2!},$$

where $\Delta\phi(t) = \phi(t+1) - \phi(t)$. Integrating from 0 to 2 we obtain

$$\int_0^2 \phi(t) dt = 2\phi(0) + 2\Delta\phi(0) + \frac{1}{3}\Delta^2\phi(0).$$

Since $\Delta\phi(0) = \phi(1) - \phi(0)$ and $\Delta^2\phi(0) = \phi(2) - 2\phi(1) + \phi(0)$, the integral is equal to

$$\int_0^2 \phi(t) dt = \frac{1}{3} [\phi(0) + 4\phi(1) + \phi(2)] = \frac{1}{3} [P(a) + 4P(m) + P(b)].$$

Using (8.73) and the fact that P agrees with f at a, m, b , we obtain

$$(8.74) \quad \int_a^b P(x) dx = \frac{b-a}{6} [f(a) + 4f(m) + f(b)].$$

Therefore, we may write

$$\int_a^b f(x) dx = \frac{b-a}{6} [f(a) + 4f(m) + f(b)] + R,$$

where $R = \int_a^b f(x) dx - \int_a^b P(x) dx$.

If f is a quadratic polynomial, P is identical to f and the error R is zero. It is a remarkable fact that we also have $R = 0$ when f is a cubic polynomial. To prove this property we use the error estimate for Lagrange interpolation given by Theorem 8-3, and we write

$$(8.75) \quad f(x) - P(x) = (x-a)(x-m)(x-b) \frac{f'''(c)}{3!},$$

where $c \in (a, b)$. When f is a cubic polynomial the third derivative f''' is constant, say $f'''(x) = C$, and the foregoing formula becomes

$$f(x) - P(x) = \frac{C}{6} (x-a)(x-m)(x-b) = \frac{C}{6} (t+h)t(t-h),$$

where $t = x - m$ and $h = (b-a)/2$. Therefore

$$R = \int_a^b [f(x) - P(x)] dx = \frac{C}{6} \int_{-h}^h (t^3 - h^2t) dt = 0,$$

since the last integrand is an odd function. This property is illustrated in Figure 8.4. The dotted curve is the graph of a cubic polynomial f that agrees with P at a, m, b . In this case $R = \int_a^b [f(x) - P(x)] dx = A_1 - A_2$, where A_1 and A_2 are the areas of the two shaded regions. Since $R = 0$ the two regions have equal areas.

We have just seen that Equation (8.74) is valid if P is a polynomial of degree ≤ 3 that agrees with f at a, m , and b . By choosing this polynomial carefully we can considerably improve the error estimate in (8.75). We have already imposed three conditions on P , namely, $P(a) = f(a)$, $P(m) = f(m)$, $P(b) = f(b)$. Now we impose a fourth condition, $P'(m) = f'(m)$. This will give P and f the same slope at $(m, f(m))$, and we can hope that this will improve the approximation of f by P throughout $[a, b]$.

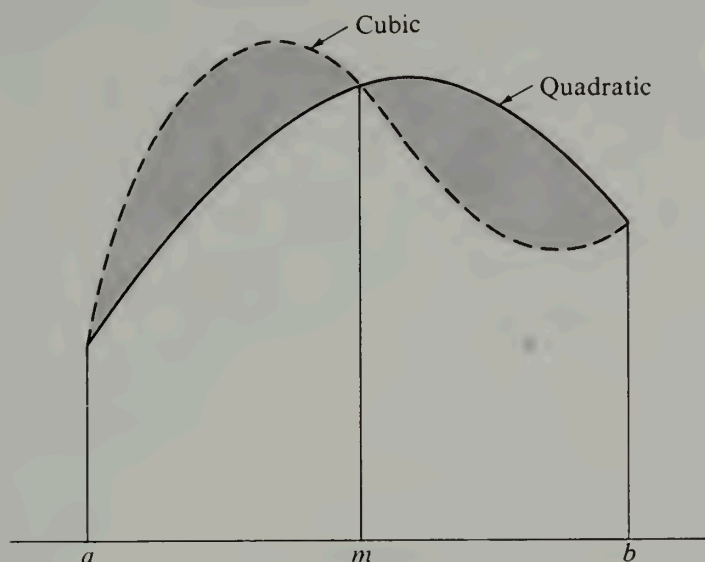


FIGURE 8.4 The two shaded regions have equal areas for every cubic interpolating polynomial.

To show that such a P can always be chosen, we let Q be the quadratic polynomial that agrees with f at a, m, b , and we let

$$P(x) = Q(x) + A(x - a)(x - m)(x - b),$$

where A is a constant to be determined. For any choice of A , this cubic polynomial P agrees with Q and hence with f at a, m, b . Now we choose A to make $P'(m) = f'(m)$. Differentiating the formula for $P(x)$ and putting $x = m$ we obtain

$$P'(m) = Q'(m) + A(m - a)(m - b).$$

Therefore if we take $A = [f'(m) - Q'(m)]/[(m - a)(m - b)]$ we also satisfy the condition $P'(m) = f'(m)$.

Next we show that for this choice of P we have

$$(8.76) \quad f(x) - P(x) = (x - a)(x - m)^2(x - b) \frac{f^{(4)}(z)}{4!}$$

for some z in (a, b) , provided that the fourth derivative $f^{(4)}$ exists in $[a, b]$. To prove (8.76) we argue as in the proof of Theorem 8-3. First we note that (8.76) is trivially satisfied for any choice of z if $x = a, m$, or b . Therefore, assume $x \neq a, x \neq m, x \neq b$, keep x fixed, and introduce a new function F defined on $[a, b]$ by the equation

$$F(t) = A(x) [f(t) - P(t)] - A(t) [f(x) - P(x)],$$

where

$$A(t) = (t - a)(t - m)^2(t - b).$$

Note that $F(t) = 0$ for $t = a, m, b$, and x . By Rolle's theorem, $F'(t)$ vanishes in each of the three open intervals determined by these four points. In addition, $F'(m) = 0$ because $A'(m) = 0$ and $f'(m) = P'(m)$. Therefore $F'(t) \neq 0$ for at least four distinct points in (a, b) . By Rolle's theorem $F''(t) = 0$ for at least three points, $F'''(t) = 0$ for at least two points, and $F^{(4)}(t) = 0$ for at least one point, say for $t = z$. From the definition of F we find

$$\begin{aligned} F^{(4)}(t) &= A(x) [f^{(4)}(t) - P^{(4)}(t)] - A^{(4)}(t) [f(x) - P(x)] \\ &= A(x) f^{(4)}(t) - 4! [f(x) - P(x)]. \end{aligned}$$

When we substitute $t = z$ in this equation we obtain (8.76).

Now it is a simple matter to prove Simpson's rule in the following form.

8-17 THEOREM. *Simpson's rule.* Assume f has a continuous fourth derivative on $[a, b]$, and let $m = (a + b)/2$. Then we have

$$(8.77) \quad \int_a^b f(x) dx = \frac{b-a}{6} [f(a) + 4f(m) + f(b)] - \frac{(b-a)^5}{2880} f^{(4)}(c)$$

for some c in $[a, b]$.

Proof. Let M_4 and m_4 denote, respectively, the maximum and minimum values of $f^{(4)}$ on $[a, b]$, and let $B(x) = -(x-a)(x-m)^2(x-b)/4!$. Since $B(x) \geq 0$ for each x in $[a, b]$, Equation (8.76) leads to the inequalities

$$m_4 B(x) \leq P(x) - f(x) \leq M_4 B(x).$$

Integrating, we find

$$(8.78) \quad m_4 \int_a^b B(x) dx \leq \int_a^b [P(x) - f(x)] dx \leq M_4 \int_a^b B(x) dx.$$

To evaluate the integral $\int_a^b B(x) dx$ we let $h = (b-a)/2$ and we have

$$\begin{aligned} \int_a^b B(x) dx &= -\frac{1}{4!} \int_a^b (x-a)(x-m)^2(x-b) dx = -\frac{1}{4!} \int_{-h}^h (t+h)t^2(t-h) dt \\ &= -\frac{2}{4!} \int_0^h t^2(t^2 - h^2) dt = \frac{1}{4!} \frac{4h^5}{15} = \frac{(b-a)^5}{2880}. \end{aligned}$$

Therefore the inequalities in (8.78) give us

$$m_4 \leq \frac{2880}{(b-a)^5} \int_a^b [P(x) - f(x)] dx \leq M_4.$$

But since $f^{(4)}$ is continuous on $[a, b]$, it assumes every value between its minimum m_4 and its maximum M_4 somewhere in $[a, b]$. Therefore

$$f^{(4)}(c) = \frac{2880}{(b-a)^5} \int_a^b [P(x) - f(x)] dx$$

for some c in $[a, b]$. Since $\int_a^b P(x) dx = \frac{1}{6}(b-a)[f(a) + 4f(m) + f(b)]$, this equation gives us (8.77).

Simpson's rule is of special interest because its accuracy is greater than might be expected from a knowledge of the function f at only three points. If the values of f are known at an odd number of equally spaced points, say at $a, a + h, \dots, a + 2nh$, it is usually simpler to apply Simpson's rule successively to each of the intervals $[a, a + 2h], [a + 2h, a + 4h], \dots$, rather than to use an interpolating polynomial of degree $\leq 2n$ over the full interval $[a, a + 2nh]$. Applying Simpson's rule in this manner, we obtain the following extension of Theorem 8-17.

8-18 THEOREM. *Extended Simpson's rule.* Assume f has a continuous fourth derivative in $[a, b]$. Let $h = (b - a)/(2n)$ and let $f_k = f(a + kh)$ for $k = 1, 2, \dots, 2n - 1$. Then we have

$$\int_a^b f(x) dx = \frac{b-a}{6n} \left(f(a) + 4 \sum_{k=1}^n f_{2k-1} + 2 \sum_{k=1}^{n-1} f_{2k} + f(b) \right) - \frac{(b-a)^5}{2880n^4} f^{(4)}(\bar{c})$$

for some $\bar{c}$ in $[a, b]$.

The proof of this theorem is requested in Exercise 9 of the next section.

8.22 Exercises

1. (a) Apply the trapezoidal rule with $n = 10$ to estimate the value of the integral

$$\log 2 = \int_1^2 \frac{dx}{x}.$$

Obtain upper and lower bounds for the error. (See Exercise 10(b) to compare the accuracy with that obtained from Simpson's rule.)

(b) What is the smallest value of n that would ensure six-place accuracy in the calculation of $\log 2$ by this method?

2. (a) Show that there is a positive number c in the interval $[0, 1]$ such that the formula

$$\int_{-1}^1 f(x) dx = f(c) + f(-c)$$

is exact for all polynomials of degree ≤ 3 .

(b) Generalize the result of part (a) for an arbitrary interval. That is, show that constants c_1 and c_2 exist in $[a, b]$ such that the formula

$$\int_a^b f(x) dx = \frac{b-a}{2} [f(c_1) + f(c_2)]$$

is exact for all polynomials of degree ≤ 3 . Express c_1 and c_2 in terms of a and b .

3. (a) Show that a positive constant c exists such that the formula

$$\int_{-1/2}^{1/2} f(x) dx = \frac{1}{3} [f(-c) + f(0) + f(c)]$$

is exact for all polynomials of degree ≤ 3 .

(b) Generalize the result of part (a) for an arbitrary interval. That is, show that constants c_1 and c_2 exist in $[a, b]$ such that the formula

$$\int_a^b f(x) dx = \frac{b-a}{3} \left[f(c_1) + f\left(\frac{a+b}{2}\right) + f(c_2) \right]$$

is exact for all polynomials of degree ≤ 3 . Express c_1 and c_2 in terms of a and b .

4. Show that positive constants a and b exist such that the formula

$$\int_0^\infty e^{-x} f(x) dx = \frac{1}{4} [a f(b) + b f(a)]$$

is exact for all polynomials of degree ≤ 3 .

5. Show that a positive constant c exists such that the formula

$$\int_{-\infty}^\infty e^{-x^2} f(x) dx = \frac{\sqrt{\pi}}{6} [f(-c) + 4f(0) + f(c)]$$

is exact for all polynomials of degree ≤ 5 .

6. Let P_n be the interpolation polynomial of degree $\leq n$ that agrees with f at $n+1$ distinct points $x_0, x_1, \dots, x_n$.

(a) Show that constants $A_0(n), A_1(n), \dots, A_n(n)$ exist, depending only on the numbers $x_0, x_1, \dots, x_n, a$, and b , and not on f , such that

$$\int_a^b P_n(x) dx = \sum_{k=0}^n A_k(n) f(x_k).$$

The numbers $A_k(n)$ are called *weights*. (They are sometimes called Christoffel numbers.)

(b) For a given set of distinct interpolation points and a given interval $[a, b]$, let $W_0(n), W_1(n), \dots, W_n(n)$ be $n+1$ constants such that the formula

$$\int_a^b f(x) dx = \sum_{k=0}^n W_k(n) f(x_k)$$

is exact for all polynomials of degree $\leq n$. Prove that

$$\sum_{k=0}^n x_k^r W_k(n) = \frac{b^{r+1} - a^{r+1}}{r+1} \quad \text{for } r = 0, 1, \dots, n.$$

This is a system of $n+1$ linear equations that can be used to determine the weights. It can be shown that this system always has a unique solution. It can also be shown that for a suitable choice of interpolation points it is possible to make all the weights equal. When the weights are all equal the integration formula is called a Chebyshev integration formula. Exercises 2 and 3 give examples of Chebyshev integration formulas. The next exercise shows that for a proper choice of interpolation points the resulting integration formula is exact for all polynomials of degree $\leq 2n+1$.

7. In this exercise you may use properties of the Legendre polynomials stated in Sections 7.17 and 7.18. Let $x_0, x_1, \dots, x_n$ be the zeros of the Legendre polynomial $P_{n+1}(x)$. These zeros are distinct and they all lie in the interval $[-1, 1]$. Let $f(x)$ be any polynomial in x of degree $\leq 2n+1$. Divide $f(x)$ by $P_{n+1}(x)$ and write

$$f(x) = P_{n+1}(x) Q(x) + R(x),$$

where the polynomials Q and R have degree $\leq n$.

- (a) Show that the polynomial R agrees with f at the zeros of P_{n+1} and that

$$\int_{-1}^1 f(x) dx = \int_{-1}^1 R(x) dx.$$

- (b) Show that $n+1$ weights $W_0(n), \dots, W_n(n)$ exist (independent of f) such that

$$\int_{-1}^1 f(x) dx = \sum_{k=0}^n W_k(n) f(x_k).$$

This gives an integration formula with $n+1$ interpolation points that is exact for all polynomials of degree $\leq 2n+1$.

(c) Take $n = 2$ and show that the formula in part (b) becomes

$$\int_{-1}^1 f(x) dx = \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right).$$

This is exact for all polynomials of degree ≤ 5 .

(d) Introduce a suitable linear transformation and rewrite the formula in part (c) for an arbitrary interval $[a, b]$.

8. This exercise describes a method of Peano for deriving the error formula in Simpson's rule.

(a) Use integration by parts repeatedly to deduce the relation

$$\int u(t)v'''(t) dt = u(t)v''(t) - u'(t)v'(t) + u''(t)v(t) - \int g(t) dt,$$

where $g(t) = u'''(t)v(t)$.

(b) Assume ϕ has a continuous fourth derivative in the interval $[-1, 1]$. Take

$$v(t) = t(1-t)^2/6, \quad u(t) = \phi(t) + \phi(-t),$$

and use part (a) to show that

$$\int_{-1}^1 \phi(t) dt = \frac{1}{3} [\phi(-1) + 4\phi(0) + \phi(1)] - \int_0^1 g(t) dt.$$

Then show that $\int_0^1 g(t) dt = \phi^{(4)}(c)/90$ for some c in $[-1, 1]$.

(c) Introduce a suitable linear transformation to deduce Theorem 8-17 from the result of part (b).

9. (a) Let $a_1, a_2, \dots, a_n$ be nonnegative numbers whose sum is 1. Assume ϕ is continuous on an interval $[a, b]$. If $c_1, c_2, \dots, c_n$ are any n points in $[a, b]$ (not necessarily distinct), prove that there is at least one point c in $[a, b]$ such that

$$\sum_{k=1}^n a_k \phi(c_k) = \phi(c).$$

[Hint. Let M and m denote the maximum and minimum of ϕ on $[a, b]$ and use the inequality $m \leq \phi(c_k) \leq M$.]

(b) Use part (a) and Theorem 8-17 to derive the extended form of Simpson's rule given in Theorem 8-18.

10. Compute $\log 2$ from the formula $\log 2 = \int_1^2 x^{-1} dx$ by using the extension of Simpson's rule with (a) $n = 2$; (b) $n = 5$. Give upper and lower bounds for the error in each case.

11. (a) Let $\phi(t)$ be a polynomial in t of degree ≤ 3 . Express $\phi(t)$ by Newton's interpolation formula and integrate to deduce the formula

$$\int_0^3 \phi(t) dt = \frac{3}{8} [\phi(0) + 3\phi(1) + 3\phi(2) + \phi(3)].$$

(b) Let P be the interpolation polynomial of degree ≤ 3 that agrees with f at the points $a, a+h, a+2h, a+3h$, where $h > 0$. Use part (a) to prove that

$$\int_a^{a+3h} P(x) dx = \frac{3h}{8} [f(a) + 3f(a+h) + 3f(a+2h) + f(a+3h)].$$

(c) Assume f has a continuous fourth derivative in $[a, b]$, and let $h = (b-a)/3$. Prove that

$$\int_a^b f(x) dx = \frac{b-a}{8} [f(a) + 3f(a+h) + 3f(a+2h) + f(b)] - \frac{(b-a)^5}{6480} f^{(4)}(c)$$

for some c in $[a, b]$. This approximate integration formula is called *Cotes' rule*.

(d) Use Cotes' rule to compute $\log 2 = \int_1^2 x^{-1} dx$ and give upper and lower bounds for the error.

12. (a) Use the vector equation $\vec{r}(t) = a \sin t \vec{i} + b \cos t \vec{j}$, where $0 < b < a$, to show that the circumference L of an ellipse is given by the integral

$$L = 4a \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 t} dt,$$

where $k = \sqrt{a^2 - b^2}/a$.

(b) Show that Simpson's rule gives the formula

$$L = \frac{\pi}{3} [a + b + \sqrt{8(a^2 + b^2)}] - \frac{a\pi^5}{23040} f^{(4)}(c)$$

for some c in $[0, \pi/2]$, where $f(t) = \sqrt{1 - k^2 \sin^2 t}$.

8.23 The Euler summation formula

The trapezoidal formula, applied to the interval $[0, n]$, where n is a positive integer, becomes

$$(8.79) \quad \int_0^n f(x) dx = \sum_{k=0}^{n-1} f(k) + \frac{1}{2} (f(n) - f(0)) - \frac{f''(c)n}{12}$$

for some c in $[0, n]$. By the mean-value theorem of differential calculus we have $f'(n) - f'(0) = nf''(b)$ for some b in $(0, n)$. Therefore the error term in (8.79) can be written as

$$-\frac{f''(c)n}{12} = -\frac{1}{12} (f'(n) - f'(0)) + \frac{n}{12} (f''(b) - f''(c)).$$

When f is a quadratic polynomial, f'' is constant, so $f''(b) = f''(c)$ and the error is exactly $-(f'(n) - f'(0))/12$. Therefore the formula

$$(8.80) \quad \int_0^n f(x) dx = \sum_{k=0}^{n-1} f(k) + \frac{1}{2} (f(n) - f(0)) - \frac{1}{12} (f'(n) - f'(0))$$

is exact when f is any polynomial of degree ≤ 2 . By substituting $f(x) = x^3$ and using the fact that

$$\sum_{k=0}^{n-1} k^3 = \frac{n^4}{4} - \frac{n^3}{2} + \frac{n^2}{4},$$

we see that (8.80) also holds for any polynomial of degree ≤ 3 . Euler discovered that if the term $[f'''(n) - f'''(0)]/720$ is added to the right of (8.80) the formula becomes exact for every polynomial of degree ≤ 5 . More generally, he found that for every integer p there is a corresponding set of correction terms, depending only on the differences of derivatives of odd order at the end points 0 and n , that make the formula exact for all polynomials of degree $\leq p$. This result is contained in the following theorem.

8-19 THEOREM. *Euler's summation formula for polynomials.* If f is a polynomial of degree $\leq p$ we have

$$(8.81) \quad \int_0^n f(x) dx = \sum_{k=0}^{n-1} f(k) - \sum_{k=1}^p \frac{B_k}{k!} (f^{(k-1)}(n) - f^{(k-1)}(0)),$$

where the numbers $B_1, B_2, B_3, \dots$, are rational numbers defined recursively as follows:

$$(8.82) \quad B_0 = 1, \quad \sum_{k=0}^{m-1} \binom{m}{k} B_k = 0 \quad \text{for } m \geq 2.$$

Note. The numbers $B_0, B_1, B_2, \dots$, are called *Bernoulli numbers*. They may be easily calculated from the recursion formula (8.82) that defines them. For example, taking $m = 2, 3, 4, 5$, we find

$$2B_1 + B_0 = 0, \quad B_1 = -\frac{1}{2},$$

$$3B_2 + 3B_1 + B_0 = 0, \quad B_2 = \frac{1}{6},$$

$$4B_3 + 6B_2 + 4B_1 + B_0 = 0, \quad B_3 = 0,$$

$$5B_4 + 10B_3 + 10B_2 + 5B_1 + B_0 = 0, \quad B_4 = -\frac{1}{30}.$$

The next six Bernoulli numbers are

$$B_5 = 0, \quad B_6 = \frac{1}{42}, \quad B_7 = 0, \quad B_8 = -\frac{1}{30}, \quad B_9 = 0, \quad B_{10} = \frac{5}{66}.$$

It can be shown that $B_{2n+1} = 0$ for $n \geq 1$. (See Exercise 2 in the next section.) Therefore the formula in (8.81) contains only derivatives of odd order.

Proof. We use a method of proof that can be modified to give an extension of the theorem applicable to functions more general than polynomials. We begin with Taylor's formula,

$$f(x+h) = f(x) + \sum_{k=1}^p \frac{f^{(k)}(x)}{k!} h^k.$$

If we take $h = 1$ and let $\Delta f(x) = f(x+1) - f(x)$, Taylor's formula becomes

$$(8.83) \quad \Delta f(x) = \sum_{k=1}^p \frac{f^{(k)}(x)}{k!}.$$

This holds for any polynomial of degree p . If we replace p by $p-r$ and replace f by its r th derivative $f^{(r)}$, Equation (8.83) gives us†

$$(8.84) \quad \Delta f^{(r)}(x) = \sum_{k=1}^{p-r} \frac{f^{(r+k)}(x)}{k!}.$$

† If f is not a polynomial but has a derivative of order $p+1$, Equation (8.83) holds with an appropriate error term $R_p(f)$ appended on the right. In Equation (8.84) the error term would be $R_{p-r}(f^{(r)})$.

Now we apply Taylor's formula (8.83) once more with p replaced by $p + 1$ and f replaced by g , where

$$g(x) = \int_0^x f(t) dt.$$

Since $\Delta g(x) = \int_x^{x+1} f(t) dt$ and $g^{(k)}(x) = f^{(k-1)}(x)$, Equation (8.83) gives us

$$\int_x^{x+1} f(t) dt = \sum_{k=1}^{p+1} \frac{f^{(k-1)}(x)}{k!},$$

or

$$(8.85) \quad \int_x^{x+1} f(t) dt - f(x) = \sum_{m=2}^{p+1} \frac{f^{(m-1)}(x)}{m!}.$$

Now let $c_0 = 1$ and let $c_1, c_2, \dots, c_p$ be p constants to be specified later. Multiply both sides of (8.84) by c_{r+1} and sum for $r = 0, 1, 2, \dots, p-1$ to obtain

$$(8.86) \quad \sum_{r=0}^{p-1} c_{r+1} \Delta f^{(r)}(x) = \sum_{r=0}^{p-1} c_{r+1} \sum_{k=1}^{p-r} \frac{f^{(r+k)}(x)}{k!}.$$

The sum on the right can be written as

$$\sum_{r=1}^p \sum_{k=1}^{p+1-r} \frac{c_r f^{(r-1+k)}(x)}{k!} = \sum_{r=1}^p \sum_{m=r+1}^{p+1} \frac{c_r f^{(m-1)}(x)}{(m-r)!},$$

where in the last step we replaced the index of summation k by $m = r + k$. The last sum is extended over all the lattice points in the triangular region shown in Figure 8.5. Therefore if we interchange the order of summation we find

$$\sum_{r=1}^p \sum_{m=r+1}^{p+1} \frac{c_r f^{(m-1)}(x)}{(m-r)!} = \sum_{m=2}^{p+1} \sum_{r=1}^{m-1} \frac{c_r f^{(m-1)}(x)}{(m-r)!} = \sum_{m=2}^{p+1} f^{(m-1)}(x) \sum_{r=1}^{m-1} \frac{c_r}{(m-r)!}.$$

Equation (8.86) now becomes

$$\sum_{r=0}^{p-1} c_{r+1} \Delta f^{(r)}(x) = \sum_{m=2}^{p+1} f^{(m-1)}(x) \sum_{r=1}^{m-1} \frac{c_r}{(m-r)!}.$$

If we add this to Equation (8.85) and remember that $c_0 = 1$, we obtain

$$(8.87) \quad \int_x^{x+1} f(t) dt - f(x) + \sum_{r=0}^{p-1} c_{r+1} \Delta f^{(r)}(x) = \sum_{m=2}^{p+1} f^{(m-1)}(x) \sum_{r=0}^{m-1} \frac{c_r}{(m-r)!}.$$

The sum on the right will vanish identically if we can choose the numbers $c_0, c_1, c_2, \dots$, so that

$$(8.88) \quad \sum_{r=0}^{m-1} \frac{c_r}{(m-r)!} = 0 \quad \text{for each } m \geq 2.$$

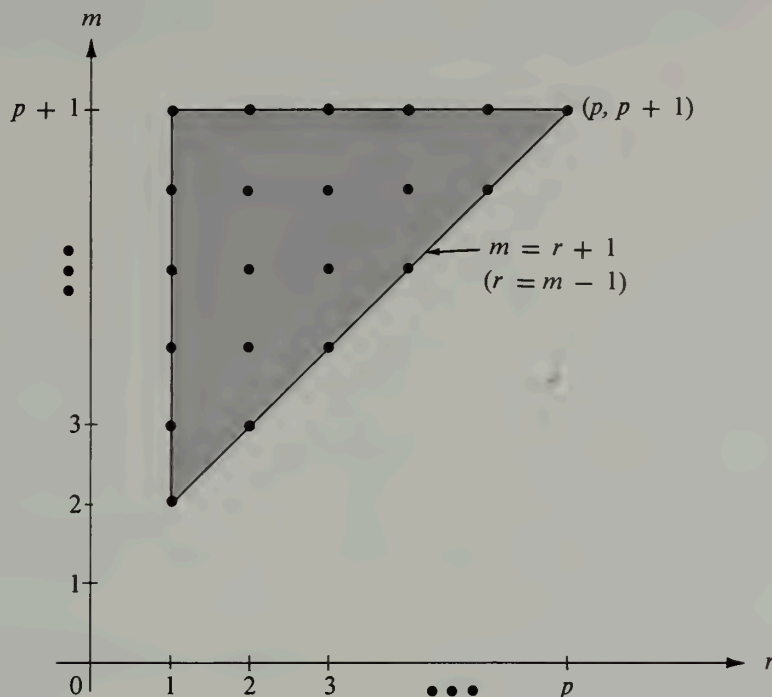


FIGURE 8.5 A lattice-point diagram illustrating the relation

$$\sum_{r=1}^p \sum_{m=r+1}^{p+1} = \sum_{m=2}^{p+1} \sum_{r=1}^{m-1}.$$

We have already chosen $c_0 = 1$. If we choose $c_r = B_r/r!$, where B_r is the r th Bernoulli number, we have

$$\frac{c_r}{(m-r)!} = \frac{B_r}{r! (m-r)!} = \frac{1}{m!} \binom{m}{r} B_r,$$

and Equation (8.88) is equivalent to

$$\sum_{r=0}^{m-1} \binom{m}{r} B_r = 0 \quad \text{for } m \geq 2.$$

But this is the defining equation for the Bernoulli numbers. Hence the right member of (8.87) vanishes and we obtain

$$(8.89) \quad \int_x^{x+1} f(t) dt = f(x) - \sum_{r=0}^{p-1} \frac{B_{r+1}}{(r+1)!} \Delta f^{(r)}(x).$$

Taking $x = 0, 1, 2, \dots, n-1$ and adding, we obtain Equation (8.81).

The foregoing argument can be applied to any function having a continuous derivative of order $p + 1$ in $[0, n]$. In place of (8.83) we use Taylor's formula with remainder,

$$\Delta f(x) = \sum_{k=1}^p \frac{f^{(k)}(x)}{k!} + R_p(f),$$

taking the integral form of the remainder (see Exercise 8, Section 7.8, in Volume I),

$$R_p(f) = \frac{1}{p!} \int_x^{x+1} (x+1-t)^p f^{(p+1)}(t) dt = \frac{1}{p!} \int_0^1 t^p f^{(p+1)}(x+1-t) dt.$$

Corresponding error terms must be added in each of Equations (8.84) and (8.85). The foregoing analysis leads to the following extension of Theorem 8-19.

8-20 THEOREM. *Euler's summation formula with error term.* Assume f has a continuous derivative of order $p + 1$ in $[0, n]$. Then we have

$$\begin{aligned} \int_0^n f(x) dx &= \sum_{k=0}^{n-1} f(k) - \sum_{k=1}^p \frac{B_k}{k!} (f^{(k-1)}(n) - f^{(k-1)}(0)) \\ &\quad + \int_0^1 \frac{B_{p+1}(t) - B_{p+1}}{(p+1)!} \sum_{r=1}^n f^{(p+1)}(r-t) dt, \end{aligned}$$

where $B_m(t)$ is the m th Bernoulli polynomial, defined by the equation

$$B_m(t) = \sum_{r=0}^m \binom{m}{r} B_r t^{m-r}.$$

Proof. The argument given in the proof of Theorem 8-19 leads once more to Equation (8.89), except that on the right we must add a sum of error terms, given by

$$\begin{aligned} \sum_{r=0}^{p-1} c_{r+1} R_{p-r}(f^{(r)}) + c_0 R_{p+1}(g) &= \sum_{r=0}^p \frac{c_r}{(p+1-r)!} \int_0^1 t^{p+1-r} f^{(p+1)}(x+1-t) dt \\ &= \int_0^1 \frac{1}{(p+1)!} \sum_{r=0}^p \binom{p+1}{r} B_r t^{p+1-r} f^{(p+1)}(x+1-t) dt. \end{aligned}$$

Since

$$\sum_{r=0}^p \binom{p+1}{r} B_r t^{p+1-r} = B_{p+1}(t) - B_{p+1},$$

Equation (8.89) becomes

$$\int_x^{x+1} f(t) dt = f(x) - \sum_{r=0}^{p-1} \frac{B_{r+1}}{(r+1)!} \Delta f^{(r)}(x) + \int_0^1 \frac{B_{p+1}(t) - B_{p+1}}{(p+1)!} f^{(p+1)}(x+1-t) dt.$$

Summing for $x = 0, 1, \dots, n-1$ we obtain Theorem 8-20.

The error term in Theorem 8-20 is not easy to estimate. If p is odd and if $f^{(p+3)}$ is

continuous and has the same sign as $f^{(p+1)}$ on $[0, n]$, it can be shown that the error term can be written as

$$- \theta \frac{B_{p+1}}{(p+1)!} (f^{(p)}(n) - f^{(p)}(0)),$$

where $0 \leq \theta \leq 1$. In other words, under the conditions stated, the absolute value of the error term does not exceed that of the next "correction term" which would normally be added if Theorem 8-20 were used with p replaced by $p+1$.

Although Euler's formula can be used to approximate integrals by sums, it is more often used to evaluate or estimate sums in terms of integrals. This is why it is usually referred to as a "summation" formula rather than an integration formula. For example, if we take $f(x) = x^4$ in Equation (8.81) we obtain

$$\sum_{k=0}^{n-1} k^4 = \int_0^n x^4 dx + B_1 n^4 + \frac{B_2}{2!} 4n^3 + \frac{B_3}{3!} 12n^2 + \frac{B_4}{4!} 24n = \frac{n^5}{5} - \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30}.$$

Euler's summation formula can be expressed in an alternate form that is exact for functions with a continuous first derivative. Because of its importance we discuss this case in detail.

8-21 THEOREM. Assume f has a continuous derivative on $[0, n]$. Then we have

$$(8.90) \quad \sum_{k=0}^n f(k) = \int_0^n f(x) dx + \int_0^n \left(x - [x] - \frac{1}{2}\right) f'(x) dx + \frac{f(0) + f(n)}{2},$$

where $[x]$ denotes the greatest integer $\leq x$.

Proof. Integration by parts gives us

$$(8.91) \quad \int_0^n \left(x - \frac{1}{2}\right) f'(x) dx = (n - \frac{1}{2})f(n) + \frac{1}{2}f(0) - \int_0^n f(x) dx.$$

Now we consider the integral $\int_0^n [x] f'(x) dx$ and write it as a sum of integrals in each of which $[x]$ has a fixed value. Thus, we have

$$\begin{aligned} \int_0^n [x] f'(x) dx &= \sum_{r=0}^{n-1} \int_r^{r+1} [x] f'(x) dx = \sum_{r=0}^{n-1} r \int_r^{r+1} f'(x) dx \\ &= \sum_{r=0}^{n-1} r(f(r+1) - f(r)) = \sum_{r=0}^{n-1} rf(r+1) - \sum_{r=0}^{n-1} rf(r) \\ &= - \sum_{r=0}^{n-1} f(r+1) + \sum_{r=0}^{n-1} (r+1)f(r+1) - \sum_{r=0}^{n-1} rf(r) \\ &= - \sum_{k=1}^n f(k) + nf(n) = - \sum_{k=0}^n f(k) + f(0) + nf(n). \end{aligned}$$

Subtracting this from Equation (8.91) we obtain

$$\int_0^n \left(x - [x] - \frac{1}{2}\right) f'(x) dx = \sum_{k=0}^n f(k) - \frac{f(0) + f(n)}{2} - \int_0^n f(x) dx,$$

which is equivalent to (8.90).

We now show how Euler's summation formula can be used to derive Stirling's formula for estimating $n!$.

8-22 THEOREM. *Stirling's formula.* If n is a positive integer we have

$$\sqrt{2\pi} n^{n+1/2} e^{-n} < n! < \sqrt{2\pi} n^{n+1/2} e^{-n} \left(1 + \frac{1}{4n}\right).$$

Proof. The proof will make use of some simple properties of functions ϕ_1 and ϕ_2 , defined as follows:

$$\phi_1(x) = \begin{cases} x - [x] - \frac{1}{2} & \text{if } x \text{ is not an integer,} \\ 0 & \text{if } x \text{ is an integer,} \end{cases} \quad \phi_2(x) = \int_0^x \phi_1(t) dt.$$

First we note that $\phi_1(x+1) = \phi_1(x)$, which means that ϕ_1 is periodic with period 1. Also, if $0 < x < 1$ we have $\phi_1(x) = x - \frac{1}{2}$, so $\int_0^1 \phi_1(t) dt = 0$. This implies that ϕ_2 is also periodic with period 1, since

$$\begin{aligned} \phi_2(x+1) &= \int_0^{x+1} \phi_1(t) dt = \int_0^1 \phi_1(t) dt + \int_1^{x+1} \phi_1(t) dt \\ &= \int_1^{x+1} \phi_1(t) dt = \int_1^{x+1} \phi_1(t-1) dt = \int_0^x \phi_1(u) du = \phi_2(x). \end{aligned}$$

Now $\phi_2(0) = 0$ so, by periodicity, $\phi_2(n) = 0$ for every integer n . Also, when $0 \leq x \leq 1$ we have $\phi_2(x) = \int_0^x (t - \frac{1}{2}) dt = x(x - 1)/2$, which shows that $-\frac{1}{8} \leq \phi_2(x) \leq 0$ for all x . The graphs of ϕ_1 and ϕ_2 are shown in Figure 8.6. Note that the strict inequalities $-\frac{1}{8} < \phi_2(x) < 0$ hold except when x is an integer or half an integer.

The error term $\int_0^n (x - [x] - \frac{1}{2})f'(x) dx$ in Euler's summation formula can now be written as $\int_0^n \phi_1(x) f'(x) dx$. Applying the formula with $f(x) = \log(x+1)$ and n replaced by $n-1$ we obtain

$$\sum_{k=0}^{n-1} \log(k+1) = \int_0^{n-1} \log(x+1) dx + \int_0^{n-1} \frac{\phi_1(x)}{x+1} dx + \frac{1}{2} \log n.$$

Since $\int \log t dt = t \log t - t$ the foregoing equation becomes

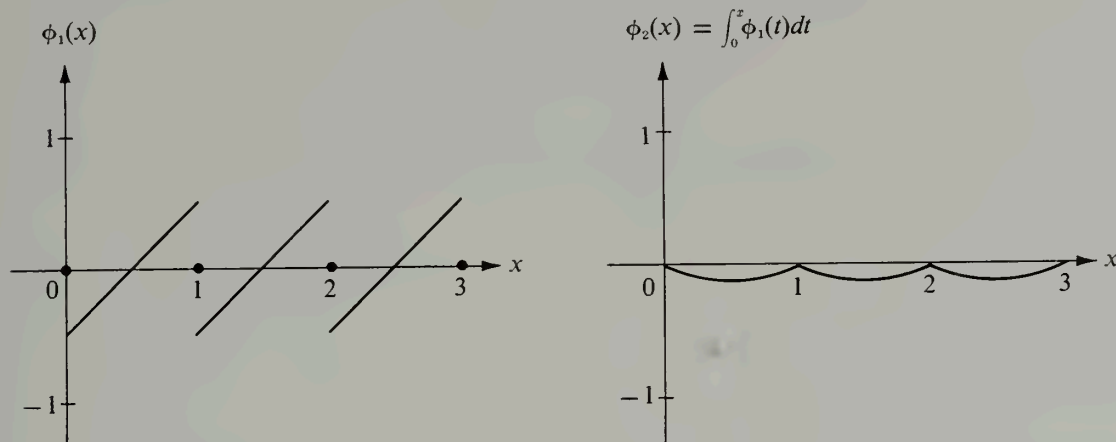
$$\log n! = n \log n - n + 1 + \int_1^n \frac{\phi_1(t-1)}{t} dt + \frac{1}{2} \log n,$$

or

$$(8.92) \quad \log n! = \left(n + \frac{1}{2}\right) \log n - n + 1 + \int_1^n \frac{\phi_1(t)}{t} dt.$$

Integrating by parts we obtain

$$\int_1^n \frac{\phi_1(t)}{t} dt = \int_1^n \frac{\phi_2'(t)}{t} dt = \left. \frac{\phi_2(t)}{t} \right|_1^n + \int_1^n \frac{\phi_2(t)}{t^2} dt = \int_1^n \frac{\phi_2(t)}{t^2} dt,$$


 FIGURE 8.6 Graphs of the periodic functions ϕ_1 and ϕ_2 .

since $\phi_2(n) = \phi_2(1) = 0$. Since $|\phi_2(t)| \leq \frac{1}{8}$ the improper integral $\int_1^\infty t^{-2}\phi_2(t) dt$ converges absolutely and we may write

$$\int_1^n \frac{\phi_2(t)}{t^2} dt = \int_1^\infty \frac{\phi_2(t)}{t^2} dt - \int_n^\infty \frac{\phi_2(t)}{t^2} dt.$$

Therefore Equation (8.92) becomes

$$(8.93) \quad \log n! = \left(n + \frac{1}{2}\right) \log n - n + C - \int_n^\infty \frac{\phi_2(t)}{t^2} dt,$$

where $C = 1 + \int_1^\infty t^{-2}\phi_2(t) dt$. Since we have $-\frac{1}{8} < \phi_2(t) < 0$ except when t is an integer or half an integer, we obtain

$$0 < - \int_n^\infty \frac{\phi_2(t)}{t^2} dt < \frac{1}{8n},$$

and Equation (8.93) gives us the inequalities

$$\exp\left((n + \frac{1}{2}) \log n - n + C\right) < n! < \exp\left((n + \frac{1}{2}) \log n - n + C + \frac{1}{8n}\right),$$

where $\exp(t) = e^t$. Using the relation $e^x < 1 + 2x$, with $x = 1/(8n)$, we may rewrite these inequalities as

$$(8.94) \quad A n^{n+1/2} e^{-n} < n! < A n^{n+1/2} e^{-n} \left(1 + \frac{1}{4n}\right).$$

where $A = e^C$. To complete the proof we need to show that $A = \sqrt{2\pi}$.

We shall deduce this from the inequality

$$(8.95) \quad \pi n < \left(\frac{2^{2n}(n!)^2}{(2n)!} \right)^2 < \frac{\pi(2n+1)}{2},$$

discovered by John Wallis (1616–1703). First we show how Wallis' inequality implies $A = \sqrt{2\pi}$; then we discuss the proof of (8.95).

If we let

$$A_n = \frac{n!}{n^{n+1/2} e^{-n}}$$

the inequality in (8.94) becomes

$$A < A_n < A \left(1 + \frac{1}{4n} \right).$$

This shows that $A_n \rightarrow A$ as $n \rightarrow \infty$. In (8.95) we write $n! = n^{n+1/2} e^{-n} A_n$ to obtain

$$\pi n < \left(\frac{2^{2n} n^{2n+1} e^{-2n} A_n^2}{(2n)^{2n+1/2} e^{-2n} A_{2n}} \right)^2 < \frac{\pi(2n+1)}{2},$$

which is equivalent to

$$\pi < \frac{A_n^4}{2A_{2n}^2} < \pi \frac{2n+1}{2n}.$$

Letting $n \rightarrow \infty$ in this last inequality we obtain

$$\pi \leq \frac{A^4}{2A^2} \leq \pi.$$

This shows that $A^2 = 2\pi$. Since $A > 0$ we have $A = \sqrt{2\pi}$, as asserted.

It remains to prove Wallis' inequality (8.95). For this purpose we introduce the numbers

$$I_n = \int_0^{\pi/2} \sin^n t \, dt,$$

where n is any nonnegative integer. We note that $I_0 = \pi/2$ and $I_1 = 1$. For $0 \leq t \leq \pi/2$ we have $0 \leq \sin t \leq 1$; hence $0 \leq \sin^{n+1} t \leq \sin^n t$. This shows that the sequence $\{I_n\}$ is monotonic decreasing. Therefore we may write

$$(8.96) \quad \frac{1}{I_{2n} I_{2n-1}} \leq \frac{1}{I_{2n}^2} \leq \frac{1}{I_{2n} I_{2n+1}}.$$

Now we shall evaluate each member of this inequality; this will lead at once to Wallis' inequality.

Integration of the identity

$$\frac{d}{dt} (\cos t \sin^{n+1} t) = (n+1) \sin^n t - (n+2) \sin^{n+2} t$$

over the interval $[0, \pi/2]$ gives us

$$0 = (n+1)I_n - (n+2)I_{n+2},$$

or

$$(8.97) \quad I_{n+2} = \frac{n+1}{n+2} I_n.$$

Using this recursion formula with n replaced by $2k-2$ we find

$$\frac{I_{2k}}{I_{2k-2}} = \frac{2k-1}{2k} = \frac{2k(2k-1)}{(2k)^2}.$$

Multiplying these equations for $k = 1, 2, \dots, n$ we find

$$\prod_{k=1}^n \frac{I_{2k}}{I_{2k-2}} = \prod_{k=1}^n \frac{2k(2k-1)}{(2k)^2} = \frac{(2n)!}{2^{2n} (n!)^2}.$$

The product on the left telescopes to I_{2n}/I_0 . Since $I_0 = \pi/2$ we obtain

$$(8.98) \quad I_{2n} = \frac{(2n)!}{2^{2n} (n!)^2} \cdot \frac{\pi}{2}.$$

In a similar way we apply the recursion formula (8.97) with n replaced by $2k-1$ and multiply the resulting equations for $k = 1, 2, \dots, n$ to obtain

$$\prod_{k=1}^n \frac{I_{2k+1}}{I_{2k-1}} = \prod_{k=1}^n \frac{(2k)^2}{2k(2k+1)} = \frac{2^{2n} (n!)^2}{(2n+1)!} = \frac{1}{2n+1} \cdot \frac{\pi}{2} \cdot \frac{1}{I_{2n}}.$$

The product on the left telescopes to $I_{2n+1}/I_1 = I_{2n+1}$, so we get

$$(8.99) \quad I_{2n} I_{2n+1} = \frac{\pi}{2(2n+1)}.$$

Since $I_{2n+1} = 2n I_{2n-1}/(2n+1)$, Equation (8.99) implies

$$I_{2n} I_{2n-1} = \frac{\pi}{4n}.$$

We use this in (8.96), together with the two relations (8.98) and (8.99). Then we multiply by $\pi^2/4$ to obtain Wallis' inequality (8.95).

8.24 Exercises

1. In Theorem 8-20 we defined the p th Bernoulli polynomial by the equation

$$B_p(t) = \sum_{r=0}^p \binom{p}{r} B_r t^{p-r},$$

where B_r is the r th Bernoulli number. The first few Bernoulli polynomials are

$$B_0(t) = 1, \quad B_1(t) = t - \frac{1}{2}, \quad B_2(t) = t^2 - t + \frac{1}{6}, \quad B_3(t) = t^3 - \frac{3}{2}t^2 + \frac{1}{2}t.$$

Prove the following properties of Bernoulli polynomials:

- (a) $B_p(1) = B_p(0) = B_p$ if $p \geq 2$.
- (b) $B'_p(t) = p B_{p-1}(t)$ if $p \geq 1$.

$$(c) \int_a^b B_p(t) dt = \frac{B_{p+1}(b) - B_{p+1}(a)}{p+1} \quad \text{if } p \geq 0.$$

$$(d) \int_0^1 B_p(t) dt = 0 \quad \text{if } p \geq 1.$$

$$(e) B_p^{(k)}(x) = k! \binom{p}{k} B_{p-k}(x) \quad \text{if } p \geq 1, \quad 0 \leq k \leq p.$$

2. (a) Use Euler's summation formula for polynomials to prove that

$$\sum_{k=0}^{n-1} k^p = \frac{B_{p+1}(n) - B_{p+1}}{p+1} \quad \text{for } p \geq 1.$$

(b) Use part (a) to prove that for every integer $n \geq 2$ we have

$$B_p(n+1) - B_p(n) = p n^{p-1} \quad \text{if } p \geq 0.$$

(c) Use part (b) to show that for every real x we have

$$B_p(x+1) - B_p(x) = p x^{p-1} \quad \text{if } p \geq 1.$$

[Hint. A polynomial with infinitely many zeros is identically zero.]

(d) Let $\Delta f(x) = f(x+1) - f(x)$. Use part (c) to show that

$$\Delta B_{2k}(1-x) = \Delta B_{2k}(x)$$

for every real x and every integer $k \geq 1$. Sum both sides for $x = 0, 1, 2, \dots, n$ and deduce that

$$B_{2k}(1-n) = B_{2k}(n)$$

for every integer $n \geq 0$.

(e) From the result of part (d) deduce that $B_{2k}(1-x) = B_{2k}(x)$ for every real x . Then use this to deduce that

$$B_p(1-x) = (-1)^p B_p(x) \quad \text{if } p \geq 1.$$

(f) Use part (e) to show that $B_{2n+1} = 0$ for $n \geq 1$.

3. Use Taylor's formula to show that

$$B_p(x+h) = \sum_{r=0}^p \binom{p}{r} B_r(x) h^{p-r} \quad \text{if } p \geq 0.$$

4. Prove Raabe's multiplication theorem:

$$B_p(mx) = m^{p-1} \sum_{r=0}^{m-1} B_p\left(x + \frac{r}{m}\right).$$

5. Evaluate the integral $\int_0^1 x B_p(x) dx$ in two ways and thereby prove that

$$\sum_{r=0}^p \binom{p}{r} \frac{B_r}{p+2-r} = \frac{B_{p+1}}{p+1}.$$

6. If we use the binomial theorem to expand $(B+x)^p$ we obtain

$$(B+x)^p = \sum_{r=0}^p \binom{p}{r} B^r x^{p-r}.$$

If we replace each power B^r by the Bernoulli number B_r , the right member becomes $B_p(x)$. This means that the definition of the Bernoulli polynomial $B_p(x)$ can be written symbolically as

$$B_p(x) = (B + x)^p,$$

with the understanding that after expanding the right member we replace each power B^r by B_r . In the same way, the recursion formula that we used to define Bernoulli numbers can be written symbolically as

$$(B + 1)^m - B^m = 0.$$

Express each of the following symbolic formulas as a relation involving Bernoulli polynomials and then prove that the formula is correct.

(a) $f(B + x + 1) - f(B + x) = f'(x)$, where f is a polynomial.

(b) If f is a polynomial and if $F'(x) = f(x)$, then

$$\sum_{k=1}^n f(k) = F(B + n + 1) - F(B + 1).$$

7. Euler's constant C is defined by the limit formula

$$C = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right).$$

(See Section 9.6 in Volume I.) Use Euler's summation formula to prove that

$$\sum_{k=1}^n \frac{1}{k} = \log n + C + \frac{1}{2n} - \frac{E(n)}{n^2},$$

where $0 \leq E(n) \leq \frac{1}{8}$. Also, show that

$$C = 1 - \int_1^{\infty} \frac{t - [t]}{t^2} dt.$$

8. (a) If $s > 0$, $s \neq 1$, use Euler's summation formula to prove that

$$\sum_{k=1}^n \frac{1}{k^s} = \frac{n^{1-s}}{1-s} + C(s) + s \int_n^{\infty} \frac{t - [t]}{t^{s+1}} dt,$$

where

$$C(s) = 1 + \frac{1}{s-1} - s \int_1^{\infty} \frac{t - [t]}{t^{s+1}} dt.$$

(b) If $s > 1$, show that $C(s) = \zeta(s)$, where ζ is the Riemann zeta function defined for $s > 1$ by the series

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}.$$

The series for $\zeta(s)$ diverges for $s \leq 1$. However, since the formula for $C(s)$ in part (a) is meaningful for $0 < s < 1$, it may be used to extend the definition of $\zeta(s)$ to the open interval $0 < s < 1$. Thus, for $s > 0$ and $s \neq 1$ we have the formula

$$\zeta(s) = 1 + \frac{1}{s-1} - s \int_1^{\infty} \frac{t - [t]}{t^{s+1}} dt.$$

9. Use Euler's summation formula to derive the following relations, where A and B are constants.

$$(a) \sum_{k=1}^n \frac{\log k}{k} = \frac{(\log n)^2}{2} + A + O\left(\frac{\log n}{n}\right) \quad \text{for } n > 1.$$

$$(b) \sum_{k=2}^n \frac{1}{k \log k} = \log(\log n) + B + O\left(\frac{1}{n \log n}\right) \quad \text{for } n > 2.$$

10. Deduce the following limit relations with the aid of Stirling's formula and/or Wallis' inequality.

$$(a) \lim_{n \rightarrow \infty} \frac{n}{(n!)^{1/n}} = e.$$

$$(b) \lim_{n \rightarrow \infty} \frac{(n!)^{2/2n}}{(2n)! \sqrt{n}} = \sqrt{\pi}.$$

$$(c) \lim_{n \rightarrow \infty} (-1)^n \binom{-\frac{1}{2}}{n} = \frac{1}{\sqrt{\pi}}.$$

SUGGESTED REFERENCES

This small list contains only a few books suggested for further reading on the general principles of numerical analysis. All these books contain further references to works of a more special nature. The list of tables given in Todd's survey (reference 7 below) is especially recommended.

Books

1. A. D. Booth, *Numerical Methods*, Academic Press, New York, 1958.
2. D. R. Hartree, *Numerical Analysis*, Oxford Univ. Press (Clarendon), London and New York, 1958.
3. F. B. Hildebrand, *Introduction to Numerical Analysis*, McGraw-Hill, New York, 1956.
4. A. S. Householder, *Principles of Numerical Analysis*, McGraw-Hill, New York, 1953.
5. W. E. Milne, *Numerical Calculus*, Princeton Univ. Press, Princeton, N. J. 1950.
6. J. B. Scarborough, *Numerical Mathematical Analysis*, Johns Hopkins Press, Baltimore, Maryland, 1958.
7. J. Todd (ed.), *Survey of Numerical Analysis*, McGraw-Hill, New York, 1962.

Tables

8. L. J. Comrie (ed.), *Chambers' Six-figure Mathematical Tables*, W. & R. Chambers, London and Edinburgh, 1949.
9. L. J. Comrie, "Interpolation and Allied Tables," 2nd rev. reprint from *Nautical Almanac* for 1937, H. M. Stationery Office, London, 1948.
10. A. J. Fletcher, J. C. P. Miller, and L. Rosenhead, *Index of Mathematical Tables*, McGraw-Hill, New York, 1946.
11. *Tables of Lagrangian Interpolation Coefficients*, Natl. Bur. Standards Columbia Press Series, Vol. 4., Columbia Univ. Press, New York, 1944.

EXISTENCE THEOREMS FOR DIFFERENTIAL EQUATIONS

9.1 Introduction

Although the study of differential equations began in the 17th century, it was not until the 19th century that mathematicians realized that relatively few differential equations could be solved by elementary means. The work of Cauchy, Liouville, and others showed the importance of establishing general theorems to guarantee the existence of solutions to certain specific classes of differential equations. Chapter 7 illustrated the use of both existence and uniqueness theorems in the study of linear differential equations. This chapter is concerned with the proofs of these theorems and related theorems.

To begin with, we consider first-order equations of the type

$$(9.1) \quad y' = f(x, y).$$

When the right-hand member, $f(x, y)$, has certain special forms the differential equation can be solved by elementary means. For example, if

$$f(x, y) = -P(x)y + Q(x),$$

where P and Q are continuous on some interval I , the differential equation in (9.1) is a first-order *linear* equation; its general solution is

$$y = e^{-\int P(x)dx} \left[\int Q(x) e^{\int P(x)dx} dx + C \right],$$

where C is an arbitrary constant. The differential equation can also be solved easily when $f(x, y)$ has the form $A(x)/B(y)$, where A and B are continuous functions. In this case the variables are separable and a solution can be found by elementary means. (See Section 4.6 in Volume I.) In Section 9.6 we shall prove that if the right-hand member $f(x, y)$ satisfies certain general conditions a solution of Equation (9.1) always exists, even though it may not be possible to express the solution in terms of the familiar functions of elementary calculus.

Existence theory for differential equations of higher order can be reduced to the first-order case by the introduction of *systems* of equations. For example, suppose we have a second-order equation of the form

$$(9.2) \quad y'' = f(x, y, y').$$

If we let $y_1 = y$ and $y_2 = y'$, this equation becomes $y_2' = f(x, y_1, y_2)$. The pair of equations

$$(9.3) \quad y_1' = y_2, \quad y_2' = f(x, y_1, y_2),$$

is an example of a system of two first-order equations for two unknown functions y_1 and y_2 . Every function y_1 that satisfies (9.2) gives rise to a pair of functions y_1, y_2 satisfying (9.3); conversely, if a pair of functions y_1, y_2 satisfies (9.3), the function y_1 is a solution of (9.2). In the same way, a third-order differential equation of the form

$$y''' = f(x, y, y', y'')$$

is equivalent to a system of three first-order equations for three unknown functions y_1, y_2, y_3 , namely,

$$y_1' = y_2, \quad y_2' = y_3, \quad y_3' = f(x, y_1, y_2, y_3).$$

More generally, any n th order equation of the form

$$y^{(n)} = f(x, y, y', y'', \dots, y^{(n-1)})$$

is equivalent to a system of n first-order equations given by

$$y_1' = y_2, \quad y_2' = y_3, \quad \dots, \quad y_{n-1}' = y_n, \quad y_n' = f(x, y_1, y_2, \dots, y_n).$$

Consider next a more general system of n first-order differential equations, say

$$(9.4) \quad y_1' = f_1(x, y_1, \dots, y_n), \quad y_2' = f_2(x, y_1, \dots, y_n), \dots, \\ y_n' = f_n(x, y_1, \dots, y_n).$$

A solution of such a system on an interval I is any n -tuple of functions $y_1, y_2, \dots, y_n$ that satisfy all the equations in (9.4) simultaneously on I . The study of such a system can be simplified considerably with the use of vector notation. We introduce a vector-valued function $\vec{y}$ defined on I by the equation

$$\vec{y}(x) = (y_1(x), y_2(x), \dots, y_n(x));$$

then we write $f_i(x, \vec{y})$ for $f_i(x, y_1, y_2, \dots, y_n)$. The system in (9.4) can now be written as

$$(9.5) \quad \vec{y}' = \vec{f}(x, \vec{y}),$$

where

$$\vec{y}' = (y_1', y_2', \dots, y_n') \quad \text{and} \quad \vec{f} = (f_1, f_2, \dots, f_n).$$

Equation (9.5) is called a first-order vector differential equation. We note that it has exactly the same form as Equation (9.1), except that it involves vector-valued functions. The existence and uniqueness theorems that we shall prove for Equation (9.1) have direct extensions for the first-order equation in (9.5). These extended theorems, in turn, give us existence and uniqueness theorems for systems and hence for n th-order equations of the form $y^{(n)} = f(x, y, y', \dots, y^{(n-1)})$. These include, in particular, the existence and uniqueness theorems for n th-order linear equations which we used in Chapter 7. Since the entire theory rests on a discussion of the first-order equation in (9.1) we turn now to this case.

*9.2 Uniqueness theorem for the differential equation $y' = f(x, y)$

In 1820 Cauchy proved the first existence and uniqueness theorems for the differential equation

$$(9.6) \quad y' = f(x, y).$$

His original proof was unnecessarily complicated and required severe restrictions on the right-hand member $f(x, y)$. For example, it required f to have partial derivatives of every order with respect to x and y . Rudolf Lipschitz (1832–1903), Guiseppe Peano (1858–1932), and others made significant improvements in Cauchy's work. For example, Peano showed in 1890 that if f is merely continuous on a closed rectangle R , then for each point (x_0, y_0) interior to R there is at least one integral curve of (9.6) passing through (x_0, y_0) . That is, there is at least one function $y = Y(x)$ with $Y(x_0) = y_0$ that satisfies (9.6) on some interval $(x_0 - c, x_0 + c)$. Moreover, he gave examples of equations having more than one integral curve passing through a given point. The equation

$$(9.7) \quad y' = 3y^{2/3}$$

has this property. Two distinct solutions passing through the origin are $Y_1(x) = 0$ and $Y_2(x) = x^3$. Since the right-hand member of (9.7) is continuous everywhere in the xy -plane, this example shows that some condition other than the continuity of f is needed to ensure the uniqueness of the solution through a given point. In 1898 W. F. Osgood (1864–1943) proved that if f is continuous in a neighborhood of (x_0, y_0) there may be a whole family of integral curves of (9.6) passing through (x_0, y_0) , with all these curves lying within a region bounded by the graphs of two extremal solutions Y_1 and Y_2 ; hence there is a unique solution through (x_0, y_0) if and only if these extremal solutions are identical. Osgood also proved that there is a unique solution through each interior point (x_0, y_0) of a rectangle R if f satisfies a condition of the form

$$(9.8) \quad |f(x, y) - f(x, z)| \leq \phi(|y - z|)$$

for every pair of points (x, y) and (x, z) in R with the same x -coordinate. Here ϕ is a function that is positive and continuous on the half-open interval $(0, a]$, where a denotes the altitude of the rectangle R . Also, as $t \rightarrow 0+$, $\phi(t)$ must approach 0 in such a way that the improper integral $\int_{0+}^a \phi(t)^{-1} dt$ diverges. That is, ϕ must satisfy the relation

$$\lim_{h \rightarrow 0+} \int_h^a \frac{1}{\phi(t)} dt = \infty.$$

For example, this condition is satisfied when $\phi(t) = t$ but not when $\phi(t) = t^{1/2}$. Further examples of permissible functions ϕ are $\phi(t) = Kt$ for $t > 0$, $\phi(t) = Kt|\log t|$ for $0 < t < 1$, $\phi(t) = Kt|\log|\log t||$ for $0 < t < 1/e$, and so on, where K is a positive constant. When $\phi(t) = Kt$, Osgood's condition (9.8) becomes

$$(9.9) \quad |f(x, y) - f(x, z)| \leq K|y - z|.$$

This is called a *Lipschitz condition* in honor of Rudolf Lipschitz, who first introduced it in 1876. The constant K is called a Lipschitz constant. A Lipschitz condition does not restrict a function very seriously. For example, if f has a bounded partial derivative D_2f on R , say $|D_2f(x, y)| \leq M$ for all (x, y) in R , it is easy to show that (9.9) is automati-

cally satisfied with the constant M as Lipschitz constant. In fact, from the mean-value theorem for derivatives we have

$$f(x, y) - f(x, z) = D_2 f(x, c)(y - z),$$

where c lies between y and z . From this we find that

$$|f(x, y) - f(x, z)| \leq M |y - z|,$$

which is a Lipschitz condition. A Lipschitz condition can be satisfied by a discontinuous function. For example, let $f(x, y) = y \sin(1/x)$ if $x \neq 0$, and let $f(0, y) = 0$. This function is discontinuous at each point of the y -axis (except at the origin), yet it satisfies a Lipschitz condition on every rectangle R , with Lipschitz constant $K = 1$.

In this section we show how a Lipschitz condition implies a uniqueness theorem for Equation (9.6). The modifications needed to prove Osgood's more general uniqueness theorem are outlined in Exercise 10 of the next section. In Section 9.6 we assume both a Lipschitz condition and continuity of f to deduce an existence theorem.

9-1 THEOREM. *Uniqueness theorem.* Assume f satisfies a Lipschitz condition

$$|f(x, y) - f(x, z)| \leq K |y - z|$$

on a rectangle R . Let (x_0, y_0) be any point interior to R . Then there is at most one function $y = Y(x)$ with $Y(x_0) = y_0$ that satisfies the differential equation

$$y' = f(x, y)$$

on an interval $I = (x_0 - c, x_0 + c)$, with the graph of Y over I lying in R .

Proof. Choose a point (x_0, y_0) interior to R , and let Y_1 and Y_2 be two functions that satisfy the differential equation $y' = f(x, y)$ on some interval $I = (x_0 - c, x_0 + c)$. Assume that the graphs of Y_1 and Y_2 over I lie in the rectangle R and that $Y_1(x_0) = Y_2(x_0) = y_0$. We shall assume that Y_1 and Y_2 do not agree everywhere on I and show that this leads to a contradiction.

If Y_1 and Y_2 do not agree everywhere on I there is at least one point x_1 in I at which $Y_1(x_1) \neq Y_2(x_1)$. Let us label the functions Y_1 and Y_2 so that $Y_2(x_1) < Y_1(x_1)$. Also, suppose that $x_0 < x_1$. (The argument is entirely analogous if $x_1 < x_0$.) If $x \in I$ let $g(x) = Y_1(x) - Y_2(x)$. At the point x_1 we have $g(x_1) > 0$. Now both Y_1 and Y_2 are solutions of the differential equation $y' = f(x, y)$, so we have

$$g'(x) = Y_1'(x) - Y_2'(x) = f[x, Y_1(x)] - f[x, Y_2(x)]$$

for each x in I . Since the points $(x, Y_1(x))$ and $(x, Y_2(x))$ are in R we may apply the Lipschitz condition to obtain the inequality

$$g'(x) \leq K |Y_1(x) - Y_2(x)| = K |g(x)|$$

for each x in I . From this we see that

$$(9.10) \quad g'(x) < (K + 1) |g(x)| \quad \text{whenever } g(x) \neq 0.$$

In particular, we have

$$(9.11) \quad g'(x_1) < (K + 1)g(x_1).$$

Now let h be the function defined on I by the equation

$$(9.12) \quad h(x) = g(x_1)e^{(K+1)(x-x_1)}.$$

We note that $h(x) > 0$ for each x in I because $g(x_1) > 0$. Since $h(x_1) = g(x_1)$ and $h'(x) = (K+1)h(x)$, we see that

$$(9.13) \quad h'(x_1) = (K+1)h(x_1) = (K+1)g(x_1) > g'(x_1),$$

because of (9.11). Now let

$$k(x) = h(x) - g(x)$$

for each x in I . The inequality in (9.13) tells us that $k'(x_1) > 0$. Therefore, by Theorem 8-3 of Volume I, there is a $\delta > 0$ such that

$$k(x) < k(x_1) \quad \text{if} \quad x_1 - \delta < x < x_1.$$

Since $k(x_1) = h(x_1) - g(x_1) = 0$, this inequality tells us that

$$(9.14) \quad k(x) < 0 \quad \text{if} \quad x_1 - \delta < x < x_1.$$

Now $k(x_0) = h(x_0) - g(x_0) = h(x_0) > 0$, so any δ for which (9.14) holds cannot exceed $x_1 - x_0$. Therefore the set of δ for which (9.14) holds is a nonempty set bounded above by $x_1 - x_0$. Let α be the least upper bound of all such δ , and let $x_2 = x_1 - \alpha$. Then x_2 satisfies $x_0 \leq x_2 \leq x_1$, and we have $k(x) < 0$ in the open interval (x_2, x_1) . Since k is continuous on I we must have $k(x_2) \leq 0$. We cannot have $k(x_2) < 0$; otherwise there would be an open interval about x_2 in which $k(x) < 0$, contradicting the definition of α . Therefore $k(x_2) = 0$. Since $k(x) < k(x_2)$ if $x_2 < x < x_1$, we must have $k'(x_2) \leq 0$. In other words, $h'(x_2) \leq g'(x_2)$, or

$$(K+1)h(x_2) \leq g'(x_2).$$

But since $h(x_2) = g(x_2)$ this gives us

$$(K+1)g(x_2) \leq g'(x_2),$$

which contradicts (9.10), since $g(x_2) \neq 0$. This contradiction completes the proof.

★9.3 Exercises

1. A function f is said to satisfy a Lipschitz condition on a plane set S if (9.9) holds for every pair of points (x, y) and (x, z) in S . The constant K is called a Lipschitz constant for S . Show that each of the following functions satisfies a Lipschitz condition on the given set. In each case determine a Lipschitz constant for the set.

(a) $f(x, y) = xy^2$; $S = \{(x, y) \mid -1 \leq x \leq 1, -1 \leq y \leq 1\}$.

(b) $f(x, y) = y^{2/3}$; $S = \{(x, y) \mid -1 \leq x \leq 1, 1 \leq y \leq 2\}$.

(c) $f(x, y) = e^{\sin(x+y)}$; $S = \text{the } xy\text{-plane}$.

(d) $f(x, y) = x^2 \arctan y + e^x$; $S = \{(x, y) \mid |x| \leq 2, y \text{ arbitrary}\}$.

(e) $f(x, y) = \begin{cases} y(1-2x) & \text{if } x \geq 0, \\ y(2x-1) & \text{if } x < 0; \end{cases} \quad S = \{(x, y) \mid |x| \leq 1, y \text{ arbitrary}\}.$

(f) $f(x, y) = P(x)y^2 + Q(x)y + R(x)$, where P , Q , and R are continuous on an interval $[a, b]$; $S = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$.

2. Consider the differential equation $y' = xy^2$. Find a solution $y = Y(x)$ defined for all real x and satisfying the initial condition stated. In each case prove that there is only one solution satisfying this initial condition.

(a) $Y(0) = 0$.

(d) $Y(0) = 1$.

(b) $Y(1) = 0$.

(e) $Y(1) = 1$.

(c) $Y(9) = 0$.

(f) $Y(2) = -1$.

3. Consider the differential equation $y' = f(x, y)$, where f is the function defined in Exercise 1(e). Find a function $y = Y(x)$ which satisfies the differential equation for all $x \neq 0$ and which also satisfies the initial condition $Y(1) = 1$. Show that this solution is unique.

4. Show that none of the following functions satisfies a Lipschitz condition on the given set.

(a) $f(x, y) = xy^2$; $S = \{(x, y) \mid -1 \leq x \leq 1, y \text{ arbitrary}\}$.

(b) $f(x, y) = y^{2/3}$; $S = \{(x, y) \mid -1 \leq x \leq 1, -1 \leq y \leq 1\}$.

(c) $f(x, y) = \begin{cases} \frac{4xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0); \end{cases} \quad S = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}.$

[Hint. Consider the difference $f(x, y) - f(x, z)$ when $y = ax, z = bx$.]

5. Let f be the function defined in Exercise 4(c). Find an infinite set of integral curves of the differential equation $y' = f(x, y)$ passing through the origin.

6. (a) Assume that f satisfies the condition

$$|f(x, y) - f(x, z)| \leq K |y - z|^\alpha$$

on a rectangle R , where K and α are constants, $K > 0$, $\alpha > 1$. Prove that $f(x, y)$ is independent of y . That is, $f(x, y) = f(x, z)$ for each pair of points (x, y) and (x, z) in R .

(b) Show that the same conclusion holds if f satisfies a condition of the form

$$|f(x, y) - f(x, z)| \leq \phi(|y - z|),$$

where $\phi(0) = \phi'(0) = 0$.

7. Let ϕ be a positive function defined and continuous on $[0, a]$ for some $a > 0$. Assume that $\phi(0) = 0$ and that $\phi'(0)$ exists. Prove that the improper integral $\int_{0+}^a \phi(t)^{-1} dt$ diverges. [Hint. The existence of $\phi'(0)$ implies that $\phi(t)/t$ approaches a finite limit as $t \rightarrow 0$. Compare the integral with $\int_{0+}^a t^{-1} dt$.]

8. Consider the differential equation

$$(9.15) \quad \frac{dy}{dx} = F(y),$$

where F is continuous and nonzero in an open interval (c, d) . Prove that for every point (x_0, y_0) with $c < y_0 < d$ there is one and only one integral curve of (9.15) given by an equation of the form $x = X(y)$ with $X(y_0) = x_0$. Also, if $c < y_1 < d$, show that the integral curve through (x_1, y_1) may be obtained from that through (x_0, y_0) by a horizontal translation. Illustrate with $F(y) = e^{-y}$, $(c, d) = (-1, 1)$.

9. (a) Let F be continuous everywhere in (c, d) , and assume that F is nonzero everywhere in (c, d) except at one point p . Choose a point y_0 in (c, d) , $y_0 \neq p$, and assume that

$$\lim_{y \rightarrow p} \int_{y_0}^y \frac{dt}{F(t)} = \infty.$$

Prove that there is one and only one integral curve of the differential equation

$$\frac{dy}{dx} = F(y)$$

passing through each point (x_0, y_0) with $c < y_0 < d$. When $y_0 = p$ the integral curve is the line $y = p$. Show that all other integral curves have this line as an asymptote. Illustrate with $F(y) = y$, $(c, d) = (-1, 1)$.

(b) Assume F is positive and continuous on (c, d) , choose y_0 in (c, d) , and assume the improper integral $\int_{c+}^{y_0} F(t)^{-1} dt$ diverges. Show that the differential equation (9.16) has a unique integral curve through each point (x_0, y_0) and that the line $y = c$ is an asymptote of this curve.

10. This exercise describes the modifications needed to extend the proof of Theorem 9-1 so that it applies to functions satisfying Osgood's condition,

$$|f(x, y) - f(x, z)| \leq \phi(|y - z|),$$

for every pair of points (x, y) and (x, z) in a rectangle R . Here ϕ is a function that is positive and continuous on the half-open interval $(0, a]$ and satisfies the limit relation

$$(9.17) \quad \lim_{h \rightarrow 0+} \int_h^a \frac{dt}{\phi(t)} = \infty,$$

where a denotes the altitude of the rectangle R .

(a) Let Y_1 and Y_2 be two solutions of $y' = f(x, y)$ whose graphs lie in R and intersect at the point (x_0, y_0) interior to R . If Y_1 and Y_2 do not agree everywhere on an open interval $I = (x_0 - c, x_0 + c)$, show that a point x_1 in I can be chosen for which $0 < |Y_1(x_1) - Y_2(x_1)| < a$. Label the functions so that $Y_1(x_1) > Y_2(x_1)$.

(b) Define g as in the proof of Theorem 9-1 and show that

$$g'(x) < 2\phi(|g(x)|) \quad \text{if } g(x) \neq 0.$$

(c) Instead of the definition given in Equation (9.12), define h to be that solution of the differential equation

$$\frac{dy}{dx} = 2\phi(y)$$

that takes the value $g(x_1)$ when $x = x_1$. Such a solution exists because of Exercise 9(b). [The condition (9.17) is used at this stage.] The graph of h has the line $y = 0$ as an asymptote. Therefore $h(x) > 0$ for all x since $h(x_1) = g(x_1) > 0$. Show that $h'(x_1) > g'(x_1)$. Then verify that the rest of the proof of Theorem 9-1 can be carried out with only trivial modifications.

9.4 The method of successive approximations

Consider the differential equation $y' = f(x, y)$, where the function f is continuous and bounded† on a closed rectangle R with center at (x_0, y_0) and edges parallel to the coordinate axes. We wish to find a function $y = Y(x)$ that satisfies the initial-value problem

$$(9.18) \quad y' = f(x, y), \quad Y(x_0) = y_0$$

on some interval I containing x_0 , with the graph of Y over I lying in R .

There are several methods available for demonstrating the existence of such a function Y . We shall use the so-called *method of successive approximations* which also has applications in many other problems. This method was first published by Liouville in 1838 in connection with the study of linear differential equations of second order. It was later extended by J. Caqué in 1864, L. Fuchs in 1870, and G. Peano in 1888 to the study of linear equations of order n . In 1890 Émile Picard (1856–1941) developed the method in

† It can be shown that continuity of f on R implies boundedness. See Theorem 4-20 of the author's *Mathematical Analysis*, Addison-Wesley Publishing Co., Reading, Massachusetts, 1957.

its most general form. In recognition of his fundamental contributions, some writers refer to the method as *Picard's method*. The method is not only of theoretical interest but can also be used to obtain numerical approximations to solutions in some cases.

The method of successive approximations begins with an initial guess at a solution of the initial-value problem (9.18). We choose any continuous function Y_0 whose graph over some interval about x_0 lies in the rectangle R . For example, we can always choose $Y_0(x) = y_0$, although this is not essential. We then substitute this guess in the right-hand member of the differential equation in (9.18), and we obtain a new differential equation,

$$y' = f[x, Y_0(x)],$$

whose right-hand member is now a continuous function of x . This equation has a unique solution Y_1 passing through the point (x_0, y_0) ; it is given by the equation

$$Y_1(x) = y_0 + \int_{x_0}^x f[t, Y_0(t)] dt.$$

We shall prove later (in Theorem 9-2) that the graph of Y_1 over some interval containing x_0 lies entirely in R . Now we repeat the process, using Y_1 in place of Y_0 . That is, we replace y by Y_1 in the right-hand member of the differential equation in (9.18) to obtain a new differential equation

$$y' = f[x, Y_1(x)].$$

This equation has a unique solution Y_2 passing through (x_0, y_0) , namely,

$$Y_2(x) = y_0 + \int_{x_0}^x f[t, Y_1(t)] dt.$$

Now we substitute Y_2 in the right-hand member of (9.18) and solve the resulting equation to determine Y_3 , and so on. This process generates a sequence of continuous functions $Y_0, Y_1, Y_2, \dots, Y_n, \dots$, where Y_{n+1} is determined from Y_n by the recursion formula

$$(9.19) \quad Y_{n+1}(x) = y_0 + \int_{x_0}^x f[t, Y_n(t)] dt \quad \text{for } n = 0, 1, 2, \dots$$

For this formula to be meaningful the graph of Y_n over some interval containing x_0 must lie in R . We shall prove later in this section that such an interval always exists. In Section 9.6 we shall prove that if f satisfies a Lipschitz condition on R the sequence of functions defined by (9.19) will converge to a continuous limit function Y given by

$$Y(x) = \lim_{n \rightarrow \infty} Y_n(x),$$

and that this limit function Y is a solution of the initial-value problem (9.18). The functions $Y_0, Y_1, Y_2, \dots$ defined by (9.19) are called *successive approximations* to Y . Before we investigate the convergence of the process we discuss some examples.

Example 1. Consider the initial-value problem $y' = 2xy$ with $y = 1$ when $x = 0$. This equation is easily solved as a separable equation or as a linear equation; it has the solution

$$y = e^{x^2}.$$

We shall show how this solution can be obtained by the method of successive approximations.

For the initial guess Y_0 we take $Y_0(x) = 1$. The recursion formula (9.19) now gives us

$$Y_1(x) = 1 + \int_0^x 2tY_0(t) dt = 1 + \int_0^x 2t dt = 1 + x^2,$$

$$Y_2(x) = 1 + \int_0^x 2tY_1(t) dt = 1 + 2 \int_0^x (t + t^3) dt = 1 + x^2 + \frac{x^4}{2},$$

$$Y_3(x) = 1 + \int_0^x 2tY_2(t) dt = 1 + 2 \int_0^x \left(t + t^3 + \frac{t^5}{2} \right) dt = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!}.$$

Using induction we may easily show that

$$Y_n(x) = \sum_{k=0}^n \frac{x^{2k}}{k!}.$$

The polynomial $Y_n(x)$ is a partial sum of the power-series expansion for e^{x^2} . Since this series converges for all real x we have

$$\lim_{n \rightarrow \infty} Y_n(x) = \sum_{k=0}^{\infty} \frac{x^{2k}}{k!} = e^{x^2}$$

for all x . Thus, in this example we are able to show directly that the successive approximations converge to a solution of the initial-value problem.

The foregoing example is simpler than most. The next example gives a more realistic idea of the labor encountered in computing the successive approximations.

Example 2. Consider the initial-value problem $y' = x^2 + y^2$ with $y = 0$ when $x = 0$. This is a nonlinear equation that does not fall under the scope of the special methods discussed in Chapter 7. We shall compute a few approximations to the solution. We choose $Y_0(x) = 0$ and determine the next three approximations as follows:

$$Y_1(x) = \int_0^x t^2 dt = \frac{x^3}{3},$$

$$Y_2(x) = \int_0^x [t^2 + Y_1^2(t)] dt = \int_0^x \left(t^2 + \frac{t^6}{9} \right) dt = \frac{x^3}{3} + \frac{x^7}{63},$$

$$Y_3(x) = \int_0^x \left[t^2 + \left(\frac{t^3}{3} + \frac{t^7}{63} \right)^2 \right] dt = \frac{x^3}{3} + \frac{x^7}{63} + \frac{2x^{11}}{2079} + \frac{x^{15}}{59535}.$$

It is now apparent that a great deal of labor will be needed to compute further approximations. For example, the next two approximations Y_4 and Y_5 will be polynomials of degrees 31 and 63, respectively.

The next example exhibits a further difficulty that can arise in the computation of the successive approximations.

Example 3. Consider the initial-value problem $y' = 2x + e^y$, with $y = 0$ when $x = 0$. We begin with the initial guess $Y_0(x) = 0$ and we find

$$Y_1(x) = \int_0^x (2t + 1) dt = x^2 + x,$$

$$Y_2(x) = \int_0^x (2t + e^{t^2+t}) dt = x^2 + \int_0^x e^{t^2+t} dt.$$

Here further progress is impeded by the fact that the last integral cannot be evaluated in terms of elementary functions. However, for a given x it is possible to calculate a numerical approximation to the integral by one of the methods described in Chapter 8 and thereby obtain an approximation to $Y_2(x)$.

Because of the difficulties displayed in the last two examples, the method of successive approximations is sometimes not very useful for the explicit determination of solutions in practice. The real value of the method is its use in establishing existence theorems.

We have already pointed out that the graph of the n th approximating function Y_n over some interval I about x_0 should lie in the rectangle R in order for the recursion formula (9.19) to be meaningful. The next theorem exhibits such an interval.

9- 2 THEOREM. Let R be the rectangle with center at (x_0, y_0) given by

$$R = \{(x, y) \mid |x - x_0| \leq a, |y - y_0| \leq b\}.$$

Let f be continuous on R and assume that

$$|f(x, y)| \leq M$$

for all (x, y) in R , where $M > 0$. Let $I = (x_0 - c, x_0 + c)$, where $c = \min \{a, b/M\}$. If the graph of Y_n over I lies in R , and if Y_{n+1} is defined on I by the recursion formula

$$(9.20) \quad Y_{n+1}(x) = y_0 + \int_{x_0}^x f[t, Y_n(t)] dt,$$

then the graph of Y_{n+1} over I also lies in R . Moreover, we have

$$(9.21) \quad |Y_{n+1}(x) - y_0| \leq M |x - x_0|$$

for every x in I .

Proof. The set $B = \{(x, y) \mid |x - x_0| \leq c, |y - y_0| \leq M |x - x_0|\}$ is a “butterfly-shaped” region bounded by the two parallel lines $x = x_0 \pm c$ and the two lines $y = y_0 \pm M(x - x_0)$, which intersect at (x_0, y_0) . Examples are shown in Figure 9.1. In Figure 9.1(a) we have $M < b/a$, so $a < b/M$ and $c = a$. In Figure 9.1(b) we have $M > b/a$, so $c = b/M$. If $M = b/a$ the lines of slope $\pm M$ through (x_0, y_0) coincide with the diagonals of R . If we prove (9.21) it follows that the point $(x, Y_{n+1}(x))$ lies in B (and hence in R) for each x in I .

The proof of (9.21) is immediate. From Equation (9.20) we obtain

$$|Y_{n+1}(x) - y_0| = \left| \int_{x_0}^x f[t, Y_n(t)] dt \right| \leq \left| \int_{x_0}^x |f[t, Y_n(t)]| dt \right|.$$

Since $|f(x, y)| \leq M$ for all (x, y) in R and since $(t, Y_n(t))$ is in R , the last integral is bounded by $M |x - x_0|$. This proves (9.21).

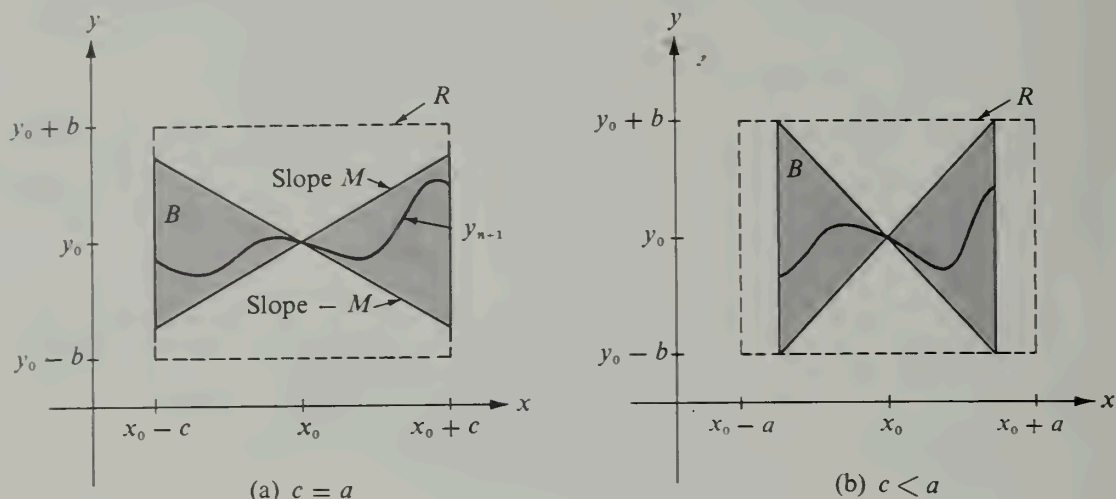


FIGURE 9.1 The region $B = \{(x, y) \mid |x - x_0| \leq c, |y - y_0| \leq M|x - x_0|\}$.

Note. Let I be the interval of Theorem 9-2. If the initial guess Y_0 is such that its graph over I lies in R we may use induction, along with Theorem 9-2, to deduce that the graphs of *all* approximating functions Y_n over I also lie in R .

9.5 Exercises

1. Consider the initial-value problem

$$y' + y = 2e^x, \quad \text{with } y = 1 \text{ when } x = 0.$$

- (a) Find the exact solution Y of this problem.
- (b) Apply the method of successive approximations, starting with the initial guess $Y_0(x) = 1$. Determine $Y_n(x)$ explicitly and show that

$$\lim_{n \rightarrow \infty} Y_n(x) = Y(x)$$

for all real x .

2. Apply the method of successive approximations to the initial-value problem

$$y' = x + y^2, \quad \text{with } y = 0 \text{ when } x = 0.$$

Take $Y_0(x) = 0$ as the initial guess and compute $Y_3(x)$.

3. Apply the method of successive approximations to the initial-value problem

$$y' = 1 + xy^2, \quad \text{with } y = 0 \text{ when } x = 0.$$

Take $Y_0(x) = 0$ as the initial guess and compute $Y_3(x)$.

4. Apply the method of successive approximations to the initial-value problem

$$y' = x^2 + y^2, \quad \text{with } y = 0 \text{ when } x = 0.$$

Start with the "bad" initial guess $Y_0(x) = 1$, compute $Y_3(x)$, and compare with the results of Example 2 in Section 9.4.

5. Consider the initial-value problem

$$y' = x^2 + y^2, \quad \text{with } y = 1 \text{ when } x = 0.$$

(a) Apply the method of successive approximations, starting with the initial guess $Y_0(x) = 1$, and compute $Y_2(x)$.

(b) Let $R = [-1, 1] \times [-1, 1]$. Find the smallest M such that $|f(x, y)| \leq M$ on R . Find an interval $I = (-c, c)$ such that the graph of every approximating function Y_n over I will lie in R .

(c) Assume the solution $y = Y(x)$ has a power-series expansion in a neighborhood of the origin. Determine the first six nonzero terms of this expansion and compare with the result of part (a).

6. Consider the initial-value problem

$$y' = 1 + y^2, \quad \text{with } y = 0 \text{ when } x = 0.$$

(a) Apply the method of successive approximations, starting with the initial guess $Y_0(x) = 0$, and compute $Y_4(x)$.

(b) Prove that every approximating function Y_n is defined on the entire real axis.

(c) Use Theorem 9-1 to show that the initial-value problem has at most one solution in any interval of the form $(-h, h)$.

(d) Solve the differential equation by separation of variables and thereby show that there is exactly one solution Y of the initial-value problem on the interval $(-\pi/2, \pi/2)$ and no solution on any larger interval. In this example, the successive approximations are defined on the entire real axis, but they converge to a limit function only on the interval $(-\pi/2, \pi/2)$.

7. We seek two functions $y = Y(x)$ and $z = Z(x)$ that simultaneously satisfy the system of equations

$$y' = z, \quad z' = x^3(y + z)$$

with initial conditions $y = 1$ and $z = 1/2$ when $x = 0$. Start with the initial guesses $Y_0(x) = 1$, $Z_0(x) = 1/2$, and use an appropriate modification of the method of successive approximations to obtain the approximating functions

$$Y_3(x) = 1 + \frac{x}{2} + \frac{3x^5}{40} + \frac{x^6}{60} + \frac{x^9}{192},$$

$$Z_3(x) = \frac{1}{2} + \frac{3x^4}{8} + \frac{x^5}{10} + \frac{3x^8}{64} + \frac{7x^9}{360} + \frac{x^{12}}{256}.$$

8. Consider the system of equations

$$y' = 2x + z, \quad z' = 3xy + x^2z,$$

with initial conditions $y = 2$ and $z = 0$ when $x = 0$. Start with the initial guesses $Y_0(x) = 2$, $Z_0(x) = 0$, use an appropriate modification of the method of successive approximations, and determine $Y_3(x)$ and $Z_3(x)$.

9. Consider the initial-value problem

$$y'' = x^2y' + x^4y, \quad \text{with } y = 5 \text{ and } y' = 1 \text{ when } x = 0.$$

Change this problem to an equivalent problem involving a system of two equations for two unknown functions $y = Y(x)$ and $z = Z(x)$, where $z = y'$. Then use an appropriate modification of the method of successive approximations, starting with initial guesses $Y_0(x) = 5$ and $Z_0(x) = 1$, and determine $Y_3(x)$ and $Z_3(x)$.

10. Let f be defined on the rectangle $R = [-1, 1] \times [-1, 1]$ as follows:

$$f(x, y) = \begin{cases} 0 & \text{if } x = 0, \\ 2y/x & \text{if } x \neq 0 \text{ and } |y| \leq x^2, \\ 2x & \text{if } x \neq 0 \text{ and } y > x^2, \\ -2x & \text{if } x \neq 0 \text{ and } y < -x^2. \end{cases}$$

- (a) Prove that f is continuous on R .
 (b) Prove that $|f(x, y)| \leq 2$ for all (x, y) in R .
 (c) Show that f does not satisfy a Lipschitz condition on R .
 (d) For each constant C satisfying $|C| \leq 1$, show that $y = Cx^2$ is a solution of the initial-value problem $y' = f(x, y)$, with $y = 0$ when $x = 0$. Show also that the graph of each of these solutions over $(-1, 1)$ lies in R .
 (e) Apply the method of successive approximations to this initial-value problem, starting with initial guess $Y_0(x) = 0$. Determine $Y_n(x)$ and show that the approximations converge to a solution of the problem on the interval $(-1, 1)$.
 (f) Repeat part (e), starting with initial guess $Y_0(x) = x$. Determine $Y_n(x)$ and show that the approximating functions converge to a solution different from any of those in part (d).
 (g) Repeat part (e), starting with the initial guess $Y_0(x) = x^3$.
 (h) Repeat part (e), starting with the initial guess $Y_0(x) = x^{1/3}$.

9.6 Existence theorem for the differential equation $y' = f(x, y)$

In this section we shall establish an existence theorem for the initial-value problem

$$(9.22) \quad y' = f(x, y), \quad \text{with } y = y_0 \text{ when } x = x_0.$$

The most satisfactory theorem of this type was first proved by Peano, who showed that if f is continuous on a closed rectangle R containing (x_0, y_0) there is at least one solution to (9.22). We shall not discuss Peano's proof here because it requires methods from advanced analysis. Instead, we shall impose a Lipschitz condition on f and give a simple proof of the following weaker version of Peano's theorem.

9-3 THEOREM. *Existence theorem.* Assume that f is continuous on the rectangle

$$R = [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b]$$

with center at (x_0, y_0) . Assume also that $|f(x, y)| \leq M$ on R , where $M > 0$, and let $I = (x_0 - c, x_0 + c)$, where $c = \min \{a, b/M\}$. If $x \in I$, let $Y_0(x) = y_0$ and define

$$(9.23) \quad Y_{n+1}(x) = y_0 + \int_{x_0}^x f[t, Y_n(t)] dt \quad \text{for } n = 0, 1, 2, \dots$$

Finally, assume f satisfies a Lipschitz condition on R , say

$$|f(x, y) - f(x, z)| \leq K |y - z|$$

for each pair of points $(x, y), (x, z)$ in R . Then:

- (i) The sequence $\{Y_n(x)\}$ converges for each x in I .
 (ii) If we let

$$(9.24) \quad Y(x) = \lim_{n \rightarrow \infty} Y_n(x)$$

for each x in I , the function Y so defined is a solution of the initial-value problem (9.22) on the interval I .

Proof. To prove (i) we shall show that the series

$$(9.25) \quad Y_0(x) + \sum_{m=0}^{\infty} [Y_{m+1}(x) - Y_m(x)]$$

converges for each x in I . This is equivalent to (i) because the partial sums of this series are given by

$$Y_0(x) + \sum_{m=0}^{n-1} [Y_{m+1}(x) - Y_m(x)] = Y_n(x).$$

To prove that the series in (9.25) converges we compare it with the convergent series

$$M \sum_{m=0}^{\infty} \frac{K^m c^{m+1}}{(m+1)!} = \frac{M}{K} \sum_{m=0}^{\infty} \frac{(Kc)^{m+1}}{(m+1)!} = \frac{M}{K} (e^{Kc} - 1).$$

The comparison is provided by the inequality

$$(9.26) \quad |Y_{m+1}(x) - Y_m(x)| \leq \frac{MK^m |x - x_0|^{m+1}}{(m+1)!} \leq \frac{MK^m c^{m+1}}{(m+1)!},$$

which is valid for each $m \geq 1$ and each x in I . We shall prove (9.26) by induction. For $m = 1$ we use (9.23) to obtain

$$Y_2(x) - Y_1(x) = \int_{x_0}^x \{f[t, Y_1(t)] - f[t, y_0]\} dt.$$

Using the Lipschitz condition we obtain

$$(9.27) \quad |Y_2(x) - Y_1(x)| \leq \left| \int_{x_0}^x |f[t, Y_1(t)] - f[t, y_0]| dt \right| \leq K \left| \int_{x_0}^x |Y_1(t) - y_0| dt \right|.$$

Now by (9.21) in Theorem 9-2 we have

$$|Y_1(t) - y_0| \leq M |t - x_0|.$$

Therefore, if $x \geq x_0$ we have $|t - x_0| = t - x_0$ for each t in $[x_0, x]$, and the inequality in (9.27) becomes

$$|Y_2(x) - Y_1(x)| \leq MK \int_{x_0}^x (t - x_0) dt = MK \frac{(x - x_0)^2}{2}.$$

If $x \leq x_0$, we have $|t - x_0| = x_0 - t$ for each t in $[x, x_0]$, and the inequality in (9.27) becomes

$$|Y_2(x) - Y_1(x)| \leq MK \int_x^{x_0} (x_0 - t) dt = MK \int_{x_0}^x (t - x_0) dt = MK \frac{(x - x_0)^2}{2}.$$

Therefore for every x in I we have

$$|Y_2(x) - Y_1(x)| \leq MK \frac{|x - x_0|^2}{2!} \leq \frac{MKc^2}{2!},$$

which proves (9.26) for $m = 1$.

Now we assume that (9.26) holds for $m = n - 1$. To prove that it holds for $m = n$ we use (9.23) again to write

$$|Y_{n+1}(x) - Y_n(x)| = \left| \int_{x_0}^x |f[t, Y_n(t)] - f[t, Y_{n-1}(t)]| dt \right| \leq K \left| \int_{x_0}^x |Y_n(t) - Y_{n-1}(t)| dt \right|$$

$$\begin{aligned} &\leq K \left| \int_{x_0}^x MK^{n-1} \frac{|t - x_0|^n}{n!} dt \right| = \frac{MK^n}{n!} \left| \int_{x_0}^x |t - x_0|^n dt \right| \\ &= \frac{MK^n |x - x_0|^{n+1}}{(n+1)!} \leq \frac{MK^n c^{n+1}}{(n+1)!}. \end{aligned}$$

This proves (9.26) for all $m \geq 1$. Therefore statement (i) is proved.

Now we define Y by Equation (9.24). Since $Y_n(x_0) = y_0$ for each n we also have $Y(x_0) = y_0$. We prove next that Y is continuous on I and that the graph of Y over I lies in R . Then we shall prove that Y satisfies the "integral equation"

$$(9.28) \quad Y(x) = y_0 + \int_{x_0}^x f[t, Y(t)] dt.$$

The continuity of Y implies continuity of the integrand in (9.28). This, in turn, enables us to differentiate the integral by the first fundamental theorem of calculus to obtain

$$Y'(x) = f[x, Y(x)],$$

which shows that Y is a solution of the differential equation in (9.22).

To prove that Y is continuous on I , we use the relation (9.23) to obtain

$$Y_{n+1}(x) - Y_{n+1}(x_1) = \int_{x_1}^x f[t, Y_n(t)] dt$$

for every choice of x and x_1 in I . This implies that

$$|Y_{n+1}(x) - Y_{n+1}(x_1)| \leq M |x - x_1|.$$

Letting $n \rightarrow \infty$ we obtain the inequality

$$(9.29) \quad |Y(x) - Y(x_1)| \leq M |x - x_1|.$$

Letting $x \rightarrow x_1$ in this inequality we find

$$\lim_{x \rightarrow x_1} Y(x) = Y(x_1).$$

Since this is true for every point x_1 in I , it follows that Y is continuous on I . When $x_1 = x_0$, inequality (9.29) gives us

$$|Y(x) - y_0| \leq M |x - x_0|.$$

If x is in I we have $M|x - x_0| < Mc \leq b$, so $|Y(x) - y_0| \leq b$, which shows that the point $(x, Y(x))$ is in R for each x in I .

It remains to prove that Y satisfies the integral equation in (9.28). From Equation (9.24) and the definition of Y_{n+1} we have

$$Y(x) = \lim_{n \rightarrow \infty} Y_{n+1}(x) = y_0 + \lim_{n \rightarrow \infty} \int_{x_0}^x f[t, Y_n(t)] dt.$$

The proof will be complete if we show that

$$(9.30) \quad \lim_{n \rightarrow \infty} \int_{x_0}^x f[t, Y_n(t)] dt = \int_{x_0}^x f[t, Y(t)] dt.$$

For this purpose we examine the difference

$$d_n = \int_{x_0}^x f[t, Y(t)] dt - \int_{x_0}^x f[t, Y_n(t)] dt = \int_{x_0}^x \{f[t, Y(t)] - f[t, Y_n(t)]\} dt.$$

We wish to prove that $d_n \rightarrow 0$ as $n \rightarrow \infty$. Using the Lipschitz condition we see that

$$(9.31) \quad |d_n| \leq K \left| \int_{x_0}^x |Y(t) - Y_n(t)| dt \right|.$$

Since we have

$$Y_n(t) = Y_0(t) + \sum_{m=0}^{n-1} [Y_{m+1}(t) - Y_m(t)]$$

and

$$Y(t) = \lim_{n \rightarrow \infty} Y_n(t) = Y_0(t) + \sum_{m=0}^{\infty} [Y_{m+1}(t) - Y_m(t)]$$

we may write

$$|Y(t) - Y_n(t)| \leq \sum_{m=n}^{\infty} |Y_{m+1}(t) - Y_m(t)| \leq M \sum_{m=n}^{\infty} \frac{K^m c^{m+1}}{(m+1)!},$$

where we have used (9.26). This inequality may be written as

$$|Y(t) - Y_n(t)| \leq M A_n,$$

where

$$A_n = \sum_{m=n}^{\infty} \frac{K^m c^{m+1}}{(m+1)!}.$$

Inequality (9.31) now becomes

$$(9.32) \quad |d_n| \leq \left| \int_{x_0}^x K M A_n dt \right| = K M A_n |x - x_0| = M |x - x_0| \sum_{m=n}^{\infty} \frac{(Kc)^{m+1}}{(m+1)!}.$$

Since the series

$$\sum_{m=0}^{\infty} \frac{(Kc)^{m+1}}{(m+1)!}$$

converges, the right-hand member of (9.32) approaches 0 as $n \rightarrow \infty$. Therefore $d_n \rightarrow 0$ as $n \rightarrow \infty$, which proves (9.30). The proof of Theorem 9-3 is now complete in all details.

Note. In the foregoing proof we established the existence of an integral curve passing through the center (x_0, y_0) of the rectangle R . There is no difficulty in extending this theorem to find an integral curve passing through any interior point (x_1, y_1) of R . In fact, every such point (x_1, y_1) is the center of some rectangle R_1 lying in R . If f satisfies the hypotheses of Theorem 9-3 in R , then f also satisfies these hypotheses on R_1 , so there exists an integral curve in R_1 passing through (x_1, y_1) . The solution will be defined on some interval of the form $(x_1 - c_1, x_1 + c_1)$.

9.7 An extension of the existence theorem

If, in Theorem 9-3, we have $M > b/a$, then $c < a$ and the interval $(x_0 - c, x_0 + c)$ in which a solution exists will be a proper subinterval of $(x_0 - a, x_0 + a)$. It is natural to ask whether further restrictions on f will ensure the existence of a solution over the full interval $(x_0 - a, x_0 + a)$. The next theorem provides an answer to this question.

9-4 THEOREM. Let f be defined and continuous on a rectangular strip R' of the form

$$R' = \{(x, y) \mid x_0 - a_1 \leq x \leq x_0 + a_2, y \text{ arbitrary}\},$$

where $a_1 > 0$ and $a_2 > 0$. Assume that f satisfies a Lipschitz condition on R' , say

$$|f(x, y) - f(x, z)| \leq K |y - z|$$

for all points (x, y) and (x, z) in R' . Assume also that f is bounded on every rectangle R_b of the form

$$R_b = \{(x, y) \mid x_0 - a_1 \leq x \leq x_0 + a_2, |y - y_0| \leq b\};$$

that is, assume that for every $b > 0$ there is a constant $M_b > 0$ such that

$$(9.33) \quad |f(x, y)| \leq M_b$$

for all (x, y) in R_b . Then there is a function $y = Y(x)$ defined on the interval $J = (x_0 - a_1, x_0 + a_2)$ such that $Y(x_0) = y_0$ and $Y'(x) = f[x, Y(x)]$ for every x in J .

Proof. As in the proof of Theorem 9-3 we obtain a solution by the method of successive approximations. Certain modifications of the proof are needed to show that the approximations converge on the full interval $(x_0 - a_1, x_0 + a_2)$.

For each x in J we define the sequence $\{Y_n(x)\}$ by the recursion formula

$$(9.34) \quad Y_0(x) = y_0, \quad Y_{m+1}(x) = y_0 + \int_{x_0}^x f[t, Y_m(t)] dt,$$

for $m = 0, 1, 2, \dots$. By induction we see that each function Y_m is continuous on J .

Let M_0 denote the maximum value of $|f(x, y_0)|$ for x in J , and let $a = \max\{a_1, a_2\}$. We show next that the inequality

$$(9.35) \quad |Y_{m+1}(x) - Y_m(x)| \leq \frac{M_0 K^m |x - x_0|^{m+1}}{(m+1)!} \leq \frac{M_0 K^m a^{m+1}}{(m+1)!}$$

is valid for each x in J and each $m \geq 0$. Inequality (9.35) is easily proved by induction. For $m = 0$ we have

$$|Y_1(x) - Y_0(x)| = \left| \int_{x_0}^x f(t, y_0) dt \right| \leq M_0 |x - x_0| \leq M_0 a.$$

The inductive step from m to $m+1$ may now be carried out exactly as in the proof of (9.26). Once we have (9.35) the convergence of the sequence $\{Y_n(x)\}$ for each x in J follows exactly as in the proof of Theorem 9-3.

Next we wish to show that the limit function Y defined on J by the equation

$$Y(x) = \lim_{n \rightarrow \infty} Y_n(x)$$

is continuous on J and satisfies the integral equation

$$(9.36) \quad Y(x) = y_0 + \int_{x_0}^x f[t, Y(t)] dt.$$

First we prove that there is a constant $B > 0$ such that

$$(9.37) \quad |Y_m(x) - y_0| \leq B$$

for each x in J and each $m \geq 1$. This inequality shows that the graph of each approximating function Y_m over J lies in the rectangle R_B given by

$$R_B = \{(x, y) \mid |x - x_0| \leq a, |y - y_0| \leq B\}.$$

To prove (9.37) we begin with the telescoping formula

$$Y_m(x) - y_0 = \sum_{r=0}^{m-1} [Y_{r+1}(x) - Y_r(x)]$$

and use (9.35) in each term on the right to obtain

$$|Y_m(x) - y_0| \leq \sum_{r=0}^{m-1} \frac{M_0 K^r a^{r+1}}{(r+1)!} \leq \frac{M_0}{K} \sum_{r=0}^{\infty} \frac{(Ka)^{r+1}}{(r+1)!} = \frac{M_0}{K} (e^{Ka} - 1).$$

This shows that (9.37) holds with

$$(9.38) \quad B = \frac{M_0}{K} (e^{Ka} - 1).$$

Letting $m \rightarrow \infty$ in (9.37) we find that $|Y(x) - y_0| \leq B$, so the graph of Y over J also lies in R_B .

The function f satisfies a Lipschitz condition on the rectangle R_B and is bounded on R_B . In fact, by (9.33) we have

$$(9.39) \quad |f(x, y)| \leq M_B$$

for all (x, y) in R_B . We now complete the proof by the same type of argument as that used in the proof of Theorem 9-3. First we note that (9.34) and (9.39) give us the inequality

$$|Y_{m+1}(x) - Y_{m+1}(x_1)| = \left| \int_{x_1}^x f[t, Y_m(t)] dt \right| \leq M_B |x - x_1|$$

for any two points x and x_1 in J . Letting $m \rightarrow \infty$ we obtain

$$|Y(x) - Y(x_1)| \leq M_B |x - x_1|.$$

Letting $x \rightarrow x_1$ we find that Y is continuous at each point x_1 in J . Now we let $m \rightarrow \infty$ in the recursion formula (9.34) to obtain

$$Y(x) = y_0 + \lim_{m \rightarrow \infty} \int_{x_0}^x f[t, Y_m(t)] dt.$$

By exactly the same argument as that used in the proof of Theorem 9-3 we deduce that

$$\lim_{m \rightarrow \infty} \int_{x_0}^x f[t, Y_m(t)] dt = \int_{x_0}^x f[t, Y(t)] dt$$

for each x in J . This shows that Y satisfies the integral equation in (9.36) everywhere on J . From this we see that $Y(x_0) = y_0$ and that $Y'(x) = f[x, Y(x)]$ for each x in J .

9.8 Another proof of the uniqueness theorem

The method of successive approximations can also be used to give a short proof of the uniqueness theorem (Theorem 9-1) if we assume that f is continuous on R . Consider the initial-value problem

$$(9.40) \quad y' = f(x, y), \quad \text{with } y = y_0 \text{ when } x = x_0.$$

Let Y be the solution constructed in Theorem 9-3 and let Z be any solution whose graph over the interval I lies in the rectangle R . (The notation is that of Theorem 9-3.) Then Z also satisfies the integral equation

$$(9.41) \quad Z(x) = y_0 + \int_{x_0}^x f[t, Z(t)] dt$$

for every x in I . Using Equations (9.41) and (9.23) we may write

$$|Z(x) - Y_{m+1}(x)| \leq \left| \int_{x_0}^x |f[t, Z(t)] - f[t, Y_m(t)]| dt \right|.$$

Since both points $(t, Z(t))$ and $(t, Y_m(t))$ lie in R , we may apply the Lipschitz condition to write

$$(9.42) \quad |Z(x) - Y_{m+1}(x)| \leq K \left| \int_{x_0}^x |Z(t) - Y_m(t)| dt \right|.$$

When $m = 0$ this inequality becomes

$$|Z(x) - Y_1(x)| \leq K \left| \int_{x_0}^x |Z(t) - y_0| dt \right| \leq bK |x - x_0|.$$

Using this in (9.42) with $m = 1$ we obtain

$$|Z(x) - Y_2(x)| \leq K \left| \int_{x_0}^x |Z(t) - Y_1(t)| dt \right| \leq bK^2 \left| \int_{x_0}^x |t - x_0| dt \right| = bK^2 \frac{|x - x_0|^2}{2}.$$

By induction we find

$$(9.43) \quad |Z(x) - Y_m(x)| \leq \frac{bK^m |x - x_0|^m}{m!} \leq \frac{bK^m c^m}{m!}$$

for each x in I and every $m \geq 1$. When $m \rightarrow \infty$ the right-hand member of (9.43) approaches 0 [since $(Kc)^m/m!$ is the general term of the convergent series for e^{Kc}] and we see that

$$\lim_{m \rightarrow \infty} Y_m(x) = Z(x)$$

for each x in I . This proves that $Z(x) = Y(x)$ for each x in I .

If f satisfies the hypotheses of Theorem 9-4 we may use the same type of argument to prove the following extension of the uniqueness theorem.

9- 5 THEOREM. *Extension of the uniqueness theorem.* Assume f satisfies the hypotheses of Theorem 9-4. Let Z be any solution of the initial-value problem (9.40) on J and assume that Z is bounded on J . Then $Z(x) = Y(x)$ for each x in J , where Y is the solution given by Theorem 9-4.

Proof. Since Z is bounded on J , its graph over J lies in a rectangle of the form

$$R_A = \{(x, y) \mid x_0 - a_1 \leq x \leq x_0 + a_2, |y - y_0| \leq A\}.$$

Let $C = \max \{A, B\}$, where B is given by (9.38). Then both rectangles R_A and R_B are contained in R_C . Now we may use exactly the same argument given in the foregoing proof of Theorem 9-1, with the rectangle R replaced by R_C , to prove that $Z(x) = Y(x)$ for every x in J .

9.9 Existence and uniqueness theorems for systems of first-order differential equations

In Section 9.1 we showed that a system of first-order equations

$$y_1' = f_1(x, y_1, \dots, y_n), \quad y_2' = f_2(x, y_1, \dots, y_n), \quad \dots, \quad y_n' = f_n(x, y_1, \dots, y_n)$$

can be expressed more briefly as a first-order vector differential equation of the form

$$\vec{y}' = \vec{f}(x, \vec{y}),$$

where $\vec{y}$ is the vector-valued function defined on an interval I by the equation

$$\vec{y}(x) = (y_1(x), y_2(x), \dots, y_n(x)),$$

and

$$\vec{f}(x, \vec{y}) = (f_1(x, \vec{y}), f_2(x, \vec{y}), \dots, f_n(x, \vec{y})),$$

where

$$f_i(x, \vec{y}) = f_i(x, y_1, y_2, \dots, y_n).$$

The derivative $\vec{y}'$ is defined by the equation

$$\vec{y}'(x) = (y_1'(x), y_2'(x), \dots, y_n'(x)).$$

In this section we shall extend the method of successive approximations and use it to prove an existence theorem and a uniqueness theorem for the initial-value problem

$$(9.44) \quad \vec{y}' = \vec{f}(x, \vec{y}), \quad \text{with } \vec{y} = \vec{y}_0 \text{ when } x = x_0.$$

In the one-dimensional case ($n = 1$) we assumed that f was continuous and satisfied a Lipschitz condition on a rectangle R of the form

$$R = \{(x, y) \mid |x - x_0| \leq a, |y - y_0| \leq b\}.$$

In the more general case we consider, instead of a rectangle R , a set S in $(n + 1)$ -space given by

$$S = \{(x, \vec{y}) \mid |x - x_0| \leq a, \|\vec{y} - \vec{y}_0\| \leq b\},$$

where the symbol $\|\vec{z}\|$ is defined for any vector $\vec{z} = (z_1, z_2, \dots, z_n)$ in n -space by the equation

$$(9.45) \quad \|\vec{z}\| = \sum_{k=1}^n |z_k|.$$

The use of $\|\vec{z}\|$ in place of the more usual Euclidean length, $|\vec{z}| = (z_1^2 + z_2^2 + \dots + z_n^2)^{1/2}$,

helps to simplify much of the discussion. The reader may easily verify that $\|\vec{z}\|$ has the fundamental "norm" properties:

- (a) $\|\vec{z}\| \geq 0$ for every vector $\vec{z}$,
- (b) $\|\alpha\vec{z}\| = |\alpha| \|\vec{z}\|$ for every vector $\vec{z}$ and every scalar α ,
- (c) $\|\vec{y} + \vec{z}\| \leq \|\vec{y}\| + \|\vec{z}\|$ for every pair of vectors $\vec{y}$ and $\vec{z}$,
- (d) $\|\vec{z}\| = 0$ implies $\vec{z} = \vec{0}$.

These properties enable us to extend all the inequalities relating absolute values that were used in the proofs in Sections 9.6 and 9.8. An existence and uniqueness theorem for (9.44) may now be stated as follows:

9- 6 THEOREM. *Existence and uniqueness of solutions to first-order vector differential equations.* Assume that each of the n components of $\vec{f}$ is continuous on the set

$$S = \{(x, \vec{y}) \mid |x - x_0| \leq a, \|\vec{y} - \vec{y}_0\| \leq b\},$$

where $a > 0, b > 0$. Assume $\vec{f}$ is bounded on S ; that is, assume there is a positive constant M such that

$$(9.46) \quad \|\vec{f}(x, \vec{y})\| \leq M$$

for all $(x, \vec{y})$ in S . Also, assume $\vec{f}$ satisfies a Lipschitz condition on S , say

$$\|\vec{f}(x, \vec{y}) - \vec{f}(x, \vec{z})\| \leq K \|\vec{y} - \vec{z}\|$$

for every pair of points $(x, \vec{y})$ and $(x, \vec{z})$ in S . Let $I = (x_0 - c, x_0 + c)$, where $c = \min\{a, b/M\}$. Then there is one and only one function $\vec{Y}$ defined on I , with $\vec{Y}(x_0) = \vec{y}_0$, such that $(x, \vec{Y}(x)) \in S$ and $\vec{Y}'(x) = \vec{f}[x, \vec{Y}(x)]$ for every x in I .

Proof. Since the proof is entirely analogous to the one-dimensional case we sketch only the principal steps. Let $\vec{Y}_0(x) = \vec{y}_0$ for each x in I , and define vector-valued functions $\vec{Y}_1, \vec{Y}_2, \dots$ on I by the recursion formula

$$(9.47) \quad \vec{Y}_{m+1}(x) = \vec{y}_0 + \int_{x_0}^x \vec{f}[t, \vec{Y}_m(t)] dt \quad \text{for } m = 0, 1, 2, \dots$$

In formulas such as this one the integral is an n -dimensional vector obtained by integrating component by component. Thus, if $\vec{F}(t) = (F_1(t), F_2(t), \dots, F_n(t))$, we define

$$\int_{x_0}^x \vec{F}(t) dt = \left(\int_{x_0}^x F_1(t) dt, \int_{x_0}^x F_2(t) dt, \dots, \int_{x_0}^x F_n(t) dt \right).$$

From this definition and Equation (9.45) it is easy to verify the relation

$$(9.48) \quad \left\| \int_{x_0}^x \vec{F}(t) dt \right\| \leq \int_{x_0}^x \|\vec{F}(t)\| dt.$$

For the recursion formula (9.47) to be meaningful we need to know that $(x, \vec{Y}_m(x)) \in S$ for each x in I . This can easily be proved by induction, with the use of (9.48). When $m = 0$ we have $(x, \vec{Y}_0(x)) = (x, \vec{y}_0)$, which is in S . Assume, then, that $(x, \vec{Y}_m(x)) \in S$ for some m and each x in I . Using (9.47) and (9.46) we obtain

$$\|\vec{Y}_{m+1}(x) - \vec{y}_0\| \leq \left| \int_{x_0}^x \|\vec{f}[t, \vec{Y}_m(t)]\| dt \right| \leq M \left| \int_{x_0}^x dt \right| = M |x - x_0|.$$

Since $|x - x_0| \leq c$ for x in I , this implies that

$$\|\vec{Y}_{m+1}(x) - \vec{y}_0\| \leq Mc \leq b,$$

which shows that $(x, \vec{Y}_{m+1}(x)) \in S$ for each x in I . Therefore the recursion formula is meaningful for every $m \geq 0$ and every x in I .

The convergence of the sequence $\{\vec{Y}_n(x)\}$ is now established exactly as in the proof of Theorem 9-3, where limits are defined component by component. In place of inequality (9.26) we use

$$\|\vec{Y}_{m+1}(x) - \vec{Y}_m(x)\| \leq \frac{MK^m |x - x_0|^{m+1}}{(m+1)!} \leq \frac{MK^m c^{m+1}}{(m+1)!}.$$

This is proved by induction in exactly the same way we proved (9.26). We then define the limit function $\vec{Y}$ by the equation

$$\vec{Y}(x) = \lim_{m \rightarrow \infty} \vec{Y}_m(x)$$

for each x in I , and verify that it satisfies the integral equation

$$\vec{Y}(x) = \vec{y}_0 + \int_{x_0}^x \vec{f}[t, \vec{Y}(t)] dt$$

exactly as in the one-dimensional case. This proves the existence theorem. The uniqueness theorem may then be proved by the same method used in Section 9.8. Instead of (9.43) we use the inequality

$$\|\vec{Z}(x) - \vec{Y}_m(x)\| \leq \frac{bK^m |x - x_0|^m}{m!} \leq \frac{bK^m c^m}{m!},$$

which is easily verified by induction.

There is, of course, a corresponding extension of Theorem 9-4, which may be stated as follows:

9-7 THEOREM. Assume the components of $\vec{f}$ are continuous on a set S' in $(n+1)$ -space of the form

$$S' = \{(x, \vec{y}) \mid x_0 - a_1 \leq x \leq x_0 + a_2, \vec{y} \text{ arbitrary}\}.$$

Assume that $\vec{f}$ satisfies a Lipschitz condition on S' and that $\vec{f}$ is bounded on every subset S_b of S' of the form

$$S_b = \{(x, \vec{y}) \mid x_0 - a_1 \leq x \leq x_0 + a_2, \|\vec{y} - \vec{y}_0\| \leq b\}.$$

Then there exists a function $\vec{y} = \vec{Y}(x)$ defined on the open interval $J = (x_0 - a_1, x_0 + a_2)$ which satisfies the differential equation $\vec{y}' = \vec{f}(x, \vec{y})$ on J with $\vec{Y}(x_0) = \vec{y}_0$. Moreover, there is exactly one such solution whose graph over J lies in a set of the type S_C for some $C > 0$.

The proof of Theorem 9-4 may easily be adapted to prove Theorem 9-7. Details are left to the reader.

9.10 Existence and uniqueness theorems for linear equations of order n

Consider a linear differential equation of order n ,

$$(9.49) \quad y^{(n)} + P_1(x)y^{(n-1)} + \cdots + P_n(x)y' = R(x),$$

where the functions $P_1, P_2, \dots, P_n$, and R are continuous on an open interval I , which may be finite or infinite. Let x_0 be a point in I and let $k_0, k_1, \dots, k_{n-1}$ be n given real numbers. We wish to prove that there is one and only one function $y = \phi(x)$ which is a solution of (9.49) on I and which satisfies the initial conditions

$$(9.50) \quad \phi(x_0) = k_0, \quad \phi'(x_0) = k_1, \quad \dots, \quad \phi^{(n-1)}(x_0) = k_{n-1}.$$

The differential equation in (9.49) is equivalent to the system

$$(9.51) \quad y_1' = y_2, \quad y_2' = y_3, \quad \dots, \quad y_{n-1}' = y_n, \quad y_n' = R(x) - L(y_1, y_2, \dots, y_n),$$

where

$$L(y_1, y_2, \dots, y_n) = P_n(x)y_1 + P_{n-1}(x)y_2 + \cdots + P_1(x)y_n.$$

We may express this as a first-order vector differential equation,

$$(9.52) \quad \vec{y}' = \vec{f}(x, \vec{y}),$$

by writing $\vec{y} = (y_1, y_2, \dots, y_n)$ and

$$\vec{f}(x, \vec{y}) = (y_2, y_3, \dots, y_n, R(x) - L(y_1, y_2, \dots, y_n)).$$

Let $J_0 = [x_0 - a_1, x_0 + a_2]$ be any finite closed subinterval of I containing the point x_0 . Let S be the set in $(n+1)$ -space given by

$$S = \{(x, \vec{y}) \mid x \in J_0, \vec{y} \text{ arbitrary}\}.$$

Each component of $\vec{f}$ is continuous on S . We show next that $\vec{f}$ satisfies a Lipschitz condition on S . For any two points $(x, \vec{y})$ and $(x, \vec{z})$ in S we have

$$\vec{f}(x, \vec{y}) - \vec{f}(x, \vec{z}) = (y_2 - z_2, \dots, y_n - z_n, -P_n(x)(y_1 - z_1) - \cdots - P_1(x)(y_n - z_n)).$$

Therefore

$$\begin{aligned} \|\vec{f}(x, \vec{y}) - \vec{f}(x, \vec{z})\| &\leq |y_2 - z_2| + \cdots + |y_n - z_n| + |P_n(x)| |y_1 - z_1| \\ &\quad + \cdots + |P_1(x)| |y_n - z_n|. \end{aligned}$$

Now each of the functions $P_1, P_2, \dots, P_n$ is continuous on J_0 . Since J_0 is a finite closed interval, each P_k is bounded on J_0 , say

$$|P_k(x)| \leq A_k$$

for all x in J_0 . If $A = \max \{A_1, A_2, \dots, A_n\}$, the inequality for $\|\vec{f}(x, \vec{y}) - \vec{f}(x, \vec{z})\|$ becomes

$$\begin{aligned} \|\vec{f}(x, \vec{y}) - \vec{f}(x, \vec{z})\| &\leq A |y_1 - z_1| + (A + 1) |y_2 - z_2| + \cdots + (A + 1) |y_n - z_n| \\ &\leq (A + 1) \|\vec{y} - \vec{z}\|. \end{aligned}$$

Therefore $\vec{f}$ satisfies a Lipschitz condition on S , with Lipschitz constant $A + 1$.

Next we show that $\vec{f}$ is bounded on every subset S_b of S of the form

$$S_b = \{(x, \vec{y}) \mid x \in J_0, \|\vec{y} - \vec{y}_0\| \leq b\},$$

where $\vec{y}_0 = (k_0, k_1, \dots, k_{n-1})$. For any point $(x, \vec{y})$ in S_b we have

$$\begin{aligned} \|\vec{f}(x, \vec{y})\| &\leq |y_2| + \dots + |y_n| + |R(x)| + |P_n(x)| |y_1| + \dots + |P_1(x)| |y_n| \\ &\leq |R(x)| + (A + 1)(|y_1| + \dots + |y_n|) = |R(x)| + (A + 1) \|\vec{y}\|. \end{aligned}$$

Now $\|\vec{y}\| = \|\vec{y} - \vec{y}_0 + \vec{y}_0\| \leq \|\vec{y} - \vec{y}_0\| + \|\vec{y}_0\| \leq b + \|\vec{y}_0\|$, since $(x, \vec{y}) \in S$. Therefore, if M_R denotes the maximum value of $|R(x)|$ for x in J_0 , the foregoing inequality for $\|\vec{f}(x, \vec{y})\|$ gives us

$$\|\vec{f}(x, \vec{y})\| \leq M_R + (A + 1)(b + \|\vec{y}_0\|).$$

This shows that $\vec{f}$ is bounded on S_b . Thus, $\vec{f}$ satisfies all the hypotheses of Theorem 9-7. Consequently, a solution $\vec{y} = \vec{Y}(x)$ of the initial-value problem

$$\vec{y}' = \vec{f}(x, \vec{y}), \quad \text{with} \quad \vec{Y}(x_0) = \vec{y}_0,$$

exists on the open interval $J = (x_0 - a_1, x_0 + a_2)$. Moreover, there is at most one such solution whose graph over J lies in a set of the form S_C for some $C > 0$. If we denote the first component of $\vec{Y}$ by ϕ_0 and use the fact that the differential equation in (9.52) is equivalent to the system in (9.51), we see that ϕ_0 is a solution of the linear equation (9.49) on J which satisfies the initial conditions (9.50). Theorem 9-7 implies that there is at most one such solution ϕ_0 that is bounded on J .

Now we must show that a solution ϕ exists over the full interval I . Choose any point x_1 in I and let $J_0 = [x_0 - a_1, x_0 + a_2]$ be any finite closed subinterval of I containing both x_0 and x_1 as interior points. We have just shown that a solution ϕ_0 exists on the open interval $J = (x_0 - a_1, x_0 + a_2)$. We now define $\phi(x_1) = \phi_0(x_1)$. For $\phi(x_1)$ to be well defined we must know that $\phi(x_1)$ depends only on x_1 and not on the choice of the interval J_0 . Let J_1 be any other finite closed interval containing both x_0 and x_1 as interior points, and let ϕ_1 be the corresponding solution defined on the interior of J_1 . Then there is a finite closed interval J_2 containing both x_0 and x_1 as interior points and lying in the interior of both J_0 and J_1 . On J_2 both solutions ϕ_0 and ϕ_1 are bounded. Therefore, by the uniqueness theorem, they must be identical on J_2 . In particular, $\phi_0(x_1) = \phi_1(x_1)$. This shows that $\phi(x_1)$ depends only on x_1 and not on the choice of the interval J_0 . Since x_1 is an arbitrary point of I we have defined the function ϕ everywhere on I . This ϕ clearly satisfies the differential equation (9.49) on I and the initial conditions (9.50). Moreover, it is unique, because any other solution ψ on I must agree with ϕ_0 on any finite closed subinterval of I and therefore must agree with ϕ on such a subinterval. Therefore ψ agrees with ϕ everywhere on I .

9.11 Contraction operators

The basic idea underlying the method of successive approximations can be used not only to establish existence theorems for solutions of differential equations but also to solve many other important problems in analysis. In this section we show how the method of successive approximations can be reformulated in a more general setting that greatly increases the scope of its applications. The use of the method in other problems is illustrated in Section 9.12.

We have seen that the method of successive approximations involves the construction of a sequence of functions $\{Y_n\}$, defined and continuous on some interval I , according to the recursion formula

$$(9.53) \quad Y_{n+1}(x) = y_0 + \int_{x_0}^x f[t, Y_n(t)] dt$$

for $n = 0, 1, 2, \dots$, with $Y_0(x) = y_0$. The integral which appears in this formula can be used to define an operator A which converts certain functions ϕ defined on I into new functions $A(\phi)$ defined on I by the equation

$$A(\phi)(x) = y_0 + \int_{x_0}^x f[t, \phi(t)] dt.$$

In operator notation, the recursion formula (9.53) becomes simply

$$Y_{n+1} = A(Y_n).$$

In the proof of Theorem 9-3 we found that the solution Y of the initial-value problem $y' = f(x, y)$, with $Y(x_0) = y_0$, satisfies the integral equation

$$Y(x) = y_0 + \int_{x_0}^x f[t, Y(t)] dt.$$

In operator notation this states that $Y = A(Y)$. In other words, the solution Y remains unaltered by the operator A . Such a function Y is called a *fixed point* of the operator A . Many important problems in analysis can be formulated so their solution depends on the existence of a fixed point for some operator. Therefore, it is worth while to try to discover properties of operators that guarantee the existence of a fixed point. We turn now to a systematic treatment of this problem.

The discussion is considerably simplified by the notation and terminology of function spaces. We begin with the collection C of all real-valued functions defined and continuous on a finite closed interval $[a, b]$. (Although the discussion may easily be extended to more general collections of functions defined on more general sets, we shall discuss only the space C .) The set C is a linear space. That is, if $\phi \in C$ and $\psi \in C$, then $(\phi + \psi) \in C$ and $c\phi \in C$ for every constant c . The space becomes a normed linear space if we define $\|\phi\|$ for each ϕ in C by the equation

$$\|\phi\| = \max_{a \leq x \leq b} |\phi(x)|,$$

where the symbol on the right stands for the maximum value of $|\phi(x)|$ for x in $[a, b]$. This is the maximum norm discussed in Chapter 8 in connection with polynomial approximation. It has the fundamental properties

- (a) $\|\phi\| \geq 0$ for every ϕ in C ,
- (b) $\|c\phi\| = |c| \|\phi\|$ for each ϕ in C and each real number c ,
- (c) $\|\phi + \psi\| \leq \|\phi\| + \|\psi\|$ for all ϕ and ψ in C ,
- (d) $\|\phi\| = 0$ implies $\phi = 0$.

If ϕ and ψ are in C we shall refer to $\|\phi - \psi\|$ as the *distance* between ϕ and ψ .

We consider now an operator A whose domain and range are both equal to the space C . That is, if $\phi \in C$ then $A(\phi)$ is again a function in C . There are many operators of this

kind that can be defined on C . The following formulas illustrate a few simple examples. In each case ϕ is an arbitrary function in C and $A(\phi)(x)$ is defined for each x in $[a, b]$ by the formula given:

$$A(\phi)(x) = \lambda\phi(x), \quad \text{where } \lambda \text{ is a fixed real number,}$$

$$A(\phi)(x) = \int_{x_0}^x \phi(t) dt, \quad \text{where } x_0 \text{ is a fixed point in } [a, b],$$

$$(9.54) \quad A(\phi)(x) = y_0 + \int_{x_0}^x f[t, \phi(t)] dt,$$

$$(9.55) \quad A(\phi)(x) = \psi(x) + \lambda \int_a^b K(x, t) \phi(t) dt.$$

In Equation (9.54), y_0 is a constant and the function f is assumed to be defined and continuous on the rectangular strip $[a, b] \times (-\infty, +\infty)$. This ensures that $A(\phi) \in C$ for each ϕ in C . In Equation (9.55), ψ is a fixed function in C and the function K is defined and continuous on the square $S = [a, b] \times [a, b]$. It can be shown that $A(\phi) \in C$ whenever $\phi \in C$.

Now let A be any operator such that $A(\phi) \in C$ whenever $\phi \in C$. We shall be interested in those operators A for which the distance $\|A(\phi) - A(\psi)\|$ is less than a fixed constant multiple of $\|\phi - \psi\|$. These are called *contraction operators*; they are defined as follows:

DEFINITION OF A CONTRACTION OPERATOR. An operator A is called a contraction operator if there is a constant α satisfying $0 \leq \alpha < 1$ such that for every pair of functions ϕ and ψ in C we have

$$(9.56) \quad \|A(\phi) - A(\psi)\| \leq \alpha \|\phi - \psi\|.$$

The constant α is called a contraction constant for A .

Note. The inequality (9.56) holds if, and only if, we have

$$|A(\phi)(x) - A(\psi)(x)| \leq \alpha \|\phi - \psi\|$$

for every x in $[a, b]$.

Example 1. Let A be the operator defined on C by the equation $A(\phi)(x) = \lambda\phi(x)$ for each x in $[a, b]$, where λ is a constant. Since

$$|A(\phi)(x) - A(\psi)(x)| = |\lambda| |\phi(x) - \psi(x)|$$

we have $\|A(\phi) - A(\psi)\| = |\lambda| \|\phi - \psi\|$. Therefore this operator is a contraction operator if, and only if, $|\lambda| < 1$, in which case $|\lambda|$ may be used as a contraction constant.

Example 2. Let A be the operator defined by (9.54), and assume that f satisfies a Lipschitz condition of the form

$$|f(x, y) - f(x, z)| \leq K |y - z|$$

on the rectangular strip $[a, b] \times (-\infty, +\infty)$. If $K(b - a) < 1$ we can easily show that A is a contraction operator with contraction constant $\alpha = K(b - a)$. In fact, for every x in $[a, b]$ we have

$$\begin{aligned}
 |A(\phi)(x) - A(\psi)(x)| &= \left| \int_{x_0}^x \{f[t, \phi(t)] - f[t, \psi(t)]\} dt \right| \leq K \left| \int_{x_0}^x |\phi(t) - \psi(t)| dt \right| \\
 &\leq K \|\phi - \psi\| \left| \int_{x_0}^x dt \right| \leq K(b - a) \|\phi - \psi\|.
 \end{aligned}$$

This shows that A is a contraction operator with contraction constant $\alpha = K(b - a)$.

The operator A of Example 2 was used in the method of successive approximations to construct a solution Y of the initial-value problem $y' = f(x, y)$, with $y = y_0$ when $x = x_0$. We have already pointed out that this solution is a fixed point of the operator A . The next theorem shows that every contraction operator has a fixed point.

9-7 THEOREM. *Fixed-point theorem for contraction operators.* Let A be a contraction operator defined on the space C of continuous functions on $[a, b]$. Then there exists one and only one function ϕ in C such that

$$(9.57) \quad A(\phi) = \phi.$$

Proof. Let ϕ_0 be any function in C and define a sequence of functions $\{\phi_n\}$ by the recursion formula

$$\phi_{n+1} = A(\phi_n) \quad \text{for } n = 0, 1, 2, \dots$$

We shall imitate the method of successive approximations and show that the series

$$(9.58) \quad \phi_0(x) + \sum_{k=0}^{\infty} [\phi_{k+1}(x) - \phi_k(x)]$$

converges for each x in $[a, b]$. Then we shall prove that the sum of this series defines a function ϕ in C satisfying (9.57).

The convergence of the series in (9.58) will be established by comparing it with the convergent geometric series

$$M \sum_{k=0}^{\infty} \alpha^k,$$

where $M = \|\phi_0\| + \|\phi_1\|$, and α is a contraction constant for A . The comparison is provided by the inequality

$$(9.59) \quad |\phi_{k+1}(x) - \phi_k(x)| \leq M\alpha^k$$

which holds for every x in $[a, b]$ and every $k \geq 1$. To prove (9.59) we note that

$$|\phi_{k+1}(x) - \phi_k(x)| = |A(\phi_k)(x) - A(\phi_{k-1})(x)| \leq \alpha \|\phi_k - \phi_{k-1}\|.$$

Therefore the inequality in (9.59) will be proved if we show that

$$(9.60) \quad \|\phi_k - \phi_{k-1}\| \leq M\alpha^{k-1}$$

for every $k \geq 1$. We now prove (9.60) by induction. For $k = 1$ we have

$$\|\phi_0 - \phi_1\| \leq \|\phi_0\| + \|\phi_1\| = M,$$

which is the same as (9.60). To prove that (9.60) holds for $k + 1$ if it holds for k we note that

$$|\phi_{k+1}(x) - \phi_k(x)| = |A(\phi_k)(x) - A(\phi_{k-1})(x)| \leq \alpha \|\phi_k - \phi_{k-1}\| \leq M\alpha^k.$$

Since this is valid for each x in $[a, b]$ we must also have

$$\|\phi_{k+1} - \phi_k\| \leq M\alpha^k.$$

This proves (9.60) by induction. Therefore the series in (9.58) converges for each x in $[a, b]$. If we let $\phi(x)$ denote its sum we have

$$\phi(x) = \lim_{n \rightarrow \infty} \phi_n(x)$$

where

$$\phi_n(x) = \phi_0(x) + \sum_{k=0}^{n-1} [\phi_{k+1}(x) - \phi_k(x)].$$

Now we must show that $\phi \in C$ and that $A(\phi) = \phi$.

To show that ϕ is continuous on $[a, b]$ we choose an arbitrary point x_1 in $[a, b]$ and show that ϕ is continuous at x_1 . For this purpose we let ϵ be a given positive number and show that there is a neighborhood $N(x_1)$ such that

$$(9.61) \quad |\phi(x) - \phi(x_1)| < \epsilon \quad \text{whenever } x \in N(x_1) \cap [a, b].$$

The idea of the proof is to choose a suitable approximating function ϕ_N and write

$$\phi(x) - \phi(x_1) = \phi(x) - \phi_N(x) + \phi_N(x) - \phi_N(x_1) + \phi_N(x_1) - \phi(x_1).$$

This gives us the inequality

$$(9.62) \quad |\phi(x) - \phi(x_1)| \leq |\phi(x) - \phi_N(x)| + |\phi_N(x) - \phi_N(x_1)| + |\phi_N(x_1) - \phi(x_1)|.$$

We show next that ϕ_N can be chosen so that each term on the right is less than $\epsilon/3$ if x is sufficiently close to x_1 .

First we note that for every x in $[a, b]$ we have

$$\phi(x) - \phi_n(x) = \sum_{k=n}^{\infty} [\phi_{k+1}(x) - \phi_k(x)].$$

Therefore, by (9.59), we have the inequality

$$(9.63) \quad |\phi(x) - \phi_n(x)| \leq \sum_{k=n}^{\infty} |\phi_{k+1}(x) - \phi_k(x)| \leq M \sum_{k=n}^{\infty} \alpha^k.$$

Since the series $\sum \alpha^k$ converges, for the given ϵ there is an N (depending only on ϵ) such that

$$M \sum_{k=n}^{\infty} \alpha^k < \frac{\epsilon}{3} \quad \text{whenever } n \geq N.$$

Therefore (9.63) shows that $|\phi(x) - \phi_n(x)| < \epsilon/3$ for every x in $[a, b]$, provided only that $n \geq N$. In particular, if we take $n = N$ we see that the first and third terms on the right of (9.62) have a sum less than $2\epsilon/3$ for any choice of x in $[a, b]$, so (9.62) becomes

$$(9.64) \quad |\phi(x) - \phi(x_1)| < \frac{2\epsilon}{3} + |\phi_N(x) - \phi_N(x_1)|.$$

Now we invoke the continuity of ϕ_N at x_1 to find a neighborhood $N(x_1)$ such that

$$|\phi_N(x) - \phi_N(x_1)| < \frac{\epsilon}{3} \quad \text{whenever } x \in N(x_1) \cap [a, b].$$

Using this inequality along with (9.64) we see that (9.61) is satisfied for this choice of $N(x_1)$. This proves that $\phi \in C$.

Next we prove that $A(\phi) = \phi$. From the contraction property of A we have

$$(9.65) \quad |A(\phi)(x) - \phi_{n+1}(x)| = |A(\phi)(x) - A(\phi_n)(x)| \leq \alpha \|\phi - \phi_n\|$$

for each x in $[a, b]$. The inequality in (9.63) shows that

$$\|\phi - \phi_n\| \leq M \sum_{k=n}^{\infty} \alpha^k.$$

Therefore $\|\phi - \phi_n\| \rightarrow 0$ as $n \rightarrow \infty$. Letting $n \rightarrow \infty$ in (9.65) we obtain

$$\lim_{n \rightarrow \infty} |A(\phi)(x) - \phi_{n+1}(x)| = 0.$$

That is, we have

$$A(\phi)(x) = \lim_{n \rightarrow \infty} \phi_{n+1}(x) = \phi(x)$$

for every x in $[a, b]$. But this means that $A(\phi) = \phi$.

Finally, we prove that the fixed point ϕ is unique. Suppose ψ is another function in C such that $A(\psi) = \psi$. Then we have

$$\|\phi - \psi\| = \|A(\phi) - A(\psi)\| \leq \alpha \|\phi - \psi\|.$$

This gives us $(1 - \alpha) \|\phi - \psi\| \leq 0$. Since $\alpha < 1$ we may divide by $1 - \alpha$ to obtain the inequality $\|\phi - \psi\| \leq 0$. But since we also have $\|\phi - \psi\| \geq 0$ this means that $\|\phi - \psi\| = 0$, and hence $\phi - \psi = 0$. The proof of the fixed-point theorem is now complete.

9.12 Applications of the fixed-point theorem

In this section we discuss two important applications of the fixed-point theorem for contraction mappings. First we show how it can be used to give a simple proof of a special case of the implicit-function theorem referred to in Chapter 4 (Section 4.16).

9-8 THEOREM. *An implicit-function theorem.* Let f be defined and continuous on a rectangular strip of the form

$$R = [a, b] \times (-\infty, +\infty).$$

Assume that the partial derivative $D_2f(x, y)$ exists and satisfies a relation of the form

$$(9.66) \quad 0 < m \leq D_2f(x, y) \leq M$$

for all (x, y) in R , where m and M are constants with $m \leq M$. Then there exists one and only one function $y = Y(x)$, continuous on $[a, b]$, such that

$$(9.67) \quad f[x, Y(x)] = 0$$

for all x in $[a, b]$. In other words, the equation $f(x, y) = 0$ serves to define y implicitly as a function of x on $[a, b]$.

Proof. Let C be the space of continuous functions on $[a, b]$, and let A denote the operator defined on C by the equation

$$A(\phi)(x) = \phi(x) - \frac{1}{M}f[x, \phi(x)]$$

for each x in $[a, b]$. Here M is the positive constant in (9.66). The function $A(\phi) \in C$ whenever $\phi \in C$. We shall prove that A is a contraction mapping. Once we know this it follows that there is a unique fixed point Y in C . For this function Y we have

$$Y(x) = Y(x) - \frac{1}{M}f[x, Y(x)]$$

for every x in $[a, b]$, which is the same as (9.67).

To show that A is a contraction mapping we consider the difference

$$(9.68) \quad A(\phi)(x) - A(\psi)(x) = \phi(x) - \psi(x) - \frac{f[x, \phi(x)] - f[x, \psi(x)]}{M}.$$

By the mean-value theorem for derivatives we have

$$f[x, \phi(x)] - f[x, \psi(x)] = D_2f[x, z(x)] [\phi(x) - \psi(x)],$$

where $z(x)$ lies between $\phi(x)$ and $\psi(x)$. Therefore (9.68) gives us

$$(9.69) \quad A(\phi)(x) - A(\psi)(x) = [\phi(x) - \psi(x)] \left(1 - \frac{D_2f[x, z(x)]}{M} \right).$$

The hypothesis (9.66) implies that

$$0 \leq 1 - \frac{D_2f[x, z(x)]}{M} \leq 1 - \frac{m}{M}.$$

Therefore (9.69) gives us the inequality

$$(9.70) \quad |A(\phi)(x) - A(\psi)(x)| \leq |\phi(x) - \psi(x)| \left(1 - \frac{m}{M} \right) \leq \alpha \|\phi - \psi\|,$$

where $\alpha = 1 - m/M$. Since $0 < m \leq M$, we have $0 \leq \alpha < 1$. Inequality (9.70) is valid for every x in $[a, b]$. Hence A is a contraction mapping. This completes the proof.

The next application of the fixed-point theorem establishes an existence theorem for the integral equation

$$(9.71) \quad \phi(x) = \psi(x) + \lambda \int_a^b K(x, t)\phi(t) dt.$$

Here K is a function defined and continuous on the square $[a, b] \times [a, b]$; ψ is a fixed function, continuous on $[a, b]$; and λ is a constant. The functions K and ψ are assumed to be known and the function ϕ is unknown. A solution of the integral equation is any continuous function ϕ that satisfies the equation on $[a, b]$.

9-9 THEOREM. *An existence theorem for integral equations.* Let K be continuous on the square $S = [a, b] \times [a, b]$, and assume that

$$(9.72) \quad |K(x, y)| \leq M,$$

for all (x, y) in S , where $M > 0$. Let ψ be a given function, continuous on $[a, b]$, and let λ be a real number satisfying the inequality

$$(9.73) \quad |\lambda| < \frac{1}{M(b-a)}.$$

Then there exists one and only one function ϕ , continuous on $[a, b]$, that satisfies the integral equation (9.71).

Proof. Let C be the space of continuous functions on $[a, b]$, and define an operator A on C by the equation

$$A(\phi)(x) = \psi(x) + \lambda \int_a^b K(x, t) \phi(t) dt.$$

It can be shown† that $A(\phi) \in C$ whenever $\phi \in C$. We shall prove that A is a contraction mapping.

Take any two functions ϕ_1 and ϕ_2 in C and consider the difference

$$A(\phi_1)(x) - A(\phi_2)(x) = \lambda \int_a^b K(x, t) [\phi_1(t) - \phi_2(t)] dt.$$

Using the inequality (9.72) we may write

$$|A(\phi_1)(x) - A(\phi_2)(x)| \leq |\lambda| M(b-a) \|\phi_1 - \phi_2\| = \alpha \|\phi_1 - \phi_2\|,$$

where $\alpha = |\lambda| M(b-a)$. Because of (9.73) we have $0 \leq \alpha < 1$, so A is a contraction mapping with contraction constant α . Therefore A has a unique fixed point ϕ in C . This function ϕ satisfies (9.71).

† See Theorem 9-35 of the author's *Mathematical Analysis*, Addison-Wesley Publishing Co., Reading, Massachusetts, 1957.

ANSWERS TO EXERCISES

Chapter 1

1.4 Exercises (page 4)

1. $A = \{1\}$; $B = \{1, 2, 3, 4, \dots\}$; $C = \{2, 4, 6, 8, \dots\}$; $D = \{1\}$; $E = \{1, -17\}$; $F = \{1, -17, -8 + \sqrt{47}, -8 - \sqrt{47}\}$
2. $A \subseteq A$; $A \subseteq B$; $A \subseteq D$; $A \subseteq E$; $A \subseteq F$; $B \subseteq B$; $C \subseteq B$; $C \subseteq C$; $D \subseteq A$; $D \subseteq B$; $D \subseteq D$; $D \subseteq E$; $D \subseteq F$; $E \subseteq E$; $E \subseteq F$; $F \subseteq F$. (Not counting "proper" inclusions)
5. (c) $A \subset C$
(d) Yes
(e) No
7. (a) False
(b) True
(c) True
(d) False
(e) True
(f) False
8. $\phi, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, S$
10. (a) False
(b) False
(c) False
(d) False
(e) True
(f) False
(g) True
(h) False
(i) True
(j) False

1.6 Exercises (page 7)

18. $A = X \cap Y' \cap Z'$, $B = X \cap Y \cap Z'$, $D = X' \cap Y \cap Z'$, $E = X' \cap Y \cap Z$, $F = X' \cap Y' \cap Z$
19. (a) (ii) is right
(b) $C \subseteq A$
21. $\phi, \{3\}, \{1, 3\}, \{2, 3\}, S$
22. (a) Yes
(b) $\{2, 3\}, S$

1.8 Exercises (page 10)

$$5. A_1 \cup A_2 \cup A_3 = (A_1 \cap A'_2 \cap A'_3) \cup (A_2 \cap A'_3) \cup A_3; \bigcup_{k=1}^n A_k = \bigcup_{k=1}^{n-1} (A_k \cap \bigcap_{j=k+1}^n A'_j) \cup A_n$$

6.	(i)	(ii)	(iii)	(iv)	(v)
(a)	$A' \cap B'$	$A \cap B'$	$A \cup B$	$(A \cap B') \cup (A' \cap B)$	$A' \cup B'$
(b)	500	200	500	300	800

1.10 Exercises (page 15)

2. $\mathcal{R} = \{\phi, A_1, A_2, A_1 \cup A_2\}$
 $\mathcal{B} = \{\phi, A_1, A_2, A_1 \cup A_2, A'_1, A'_2, A'_1 \cap A'_2, S\}$
3. $\mathcal{R} = \{\phi, A_1, A_2, A_3, A_1 \cup A_2, A_2 \cup A_3, A_1 \cup A_3, A_1 \cup A_2 \cup A_3\}$
 $\mathcal{B} = \{\phi, A_1, A_2, A_3, A_1 \cup A_2, A_2 \cup A_3, A_1 \cup A_3, A_1 \cup A_2 \cup A_3, A'_1, A'_2, A'_3, A'_1 \cap A'_2, A'_2 \cap A'_3, A'_1 \cap A'_3, A'_1 \cap A'_2 \cap A'_3, S\}$ (if $n > 3$)

1.13 Exercises (page 21)

1. $\{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$
2. 1326
3. 54
4. $\{H, T\} \times \{H, T\} \times \{1, 2, 3, 4, 5, 6\}$; 24 outcomes
5. $52!/(13!)^4$
6. 36
- (a) 18
- (b) 12
- (c) 24
7. (a) $13 \cdot 12 \cdot 11 \cdot 72 = 123552$ (not including triplets or quadruplets)
- (b) 5148
- (c) 36 (not including 10JQKA)
- (d) 4
8. (a) $\frac{2 \cdot 98!}{(49!)^2}$
- (b) $\frac{98!}{48! \cdot 50!}$
9. 16
10. n^k

1.15 Exercises (page 25)

1. (a) $f(k) = 2k$
- (b) $f(k) = 3^k$
- (c) $f(k) = p_k$, where p_k is the k th prime ≥ 2
- (d) one such function is $f(k) = (g(k), h(k))$, where

$$g(k) = \frac{m^2(k) + 3m(k)}{2} - k + 2, \quad h(k) = k - \frac{m^2(k) + m(k)}{2},$$

and

$$m(k) = \left\lceil \frac{\sqrt{8k - 7} - 1}{2} \right\rceil$$

where $[x]$ denotes the greatest integer $\leq x$

- (e) $f(k) = 2^{g(k)} 3^{h(k)}$, where $g(k)$ and $h(k)$ are as defined in part (d)

1.18 Exercises (page 30)

1. $A \subseteq B'$
2. $x \in A' \cap B' \cap C'$
3. $x \in A \cap B' \cap C'$
4. $x \in A \cup B \cup C$
5. $x \in (A \cap B' \cap C') \cup (A' \cap B \cap C') \cup (A' \cap B' \cap C)$
6. $x \in (A' \cap B') \cup (B' \cap C') \cup (A' \cap C')$
7. $x \in (A \cap B) \cup (A \cap C) \cup (B \cap C)$
8. $x \in (A \cap B \cap C') \cup (A \cap B' \cap C) \cup (A' \cap B \cap C)$
9. $x \in (A \cap B \cap C)'$
10. $x \in A \cap C \cap B'$
11. $x \in A \cap B \cap C$
12. $x \in A \cup B \cup C$
15. (a) $1 - a$
 (b) $1 - b$
 (c) $a + b - c$
 (d) $1 - c$
 (e) $1 - a + c$
 (f) $a - c$

1.20 Exercises (page 34)

2. (a) $5/14$
 (b) $45/91$
 (c) $10/91$
 (d) $36/91$
3. (a) $23/36$
 (b) $1/6$
 (c) $1/3$
 (d) $5/36$
 (e) $13/36$
 (f) $2/3$
4. $1/4$
5. (a) $4 \binom{13}{5} / \binom{52}{5}$
 (b) $40 / \binom{52}{5}$
 (c) $4 / \binom{52}{5}$
6. $9/47$
7. $8/47$
8. $4/47$
9. $1/3$
11. $P_0 = 1 - P(A) - P(B) + P(A \cap B)$, $P_1 = P(A) + P(B) - 2P(A \cap B)$, $P_2 = P(A \cap B)$
13. (a) 5 to 9
 (b) 45 to 46
 (c) 10 to 81
 (d) 36 to 55

1.23 Exercises (page 39)

2. (a) $P(A) = \frac{3}{10}$; $P(B|A) = \frac{6}{11}$; $P(A \cap B) = \frac{9}{55}$

4. $\binom{98}{48} / \left[\binom{100}{50} - \binom{98}{50} \right]$
 5. $4/11$
 6. (a) $1 - \frac{26! \cdot 34!}{21! \cdot 39!} = 1 - \frac{\binom{34}{13}}{\binom{39}{13}} = 1 - \frac{\binom{26}{5}}{\binom{39}{5}}$
 (b) $1 - \frac{\binom{34}{13} + 5\binom{34}{12}}{\binom{39}{13}}$
 (c) $3\binom{26}{5} / \left(\binom{39}{5} \binom{39}{13} \right)$
 9. $1/2$
 15. (a) $P(A) = P(B) = P(C) = \frac{1}{2}$; $P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4}$; $P(A \cap B \cap C) = 0$

1.26 Exercises (page 45)

1. (a) $P(H, H) = p_1 p_2$; $P(H, T) = p_1(1 - p_2)$; $P(T, H) = (1 - p_1)p_2$; $P(T, T) = (1 - p_1)(1 - p_2)$
 (b) Yes
 (c) No
 (d) H_1 and H_2 , H_1 and T_2 , H_2 and T_1 , T_1 and T_2
 2. (a) $319/512$
 (b) $11/1024$
 (c) 6
 3. $\binom{10}{3} 5^7 / 6^{10} = \frac{390625}{2519424}$
 4. (a) $5/16$
 (b) $1/2$
 (c) $3/16$
 5. (a) $(5!)^2 / 10! = 1/252$
 (b) $1/2$
 6. (a) $36p^{10} - 80p^9 + 45p^8$
 (b) $7/128$
 7. It is advantageous to bet even money
 8. $\binom{n}{k} \frac{w^k b^{n-k}}{(w+b)^n}$
 9. $\binom{8}{3} 17^5 / 18^8 = \frac{9938999}{1377495072}$
 10. $193/512$
 11. $1 - (19/20)^{10} = 0.4013$
 12. $193/512$

1.27 Miscellaneous exercises on probability (page 47)

1. $1/6$
 2. (a) $2/19$
 (b) $1/10$
 3. (a) $1/4$
 (b) $7/12$
 4. (a) $15/34$
 (b) $13/51$
 (c) $13/165$

5. (a) $5/9$
 (b) $2/9$
 (c) $2/9$
 (d) No
6. $p^3 + 6p\left(\frac{1-p}{2}\right)^2$
7. $np(1-p)^{n-1} + np^{n-1}(1-p)$
8. $\frac{n}{2^{n-1}}\left(1 - \frac{n}{2^{n-1}}\right)^{n-1}$

Chapter 2

2.9 Exercises (page 63)

1. $1/3$
2. $2\sqrt{3} - \frac{38}{3}$
3. $\pi^2/4$
4. 2π
5. 6
6. $t^{-3}(e^{t^2} - e^t) + t^{-2} - t^{-1}$
8. $1/6$
9. $\pi/2$
10. $(\log 2)/6$

2.13 Exercises (page 76)

1. $-3\pi/2$
2. $\frac{3}{2} + \cos 1 + \sin 1 - \cos 2 - 2 \sin 2$
3. $e - \frac{1}{e}$
4. $\frac{7}{3} \log 2$
5. $\pi^2 - \frac{40}{9}$
6. 6
7. $80/3$
8. (a) $8/3$
 (b) 2
 (c) 320π
9. $\int_0^1 \left[\int_x^1 f(x, y) dy \right] dx$
10. $\int_0^4 \left[\int_{x/2}^{\sqrt{x}} f(x, y) dy \right] dx$
11. $\int_1^2 \left[\int_1^{y^2} f(x, y) dx \right] dy$
12. $\int_0^1 \left[\int_{2-y}^{1+\sqrt{1-y^2}} f(x, y) dx \right] dy$

13. $\int_{-1}^0 \left[\int_{-\sqrt{4y+4}}^{\sqrt{4y+4}} f(x, y) dx \right] dy + \int_0^8 \left[\int_{-\sqrt{4y+4}}^{2-y} f(x, y) dx \right] dy$
14. $\int_0^1 \left[\int_{e^y}^e f(x, y) dx \right] dy$
15. $\int_{-1}^0 \left[\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) dx \right] dy + \int_0^1 \left[\int_{-\sqrt{1-y}}^{\sqrt{1-y}} f(x, y) dx \right] dy$
16. $\int_0^1 \left[\int_{y^{1/2}}^{y^{1/3}} f(x, y) dx \right] dy$
17. $\int_{-1}^0 \left[\int_{-2\arcsin y}^{\pi} f(x, y) dx \right] dy + \int_0^1 \left[\int_{\arcsin y}^{\pi - \arcsin y} f(x, y) dx \right] dy$
18. $\int_{-2}^0 \left[\int_{2x+4}^{4-x^2} f(x, y) dy \right] dx$
19. $\int_0^1 \left[\int_x^{2-x} (x^2 + y^2) dy \right] dx = \frac{4}{3}$
20. $y = 0, y = x \tan c, x^2 + y^2 = a^2, x^2 + y^2 = b^2$
21. (a) $\int_1^8 \left[\int_{y^{1/3}}^y f(x, y) dx \right] dy$
(b) $4e^8 + 2e/3$
22. $m = 2; n = 1$

2.15 Exercises (page 81)

1. $\bar{x} = -\frac{1}{2}, \bar{y} = \frac{8}{5}$
2. $\bar{x} = 1, \bar{y} = 0$
3. $\bar{x} = \frac{18}{13}, \bar{y} = \frac{50}{39}$
4. $\bar{x} = \pi/2, \bar{y} = \pi/8$
5. $\bar{x} = (\sqrt{2} + 1) \left(\frac{\pi\sqrt{2}}{4} - 1 \right) = \frac{\pi}{2} + \frac{\pi\sqrt{2}}{4} - 1 - \sqrt{2}, \bar{y} = \frac{\sqrt{2} + 1}{4}$
6. $\bar{x} = \frac{2a^2 \log a - a^2 + 1}{4(a \log a - a + 1)}, \bar{y} = \frac{a(\log a)^2}{2(a \log a - a + 1)} - 1$
7. $\bar{x} = \bar{y} = 1/5$
8. $\bar{x} = \bar{y} = 256/(315\pi)$
9. $\frac{26}{3} - \frac{15}{2} \log 3$
10. $\bar{x} = \frac{2}{3} |\overrightarrow{AB}|, \bar{y} = \frac{2}{3} |\overrightarrow{AD}|$; assuming the x - and y -axes are chosen along sides AB and AD , respectively
11. $I_x = \frac{5\pi}{12}, I_y = \frac{2\pi^3}{3} - \pi$
12. $I_x = \frac{1}{12} b^3(a - c), I_y = \frac{1}{12} b(a^3 - c^3)$
13. $I_x = I_y = (1 - 5\pi/16) r^4$
14. $I_x = I_y = 9/8$
15. $I_x = \frac{1}{64} [(4a - 1)e^{4a} - 1], I_y = \frac{1}{32} [(a^3 - 3a^2 + 6a - 6)e^{2a} + 6]$
16. $I_x = 72/105, I_y = 148/45$
19. $\frac{1}{3}h [\sqrt{2} + \log(1 + \sqrt{2})]$
20. $h^2 + \frac{1}{2}r^2$
21. (a) $(\frac{13}{6}, 1)$
(b) $(\frac{7}{3}, \frac{9}{2})$
(c) $(\frac{11}{4}, \frac{11}{4})$
(d) $(\frac{19}{8}, \frac{13}{8})$
22. $h = 2\sqrt{3}$

2.18 Exercises (page 89)

1. $\int_0^{2\pi} \left[\int_0^a f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
2. $\int_{-\pi/2}^{\pi/2} \left[\int_0^{2 \cos \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
3. $\int_0^{2\pi} \left[\int_a^b f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
4. $\int_0^{\pi/2} \left[\int_0^{g(\theta)} f(r \cos \theta, r \sin \theta) r dr \right] d\theta$, where $g(\theta) = 1/(\cos \theta + \sin \theta)$
5. $\int_0^{\pi/4} \left[\int_0^{\tan \theta \sec \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta + \int_{\pi/4}^{3\pi/4} \left[\int_0^{\csc \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
 $+ \int_{3\pi/4}^{\pi} \left[\int_0^{\tan \theta \sec \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
6. $\frac{3}{8}\pi a^4$
7. $\frac{1}{6}a^3[\sqrt{2} + \log(1 + \sqrt{2})]$
8. $\sqrt{2} - 1$
9. $\pi a^4/8$
10. $\int_0^{\pi/4} \left[\int_0^{\sec \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta + \int_{\pi/4}^{\pi/2} \left[\int_0^{\csc \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
11. $\int_{\pi/4}^{\pi/3} \left[\int_0^{2 \sec \theta} f(r) r dr \right] d\theta$
12. $\int_0^{\pi/2} \left[\int_{g(\theta)}^1 f(r \cos \theta, r \sin \theta) r dr \right] d\theta$, where $g(\theta) = 1/(\cos \theta + \sin \theta)$
13. $\int_0^{\pi/4} \left[\int_{\tan \theta \sec \theta}^{\sec \theta} f(r \cos \theta, r \sin \theta) r dr \right] d\theta$
14. $\pi^4/3$
15. (a) $1 + 2u$
(c) $14/3$
(d) $2 + \frac{2}{\sqrt{3}} \left(\arctan \frac{1}{\sqrt{3}} - \arctan \frac{5}{\sqrt{3}} \right)$
16. (a) $4(u^2 + v^2)$
(c) 0
17. $\frac{\pi}{1-p} [(p^2 + r^2)^{1-p} - p^{2(1-p)}]$ if $p \neq 1$; $\pi \log(1 + r^2)$ if $p = 1$.
 $I(p, r)$ tends to a finite limit when $p > 1$

2.20 Exercises (page 96)

1. $1/364$
2. $\log \sqrt{2} - \frac{5}{16}$
3. $1/48$
4. $\frac{4}{3}\pi abc$
5. $\pi/6$
6. $\int_0^1 \left\{ \int_0^x \left[\int_0^{1-x} f(x, y, z) dy \right] dz + \int_x^1 \left[\int_{z-x}^{1-x} f(x, y, z) dy \right] dz \right\} dx$
7. $\int_0^1 \left\{ \int_{-z}^z \left[\int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f(x, y, z) dy \right] dx \right\} dz$
8. $\int_0^1 \left\{ \int_0^{x^2} \left[\int_0^1 f(x, y, z) dy \right] dz + \int_{x^2}^{1+x^2} \left[\int_{\sqrt{z-x^2}}^1 f(x, y, z) dy \right] dz \right\} dx$
10. $16\pi/3$
11. $1/6$
12. $\frac{1}{60}\pi a^2 h(3a^2 + 2h^2)$

13. $\frac{4}{3}\pi a^3$
14. $\frac{4}{3}\pi(b^3 - a^3)$
15. $\frac{4}{3}\pi R^3(a^2 + b^2 + c^2)^{-1/2}$
18. $\frac{2}{3}\pi(5\sqrt{5} - 4)$
19. $32/9$
20. $\frac{4}{3}\pi(b^5 - a^5)$
21. On the axis at distance $\frac{2}{3}h$ from the base
22. On the axis at distance $\frac{1}{3}h$ from the base
23. On the axis of symmetry at distance $\frac{3}{8} \cdot \frac{b^4 - a^4}{b^3 - a^3}$ from the "cutting plane" of the hemispheres
24. $\bar{x} = \bar{y} = \bar{z} = \frac{7}{12}h$ (assuming the specified corner is at the origin)
25. $\frac{3}{20}M(a^2 + 4h^2)$
26. $\frac{2}{5}MR^2$
27. $\frac{3}{5}Ma^2$
28. $2^{1/4}$
30. (a) $\frac{4}{3} \cdot \frac{a^2b^2}{a+b}$

Chapter 3

3.3 Exercises (page 106)

1. (b) $X \leq b$.
2. (a) $\{\omega \mid X(\omega) \in (a, b], Y(\omega) \in (c, d]\}$
(c) $X \leq a, Y \leq d$
(d) $P(a < X \leq b, c < Y \leq d) = P(X \leq b, Y \leq d) - P(X \leq a, Y \leq d) - P(X \leq b, Y \leq c) + P(X \leq a, Y \leq c)$
3. (a) $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}, \{(5, 6), (6, 5)\}, \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), (5, 6), (6, 5)\}$
(b) $P(X = 7) = \frac{1}{6}; P(X = 11) = \frac{1}{18}; P(X = 7 \text{ or } X = 11) = \frac{2}{9}$
4. $Y = X_1 + X_2 + X_3 + X_4; P(Y = 0) = \frac{1}{16}; P(Y = 1) = \frac{1}{4}; P(Y \leq 1) = \frac{5}{16}$
5. $Y = 7X$ if $0 \leq X \leq 100$; $Y = 10X - 300$ if $X > 100$
6. (a) $Z = Y - 1$
(b) $U = Y_1 + Y_2 - 1$

3.6 Exercises (page 118)

2.	t	2	3	4	5	6	7	8	9	10	11	12
	$p_x(t)$	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

3. (b) $p(-2) = \frac{1}{2}, p(0) = p(2) = \frac{1}{4}$
(c) $0, \frac{3}{4}, \frac{1}{4}, \frac{1}{4}, 1, 0, \frac{1}{4}, \frac{1}{4}$
5. (a) $c = 1/3$
(b) $1/9, 1/3, 2/3, 8/9$
(c) $F(t) = 0$ for $t < 0$, $F(t) = 1/9$ for $0 \leq t < 1$, $F(t) = 1/3$ for $1 \leq t < 2$, $F(t) = 1$ for $t \geq 2$
(d) No such t
(e) $t = 2$

6. (a)

k	0	1	2	3	4
$p(k)$	$\frac{16}{81}$	$\frac{32}{81}$	$\frac{8}{27}$	$\frac{8}{81}$	$\frac{1}{81}$

 $p(t) = 0$ for $t \neq 0, 1, 2, 3, 4$
- (b) $F(t) = 0$ for $t < 0$, $F(t) = 16/81$ on $[0, 1)$, $F(t) = 16/27$ on $[1, 2)$, $F(t) = 8/9$ on $[2, 3)$, $F(t) = 80/81$ on $[3, 4)$, $F(t) = 1$ for $t \geq 4$
- (c) $8/27, 56/81$
7. (b) $P(X = 0) = (1 - p)^2$; $P(X = 1) = 2p(1 - p)$
8. $p_X(k) = \frac{2k}{n(n+1)}$; $F_X(t) = \frac{[t]([t] + 1)}{n(n+1)}$ for $0 \leq t \leq n$, where $[t]$ denotes the greatest integer $\leq t$; $F_X(t) = 0$ for $t < 0$; $F_X(t) = 1$ for $t > n$
9. $p(k) = e^{-c} \frac{c^k}{k!}$; $k = 0, 1, 2, 3, \dots$; $c \geq 0$
 $p(t) = 0$ for $t \neq 0, 1, 2, 3, \dots$
10. (a) $p_X(t) = \frac{1}{2}$ at $t = -1$ and $t = +1$; $p_X(t) = 0$ elsewhere
 $F_X(t) = 0$ for $t < -1$; $F_X(t) = \frac{1}{2}$ for $-1 \leq t < 1$; $F_X(t) = 1$ for $t \geq 1$
11. $P(A) = 2/3$; $P(B) = 2/3$; $P(A \cap B) = 1/3$; $P(B|A) = 1/2$; $P(A \cup B) = 1$

3.8 Exercises (page 128)

1. (a) $c = 1$; $f(t) = 1$ if $0 \leq t \leq 1$; $f(t) = 0$ otherwise
 (b) $0, 1/3, 1/3$
2. $c = \frac{1}{2}$; $F(t) = 0$ if $t < -\pi/2$; $F(t) = \frac{1}{2} \cos t$ if $-\pi/2 \leq t < 0$; $F(t) = 1 - \frac{1}{2} \cos t$ if $0 \leq t < \pi/2$; $F(t) = 1$ if $t \geq \pi/2$
3. (a) $f(t) = 0$ if $t < \frac{1}{4}$; $f(t) = 8t - 1$ if $\frac{1}{4} \leq t < \frac{1}{2}$; $f(t) = 7 - 8t$ if $\frac{1}{2} \leq t < \frac{3}{4}$; $f(t) = 0$ if $t \geq \frac{3}{4}$
 (b) $F(t) = 0$ if $t < \frac{1}{4}$; $F(t) = 4t^2 - t$ if $\frac{1}{4} \leq t < \frac{1}{2}$; $F(t) = -4t^2 + 7t - 2$ if $\frac{1}{2} \leq t < \frac{3}{4}$; $F(t) = 1$ if $t \geq \frac{3}{4}$
 (c) $1, 1, 1/2, 0, 5/16$
4. (a) $0, 5/6, 2/3, 1/2$
 (b) $t = 1$
5. (a)

k	5	10	15	20	25	30
$P(X \geq k)$	$\frac{5}{6}$	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$	0
- (b) Let each Styx train arrive 5 minutes before a Lethe train
7. Same as in Exercise 6
8. $1/2$; 0
12. $F_Y(t) = 0$ if $t < 0$; $f_Y(t) = 0$ if $t \leq 0$; $f_Y(t) = e^{-t/2}/\sqrt{2\pi t}$ if $t > 0$

3.11 Exercises (page 134)

2. (a) $P(X = x_1) = P(X = x_2) = P(Y = y_1) = P(Y = y_2) = \frac{1}{2}(p + q)$
 (b) $p = q = \frac{1}{2}$
3. (a) $F(x, y) = \left(\frac{x-a}{b-a}\right)\left(\frac{y-c}{d-c}\right)$ if $a \leq x \leq b$ and $c \leq y \leq d$,
 $F(x, y) = \frac{x-a}{b-a}$ if $a \leq x \leq b$ and $y > d$, $F(x, y) = \frac{y-c}{d-c}$ if $x > b$ and $c \leq y \leq d$,
 $F(x, y) = 1$ if $x > b$ and $y > d$, $F(x, y) = 0$ otherwise
 (b) $F_X(x) = (x-a)/(b-a)$ if $a \leq x \leq b$; $F_X(x) = 0$ if $x < a$; $F_X(x) = 1$ if $x > b$.
 $F_Y(y) = (y-c)/(d-c)$ if $c \leq y \leq d$; $F_Y(y) = 0$ if $y < c$; $F_Y(y) = 1$ if $y > d$
 (c) X and Y are independent

5. $P(Y = 0) = \frac{1}{2}$; $P(Y = 1) = P(Y = 2) = \frac{1}{4}$
7. $1/5, 4/5$
8. $1/3$
9. (b) $f(x, y) = \frac{1}{2}$ if $(x, y) \in Q$; $f(x, y) = 0$ if $(x, y) \notin Q$;
 $f_X(x) = 1 - |x|$ if $|x| \leq 1$; $f_X(x) = 0$ if $|x| > 1$;
 $f_Y(y) = 1 - |y|$ if $|y| \leq 1$; $f_Y(y) = 0$ if $|y| > 1$.
 X and Y are not independent
10. $g(u, v) = f(u + a, v + b)$

3.13 Exercises (page 140)

1. (a) $F_Y(t) = 0$ if $t < 1$; $F_Y(t) = (t - 1)/3$ if $1 \leq t \leq 4$; $F_Y(t) = 1$ if $t > 4$; $f_Y(t) = 1/3$ if $1 \leq t \leq 4$; $f_Y(t) = 0$ otherwise
- (b) $F_Y(t) = 0$ if $t < -2$; $F_Y(t) = (t + 2)/3$ if $-2 \leq t \leq 1$; $F_Y(t) = 1$ if $t > 1$; $f_Y(t) = 1/3$ if $-2 \leq t \leq 1$; $f_Y(t) = 0$ otherwise
- (c) $F_Y(t) = 0$ if $t < 0$; $F_Y(t) = t^{1/2}$ if $0 \leq t \leq 1$; $F_Y(t) = 1$ if $t > 1$; $f_Y(t) = (2t)^{-1/2}$ if $0 \leq t \leq 1$; $f_Y(t) = 0$ otherwise
- (d) $F_Y(t) = e^t$ if $t \leq 0$; $F_Y(t) = 1$ if $t > 0$; $f_Y(t) = e^t$ if $t \leq 0$; $f_Y(t) = 0$ if $t > 0$
- (e) $F_Y(t) = e^{t/2}$ if $t \leq 0$; $F_Y(t) = 1$ if $t > 0$; $f_Y(t) = \frac{1}{2} e^{t/2}$ if $t \leq 0$; $f_Y(t) = 0$ if $t > 0$
- (f) $F_Y(t) = 0$ if $t < 1$; $F_Y(t) = \log t$ if $1 \leq t \leq e$; $F_Y(t) = 1$ if $t > e$; $f_Y(t) = 1/t$ if $1 \leq t \leq e$; $f_Y(t) = 0$ otherwise
2. Let ψ be the inverse of ϕ , defined on the open interval (a, b) .
 Then $F_Y(t) = 0$ if $t \leq a$; $F_Y(t) = F_X[\psi(t)]$ if $a < t < b$; $F_Y(t) = 1$ if $t \geq b$; $f_Y(t) = f_X[\psi(t)] \psi'(t)$ if $a < t < b$; $f_Y(t) = 0$ otherwise
3. (a) $f_Y(t) = 0$ if $t \leq 0$; $f_Y(t) = (2\pi t)^{-1/2} e^{-t/2}$ if $t > 0$
- (b) $f_Y(t) = 0$ if $t < 0$; $f_Y(t) = 4t(2\pi)^{-1/2} e^{-t/2}$ if $t \geq 0$
- (c) $f_Y(t) = 0$ if $t \leq 0$; $f_Y(t) = (2\pi t^2)^{-1/2} e^{-(\log t)^2/2}$ if $t > 0$
- (d) $f_Y(t) = (2\pi)^{-1/2} \sec^2 t e^{-(\tan^2 t)/2}$ if $|t| < \pi/2$; $f_Y(t) = 0$ if $|t| \geq \pi/2$
4. (a) $f(x, y) = 1$ if $(x, y) \in [0, 1] \times [0, 1]$; $f(x, y) = 0$ otherwise
- (e) $f_V(v) = 1 + v$ if $-1 \leq v < 0$; $f_V(v) = 1 - v$ if $0 \leq v \leq 1$; $f_V(v) = 0$ if $|v| > 1$
- (f) U and V are not independent
5. (a) $f(x, y) = e^{-(x+y)}$ if $x > 0, y > 0$; $f(x, y) = 0$ otherwise
- (c) $g(u, v) = ue^{-u}$ if $u > 0, 0 < v < 1$; $g(u, v) = 0$ otherwise
- (d) $f_U(u) = ue^{-u}$ if $u > 0$; $f_U(u) = 0$ if $u \leq 0$
- (e) $f_V(v) = 1$ if $0 < v < 1$; $f_V(v) = 0$ otherwise
6. (b) $g(u, v) = (v/2\pi)e^{-(1+u^2)v^2/2}$ if $v \geq 0$
- (c) $f_U(u) = [\pi(1 + u^2)]^{-1}$
7. $f_Z(t) = \pi^{-1/2} e^{-t^2}$

3.15 Exercises (page 146)

1. $E(X) = 7/2$, $\text{Var}(X) = 35/12$
7. (a) $E(X) = \text{Var}(X) = \lambda$
- (b) None
- (c) $E(X) = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$
- (d) $E(X) = m$, $\text{Var}(X) = \sigma^2$
8. (a) $C(r) = (r - 1)/2$
- (b) $F_X(t) = \frac{1}{2} |t|^{1-r}$ if $t < -1$; $F_X(t) = 1/2$ if $-1 \leq t \leq 1$; $F_X(t) = 1 - \frac{1}{2} |t|^{1-r}$ if $t > 1$
- (c) $P(X < 5) = 1 - 5^{1-r}/2$; $P(5 < X < 10) = (5^{1-r} - 10^{1-r})/2$
- (d) X has a finite expectation when $r > 2$; $E(X) = 0$
- (e) Variance is finite for $r > 3$; $\text{Var}(X) = (r - 1)/(r - 3)$

9. $E(X) = E(Y) = -1/37$; $E(Z) = -1767/50653$; $E(X + Y + Z) = -4505/50653$
10. $E(X) \rightarrow \infty$ as $n \rightarrow \infty$
12. (a) $(2/\pi)^{1/2}$
 (b) $e^{1/2}$
 (c) $e^2 - e$
 (d) $(\pi/2)^{1/2}$

3.19 Exercises (page 154)

4. 251
5. 0
6. Chebyshev's inequality gives $1/9$; tables give 0.0027
8. (b) 0.6826
9. (b) 0.0796
10. (a) 0.0090
 (b) 0.0179

Chapter 4

4.3 Exercises (page 158)

2. All open except (d), (e), (h), and (j)
3. All open except (d)
5. (e) One example is the collection of all neighborhoods $N(\vec{0}; 1/k)$, where $k = 1, 2, 3, \dots$
6. (a) Both
 (b) Both
 (c) Closed
 (d) Open
 (e) Closed
 (f) Neither
 (g) Closed
 (h) Neither
 (i) Closed
 (j) Closed
 (k) Neither
 (l) Closed
8. (e) One example is the collection of all sets of the form $S_k = \{\vec{X} \mid |\vec{X}| \leq 1 - 1/k\}$ for $k = 1, 2, 3, \dots$. Their union is the open set $N(\vec{0}; 1)$

4.5 Exercises (page 164)

1. $f'(\vec{X}; \vec{Y}) = 4|\vec{X}|^2 \vec{X} \cdot \vec{Y}$
2. (a) All points on the straight line $2x + 3y = 3/26$
 (b) All points on the two lines $x^2 - 3y^2 = 0$
7. $(1 - m^2)/(1 + m^2)$; No
9. $f(0, 0) = 1$
11. (a) Continuous for all (x, y) ; $\frac{\partial f}{\partial x} = 4x^3 - 8xy^2$; $\frac{\partial f}{\partial y} = 4y^3 - 8x^2y$
 (b) Continuous for $(x, y) \neq (0, 0)$; $\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2}$; $\frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2}$

- (c) Continuous whenever $y \neq 0$; $\frac{\partial f}{\partial x} = -\frac{2x}{y} \sin(x^2)$; $\frac{\partial f}{\partial y} = -\frac{1}{y^2} \cos(x^2)$
- (d) Continuous whenever $y \neq 0$ and $\frac{x^2}{y} \neq \frac{\pi}{2} + \pi k$ ($k = 0, 1, 2, \dots$); $\frac{\partial f}{\partial x} = \frac{2x}{y} \sec^2 \frac{x^2}{y}$;
 $\frac{\partial f}{\partial y} = -\frac{x^2}{y^2} \sec^2 \frac{x^2}{y}$
- (e) Continuous whenever $x \neq 0$; $\frac{\partial f}{\partial x} = -\frac{y}{x^2 + y^2}$; $\frac{\partial f}{\partial y} = \frac{1}{x^2 + y^2}$
- (f) Continuous for $(x, y) \neq (0, 0)$; $\frac{\partial f}{\partial x} = \frac{|y|}{x^2 + y^2}$; $\frac{\partial f}{\partial y} = -\frac{x^2}{|y|(x^2 + y^2)}$
- (g) Continuous whenever $xy \neq 1$; $\frac{\partial f}{\partial x} = \frac{1 + y^2}{1 + x^2 + y^2 + x^2 y^2}$; $\frac{\partial f}{\partial y} = \frac{1 + x^2}{1 + x^2 + y^2 + x^2 y^2}$
- (h) Continuous whenever $(x, y) \neq (0, 0)$; $\partial f / \partial x = y^2 / (x^2 + y^2)^{3/2}$; $\partial f / \partial y = -xy / (x^2 + y^2)^{3/2}$
- (i) Continuous whenever $(x, y) \neq (0, 0)$; $\partial f / \partial x = y^2 x^{y^2-1}$; $\partial f / \partial y = 2yx^{y^2} \log x$
- (j) Continuous whenever $y \neq 0$, and $0 \leq x \leq y$ or $y \leq x \leq 0$; $\frac{\partial f}{\partial x} = -\frac{1}{\sqrt{x(y-x)}}$;
 $\frac{\partial f}{\partial y} = \frac{\sqrt{x}}{2y\sqrt{y-x}}$

12. $n = -3/2$

13. $a = b = 1$

4.7 Exercises (page 167)

1. (c) $f'(\vec{X}; \vec{Y}) = 2(x_1 y_1 + x_2 y_2 + \dots + x_n y_n)$; $\vec{Z} = \vec{X} + \frac{1}{2} \vec{Y}$
3. One example is $S = N(\vec{0}; 1) \cup N(3\vec{i}; 1)$, with $f = 0$ on $N(\vec{0}; 1)$, $f = 1$ on $N(3\vec{i}; 1)$
4. One example is $f(\vec{X}) = \vec{X} \cdot \vec{Y}$, where $\vec{Y}$ is a fixed vector

4.10 Exercises (page 172)

1. (a) $(2x + y^3 \cos xy)\vec{i} + (2y \sin xy + xy^2 \cos xy)\vec{j}$
 (b) $e^x \cos y \vec{i} - e^x \sin y \vec{j}$
 (c) $2xy^3 z^4 \vec{i} + 3x^2 y^2 z^4 \vec{j} + 4x^2 y^3 z^3 \vec{k}$
 (d) $2x \vec{i} - 2y \vec{j} + 4z \vec{k}$
 (e) $\frac{2x}{x^2 + 2y^2 - 3z^2} \vec{i} + \frac{4y}{x^2 + 2y^2 - 3z^2} \vec{j} - \frac{6z}{x^2 + 2y^2 - 3z^2} \vec{k}$
 (f) $y^z x^{yz-1} \vec{i} + zy^{z-1} x^{yz} \log x \vec{j} + y^z x^{yz} \log x \log y \vec{k}$
2. (a) $-2/\sqrt{6}$
 (b) $1/\sqrt{6}$
3. $(1, 0)$, in the direction of $\vec{i}$; $(-1, 0)$, in the direction of $-\vec{i}$
4. $2\vec{i} + 2\vec{j}$; $14/5$
5. $(a, b, c) = (6, 24, -8)$ or $(-6, -24, 8)$
6. The set of points (x, y) on the line $5x - 3y = 6$; $\nabla f(\vec{X}) = 5\vec{i} - 3\vec{j}$

4.13 Exercises (page 179)

1. (a) $-2/3$
 (b) $x^2 - y^2$
 (c) 0
2. (a) $(1 + 3x^2 + 3y^2)(x\vec{i} + y\vec{j}) - (x^2 + y^2)^{1/2} \vec{k}$, or any scalar multiple thereof
 (b) $\cos \theta = -[1 + (1 + 3(x^2 + y^2))^2]^{-1/2}$; $\cos \theta \rightarrow -\frac{1}{2}\sqrt{2}$ as $(x, y, z) \rightarrow (0, 0, 0)$

3. $U(x, y) = \frac{1}{2} \log(x^2 + y^2)$; $V(x, y) = \arctan(y/x)$
 4. (b) No
 6. $x/x_0 + y/y_0 + z/z_0 = 3$
 7. $x + y + 2z = 4$, $x - y - z = -1$
 8. $c = \pm 3$
 11. (b) $F''(t) = \frac{\partial^2 f}{\partial x^2} [X'(t)]^2 + 2 \frac{\partial^2 f}{\partial x \partial y} X'(t) Y'(t) + \frac{\partial^2 f}{\partial y^2} [Y'(t)]^2 + \frac{\partial f}{\partial x} X''(t) + \frac{\partial f}{\partial y} Y''(t)$
 12. (a) $F'(t) = 4t^3 + 2t$; $F''(t) = 12t^2 + 2$
 (b) $F'(t) = (2\cos^2 t - 1)e^{\cos t \sin t} \cos(\cos t \sin^2 t) + (3 \sin^3 t - 2 \sin t)e^{\cos t \sin t} \sin(\cos t \sin^2 t)$;
 $F''(t) = (5\cos^6 t - 3\cos^4 t - 4\cos^3 t - \cos^2 t - 4\cos t)e^{\cos t \sin t} \cos(\cos t \sin^2 t)$
 $+ (14\sin^3 t - 12\sin^5 t - 4\sin t + 7\cos t - 9\cos^3 t)e^{\cos t \sin t} \sin(\cos t \sin^2 t)$
 (c) $F'(t) = \frac{2e^{2t} \exp(e^{2t})}{1 + \exp(e^{2t})} + \frac{2e^{-2t} \exp(e^{-2t})}{1 + \exp(e^{-2t})}$, where $\exp(u) = e^u$;
 $F''(t) = \frac{4[1 + e^{2t} + \exp(e^{2t})]e^{2t} \exp(e^{2t})}{[1 + \exp(e^{2t})]^2} - \frac{4[1 + e^{-2t} + \exp(e^{-2t})]e^{-2t} \exp(e^{-2t})}{[1 + \exp(e^{-2t})]^2}$

4.15 Exercises (page 185)

1. (b) $\frac{\partial f}{\partial x} = -2x \sin(x^2 + y^2) \cos[\cos(x^2 + y^2)]e^{\sin[\cos(x^2 + y^2)]}$
 2. $G(x, y) = x - y$
 4. (a) $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial s}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial t}$
 (c) $\frac{\partial^2 F}{\partial s \partial t} = \frac{\partial^2 f}{\partial x^2} \frac{\partial X}{\partial s} \frac{\partial X}{\partial t} + \frac{\partial^2 f}{\partial x \partial y} \left(\frac{\partial X}{\partial s} \frac{\partial Y}{\partial t} + \frac{\partial X}{\partial t} \frac{\partial Y}{\partial s} \right) + \frac{\partial^2 f}{\partial y^2} \frac{\partial Y}{\partial s} \frac{\partial Y}{\partial t} + \frac{\partial f}{\partial x} \frac{\partial^2 X}{\partial s \partial t} + \frac{\partial f}{\partial y} \frac{\partial^2 Y}{\partial s \partial t}$
 5. (a) $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} + t \frac{\partial f}{\partial y}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} + s \frac{\partial f}{\partial y}$; $\frac{\partial^2 F}{\partial s^2} = \frac{\partial^2 f}{\partial x^2} + 2t \frac{\partial^2 f}{\partial x \partial y} + t^2 \frac{\partial^2 f}{\partial y^2}$;
 $\frac{\partial^2 F}{\partial t^2} = \frac{\partial^2 f}{\partial x^2} + 2s \frac{\partial^2 f}{\partial x \partial y} + s^2 \frac{\partial^2 f}{\partial y^2}$; $\frac{\partial^2 F}{\partial s \partial t} = \frac{\partial^2 f}{\partial x^2} + (s + t) \frac{\partial^2 f}{\partial x \partial y} + st \frac{\partial^2 f}{\partial y^2} + \frac{\partial f}{\partial y}$
 (b) $\frac{\partial F}{\partial s} = t \frac{\partial f}{\partial x} + \frac{1}{t} \frac{\partial f}{\partial y}$; $\frac{\partial F}{\partial t} = s \frac{\partial f}{\partial x} - \frac{s}{t^2} \frac{\partial f}{\partial y}$; $\frac{\partial^2 F}{\partial s^2} = t^2 \frac{\partial^2 f}{\partial x^2} + 2 \frac{\partial^2 f}{\partial x \partial y} + \frac{1}{t^2} \frac{\partial^2 f}{\partial y^2}$;
 $\frac{\partial^2 F}{\partial t^2} = s^2 \frac{\partial^2 f}{\partial x^2} - 2 \frac{s^2}{t^2} \frac{\partial^2 f}{\partial x \partial y} + \frac{s^2}{t^4} \frac{\partial^2 f}{\partial y^2} + \frac{2s}{t^3} \frac{\partial f}{\partial y}$; $\frac{\partial^2 F}{\partial s \partial t} = st \frac{\partial^2 f}{\partial x^2} - \frac{s}{t^3} \frac{\partial^2 f}{\partial y^2} + \frac{\partial f}{\partial x} - \frac{1}{t^2} \frac{\partial f}{\partial y}$
 (c) $\frac{\partial F}{\partial s} = \frac{1}{2} \frac{\partial f}{\partial x} + \frac{1}{2} \frac{\partial f}{\partial y}$; $\frac{\partial F}{\partial t} = -\frac{1}{2} \frac{\partial f}{\partial x} + \frac{1}{2} \frac{\partial f}{\partial y}$; $\frac{\partial^2 F}{\partial s \partial t} = -\frac{1}{4} \frac{\partial^2 f}{\partial x^2} + \frac{1}{4} \frac{\partial^2 f}{\partial y^2}$;
 $\frac{\partial^2 F}{\partial s^2} = \frac{1}{4} \frac{\partial^2 f}{\partial x^2} + \frac{1}{2} \frac{\partial^2 f}{\partial x \partial y} + \frac{1}{4} \frac{\partial^2 f}{\partial y^2}$; $\frac{\partial^2 F}{\partial t^2} = \frac{1}{4} \frac{\partial^2 f}{\partial x^2} - \frac{1}{2} \frac{\partial^2 f}{\partial x \partial y} + \frac{1}{4} \frac{\partial^2 f}{\partial y^2}$
 6. $a = b = 1$; $c = d = 0$
 7. $\frac{\partial F}{\partial x} = \frac{1}{2} \frac{\partial f}{\partial u} + \frac{1}{2} \frac{\partial f}{\partial v}$; $\frac{\partial F}{\partial y} = -\frac{1}{2} \frac{\partial f}{\partial u} + \frac{1}{2} \frac{\partial f}{\partial v}$
 8. $\frac{\partial^2 \phi}{\partial r^2} = \cos^2 \theta \frac{\partial^2 f}{\partial x^2} + \cos \theta \sin \theta \left(\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y \partial x} \right) + \sin^2 \theta \frac{\partial^2 f}{\partial y^2}$;
 $\frac{\partial^2 \phi}{\partial r \partial \theta} = -r \cos \theta \sin \theta \frac{\partial^2 f}{\partial x^2} + r \cos^2 \theta \frac{\partial^2 f}{\partial x \partial y} - r \sin^2 \theta \frac{\partial^2 f}{\partial y \partial x} + r \cos \theta \sin \theta \frac{\partial^2 f}{\partial y^2}$
 $- \sin \theta \frac{\partial f}{\partial x} + \cos \theta \frac{\partial f}{\partial y}$;
 9. $\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial r} + \frac{\partial f}{\partial z} \frac{\partial Z}{\partial r}$; $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial s} + \frac{\partial f}{\partial z} \frac{\partial Z}{\partial s}$;
 $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial t} + \frac{\partial f}{\partial z} \frac{\partial Z}{\partial t}$

10. (a) $\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + 2 \frac{\partial f}{\partial z}$; $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} - 2 \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} + 3 \frac{\partial f}{\partial y} - \frac{\partial f}{\partial z}$
 (b) $\frac{\partial F}{\partial r} = 2r \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \right)$; $\frac{\partial F}{\partial s} = 2s \left(\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} - \frac{\partial f}{\partial z} \right)$; $\frac{\partial F}{\partial t} = 2t \left(\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \right)$
11. $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial s} + \frac{\partial f}{\partial z} \frac{\partial Z}{\partial s}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial t} + \frac{\partial f}{\partial z} \frac{\partial Z}{\partial t}$
12. (a) $\frac{\partial F}{\partial s} = 2s \frac{\partial f}{\partial x} + 2s \frac{\partial f}{\partial y} + 2t \frac{\partial f}{\partial z}$; $\frac{\partial F}{\partial t} = 2t \frac{\partial f}{\partial x} - 2t \frac{\partial f}{\partial y} + 2s \frac{\partial f}{\partial z}$
 (b) $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + t \frac{\partial f}{\partial z}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} + s \frac{\partial f}{\partial z}$
13. $\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial r}$; $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial s}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial X}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial Y}{\partial t}$
14. (a) $\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x}$; $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial y}$
 (b) $\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} + 2r \frac{\partial f}{\partial y}$; $\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x} + 2s \frac{\partial f}{\partial y}$; $\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x} + 2t \frac{\partial f}{\partial y}$
 (c) $\frac{\partial F}{\partial r} = \frac{1}{s} \frac{\partial f}{\partial x}$; $\frac{\partial F}{\partial s} = \frac{-r}{s^2} \frac{\partial f}{\partial x} + \frac{1}{t} \frac{\partial f}{\partial y}$; $\frac{\partial F}{\partial t} = \frac{-s}{t^2} \frac{\partial f}{\partial y}$
15. $a_k(\vec{T}) = D_k f[\vec{U}(\vec{T})]$ for $k = 1, 2, 3, \dots, n$

4.18 Exercises (page 194)

- $n = 2$
- $\partial f / \partial x = -1/(2y + 2z + 1)$; $\partial f / \partial y = -2(y + z)/(2y + 2z + 1)$; $\partial^2 f / (\partial x \partial y) = 2/(2y + 2z + 1)^3$
- $\partial^2 z / (\partial x \partial y) = [\sin(x + y) \cos^2(y + z) + \sin(y + z) \cos^2(x + y)] / \cos^3(y + z)$
- $\frac{\partial f}{\partial x} = -\frac{D_1 F + 2x D_2 F}{D_1 F + 2z D_2 F}$; $\frac{\partial f}{\partial y} = -\frac{D_1 F + 2y D_2 F}{D_1 F + 2z D_2 F}$
- $D_1 F = f'[x + g(y)]$; $D_2 F = f'[x + g(y)]g'(y)$; $D_{1,1} F = f''[x + g(y)]$;
 $D_{1,2} F = f''[x + g(y)]g'(y)$; $D_{2,2} F = f''[x + g(y)][g'(y)]^2 + f'[x + g(y)]g''(y)$
- $\partial X / \partial v = (1 + xu)/(x - y)$; $\partial Y / \partial u = (1 - yv)/(x - y)$; $\partial Y / \partial v = (1 + yu)/(y - x)$
- $\partial X / \partial y = -(1 + xu)(1 + u)$; $\partial V / \partial u = (1 - yv)/(1 + yu)$; $\partial V / \partial y = (1 - x)/(1 + u)$
- $\frac{\partial X}{\partial v} = \frac{\partial(F, G)}{\partial(y, v)} / \frac{\partial(F, G)}{\partial(x, y)}$; $\frac{\partial Y}{\partial u} = \frac{\partial(F, G)}{\partial(u, x)} / \frac{\partial(F, G)}{\partial(x, y)}$; $\frac{\partial Y}{\partial v} = \frac{\partial(F, G)}{\partial(v, x)} / \frac{\partial(F, G)}{\partial(x, y)}$
- $\vec{T} = \pm \frac{1}{\sqrt{751}} (24 \vec{i} - 4\sqrt{7} \vec{j} + 3\sqrt{7} \vec{k})$
- $2\vec{i} + \vec{j} + \sqrt{3} \vec{k}$, or any scalar multiple thereof

4.20 Exercises (page 199)

- $f(x, y, z) = (x^2 + y^2 + z^2)/2 + C$
- $f(x, y, z) = x^2/2 - y^2/2 + xz - yz + C$
- $\vec{V}$ is not a gradient
- $\vec{V}$ is not a gradient
- $\vec{V}$ is not a gradient
- $f(x, y, z) = y^2 \sin x + xz^3 - 4y + 2z + C$
- $f(x, y, z) = x + 2x^2y - x^3z^2 + 2y - z^3 + C$
- $f(x, y) = x^2e^y + xy - y^2 + C$
- $f(x, y) = x \sin y + y \cos x + (x^2 + y^2)/2 + C$
- $f(x, y) = x \sin(xy) + C$

11. (b) $f(x, y) = \frac{ar^{n+1}}{n+1} + C$ if $n \neq -1$; $f(x, y) = a \log r + C$ if $n = -1$
 12. $9x^2 - 18xy + 16y^2 - 7 = 0$
 13. $x^2 - y^2 + 2xz - 2yz + 9 = 0$
 15. (a) $v(x, y) = e^x \sin y$
 (b) $v(x, y) = -\sin x \sinh y$
 (c) $v(x, y) = \arctan(y/x)$ if $x > 0$; $v(x, y) = \pi/2$ if $x = 0, y > 0$; $v(x, y) = -\pi/2$ if $x = 0, y < 0$; $v(x, y) = \arctan(y/x) + \pi$ if $x < 0, y > 0$; $v(x, y) = \arctan(y/x) - \pi$ if $x < 0, y < 0$

4.23 Exercises (page 209)

1. Absolute minimum at (0, 1)
2. Saddle point at (0, 1)
3. Saddle point at (0, 0)
4. Absolute minimum at each point of the line $y = x + 1$
5. Saddle point at (1, 1)
6. Absolute minimum at (1, 0)
7. Saddle point at (0, 0)
8. Saddle point at (0, 0); relative maximum at (2, 3)
9. Saddle point at (0, 0); relative minimum at (1, 1)
10. Saddle points at $(n\pi + \pi/2, 0)$, where n is any integer
11. Absolute minimum at (0, 0); saddle point at $(-1/4, -1/2)$
12. Absolute minimum at $(-1/26, -3/26)$; absolute maximum at (1, 3)
13. Absolute maximum at $(\pi/3, \pi/3)$; absolute minimum at $(2\pi/3, 2\pi/3)$; relative maximum at (π, π) ; relative minimum at (0, 0); saddle points at $(0, \pi)$ and $(\pi, 0)$
14. Saddle point at (1, 1)
15. Absolute maximum at each point of the circle $x^2 + y^2 = 1$; absolute minimum at (0, 0)
17. (c) Relative maximum at (2, 2); no relative minima; saddle points at (0, 3), (3, 0), and (3, 3)
18. Maximum value is $1/4$; no minimum
19. Maximum is 2; minimum is 1
20. (a) Maximum is $\frac{\sqrt{a^2 + b^2}}{ab}$ at $(b(a^2 + b^2)^{-1/2}, a(a^2 + b^2)^{-1/2})$; minimum is $-\frac{\sqrt{a^2 + b^2}}{ab}$ at $(-b(a^2 + b^2)^{-1/2}, -a(a^2 + b^2)^{-1/2})$
 (b) Minimum is $a^2b^2/(a^2 + b^2)$ at $\left(\frac{ab^2}{a^2 + b^2}, \frac{a^2b}{a^2 + b^2}\right)$; no maximum
21. Maximum is $1 + \sqrt{2}/2$ at the points $(n\pi + \pi/8, n\pi - \pi/8)$, where n is any integer; minimum is $1 - \sqrt{2}/2$ at $(n\pi + 5\pi/8, n\pi + 3\pi/8)$, where n is any integer
22. Maximum is 3 at $(1/3, -2/3, 2/3)$; minimum is -3 at $(-1/3, 2/3, -2/3)$
23. (0, 0, 1) and (0, 0, -1)
24. 1
25. (1, 0, 0), (0, 1, 0), (-1, 0, 0), (0, -1, 0)
26. $\frac{a^ab^bc^c}{(a+b+c)^{a+b+c}}$ at $\left(\frac{a}{a+b+c}, \frac{b}{a+b+c}, \frac{c}{a+b+c}\right)$
27. Let $x^* = \frac{1}{n} \sum_{i=1}^n x_i$, $y^* = \frac{1}{n} \sum_{i=1}^n y_i$, $u_i = x_i - x^*$. Then $a = \left(\sum_{i=1}^n y_i u_i\right) / \left(\sum_{i=1}^n u_i^2\right)$, and $b = y^* - ax^*$

28. Let $x^* = \frac{1}{n} \sum_{i=1}^n x_i$, $y^* = \frac{1}{n} \sum_{i=1}^n y_i$, $z^* = \frac{1}{n} \sum_{i=1}^n z_i$, $u_i = x_i - x^*$, $v_i = y_i - y^*$, and let

$$\Delta = \begin{vmatrix} \sum u_i^2 & \sum u_i v_i \\ \sum u_i v_i & \sum v_i^2 \end{vmatrix},$$

where the sums are for $i = 1, 2, \dots, n$. Then

$$a = \frac{1}{\Delta} \begin{vmatrix} \sum u_i z_i & \sum u_i v_i \\ \sum v_i z_i & \sum v_i^2 \end{vmatrix}, \quad b = \frac{1}{\Delta} \begin{vmatrix} \sum v_i z_i & \sum u_i v_i \\ \sum u_i z_i & \sum u_i^2 \end{vmatrix}, \quad c = z^* - ax^* - by^*$$

29. $m^2 = \frac{A + C - \sqrt{(A - C)^2 + 4B^2}}{2(AC - B^2)}$

31. $abc\sqrt{3}/2$

32. $5 \log r + 3 \log \sqrt{3}$

4.26 Miscellaneous exercises (page 217)

2. One example: $f(x, y) = 3x$ when $x = y$, $f(x, y) = 0$ otherwise

3. $D_1 f(0, 0) = 0$; $D_2 f(0, 0) = -1$; $D_{2,1} f(0, 0) = 0$; $D_{1,2} f(0, 0)$ does not exist

$$\begin{aligned} 5. F'''(t) &= \frac{\partial^3 f}{\partial x^3} [X'(t)]^3 + 3 \frac{\partial^3 f}{\partial x^2 \partial y} [X'(t)]^2 Y'(t) + 3 \frac{\partial^3 f}{\partial x \partial y^2} X'(t) [Y'(t)]^2 \\ &\quad + \frac{\partial^3 f}{\partial y^3} [Y'(t)]^3 + 3 \frac{\partial^2 f}{\partial x^2} X'(t) X''(t) + 3 \frac{\partial^2 f}{\partial x \partial y} [X''(t) Y'(t) + X'(t) Y''(t)] \\ &\quad + 3 \frac{\partial^2 f}{\partial y^2} Y'(t) Y''(t) + \frac{\partial f}{\partial x} X'''(t) + \frac{\partial f}{\partial y} Y'''(t), \end{aligned}$$

assuming the mixed partial derivatives are independent of the order of differentiation

6. 8

8. $\partial x / \partial u = \partial x / \partial v = 0$

11. A sphere with center at the origin and radius $\sqrt{2}$

13. $9/4$ at the points $(-1/2, \pm \sqrt{3}/2)$; $-1/4$ at $(1/2, 0)$

14. $2\sqrt{b-1}$ if $b > 2$; $|b|$ if $b \leq 2$

15. $f(x) = x^2$

16. $(4 \pm \sqrt{5})/\sqrt{2}$

17. Angle is $\pi/3$; width across the bottom is $c/3$; maximum area is $c^2/(4\sqrt{3})$

20. (a) $\phi'(t) = A'(t) \int_c^{B(t)} f[A(t), y] dy + B'(t) \int_a^{A(t)} f[x, B(t)] dx$

(b) $\phi'(t) = 2t e^{t^2} (2e^{t^2} - e^a - e^c)$

Chapter 5

5.3 Exercises (page 223)

1. $-14/15$

2. $4/3$

3. 0

4. $-2\pi a^2$

5. $1/35$

6. 40

8. $\pi a^3/4$

9. $2a^3$

10. $a = (3c/2)^{1/2}$

5.6 Exercises (page 227)

1. $-369/10$
2. $23/6$
3. -2π
4. 0
5. $1 + \sqrt{2}$
6. $256a^3/15$
7. $2\pi^2a^3(1 + 2\pi^2)$
8. $[(2 + t_0^2)^{3/2} - 2\sqrt{2}]/3$
9. (a) $-2\sqrt{2}\pi$
(b) $-\pi$
10. 5
11. $\bar{x} = \frac{6ab^2}{3a^2 + 4\pi^2b^2}; \bar{y} = -\frac{6\pi ab^2}{3a^2 + 4\pi^2b^2}$
12. $2\pi/3$
13. $\frac{600 - 36\sqrt{2} - 49\log(9 - 4\sqrt{2})}{64[6\sqrt{2} + \log(3 + 2\sqrt{2})]}$
14. (b) $4a^3$
15. $I_x = (a^2 + b^2)^{1/2}[\pi a^4 + (4\pi^3 - \pi/2)a^2b^2 + 32\pi^5b^4/5]$
 $I_y = (a^2 + b^2)^{1/2}[\pi a^4 + (4\pi^3 + \pi/2)a^2b^2 + 32\pi^5b^4/5]$

5.9 Exercises (page 233)

2. (b) $(2e^{2\pi} - 5e^\pi - 5\pi - 3)/10$
3. (b) 3
4. $8/5$
6. $-4b^2 + 8\pi b - 4$; minimum occurs when $b = \pi$

5.11 Exercises (page 236)

3. $\phi(x, y) = \arctan(y/x)$ if $x > 0$; $\phi(x, y) = \pi/2$ if $x = 0$ and $y > 0$; $\phi(x, y) = -\pi/2$ if $x = 0$ and $y < 0$; $\phi(x, y) = \arctan(y/x) + \pi$ if $x < 0$ and $y > 0$; $\phi(x, y) = \arctan(y/x) - \pi$ if $x < 0$ and $y < 0$

5.14 Exercises (page 241)

1. $x^2/2 + 2xy + y^2/2 = C$
2. $x^2y = C$
3. $x^3/3 - xy - y/2 + (\sin 2y)/4 = C$
4. $\cos 2x \sin 3y = C$
5. $x^3y + 4x^2y^2 - 12e^y + 12ye^y = C$
6. $\int Q(x)e^{\int P(x)dx} dx - ye^{\int P(x)dx} = C$
8. (a) $x + y = Cy^2$
(b) $y^3/x^3 - 3\log|x| = C$
9. (a) $6(xy)^{1/2} - (y/x)^{3/2} = C$; $(x^5y)^{-1/2}$ is an integrating factor
(b) $x + e^{-x} \sin y = C$; $e^{-x} \cos y$ is an integrating factor
10. $x^3y^4 + x^4y^5 = C$, $10x^3y^4 + x^5y^5 = C$, respectively; x^2y^3 is a common integrating factor

5.17 Exercises (page 248)

1. (a) -4
 (b) 4
 (c) 8
 (d) 4π
 (e) $3\pi/2$
2. 0
3. $n = 3$
4. $-\pi$
9. $g(x, y) = \pm [P^2(x, y) + Q^2(x, y)]^{1/2}$

5.19 Exercises (page 252)

1. (b) 0
2. $0, 2\pi, -2\pi$
3. As many as three
4. As many as seven
5. (a) -3
6. 2π

5.21 Exercises (page 259)

1. (a) $\operatorname{div} \vec{F}(x, y, z) = 2x + 2y + 2z$; $\operatorname{curl} \vec{F}(x, y, z) = \vec{0}$
 (b) $\operatorname{div} \vec{F}(x, y, z) = 0$; $\operatorname{curl} \vec{F}(x, y, z) = 2\vec{i} + 4\vec{j} + 6\vec{k}$
 (c) $\operatorname{div} \vec{F}(x, y, z) = -x \sin y$; $\operatorname{curl} \vec{F}(x, y, z) = \vec{i} + \vec{j}$
 (d) $\operatorname{div} \vec{F}(x, y, z) = ye^{xy} - x \sin(xy) - 2xz \sin(xz^2)$;
 $\operatorname{curl} \vec{F}(x, y, z) = z^2 \sin(xz^2) \vec{j} - [xe^{xy} + y \sin(xy)] \vec{k}$
 (e) $\operatorname{div} \vec{F}(x, y, z) = 2x \sin y + 2y \sin(xz) - xy \sin z \cos(\cos z)$;
 $\operatorname{curl} \vec{F}(x, y, z) = [x \sin(\cos z) - xy^2 \cos(xz)] \vec{i} - y \sin(\cos z) \vec{j} + [y^2 z \cos(xz) - x^2 \cos y] \vec{k}$
2. $\vec{0}$
4. $n = -3$
5. No such vector field
10. One such field is $\mu(x, y, z) = (xyz)^{-2}$

5.24 Exercises (page 263)

1. (a), (c), and (e) are false; all others true
4. $\operatorname{div}(\vec{V} \times \vec{r}) = 0$; $\operatorname{curl}(\vec{V} \times \vec{r}) = (c + 1)\vec{V}$
6. $16(a + b)$

5.26 Exercises (page 267)

1. $(3x - 2z)\vec{j} - x\vec{k}$ is one such field
2. $(x^2/2 - xy - yz + z^2/2)\vec{j} + (x^2/2 - xz)\vec{k}$ is one such field
3. $(x^2y/2 + z^2/2)\vec{j} + \nabla f(x, y)$ for some f independent of z
5. $\vec{G}(x, y, z) = \frac{yz}{r(x^2 + y^2)} \vec{i} - \frac{xz}{r(x^2 + y^2)} \vec{j}$ satisfies $\operatorname{curl} \vec{G} = r^{-3} \vec{r}$ at all points not on the z -axis
6. $f(r) = Cr^{-3}$
9. $\vec{F}(x, y, z) = -\frac{1}{2}(z^3 \vec{i} + x^3 \vec{j} + y^3 \vec{k})$, $\vec{G}(x, y, z) = \frac{1}{2} \nabla(x^3y + y^3z + z^3x)$

Chapter 6

6.3 Exercises (page 282)

1. $(a_2b_3 - a_3b_2)(x - x_0) + (a_3b_1 - a_1b_3)(y - y_0) + (a_1b_2 - a_2b_1)(z - z_0) = 0$;
 $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = (a_2b_3 - a_3b_2)\vec{i} + (a_3b_1 - a_1b_3)\vec{j} + (a_1b_2 - a_2b_1)\vec{k}$
2. $x^2/a^2 + y^2/b^2 = z$; $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = -2bu^2 \cos v \vec{i} - 2au^2 \sin v \vec{j} + abu \vec{k}$
3. $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$;
 $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = abc \sin u \left(\frac{\sin u \cos v}{a} \vec{i} + \frac{\sin u \sin v}{b} \vec{j} + \frac{\cos u}{c} \vec{k} \right)$
4. $z = f(\sqrt{x^2 + y^2})$; $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = -uf'(u) \cos v \vec{i} - uf'(u) \sin v \vec{j} + u \vec{k}$
5. $y^2/a^2 + z^2/b^2 = 1$; $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = b \sin v \vec{j} + a \cos v \vec{k}$
6. $(\sqrt{x^2 + y^2} - a)^2 + z^2 = b^2$;
 $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = b(a + b \cos u)(\cos u \sin v \vec{i} + \cos u \cos v \vec{j} + \sin u \vec{k})$
7. $|abc| \cosh v \left[\left(\frac{\sin^2 u}{a^2} + \frac{\cos^2 u}{b^2} \right) \cosh^2 v + \frac{\sinh^2 v}{c^2} \right]^{1/2}$
8. $\sqrt{128v^2 + 4}$
9. $|u - v| \sqrt{36u^2v^2 + 9(u + v)^2 + 4}$
10. $\sqrt{2u^4 + u^2}$

6.5 Exercises (page 287)

2. $\pi a^2 \sqrt{3}$
3. $(2\pi - 4)a^2$
4. 4
5. (a) A circular paraboloid
 (b) $-2u^2 \cos v \vec{i} - 2u^2 \sin v \vec{j} + u \vec{k}$
 (c) $n = 6$
6. $\sqrt{2} \pi a^2/4$
7. $2\pi\sqrt{6}$
8. $2\pi a^2(3\sqrt{3} - 1)/3$
9. $4\pi^2 ab$

6.9 Exercises (page 296)

1. $4\pi/3$
3. $\bar{x} = \bar{y} = \bar{z} = a/2$
4. $1/2$
7. 0
8. 0
9. 0
10. $\pi \sqrt{2}$
12. On the axis of the cone, at a distance $\frac{1}{4}\alpha(1 - \cos \alpha)/[1 - \cos(\alpha/2)]$ from the center of the sphere.

6.12 Exercises (page 303)

1. 0
2. $-\pi$
3. -4
4. $4/3$
14. (c) $3\pi/2$

6.16 Exercises (page 313)

1. 3
2. (a) 144π
(b) -16π
(c) 128π
3. (a) $3|V|$
(b) $9|V|\bar{z}$
(c) $|V|\bar{x}$
(d) $4I_z$
15. 8π

Chapter 7

7.3 Exercises (page 323)

1. $x = \frac{\sqrt{6}}{2k} \int \frac{dy}{\sqrt{A^3 - y^3}} + B$
2. $x = \frac{\sqrt{6}}{2k} \int \frac{dy}{\sqrt{A^3 + y^3}} + B$
3. $y = \log|Ax + B|$
4. $y = -x + \log|1 + Ae^{2x}| + B$
5. $x = \int \frac{dy}{\sqrt{y^4 - 2y^2 + A}} + B$
6. $y = Ae^{x^2/2} - 4x + B$
7. $y = x + A[x\sqrt{1 - x^2} + \arcsin x] + B$
8. $y = Ax + (A^2 + 1)\log|x - A| + B$; singular solution $y = C - \frac{1}{2}x^2$
9. $x = \frac{\sqrt{2}}{k} \int \frac{dy}{\sqrt{A^2 - y^4}} + B$
10. $x = \frac{\sqrt{2}}{k} \int \frac{dy}{\sqrt{y^4 + A}} + B$
12. $Ax^2 + y^2 = B$
13. $\log x = \frac{Ay}{x} + A^2 \log\left(\frac{y}{x} - A\right) + B$, and $y = Cx$
14. $y = Ax \tan(B - A \log x)$, or $y = Ax \tanh(B + A \log x)$, or $y = Cx$
15. $y = x^2(A \log x + B)^2$
16. $f(x) = \arcsin(e^x/\sqrt{2}) - \pi/4$ ($x \leq \log \sqrt{2}$)
17. $y = \frac{a^{-1/k}(a-x)^{1+1/k}}{2(1+1/k)} - \frac{a^{1/k}(a-x)^{1-1/k}}{2(1-1/k)} + \frac{ak}{k^2-1}$ if $k > 1$;
 $y = \frac{(a-x)^2}{4a} - \frac{a}{2} \log\left(1 - \frac{x}{a}\right) - \frac{a}{4}$ if $k = 1$

$$18. s(t) = \frac{1}{2}gt^2 - ct + c\left(t - \frac{w}{k}\right)\log\left(1 - \frac{kt}{w}\right)$$

$$19. s(t) = ct + c\left(\frac{w}{k} - t\right)\log\left(1 - \frac{kt}{w}\right)$$

$$20. s(t) = \frac{wv_0}{k} \log \frac{w}{w - kt}$$

7.7 Exercises (page 333)

$$1. y = e^x(A \cos \sqrt{2}x + B \sin \sqrt{2}x)$$

$$2. y = Ae^x + Be^{-3x}$$

$$3. y = e^x(A \cos x + B \sin x)$$

$$4. y = e^x(A \cos 2x + B \sin 2x)$$

$$5. y = (A + Bx)e^{-x}$$

$$6. y = (A + Bx)e^x$$

$$7. y = \frac{5}{3} - \frac{2}{3}e^{-3x/2}$$

$$8. y = -\cos(5x - 15)$$

$$9. y = \frac{a}{2}e^{b(x-1)} + \frac{b}{2}e^{a(x-1)}, \text{ where } a = 2 - \sqrt{5}, b = 2 + \sqrt{5}$$

$$10. y = 2e^{-2x}(\cos x + \sin x)$$

$$11. u(x) = 6(e^{4x} - e^{-x})/5; \quad v(x) = e^x - e^{-5x}$$

$$12. u(x) = \frac{1}{2}e^{2x-\pi} \sin 5x; \quad v(x) = \frac{5}{6}e^{-2x-\pi} \sin 3x$$

$$15. u(x) = e^{-x^2}; \quad Q(x) = 4x^2 + 2$$

$$16. k = n^2\pi^2; \quad f_k(x) = C \sin n\pi x \quad (n = 1, 2, 3, \dots)$$

$$17. (a) y'' - y = 0$$

$$(b) y'' - 4y' + 4y = 0$$

$$(c) y'' + y' + \frac{5}{4}y = 0$$

$$(d) y'' + 4y = 0$$

$$(e) y'' - y = 0$$

7.9 Exercises (page 340)

$$1. y = Ae^x + Be^{-x} - x$$

$$2. y = e^x(A \cos \sqrt{2}x + B \sin \sqrt{2}x) - \frac{8}{27} + \frac{2}{9}x + \frac{1}{3}x^2 + \frac{1}{3}x^3$$

$$3. y = Ae^x + B - 2x - x^2 - \frac{1}{3}x^3$$

$$4. y = Ae^{-x} + B + \frac{1}{3}x^3$$

$$5. y = Ae^x + Be^{4x} + \frac{9}{32} + \frac{1}{8}x + \frac{1}{4}x^2$$

$$6. y = Ae^{2x} + Be^{-3x} - \frac{7}{12} + \frac{1}{2}x - x^2 - \frac{1}{3}x^3$$

$$7. y = (A + Bx - \log x)e^{-x}$$

$$8. y = A \sin x + (B + \log|\csc x + \cot x|) \cos x - 2$$

$$9. y = Ae^x + Be^{-x} + (e^x - e^{-x})\log(1 + e^x) - xe^x - 1$$

$$10. y = (A + \frac{1}{3}x)e^x - \frac{1}{6} + \frac{1}{3}e^{-x} + (B + \frac{1}{2})e^{-2x} - \frac{1}{3}(e^x + e^{-2x})\log(1 + e^x)$$

$$11. y = \begin{cases} (A + Bx)e^{-3x} + \frac{1}{9} & \text{when } 1 < x < 2, \\ (a + bx)e^{-3x} & \text{when } x < 1 \text{ or } x > 2 \end{cases}$$

$$13. y = (A + \frac{1}{4}x)e^{2x} + Be^{-2x}$$

$$14. y = (x - \frac{4}{3})e^x + Ae^{-2x} + B$$

$$15. y = \frac{1}{4}e^{2x} + (A + \frac{1}{3}x - \frac{1}{9})e^x + Be^{-2x}$$

$$16. y = (A + Bx + \frac{1}{3}x^3)e^x + x + 2$$

$$18. y = A \cos 2x + B \sin 2x + x \cos x + \frac{2}{3} \sin x$$

$$19. y = A \sin x + (B - \frac{1}{2}x) \cos x$$

$$20. y = A + Be^{3x} - \frac{1}{3}e^{2x}(3 \sin x + \cos x)$$

21. $y = A \sin x + B \cos x + \frac{1}{40} e^{2x} (3 \sin 3x - \cos 3x)$
 22. (a) $1/(2\pi\sqrt{2})$
 (b) $R < \sqrt{2}$
 24. $y = (A + Bx^3)e^x + (x^2 - 2x + 2)e^{2x}$
 25. $y = Ae^{4x} \int e^{-4x-x^{3/3}} dx + Be^{4x}$
 26. $y = Ax^{1/2} + Bx^{-1/2}$
 27. $y = Ae^x + Bx^2e^{-x} - x$
 28. $y = A(x^2 - 2) + B/x$
 29. $y = x^{-2}[A + B(x-1)^3 + \frac{1}{5}x^3 + \frac{2}{3}x^2 - \frac{7}{6}x + \frac{1}{2} - (x-1)^3 \log|x-1|]$
 30. $a = 1, -1$; $y = [Ae^{g(x)} + Be^{-g(x)}]/x$

7.11 Exercises (page 347)

7. (b) $y = \sum_{i=0}^{n-1} k_i u_i$

7.13 Exercises (page 350)

1. $y = c_1 + c_2e^{-x} + c_3e^{3x}$
 2. $y = c_1 + c_2e^x + c_3e^{-x}$
 3. $y = c_1 + (c_2 + c_3x)e^{-2x}$
 4. $y = (c_1 + c_2x + c_3x^2)e^x$
 5. $y = (c_1 + c_2x + c_3x^2 + c_4x^3)e^{-x}$
 6. $y = c_1e^{2x} + c_2e^{-2x} + c_3\cos 2x + c_4\sin 2x$
 7. $y = e^{\sqrt{2}x}(c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x) + e^{-\sqrt{2}x}(c_3 \cos \sqrt{2}x + c_4 \sin \sqrt{2}x)$
 8. $y = c_1e^x + e^{-x/2}(c_2 \cos \frac{1}{2}\sqrt{3}x + c_3 \sin \frac{1}{2}\sqrt{3}x)$
 9. $y = e^{-x}[(c_1 + c_2x)\cos x + (c_3 + c_4x)\sin x]$
 10. $y = (c_1 + c_2x)\cos x + (c_3 + c_4x)\sin x$
 11. $y = c_1 + c_2x + (c_3 + c_4x)\cos \sqrt{2}x + (c_5 + c_6x)\sin \sqrt{2}x$
 12. $y = c_1 + c_2x + (c_3 + c_4x)\cos 2x + (c_5 + c_6x)\sin 2x$
 13. $f(x) = \frac{1}{2m^2} (e^{mx} - \cos mx - \sin mx)$
 16. (a) $y^{(4)} - 5y'' + 4y = 0$
 (b) $y''' + 6y'' + 12y' + 8y = 0$
 (c) $y^{(4)} - 2y''' + y'' = 0$
 (d) $y^{(4)} - 2y''' + y'' = 0$
 (e) $y^{(5)} - 2y^{(4)} + y''' = 0$
 (f) $y^{(4)} + 8y''' + 33y'' + 68y' + 52y = 0$
 (g) $y^{(4)} - 2y'' + y = 0$
 (h) $y^{(6)} + 4y'' = 0$

7.16 Exercises (page 358)

1. $y_1 = -2x - x^2 - \frac{1}{3}x^3$
 2. $y_1 = \frac{1}{4}xe^{2x}$
 3. $y_1 = (x - \frac{4}{3})e^x$
 4. $y_1 = \frac{1}{3}\sin x$
 5. $y_1 = \frac{1}{2}x^2e^x + e^{2x}$
 6. $y_1 = \frac{1}{2}xe^x$
 7. $y_1 = x \cosh x$

8. $y_1 = \frac{1}{24}x^4 e^{-x}$
 9. $50y_1 = (11 - 5x)e^x \sin 2x + (2 - 10x)e^x \cos 2x$
 10. $y_1 = -(\frac{5}{8}x + \frac{3}{8}x^2 + \frac{1}{12}x^3)e^{-x}$
 12. $y_1 = \frac{x^m e^{\alpha x}}{p_A^{(m)}(\alpha)}$
 15. (b) $2D$
 (c) $3D^2$
 (d) nD^{n-1}
 16. $y = Ae^x + Be^{-x} + \frac{1}{2}e^x \int \frac{e^{-x}}{x} dx - \frac{1}{2}e^{-x} \int \frac{e^x}{x} dx$
 17. $y = (A + \frac{1}{2}x) \sin 2x + (B + \frac{1}{4} \log |\cos 2x|) \cos 2x$
 18. $y = Ae^x + Be^{-x} + \frac{1}{2} \sec x$
 19. $y = (A + Bx)e^x + e^{ex} - xe^x \int e^{ex} dx + e^x \int xe^{ex} dx$
 20. $y = -\frac{1}{8} \log |x| + \frac{1}{3}e^x \int \frac{e^{-x}}{x} dx - \frac{1}{4}e^{2x} \int \frac{e^{-2x}}{x} dx$
 $+ \frac{1}{24}e^{4x} \int \frac{e^{-4x}}{x} dx + Ae^x + Be^{2x} + Ce^{4x}$

7.18 Exercises (page 366)

2. $f(x) = u_1(x)$ ($\alpha = 1$)
 3. (a) $A = (a - b)/2$, $B = (a + b)/2$
 (b) $\frac{d}{dt} \left[(t^2 - 1) \frac{dy}{dt} \right] - \alpha(\alpha + 1)y = 0$, where $\alpha = 1$ or -2 , and $x = (t + 1)/2$
 4. $u_1(x) = 1 + \sum_{m=1}^{\infty} (-1)^m 2^m \frac{\alpha(\alpha - 2) \cdots (\alpha - 2m + 2)}{(2m)!} x^{2m}$ for all x ;
 $u_2(x) = x + \sum_{m=1}^{\infty} (-1)^m 2^m \frac{(\alpha - 1)(\alpha - 3) \cdots (\alpha - 2m + 1)}{(2m + 1)!} x^{2m+1}$ for all x
 5. $u_1(x) = 1 + \sum_{m=1}^{\infty} \frac{(-1)^m}{(3m + 2)(3m - 1) \cdots 8 \cdot 5} x^{3m}$ for all x ;
 $u_2(x) = x^{-2} \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{3^n n!} x^{3n} \right)$ for all $x \neq 0$
 6. $y = x^2 \left(\frac{1}{6} + \sum_{n=1}^{\infty} \frac{(\alpha - 2)(\alpha - 3) \cdots (\alpha - n - 1)}{n!(n + 3)!} x^n \right)$ for all x
 11. (b) $f(x) = \frac{1}{5}P_0(x) + \frac{4}{7}P_2(x) + \frac{8}{35}P_4(x)$
 15. (b) $\frac{2n}{4n^2 - 1}$

7.20 Exercises (page 375)

5. $J_{-3/2}(x) = -\left(\frac{2}{\pi x}\right)^{1/2} \left(\frac{\cos x}{x} + \sin x \right)$

9. (a) $y = x^{1/2} [c_1 J_{1/3}(\frac{2}{3}x^{3/2}) + c_2 J_{-1/3}(\frac{2}{3}x^{3/2})]$
 (b) $y = x^{1/2} [c_1 J_{1/4}(\frac{1}{2}x^2) + c_2 J_{-1/4}(\frac{1}{2}x^2)]$
 (c) $y = x^{1/2} [c_1 J_\alpha(2\alpha x^{1+m/2}) + c_2 J_{-\alpha}(2\alpha x^{1+m/2})]$, where $\alpha = 1/(m+2)$, provided that $1/(m+2)$ is not an integer; otherwise replace the appropriate J by K
 (d) $y = x^{1/2} [c_1 J_\alpha(\frac{1}{2}x^2) + c_2 J_{-\alpha}(\frac{1}{2}x^2)]$, where $\alpha = \sqrt{2}/8$
 10. $y = g_\alpha$ satisfies $x^2 y'' + (1 - 2c)xy' + (a^2 b^2 x^{2b} + c^2 - \alpha^2 b^2)y = 0$
 (a) $y = x^{-5/2} [c_1 J_5(2x^{1/2}) + c_2 K_5(2x^{1/2})]$
 (b) $y = x^{-5/2} [c_1 J_{5/2}(x) + c_2 J_{-5/2}(x)]$
 (c) $y = x^{-5/2} [c_1 J_1(\frac{2}{3}x^{5/2}) + c_2 K_1(\frac{2}{3}x^{5/2})]$
 (d) $y = x [c_1 J_0(2x^{1/2}) + c_2 K_0(2x^{1/2})]$

★7.25 Exercises (page 385)

1. (a) $2i$
 (b) $-i$
 (c) $\frac{1}{2} - \frac{1}{2}i$
 (d) $18 + i$
 (e) $-\frac{1}{3} + \frac{3}{3}i$
 (f) $1 + i$
 (g) 0
 (h) $1 + i$
 2. (a) $\sqrt{2}$
 (b) 5
 (c) 1
 (d) 1
 (e) $\sqrt{2}$
 (f) $\sqrt{65}$
 4. (a) $2e^{i\pi/2}$
 (b) $3e^{-i\pi/2}$
 (c) $e^{\pi i}$
 (d) e^{0i}
 (e) $2\sqrt{3}e^{5\pi i/6}$
 (f) $e^{i\pi/4}$
 (g) $2\sqrt{2}e^{i\pi/4}$
 (h) $2\sqrt{2}e^{-i\pi/4}$
 (i) $\frac{1}{2}\sqrt{2}e^{-i\pi/4}$
 (j) $\frac{1}{2}e^{-i\pi/2}$
 8. (b) $z = 2n\pi i$ where n is any integer
 10. (c) $\frac{1}{2}\sqrt{3} + \frac{1}{2}i, -\frac{1}{2}\sqrt{3} + \frac{1}{2}i, -i$
 (d) $a + bi, -a - bi, -b + ai, b - ai$, where $a = \frac{1}{2}\sqrt{2 + \sqrt{2}}$ and $b = \frac{1}{2}\sqrt{2 - \sqrt{2}}$
 (e) $a - bi, -a + bi, b + ai, -b - ai$, where a and b are as in (d)

Chapter 8

8.4 Exercises (page 394)

2. (a) No
 (b) Yes

- (c) Yes
- (d) No
- (e) Yes
- (f) No
- (g) Yes
- 3. (a) Neither
- (b) Seminorm
- (c) Seminorm
- (d) Neither
- (e) Seminorm
- (f) Norm
- (g) Neither
- (h) Neither
- 7. (a), (b), (c)

8.7 Exercises (page 401)

1. (a) $P(x) = \frac{1}{6}(x^2 + 13x + 12)$
- (b) $P(x) = \frac{1}{2}(x^2 - 5x + 6)$
- (c) $P(x) = -\frac{1}{6}(x^3 - 6x^2 + 5x - 6)$
- (d) $P(x) = 2x^3 + x^2 - x - 2$
- (e) $P(x) = -5x^3 - x^2 + 10x - 5$
2. $P(x) = \frac{1}{640}(9x^4 - 196x^2 + 640)$
4. (a) $Q(x) = 2x^3 + 3x^2 - x - 3$
- (b) $Q(x) = 4x^3 + 7x^2 - 3x - 7$
5. (a) $P(32) = \frac{31}{21}; f(32) - P(32) = \frac{43}{42}$
- (b) $P(32) = \frac{992}{255}; f(32) - P(32) = -\frac{709}{510}$
- (c) $P(32) = \frac{43}{15}; f(32) - P(32) = -\frac{11}{30}$
- (d) $P(32) = -\frac{403}{1530}; f(32) - P(32) = \frac{2114}{765}$
7. (a) $L_0(x) = \frac{1}{72}(u-1)(u-3)(u-4)(u-6); L_1(x) = -\frac{1}{30}u(u-3)(u-4)(u-6);$
 $L_2(x) = \frac{1}{18}u(u-1)(u-4)(u-6); L_3(x) = -\frac{1}{24}u(u-1)(u-3)(u-6);$
 $L_4(x) = \frac{1}{180}u(u-1)(u-3)(u-4)$
- (b) $P(2.6) = 20$
8. (b) $x \geq 1.581$
- (c) $h \leq 0.0006$
14. (d) Let $B_0(x) = 1$ and let $B_n(x) = (x - x_0)(x - x_0 - nh)^{n-1}/n!$ for $n \geq 1$; the one and only polynomial P of degree $\leq n$ satisfying the conditions $P(x_0) = c_0, P'(x_1) = c_1, P''(x_2) = c_2, \dots, P^{(n)}(x_n) = c_n$ is given by $P(x) = c_0B_0(x) + \dots + c_nB_n(x)$

8.11 Exercises (page 411)

4. (b) $\phi_{k+1}(x) = \frac{\sqrt{(2k+1)(2k+3)}}{k+1} x\phi_k(x) - \frac{k}{k+1} \sqrt{\frac{2k+3}{2k-1}} \phi_{k-1}(x)$, where $\phi_k = P_k/||P_k||$

5. (b) $\sum_{k=0}^m P_k^2(x) = \frac{P_m}{P_{m+1}} [P_m(x)P'_{m+1}(x) - P_{m+1}(x)P'_m(x)]$
6. (a) Let $a_0 = \frac{1}{2} \int_{-1}^1 f(t) dt$, $a_1 = \frac{3}{2} \int_{-1}^1 t f(t) dt$, $a_2 = \frac{5}{2} \int_{-1}^1 \frac{3t^2 - 1}{2} f(t) dt$,
 $a_3 = \frac{7}{2} \int_{-1}^1 \frac{5t^3 - 3t}{2} f(t) dt$. Then for $k = 1$: $A_0 = a_0$, $A_1 = a_1$; for $k = 2$:
 $A_0 = a_0 - a_2/2$, $A_1 = a_1$, $A_2 = 3a_2/2$; for $k = 3$: $A_0 = a_0 - a_2/2$, $A_1 = a_1 - 3a_3/2$,
 $A_2 = 3a_2/2$, $A_3 = 5a_3/2$
- (b) $\|f - T_2\|^2 = \int_{-1}^1 f^2(t) dt - 2A_0^2 - \frac{2}{3}A_1^2 - \frac{2}{5}A_2^2 - \frac{4}{3}A_0A_2$,
 $\|f - T_3\|^2 = \int_{-1}^1 f^2(t) dt - 2A_0^2 - \frac{2}{3}A_1^2 - \frac{2}{5}A_2^2 - \frac{2}{7}A_3^2 - \frac{4}{3}A_0A_2 - \frac{4}{5}A_1A_3$
7. $\frac{1}{2} + \frac{5}{8} P_2(x) - \frac{3}{16} P_4(x) = \frac{15}{128}(1 + 14x^2 - 7x^4)$
8. (c) $S_0(x) = \frac{1}{2}$; $S_1(x) = \frac{1}{2} + \frac{1}{2}x$; $S_2(x) = -\frac{3}{4} + \frac{1}{2}x + \frac{15}{4}x^2$; $S_3(x) = -\frac{3}{4} - \frac{15}{4}x + \frac{15}{4}x^2 + \frac{35}{4}x^3$
11. (a) $\|P - f\|^2 = \frac{n-1}{n} - \frac{\log^2 n}{n-1}$
- (b) $P(x) = \left(\frac{12}{(n-1)^2} - \frac{6(n+1)}{(n-1)^3} \log n \right) x + \frac{4(n^3-1)\log n}{(n-1)^4} - \frac{6(n+1)}{(n-1)^2}$,
 $\|P - f\|^2 = 36 \log 2 - 28 \log^2 2 - \frac{23}{2} = 0.0007$ when $n = 2$
12. (a) $\|P - f\|^2 = \frac{1}{2n} [(n-2)e^{2n} + 4e^n - n - 2]$
- (b) $P(x) = (18 - 6e)x + 4e - 10$; $\|P - f\|^2 = 20e - \frac{7}{2}e^2 - \frac{57}{2} = 0.0038$

8.14 Exercises (page 420)

6. $Q(x) = -\frac{5}{2}x^4 + \frac{35}{16}x^3 - \frac{25}{32}x^2 + \frac{25}{256}x - \frac{1}{512}$
11. (b) $U_0(x) = 1$, $U_1(x) = 2x$, $U_2(x) = 4x^2 - 1$, $U_3(x) = 8x^3 - 4x$, $U_4(x) = 16x^4 - 12x^2 + 1$,
 $U_5(x) = 32x^5 - 32x^3 + 6x$

8.17 Exercises (page 427)

1. 10.067
 0.08367
 10.318 0.01733; $P(2.5) = 25.261$
 0.17033
 10.829
2. (a) 102
 -30
 12 5; $P(2) = -8$
 -5
 -3
 (b) $Q(2) = -2.23$

5. (b) $f(x_0, x_1, \dots, x_n) = \sum_{k=0}^n u(x_0, x_1, \dots, x_k) v(x_k, x_{k+1}, \dots, x_n)$

8.19 Exercises (page 432)

4. (b)	8	-5040	13068	-13132	6769	-1960	322	-28	1		
	9	40320	-109584	118124	-67284	22449	-4536	546	-36	1	
	10	-3628800	1026576	-1172700	723680	-269325	63273	-9450	870	-45	1

(c) $1 + 2x + 2x^2 - 3x^3 + x^4$

5. (c)	8	1	127	966	1701	1050	266	28	1		
	9	1	255	3025	7770	6951	2646	462	36	1	
	10	1	511	9330	34105	42525	22827	5880	750	45	1

(d) $-1 + 6x^{(1)} + 16x^{(2)} + 9x^{(3)} + x^{(4)}$

7. (a) $\frac{4}{3}n^3 + \frac{11}{2}n^2 + \frac{61}{6}n$

(b) $\frac{1}{4}n^4 + \frac{5}{6}n^3 + \frac{3}{4}n^2 + \frac{1}{6}n$

(c) $\frac{1}{4}n^4 + \frac{3}{2}n^3 + \frac{11}{4}n^2 + \frac{3}{2}n$

(d) $\frac{1}{3}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$

8.22 Exercises (page 441)

1. (a) $0.693773 - \epsilon$, where $0.000208 \leq \epsilon \leq 0.001667$. This gives the inequalities $0.6921 < \log 2 < 0.6936$

(b) $n = 578$

2. (a) $c = \sqrt{3}/3$

(b) $c_1 = \frac{a+b}{2} + \frac{b-a}{2} \frac{\sqrt{3}}{3}$, $c_2 = \frac{a+b}{2} - \frac{b-a}{2} \frac{\sqrt{3}}{3}$

3. (a) $c = \sqrt{2}/2$

(b) $c_1 = \frac{a+b}{2} + \frac{b-a}{2} \frac{\sqrt{2}}{2}$, $c_2 = \frac{a+b}{2} - \frac{b-a}{2} \frac{\sqrt{2}}{2}$

4. $a = 2 + \sqrt{2}$, $b = 2 - \sqrt{2}$

5. $c = \sqrt{\frac{3}{2}}$

7. (d) $\int_a^b f(x) dx = \frac{b-a}{18} \left[5f\left(\frac{b+a}{2} - \frac{b-a}{2} \sqrt{\frac{3}{5}}\right) + 8f\left(\frac{b+a}{2}\right) + 5f\left(\frac{b+a}{2} + \frac{b-a}{2} \sqrt{\frac{3}{5}}\right) \right]$

10. (a) $\log 2 = 0.693254 - \epsilon$, where $0.000016 \leq \epsilon \leq 0.000521$; this leads to the inequalities $0.69273 < \log 2 < 0.69324$

- (b) $\log 2 = 0.69315023 - \epsilon$, where $0.00000041 \leq \epsilon \leq 0.00001334$; this leads to the inequalities $0.693136 < \log 2 < 0.693149$

11. (d) $\log 2 = 0.693750 - \epsilon$, where $0.000115 \leq \epsilon \leq 0.003704$; this leads to the inequalities $0.69004 < \log 2 < 0.69364$

Chapter 9***9.3 Exercises (page 462)**

1. (a) $K = 2$

(b) $K = \frac{2}{3} \sqrt[3]{2}$

- (c) $K = e$
 (d) $K = 4$
 (e) $K = 3$
 (f) $K = 2 \|P\| \max\{|d|, |c|\} + \|Q\|$, where $\|P\| = \max_{a \leq x \leq b} |P(x)|$ and $\|Q\| = \max_{a \leq x \leq b} |Q(x)|$
2. (a) $y = 0$
 (b) $y = 0$
 (c) $y = 0$
 (d) $y(2 - x^2) = 2$
 (e) $y(3 - x^2) = 2$
 (f) $y(2 - x^2) = 2$
3. $y = e^{x-x^2}$ if $x \geq 0$; $y = e^{x^2-x}$ if $x \leq 0$
5. $(3x^2 - y^2)^2 = Cy$

9.5 Exercises (page 468)

1. (a) $Y(x) = e^x$
 (b) $Y_n(x) = 2e^x - \sum_{k=0}^n \frac{x^k}{k!}$ if n is odd; $Y_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$ if n is even
2. $Y_3(x) = \frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160} + \frac{x^{11}}{4400}$
3. $Y_3(x) = x + \frac{x^4}{4} + \frac{x^7}{14} + \frac{x^{10}}{160}$
4. $Y_3(x) = \frac{x^3}{3} + \frac{4x^7}{63} + \frac{8x^9}{405} + \frac{184x^{11}}{51975} + \frac{4x^{13}}{12285}$
5. (a) $Y_2(x) = 1 + x + x^2 + \frac{2x^3}{3} + \frac{x^4}{6} + \frac{2x^5}{15} + \frac{x^7}{63}$
 (b) $M = 2$; $c = \frac{1}{2}$
 (c) $Y(x) = 1 + x + x^2 + \frac{4x^3}{3} + \frac{7x^4}{6} + \frac{6x^5}{5} + \dots$
6. (a) $Y_4(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \frac{38x^9}{2835} + \frac{134x^{11}}{51975} + \frac{4x^{13}}{12285} + \frac{x^{15}}{59535}$
 (d) $Y(x) = \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \frac{62x^9}{2835} + \dots$ for $|x| < \frac{\pi}{2}$
8. $Y_3(x) = 2 + x^2 + x^3 + \frac{3x^5}{20} + \frac{x^6}{10}$; $Z_3(x) = 3x^2 + \frac{3x^4}{4} + \frac{6x^5}{5} + \frac{3x^7}{28} + \frac{3x^8}{40}$
9. $Y_3(x) = 5 + x + \frac{x^4}{12} + \frac{x^6}{6} + \frac{2x^7}{63} + \frac{x^9}{72}$;
 $Z_3(x) = 1 + \frac{x^3}{3} + x^5 + \frac{2x^6}{9} + \frac{x^8}{8} + \frac{11x^9}{324} + \frac{7x^{11}}{264}$
10. (e) $Y_n(x) = 0$; $\lim_{n \rightarrow \infty} Y_n(x) = 0$
 (f) $Y_n(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x \leq 0 \end{cases}$; $\lim_{n \rightarrow \infty} Y_n(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x \leq 0 \end{cases}$
 (g) $Y_n(x) = \frac{2x^n}{3^n}$; $\lim_{n \rightarrow \infty} Y_n(x) = 0$
 (h) $Y_n(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x \leq 0 \end{cases}$; $\lim_{n \rightarrow \infty} Y_n(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x \leq 0 \end{cases}$

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